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Engineering and Design
Risk Assessment for Flood Risk Management and Coastal Storm Risk
Management Studies

FOR THE COMMANDER:


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Purpose. This engineer manual describes and provides procedures for risk and uncertainty for U.S. Army Corps of Engineers flood risk management and coastal storm risk management studies.

Applicability. The guidance presented and procedures described in this manual apply to all Headquarters, U.S. Army Corps of Engineers elements, major subordinate commands, districts, laboratories, and separate field operating activities having Civil Works responsibilities.

Distribution statement. Approved for public release; distribution is unlimited.

Proponent and exception authority. The proponent of this manual is the Headquarters, U.S. Army Corps of Engineers, Engineering and Construction Division. The proponent has the authority to approve exceptions or waivers to this manual that are consistent with controlling law and regulations. Only the proponent of a publication or form may modify it by officially revising or rescinding it.

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Summary of Change

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Risk Assessment for Flood Risk Management and Coastal Storm Risk Management Studies

This major revision, dated 29 September 2025:

- Updates the risk-based analysis for flood damage reduction studies and adds the risk-based analysis for coastal storm risk management studies.
- Updates every chapter to reflect current standard engineering design and construction practices and incorporates information from expired or rescinded publications. This includes the following content recategorization:
 - Chapter 1: Introduction (Revised)
 - Chapter 2: Principles of Risk Analysis in Flood Risk Management Studies (New)
 - Chapter 3: Uncertainty of Discharge-Frequency Function (Revised; previously Chapter 4)
 - Chapter 4: Uncertainty in the Discharge-Stage Transformation (Revised; previously Chapter 5)
 - Chapter 5: Uncertainty in the Stage-Damage Transformation (Revised; previously Chapter 6)
 - Chapter 6: Evaluating Flood Risk with Levee or Floodwall Systems (New)
 - Chapter 7: Evaluating Flood Risk for Reservoirs or Diversions (New)
 - Chapter 8: Evaluating Flood Risk for Channel Modifications (New)
 - Chapter 9: Evaluating Flood Risk for Nonstructural Measures (New)
 - Chapter 10: Assessing System-Wide Risk (New)
 - Chapter 11: Understanding and Estimating Potential Life Loss (New)
 - Chapter 12: Computing Flood Hazard and Exposure in Coastal Systems (New)
- Revises Appendix A – References
- Adds Appendixes B through D
- Adds in the new “Associated publications” paragraph on page 1.
- Adds a Glossary of Terms

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Glossary of Terms

Chapter 1

Introduction

1–1. Purpose

This engineer manual describes and provides procedures for risk and uncertainty for U.S. Army Corps of Engineers flood risk management and coastal storm risk management studies.

1–2. Distribution statement

Approved for public release; distribution is unlimited.

1–3. References

See Appendix A.

1–4. Records management (recordkeeping) requirements

The records management requirement for all record numbers, associated forms, and reports required by this publication are addressed in the Army Records Retention Schedule. Detailed information for all related record numbers is located on the U.S. Army Corps of Engineers (USACE) Records Management Site <https://usace.dps.mil/sites/INTRA-CIOG6/SitePages/Records-Management.aspx>. If any record numbers, forms, and reports are not current, addressed, and/or published correctly, see DA Pam 25-403 for guidance.

1–5. Associated publications

This section contains no entries.

1–6. General

a. The goal of flood risk management and coastal storm risk management (CSRM) is to reduce risk to property and lives in a cost-effective manner. Flood risk management and CSRM studies identify plans and designs that achieve this goal. This manual provides guidance and describes procedures for risk assessments for these studies. The procedures apply to all levels of planning studies and to other USACE studies, including, but not limited to, studies to support levee system evaluation, review of various permit applications, and regulation of discharge by reservoirs and other water management facilities.

b. USACE policy is to explicitly consider uncertainty of inputs in risk assessments, and to weigh that uncertainty in plan selection and decision making. This manual provides guidance and describes procedures for integrating uncertainty analysis into risk assessments to each study phase. Appendix B presents a range of efforts for each element of executing a risk assessment for a flood risk management study. This manual

also defines methods of reporting risk and the associated uncertainty for USACE projects.

c. The procedures described herein apply a risk framework to the assessment of proposed flood risk management and CSRM plans. Quantitative and qualitative methods of representing the likelihood and consequences of exceedance of the capacity of selected measures are also included. The procedures are an extension and expansion of the traditional plan formulation and evaluation regulations described in other USACE guidance materials, in particular ER 1105-2-100 and ER 1105-2-101, and thus do not supersede guidance presented therein.

Chapter 2

Principles of Risk Analysis in Flood Risk Management Studies

2–1. Overview

a. As noted in ER 1105-2-101, no project or action that is proposed, evaluated, adopted, and implemented can completely mitigate or eliminate flood risks. Further, the information used to estimate flood risk, formulate and evaluate plans, and determine the results of the analyses is uncertain. All measured or estimated values in project development are to various degrees inaccurate—reflecting both inherent natural variability in flooding phenomena (for example, magnitude and location of rainfall events) and lack of knowledge in estimating various parameters (such as estimating Manning’s *n* value) relevant to project works and their performance. Pursuing the management of flood risk within a risk framework is an explicit means of better understanding both flooding and associated consequences and the uncertainty in their estimation, and thus should support development of robust strategies for managing flood risk. Examples of a coastal storm risk assessment, hydrologic engineering studies design, and managed overtopping of levee systems are included in Appendixes B, C, and D respectively.

b. Evaluating and reducing risk requires a decision framework that explicitly evaluates the level of risk if no action is taken and recognizes the monetary and nonmonetary costs and benefits of reducing risks when making decisions. This decision framework requires separating the whole of risk analysis into its component tasks by assessing the risk and related uncertainties for the purpose of successful management of the risk, facilitated by effective communication about the risks. In this way, the risk framework comprises three tasks: risk assessment, risk management, and risk communication (see the Glossary of Terms for definitions of these terms). This manual focuses on the risk assessment methodology and techniques for estimating various relevant uncertainties.

c. The procedures described in this manual are an extension and expansion of the plan formulation and evaluation procedures described in ER 1105-2-100; nothing herein is intended to supersede that guidance. In addition, the procedures align with the risk assessment requirements defined in ER 1105-2-101. The results of flood risk management plan evaluation following the guidelines in this manual include assessments of economic contribution (benefit), life safety risk and probability of infrastructure capacity exceedance, along with assessments of the confidence in those indices, considering the uncertainty in inputs to risk computations.

d. In this manual, the terms “frequency” and “probability” are used interchangeably. “Stage” and “elevation” are also used interchangeably and indicate a vertical distance from a selected vertical datum, such as the North American Vertical Datum of 1988 (NAVD88) (NOAA 2024g).

2–2. Definition of flood risk

a. The following definitions are specified in ER 1105-2-101 and reproduced here for ease of reference.

b. Risk is broadly defined as a situation or event where something of value is at stake and its gain or loss is uncertain. Risk is typically expressed as a combination of the likelihood and consequence of an event. Consequences are measured in terms of harm to people, cost, time, environmental harm, property damage, and other metrics. Choosing the appropriate risk metrics and actively using them in decision making is critical to effective risk management in support of a vibrant economy, thriving ecosystems, and sustainable communities. Flood risk explicitly considers the probability and consequences of subjecting people and property to the entire range of likely flood events, given risk management provided by any structural or nonstructural measures.

c. Flood risk can be conceptualized as a function of the hazard, performance, exposure, vulnerability, and consequences, as depicted in Figure 2–1.

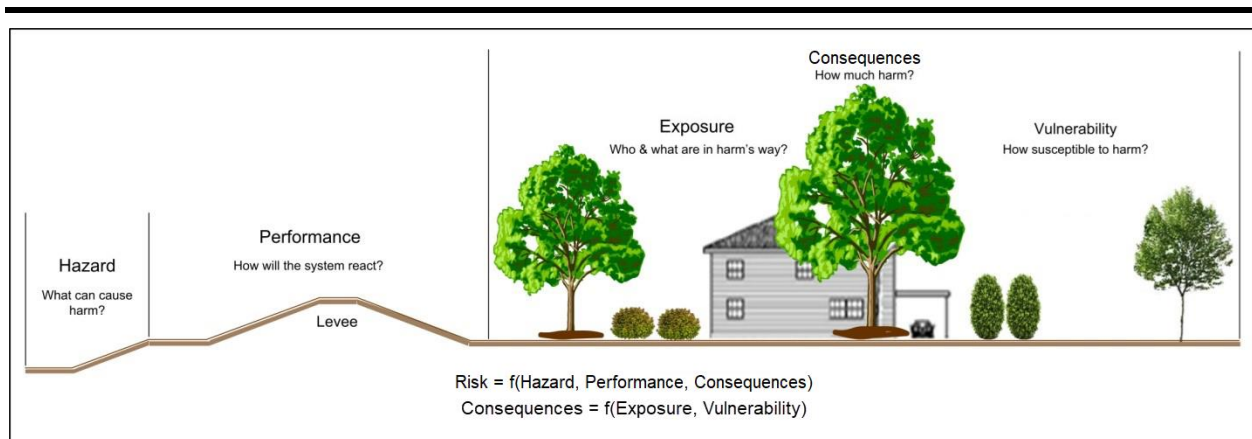


Figure 2–1. Flood risk conceptualized

(1) The “hazard” is what causes the harm: in this case, a flood. The flood hazard is described in terms of frequency, stage, velocity, extent, and depth.

(2) “Performance” is the system’s reaction to the hazard. In Figure 2–1, performance refers to the system features and the capability to contain or manage the flood hazard for the full range of possible events and for a specific event or load.

(3) “Exposure” is a factor in evaluating consequences and describes who and what may be harmed by the flood hazard. Exposure incorporates a description of where the flooding occurs at a given frequency and what exists in that area. Tools such as flood inundation maps provide information on the extent and depth of flooding; structure inventories, population data, crop data, and habitat acreage provide information on the population and property that may be affected by the flood hazard.

(4) “Consequences” are the harm that results from a single occurrence of the hazard. Consequences are measured in terms of metrics, such as economic damage, acreage of habitat lost, crop values damaged, and lives lost.

(5) “Vulnerability” is a factor in evaluating consequences and describes the susceptibility to harm of human beings, property, and the environment exposed to the hazard. Depth-damage functions, depth-mortality functions, and other similar relationships can be used to describe vulnerability.

2–3. Computation of flood risk

a. The following section describes a basic approach to computing flood risk. Flood risk can be evaluated in terms of both economic risk and life risk. Aside from the type of consequences, the approach is similar for calculation of expected annual damage (EAD) and expected annual life loss (EALL). The description here applies to either.

(1) Economic risk is the combination of harm to property, infrastructure, and other assets and economic systems (all measured in monetary terms) and the likelihood of that harm occurring. A commonly used metric of economic risk is EAD when computed on an annual basis over the period of analysis. See Chapter 5 for additional information about computing consequences in support of EAD.

(2) Life risk is the combination of likelihood and potential life loss due to flooding. Potential life loss includes consideration of those in harm’s way from flooding and how susceptible to harm they are. A commonly used metric of life risk is EALL. See Chapter 11 for additional information about computing consequences in support of EALL.

b. With a long record of historical hydrologic and hydraulic conditions, annual flood risk (as EAD) could be approximated directly by computing the consequence of interest for each year, then averaging over the period of record. However, adequate data for this type of period of record analysis is seldom available. Even if some data is available, relevant data sets often lack sufficient large, rare events to assess the consequences properly. Accordingly, flood risk is commonly computed by first analyzing available hydrologic and hydraulic data to derive a probabilistic model of the annual likelihood of high flow, stage, and subsequent flood inundation. This model is then used to derive a related probabilistic model of the consequence of interest; a model from which the mean (similarly, long-term average) annual consequence can be computed. See equation 2–1.

c. The expected annual consequence may be defined mathematically as:

$$E[X] = \overbrace{\int_{-\infty}^{\infty} x f_x(x) dx}^{(a)} = \overbrace{\int_{-\infty}^{\infty} (1 - F_x(x)) dx}^{(b)} \quad (2-1)$$

where:

X = the consequence, such as inundation damage, life loss, or emergency costs

$E[X]$ = expected value of the consequence

x = a value of the consequence

$f_X(x)$ = the probability density function (PDF) of X

$F_X(x)$ = the cumulative distribution function (CDF) of X ,

$F_X(x) = P(X \leq x)$ and $(1 - F_X(x)) = P(X > x)$

(1) Equation 2–1 integrates over the complete probability domain and thus captures the complete range of consequence in a single value, the expected annual consequence (EAD or EALL). The first integration reflected in equation 2–1(a) is similar to calculating the probability-weighted average consequence, and the second integration in equation 2–1(b) allows computation of the expected value as the area beneath the cumulative distribution (with probability on the horizontal axis). Thinking about the equation as calculating a probability-weighted average consequence sheds light on the importance of more frequent consequences compared to less frequent consequences: In their contribution to the average, a very high-frequency event with very low consequences could matter as much as, if not more than, a very low-frequency event with very high consequences.

(2) A graphical example of the application of equation 2–1 is provided in Figure 2–2(a). A common approach for carrying out integration is based on the discrete trapezoidal areas under the curve. For an interval of exceedance probability, the area under the curve is calculated as a trapezoid where the top of the trapezoid is a straight line between the two function values. This calculation is repeated on intervals of exceedance probability between 0 and 1. As shown in Figure 2–2(b), the accuracy of the EAD estimate can be improved by selecting smaller frequency intervals.

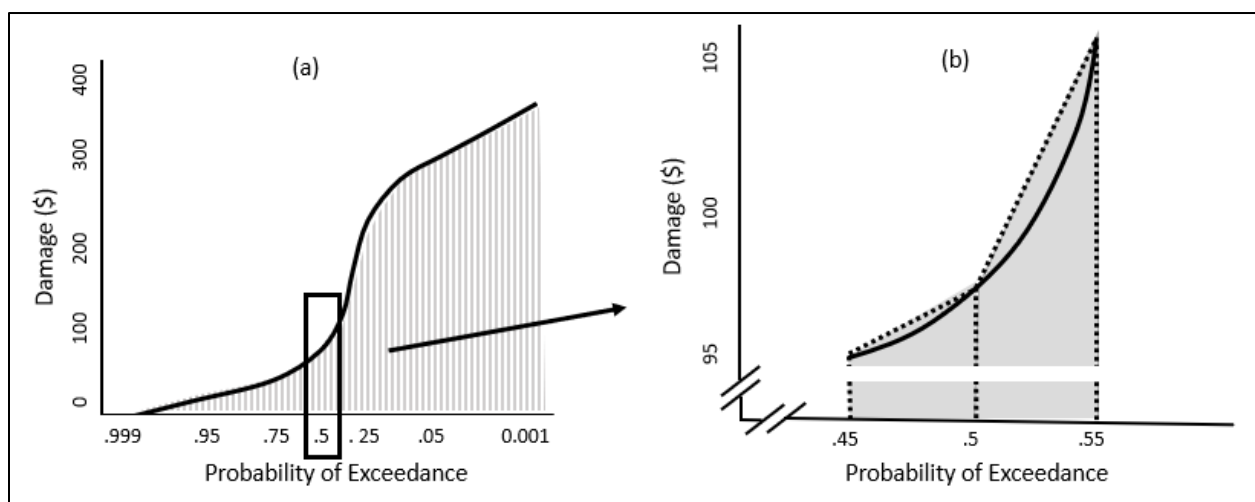


Figure 2–2. Trapezoidal rule for frequency curve integration

d. The relationship between exceedance probability and consequence used in equation 2-1 is typically developed by combining a series of interrelated functions, including a discharge-frequency function, stage-discharge function, and stage-damage function. This approach is referred to as curve combination. The EAD or $E[X=\text{damage}]$ is the mean of the resulting damage frequency curve, as described by equation 2-1.

e. An example calculation of risk using the relationship between the functions for natural riverine conditions is provided in Figure 2-3. The example focuses on economic risk, but the same process can be applied for all types of consequences. The steps for calculating economic risk from the three functions is described in more detail below.

(1) The process begins in Figure 2-3(a) with a discharge versus frequency function. The word “frequency” is used interchangeably with “probability” here. The discharge-frequency function is often obtained from a frequency analysis of the annual peak flows. See Chapter 3 to learn more about the development of a discharge-frequency function. In this example, the peak flow of the 0.05 annual exceedance probability (AEP) event is 4,000 cubic feet per second (cfs).

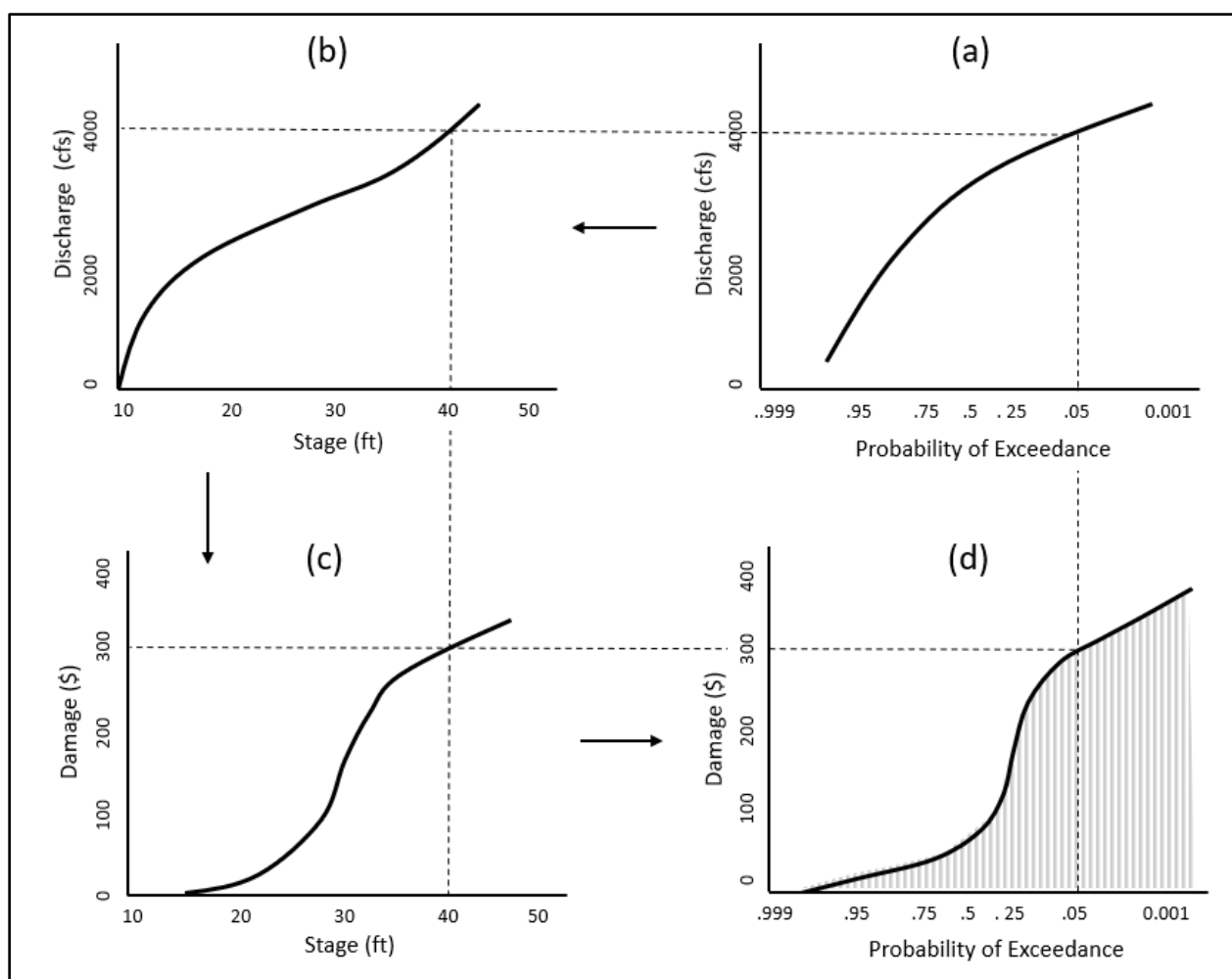


Figure 2-3. Schematic of economic risk computation (expected annual damage)

(2) Flood consequences are principally correlated with flood stage, so the next step of the process is to translate each flow in Figure 2–3(a) to a stage, using the relationship shown in Figure 2–3(b). Chapter 4 describes development of the stage-discharge function. In the example, the peak flow of the 0.05 AEP event is 4,000 cfs and has a corresponding stage of 40 feet. Thus, the 0.05 AEP event has a stage of 40 feet.

(3) The next step of the process is to estimate the consequence for a given stage using the stage versus consequence function shown in Figure 2–3(c). The stage-consequence function is typically described as either stage-damage or stage-life loss. See Chapter 5 to learn more about stage-damage functions and Chapter 11 to learn more about life loss. In the example, the peak flow of the 0.05 AEP event is 4,000 cfs, has a stage of 40 feet, and causes damage of \$300. Thus the 0.05 AEP event has a damage of \$300.

(4) The above steps are performed over the full range of frequencies to develop the damage-frequency function shown in Figure 2–3(d). The damage-frequency function is then integrated to calculate EAD or EALL using equation 2–1(b) with a graphical application of the equation, as described in paragraph 2–3.c(2). An example of the application is provided in Figure 2–4 and Table 2–1. In Figure 2–4, an example damage-frequency function is plotted with shaded trapezoids filling the space under the curve. Table 2–1 contains the corresponding EAD calculation that results from summing together the area of the trapezoids.

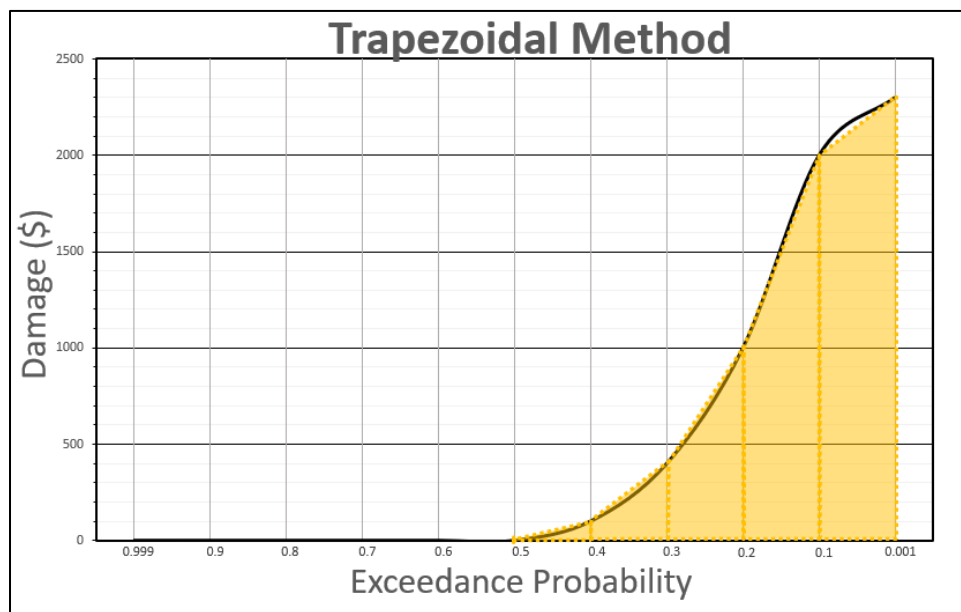


Figure 2–4. Example integration of the of a damage frequency function using the trapezoidal method

Table 2–1
Example trapezoidal method calculation

Exceedance Probability	Damage (\$)	Area Under Curve (\$)
0.999	0	0
0.9	0	0
0.8	0	0
0.7	0	0
0.6	0	0
0.5	0	0
0.4	100	5
0.3	400	25
0.2	1,000	70
0.1	2,000	150
0.001	2,300	212
–	Sum (EAD)	462

f. Expected annual damage is calculated using data that typically reflect the conditions at a specific point in time. However, risk assessments are typically required to cover a period of 50 or more years. Over such a long period of time, risk changes as the risk components change. For example, changes in rainfall intensity, changes in surface runoff, and changes in population density over time result in changes in risk. Thus, to accurately describe risk for the entire period of analysis, EAD needs to be calculated for at least two analysis years: a base year that reflects existing conditions and a future year that reflects the most likely future conditions.

g. Assessing flood risk as adjusted by flood risk management measures involves adding steps to the method described in paragraph 2–3.b to 2–3.e and Figure 2–3. In some cases, the effect of the measure involves modifying one of the three relationships in Figure 2–3(a-c). In other cases, the measure requires including an additional relationship. Several examples follow. Note that the examples do not include all possible measures, and the examples focus on the role of the measures in the assessment of risk. For more comprehensive information on the evaluation of flood risk management measures, see Chapter 6 through Chapter 9.

(1) *Channel capacity measures.* These measures typically involve increasing the depth and/or width of a channel and decreasing channel roughness so the stage in the channel is lower for a given discharge. The change in flood risk can be captured by modifying the stage-discharge function (Figure 2–3(b)).

(2) *Nonstructural measures.* These measures have less damage for a given flood stage. The change in flood risk can be captured by modifying the stage-damage function (Figure 2–3(c)), as shown in Chapter 9.

(3) *Reservoir and diversions.* These are flood risk management measures that reduce the peak stage through storage attenuation or by diverting water away from damage areas. When storage or diversion is used to alter flows, the unregulated discharge-frequency function is either replaced by a regulated discharge-frequency function or transformed into a regulated function by including a regulated-unregulated transform function between the combination of the curves in Figure 2–3(a-b). The transform function is developed using either historical regulation data or a simulation model of the system. Derivation and application of this transform function and its uncertainty are described in more detail in Chapter 7.

(4) *Regulated-unregulated transformation.* As illustrated in Figure 2–5, the transform function in Figure 2–5(c) represents the reservoir’s capability to reduce downstream flows (outflows from the reservoir) by storing high inflows that would otherwise cause flooding. The curve is approximately flat after inflows of about 2,000 cfs, representing storage of the inflows above that threshold. As peak inflows increase above approximately 3,500 cfs to 4,000 cfs, the retention capacity of the reservoir is exceeded, so for larger events, outflows match inflows. The regulated-unregulated transform function, together with the unregulated discharge-frequency function, produce a regulated discharge-frequency function where the peak outflows have been reduced for a substantial range of recurrence intervals/peak inflows, as shown in Figure 2–5(d).

(5) *Levee measure.* A levee is a lateral structure that holds water in the channel to reduce interior floodplain water levels across a range of water levels in a channel. Inclusion of a levee modifies the method for assessing risk by requiring a system response curve between Figure 2–3(c-d). A system response curve is a representation of the performance of a levee and thus the performance component of risk shown in Figure 2–1. This curve represents the probability of levee failure conditioned on the applied hazard (channel stage). For a given hazard loading, the system response curve is used to weight the damage that would occur should the levee fail by the associated conditional probability of failure. In other words, the stage-damage function is combined with the system response curve to develop a probability-weighted stage-damage function, as in Figure 2–6.

(6) *Stage-damage function.* Figure 2–6(a) shows a stage-damage function, which is the same type of function found in Figure 2–3(c). Figure 2–6(b) shows a system response curve—the relationship between the probability of levee failure and the hazard (such as stage in the channel). Damage and the probability of levee failure are matched from Figure 2–6(a, b) according to stage. Damage is multiplied by the probability of levee failure to produce a relationship between probability-weighted damage and the stage of the channel. The resulting probability-weighted stage-damage function is plotted in Figure 2–6(c), and then used in the Figure 2–3 computation.

(7) *Nonfailure damage.*

(a) Damage is sometimes expected in the floodplain despite a levee measure operating successfully, which is considered a nonfailure scenario. Damage can happen in a nonfailure scenario if the levee is outflanked, if inundation occurs from a source

separate from the leveed channel before the levee is overtopped, or if overtopping does not result in levee failure. Damage from the nonfailure scenario needs to be combined with damage from the failure scenario in developing the probability-weighted stage-damage function.

(b) The combination of damages takes place using the same approach as illustrated in Figure 2–6 with an application of total probability theorem. The failure scenario and nonfailure scenario are taken to be mutually exclusive and collectively exhaustive events, which means that the complement to the system response curve reflects the nonfailure scenario, and the system response curve can be used to calculate a probability-weighted average of failure damage and nonfailure damage.

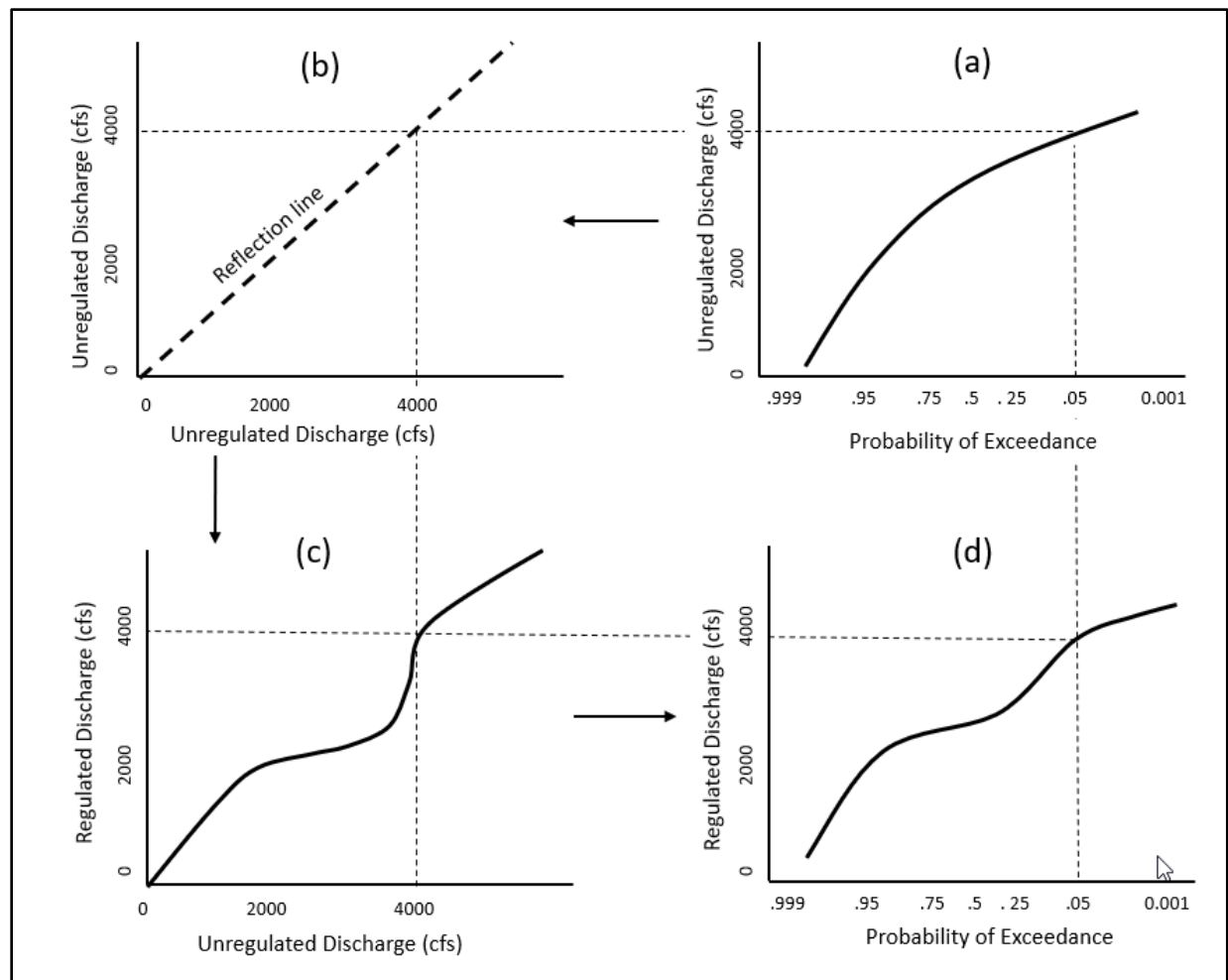


Figure 2–5. Inclusion of a regulated-unregulated transform function in the calculation of economic risk

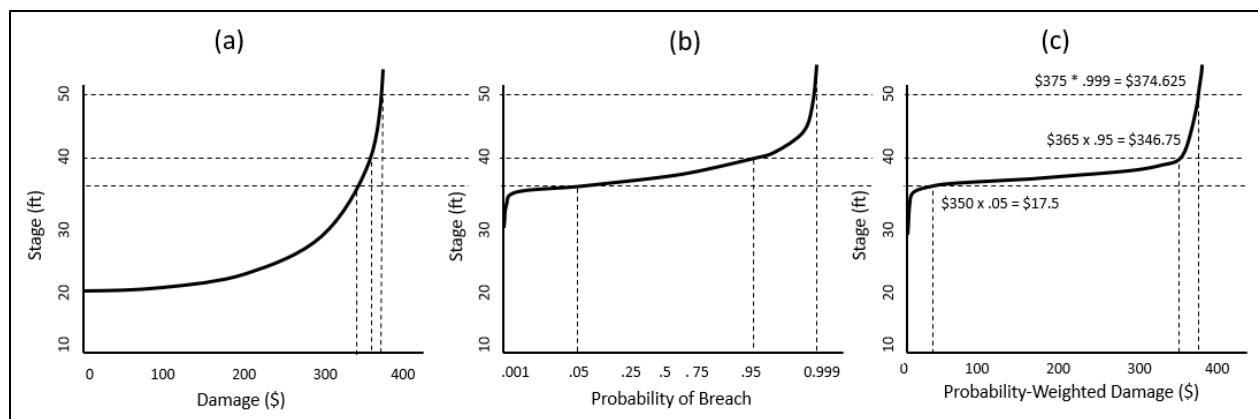


Figure 2-6. Inclusion of a system response curve in the calculation of economic risk

h. A stage-frequency function may be used to replace Figure 2-3(a-b) when there is no unique correspondence between flow and stage, such as in locations highly controlled by backwater or tidal conditions, or in the case of ice jams that affect stage and cause flooding. This procedure is demonstrated in Figure 2-7. Care should be taken in using this alternate approach because hydrologic and hydraulic uncertainties are collapsed into just hydrologic (frequency) uncertainty as defined by the period of record (discussed further in paragraph 3-5.c). Beginning the analysis with a stage-frequency function should be avoided when stage does have a direct relationship with flow. When needed, stage-frequency functions should be developed as described in paragraph 3-2.b and EM 1110-2-1415.

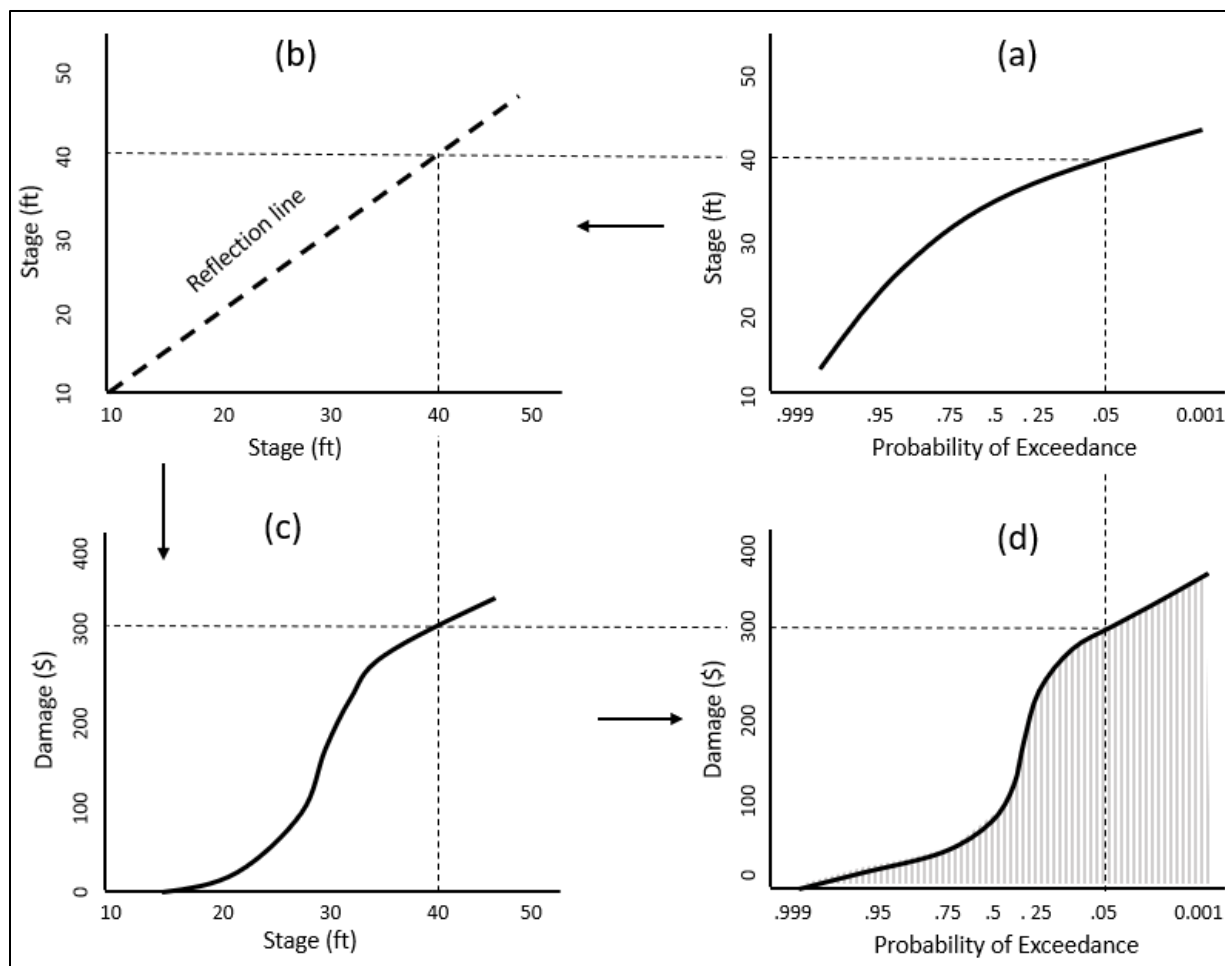


Figure 2-7. Risk assessment using stage-frequency function

2-4. Deterministic system performance metrics

a. Expected annual damage. Although EAD is often a study focus, system performance metrics that describe the probability of flooding without considering the consequence portion of the flood risk calculation are also useful. System performance metrics include AEP, long-term exceedance probability (LTEP), and assurance, a version of which is also known as conditional nonexceedance probability (CNP). AEP and LTEP can be calculated deterministically as described in the following sections, while assurance is a statistic that requires the incorporation of uncertainty into the risk assessment. Assurance is described in paragraph 2-7.e.

b. System performance.

(1) System performance is typically evaluated based on a threshold at a specific location, and should represent the weakest component of the system. Examples of locations at which to evaluate system performance include a consequence area index point, a specific grid cell, or a system response curve location. Typically, the threshold is a stage relevant to flooding. However, other examples of threshold variables include

flow, duration and floodplain, or flooding extent. The threshold variable and value are typically chosen based on local conditions.

(2) For example, when lateral structures such as levees or floodwalls are not considered, the performance of the system could be based on the stage that causes a damage value of interest. Often, analysts use the stage that causes 5% of the damage of the 0.01 AEP event. When a levee or similar structure is considered, evaluation of system performance depends on whether a system response curve is included. If a system response curve is not included in the assessment, then the top elevation of a levee can be used as a threshold. Alternatively, use of a system response curve replaces that fixed threshold by considering the joint probability of hazard loading and levee breach for the entire range of channel stages having a nonzero probability of breach.

c. *Annual exceedance probability.* In situations where structural performance (system response curve) is not part of the assessment, AEP is the probability that a specific threshold at a given location is exceeded in any given year. For example, for a threshold defined as a stage of 20 feet, AEP is the annual frequency of stage reaching at least 20 feet at that location, so the exceedance frequency of 20 feet on the stage-frequency curve. In situations where structural response is considered, AEP includes the probability of failure as described by the system response curve. AEP is the joint probability of hazard loading and structural failure.

d. *Long-term exceedance probability.* LTEP is the probability of exceedance at least once within a given interval of time. For example, the 30-year LTEP refers to the probability that a threshold is exceeded in at least one year during a 30-year interval. LTEP accounts for the repeated annual exposure to flood risk over time. LTEP is calculated directly from an AEP estimate using a simplification of the binomial theorem, as follows:

$$LTEP = 1 - (1 - AEP)^N \quad (2-2)$$

where:

N is the number of years. For example, if the AEP of a levee is 0.01, the probability of exceedance during a thirty-year period is $1 - (1 - 0.01)^{30} = 0.26$. Assuming in this example that the AEP of the levee includes the probability of levee failure, then so does the LTEP because LTEP is a direct function of AEP.

2-5. Uncertainty in risk assessments

a. Risk assessments require explicit consideration of the uncertainties associated with key hydrologic, hydraulic, and geotechnical variables and the character of floodplain occupancy. The risk assessment and study effort should concentrate on the uncertainties of those variables that have a significant impact on study conclusions and recommendations. This focus can be accomplished by determining the best estimate of each of the functions used for risk computation, and then describing the uncertainty over the range of the function.

b. Distributions of uncertainty can be developed for all functions used in a risk analysis. Methods for estimating uncertainties are presented in subsequent chapters of this manual. Figure 2–8 illustrates an example of uncertainties over the range of a stage-discharge function. The uncertainty is described as a probability density function (PDF, or bell-shaped curve) for every discharge value, positioned on the most likely value. The PDF exists on a Z-axis and so is viewed as coming out of the page. Procedures for incorporating uncertainty are described in paragraph 2–6.

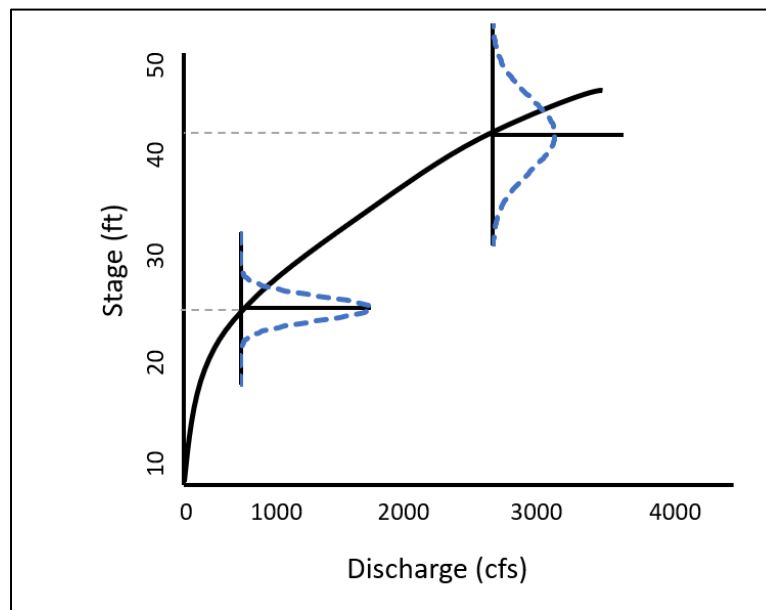


Figure 2–8. Illustration of distribution of estimates of channel stage for specified discharge

2–6. Risk and uncertainty procedures

a. With a quantitative description of uncertainty about the best estimates, the risk computation procedure illustrated in Figure 2–3 can be adapted to incorporate that uncertainty. The result is a risk and uncertainty assessment that allows calculation of metrics, such as EAD with uncertainty, AEP with uncertainty, and assurance.

b. Practically, computations for a risk and uncertainty assessment may be accomplished with Monte Carlo simulation. A Monte Carlo analysis replaces a probability distribution by a very large, random sample of values from that distribution, and evaluates each sample member to form a new sample of the analysis result. USACE typically employs either of two Monte Carlo approaches in the agency’s risk assessment software: “curve sampling” or “event sampling.” The event-sampling approach to assessing risk with uncertainty is described in paragraph 2–8, and the current section describes the curve-sampling approach to risk and uncertainty procedures and metrics.

c. The curve-sampling approach uses random sampling to perturb each summary function (or, curve) in the risk computation within its uncertainty, producing a “realization” of each function. Each entire function (discharge-frequency, discharge-stage and stage-damage) is replaced by the randomly sampled function in the risk computation shown in Figure 2–3. After sampling, the realizations of each function are combined to yield a realization of the damage-frequency function, which is integrated to compute EAD. Figure 2–9 demonstrates one realization of curve sampling, which produces one estimate of EAD.

d. With repeated random sampling (a Monte Carlo analysis), the distributions of uncertainty are reproduced to explicitly capture the uncertainty in the risk computation and risk metrics. Thousands of realizations are needed to fully reproduce the uncertainty around each function and thus around EAD and system performance metrics (for example, AEP). Figure 2–10 shows a sample of EAD estimates resulting from the Monte Carlo analysis, describing the impact of the knowledge uncertainties on EAD.

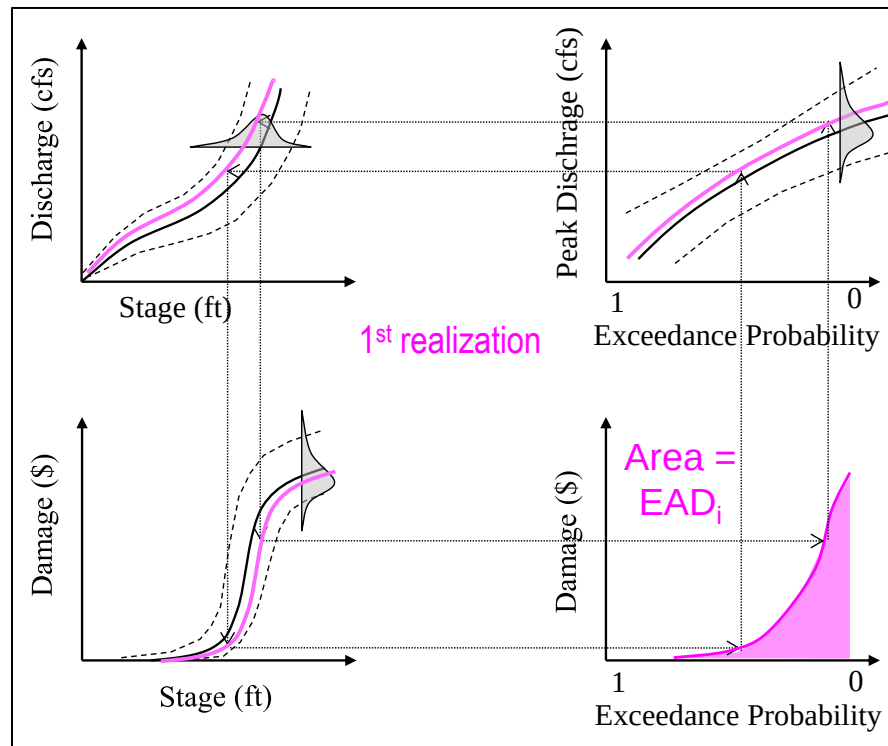


Figure 2–9. Risk computation including uncertainty with curve sampling approach

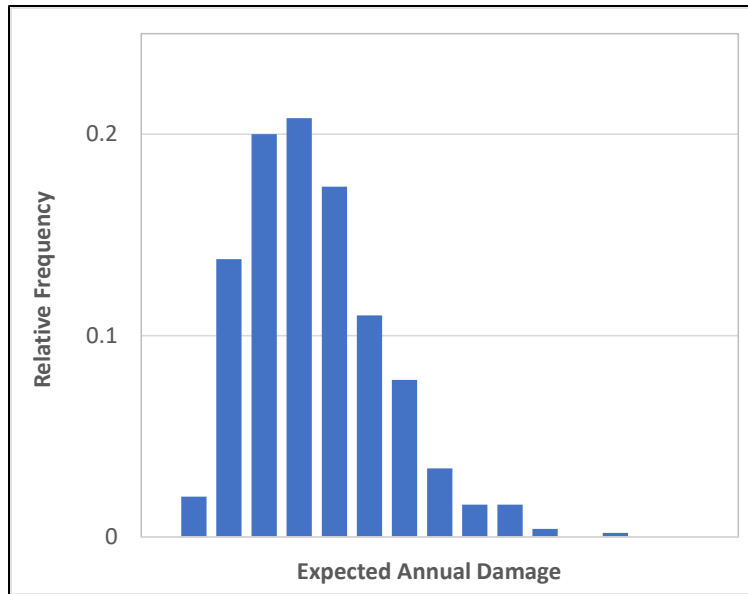


Figure 2–10. Sample of expected annual discharge from Monte Carlo incorporating knowledge uncertainty

2–7. Risk and uncertainty metrics

a. In addition to describing uncertainty about economic metrics such as EAD, explicit inclusion of uncertainty in a risk assessment also enables an analyst to describe the uncertainty around system performance metrics.

b. AEP of a given threshold can be described using statistical summaries of a random sample of AEP estimates generated from Monte Carlo simulation (one estimate from each realization). The mean, median, and assurance (percent less than some value) of AEP from that sample are often reported and used as metrics. In Figure 2–12, an initial stage-frequency function is represented by the green line, while the thinner black lines represent randomly sampled realizations of stage-frequency that result from combining sampled discharge-frequency and stage-discharge functions.

c. In the figure, an example stage of interest of 64 feet is shown. Each Monte Carlo realization has a different value of the AEP of 64 feet, shown as blue dots. The mean AEP is the average of the sample of AEP of 64 feet across all stage-frequency realizations. Figure 2–12 shows the sample of these AEP values as a histogram. The sample is compared to quantiles of interest to describe assurance of AEP of a given threshold. For example, the assurance that AEP is less than 0.01 is calculated as the fraction of realizations for which AEP is less than 0.01, as shown for 64 feet by the solid bars in the histogram in Figure 2–12, and equal to 44%.

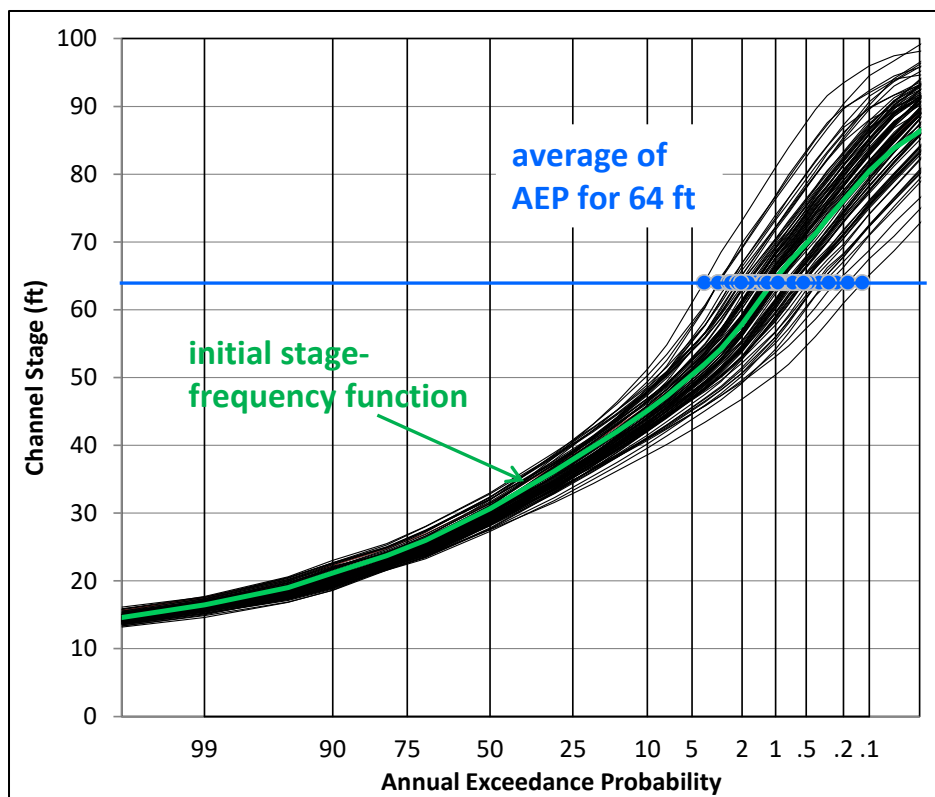


Figure 2–11. Example of mean annual exceedance probability for a stage threshold

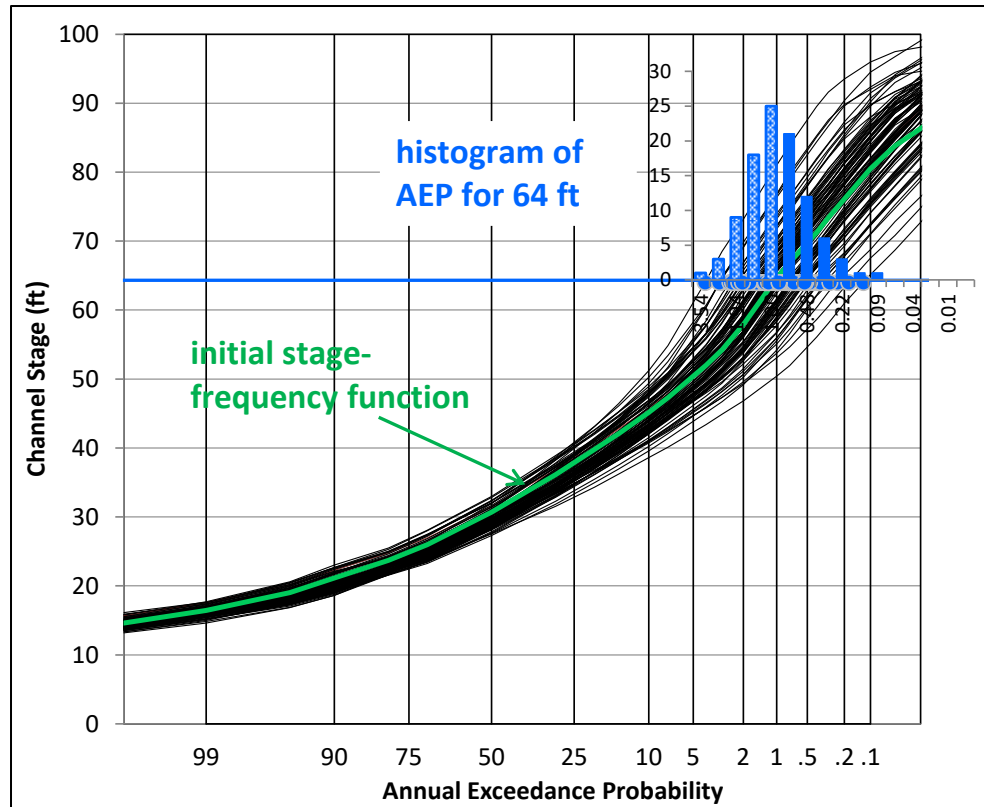


Figure 2-12. Example of distribution of annual exceedance probability for a stage threshold

d. LTEP described in paragraph 2-4.d is commonly calculated using a central measure of the AEP distribution, such as the mean or median. Using a simplification of the binomial distribution PDF to describe the likelihood of exceeding some threshold at least once in a given period of time, $LTEP = 1 - (1 - p)^N$, where p is the AEP of the threshold at the location of interest, $1 - p$ is the annual nonexceedance probability, and N is the number of years. LTEP can be viewed as the complement to the probability of no exceedance occurring over the period of N years. A common example is that the LTEP of the 0.01 event occurring at least once in 30 years is $LTEP = 1 - (1 - 0.01)^{30} = 0.26$.

e. The term “assurance” describes the uncertainty that the river-infrastructure system contains the hazard at a given likelihood or stage, either the likelihood that a quantile (such as 1% stage) is below some target stage threshold, or the likelihood that the AEP of a target stage threshold is below some probability (such as 1%). Assurance is calculable only when risk assessment is carried out with uncertainty.

(1) Assurance of a target stage reflects the chance that a stage quantile (the stage with a given exceedance probability; for example, 1% stage) is less than the target stage or top of levee (TOL). In the example of a levee, assurance is the chance that a probabilistic channel stage, such as the 0.01 AEP quantile, does not exceed (overtop) the levee, given all uncertainties. Assurance is the term selected to replace CNP. Figure 2-13 shows random realizations of the stage-frequency function that were generated by

sampling and combining discharge-frequency and stage-discharge functions, and the sample of 0.01 AEP stage estimates is shown as the vertically aligned histogram at the 0.01 exceedance probability.

(2) Assume, for example, that the target threshold is a stage of 70 feet. Then assurance of the 0.01 AEP stage for 70 feet reflects the fraction of the 0.01 AEP stages from all realizations that are less than 70 feet, represented by the solid bars of the histogram of 0.01 stages. This fraction, 86% in this example, reflects the likelihood that the 0.01 AEP stage is below the target stage of 70 feet, given the uncertainty in both discharge-frequency and discharge-stage. In other words, given the uncertainties, there is 86% chance that the 0.01 AEP stage is below 70 feet.

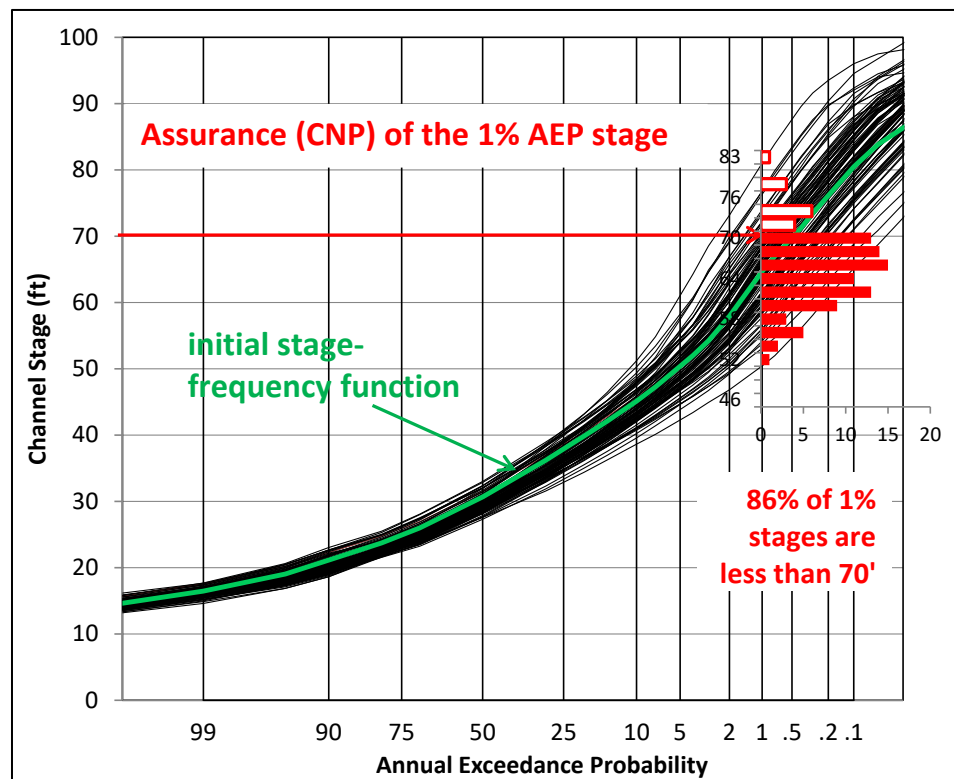


Figure 2-13. Example of assurance for a stage threshold

(3) Assurance of AEP is the likelihood that AEP falls below a given value, such as 0.01. Assurance of AEP provides the same result as assurance of a target threshold when no system response curve is included in a risk assessment. Figure 2-14 shows the many random realizations of stage-frequency curves, with a histogram depicting the values of AEP for stage = 70 feet. In the figure, assurance of 0.01 AEP for the target stage of 70 feet is 86%, because 86% of the 0.01 event stages are less than 70 feet. In other words, given the uncertainties, there is 86% chance that the AEP of 70 feet is below 0.01. Note, as shown in Figure 2-15, which includes both histograms, that this assurance that the AEP of 70 feet is less than 0.01 is the same as the assurance that the 0.01 stage is less than 70 feet. This result follows from the fact that any realization's stage-frequency curve that has a 0.01 stage less than 70 feet must also have an AEP of

70 feet that is less than 0.01. However, when a system response curve is included in the computation, assurance of AEP does not equal assurance of a target threshold.

(4) Rather than stating the likelihood that AEP is less than some specified quantile, such as 0.01 for a given stage, a similar metric might take the 90% as given, and find the AEP that has 90% assurance at a given stage. Figure 2–16 shows that the AEP that has 90% assurance for a stage of 70 feet is AEP = 0.012.

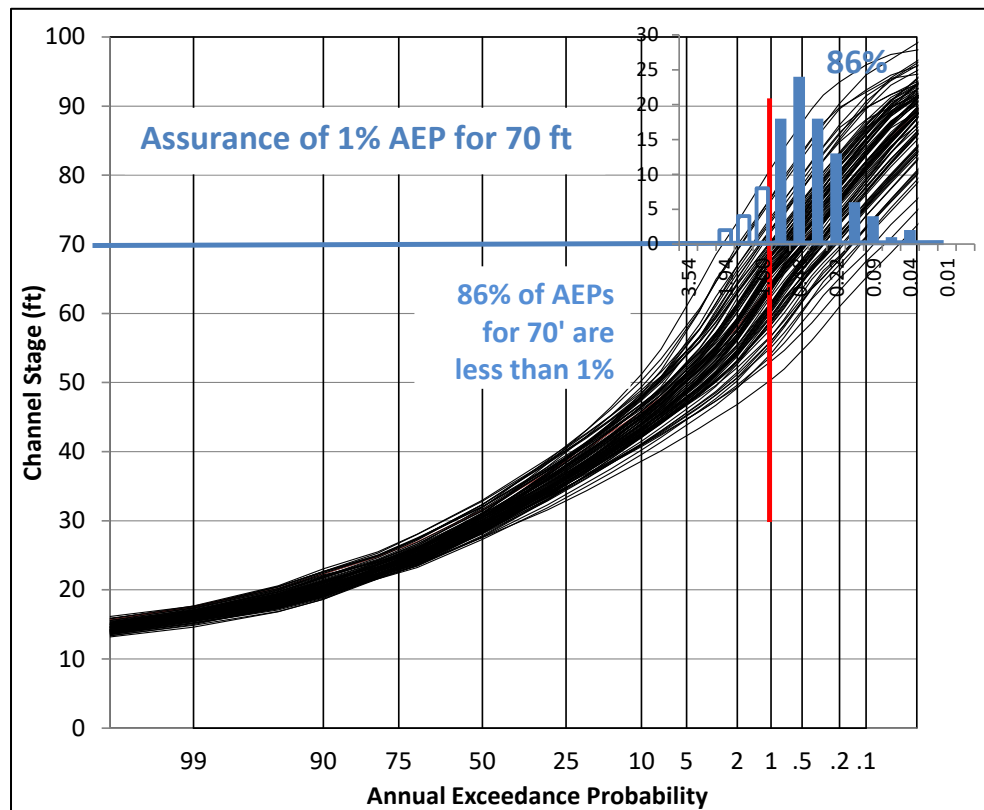


Figure 2–14. Histogram of annual exceedance probability and assurance of AEP < 0.01 for stage = 70 feet

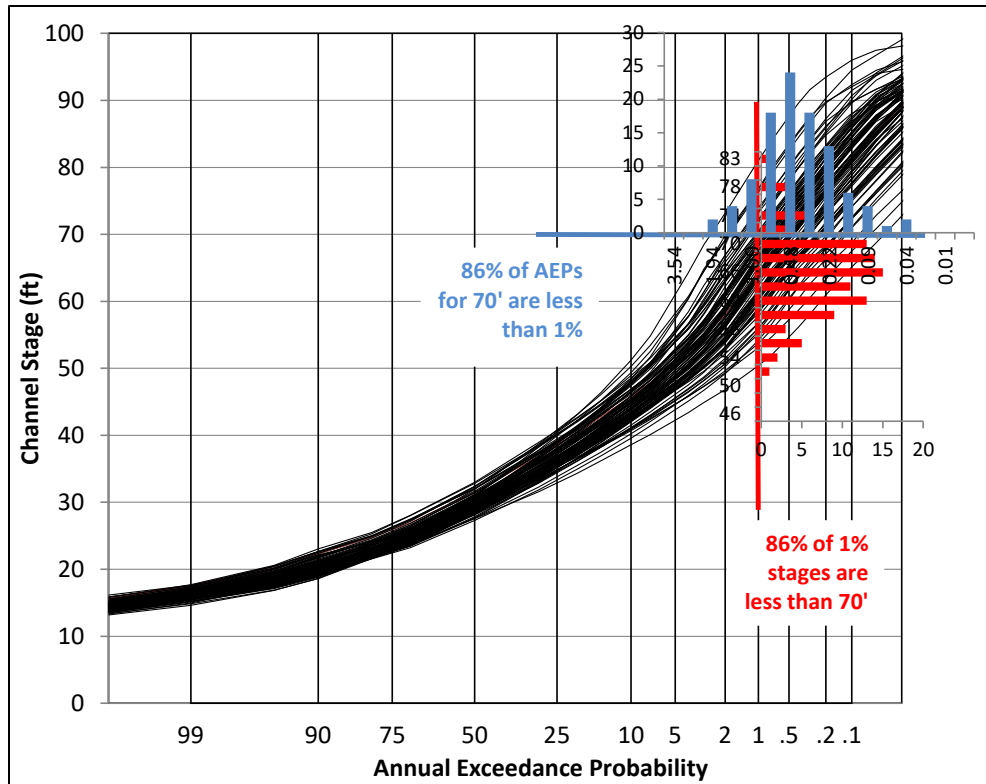


Figure 2-15. Example of relationship between assurance of annual exceedance probability and assurance of target stage

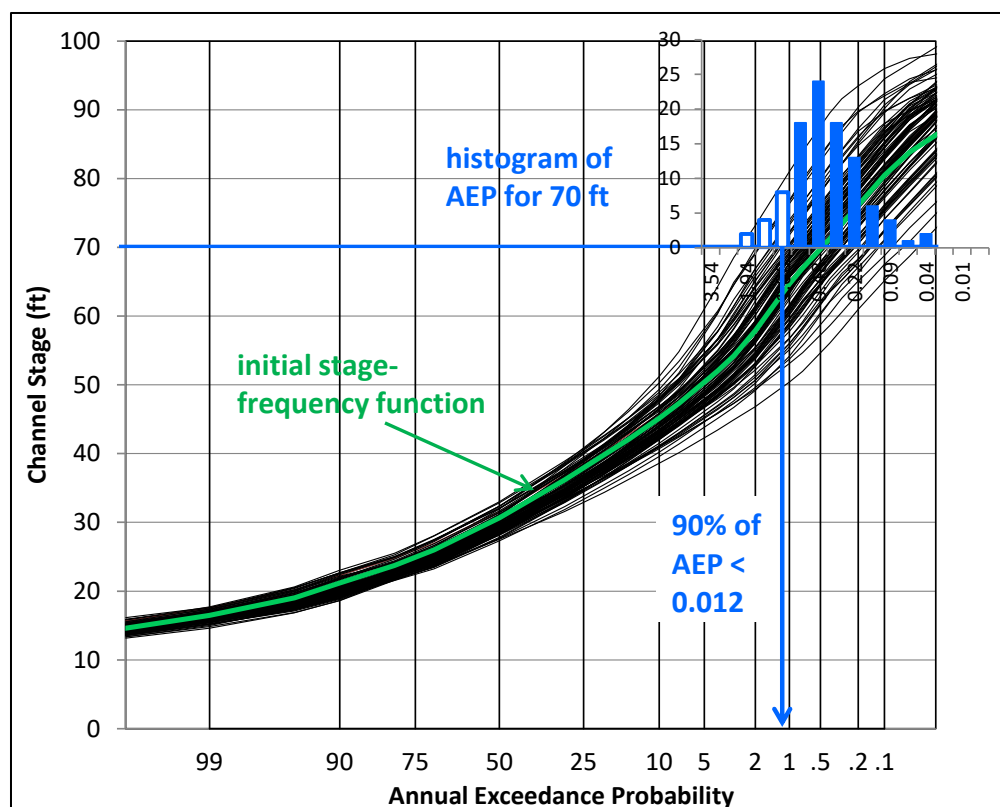


Figure 2-16. Annual exceedance probability with 90% assurance for a given stage

(5) When a system response curve is included in the assessment of a levee, then assurance includes the chance that water may move past the levee and inundate the floodplain, given the joint probability of hazard loading (discharge or stage magnitude) and structural failure (levee breach). When using the curve sampling approach to incorporating uncertainty, the two types of assurance (of a threshold or an AEP) incorporate a system response curve differently, and the results are no longer the same.

(a) The assurance of a target threshold, such as top of a levee, incorporates the system response curve after all realizations are complete. Figure 2-17 shows this computation. Realizations of a channel stage quantile (in this case, the 0.01 stage) are discretized into ranges, as shown in the histogram for the 0.01 stages (which provides the likelihood of 0.01 stage in each range). As in Figure 2-13, 86% of the 0.01 stages are below 70 feet. For each range, the likelihood of stage in that range is multiplied by the conditional likelihood that the levee does not fail in that stage range (as represented by the system response curve). The product of those probabilities is shown as the new histogram on the right.

(b) Reducing each histogram bar in this way and summing the total provides the assurance of the target stage (70 feet in this example, which is the TOL), given the likelihood of levee failure reflected in the system response curve. In this example, the assurance of the stage 70 feet being less than the 0.01 stage is reduced from 86% to 46% with the system response curve shown. The assurance with a system response

curve included must logically be less than the assurance without a system response curve, as excluding the system response curve reflects an assumption of no chance of levee failure below the top of the levee. The same example is also shown in Figure 2–19(a) and (b), with the sampled stage-frequency curves included in the image.

(c) The assurance of AEP is also affected by including a system response curve. In the Monte Carlo simulation approach to incorporating uncertainty, each AEP realization reflects the joint probability of loading and failure, so the system response curve is included within each realization, rather than after all realizations are complete. The following example shows that assurance of AEP is different from assurance of a target stage when we include a system response curve in the risk assessment. The computation of AEP given system response is depicted in Figure 2–18, where for each range of stage, the likelihood of stage in the range is multiplied by the likelihood of the levee failing in that range to calculate the joint probability of occurrence and failure in each range.

(d) In the realization shown, the AEP, also known as the annual probability of failure (APF), is then the sum of those probabilities, which increases from 1% to 3.5% with the inclusion of the system response function. The example is shown for all realizations in Figure 2–19(c) and (d), as histograms of AEPs of the stage 70 feet, with and without inclusion of the system response curve, plotted with the sampled stage-frequency curves. Solid blue bars represent the relative frequency of AEPs that are less than or equal to 0.01. Note that the distribution of AEPs in Figure 2–19(c) is the same distribution of AEPs represented by the blue horizontal histogram in Figure 2–14, and matches the stage-frequency curves plotted behind. As shown in Figure 2–19(d), the relative frequency of AEPs less than 0.01 is lower when a system response curve is included (and the histogram of AEP values no longer matches stage-frequency curves plotted behind).

(e) The lower AEPs, and so lower assurance of AEP, reflects the higher chance that a leveed area could be inundated due to the imperfection of a lateral structure (levee) captured by the system response curve. Further observe in Figure 2–19(b) and Figure 2–19(d) (with system response) that the fraction of AEPs that are less than 0.01 at the target stage of 70 feet (46%) is no longer equal to the fraction of nonfailure 1% stages (34%). Figure 2–20 provides a closer look at the histograms in Figure 2–19 (c) and (d), except with AEP on a linear axis.

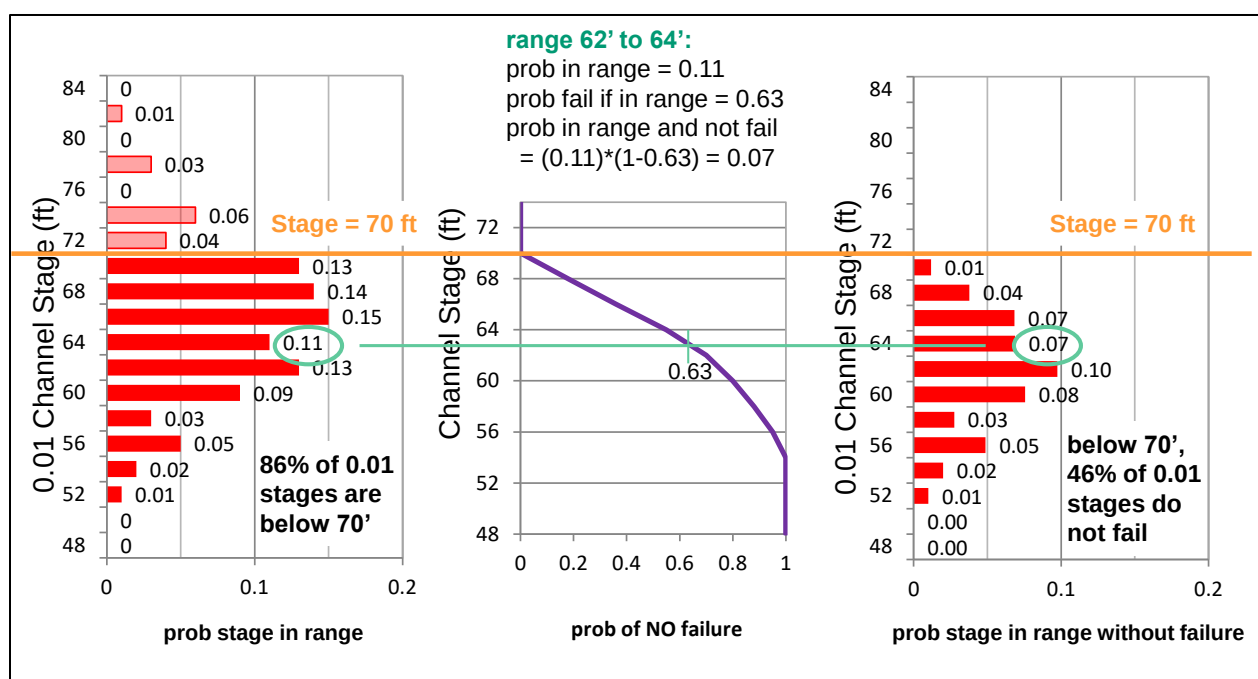


Figure 2-17. Computation of assurance of target stage with system response curve

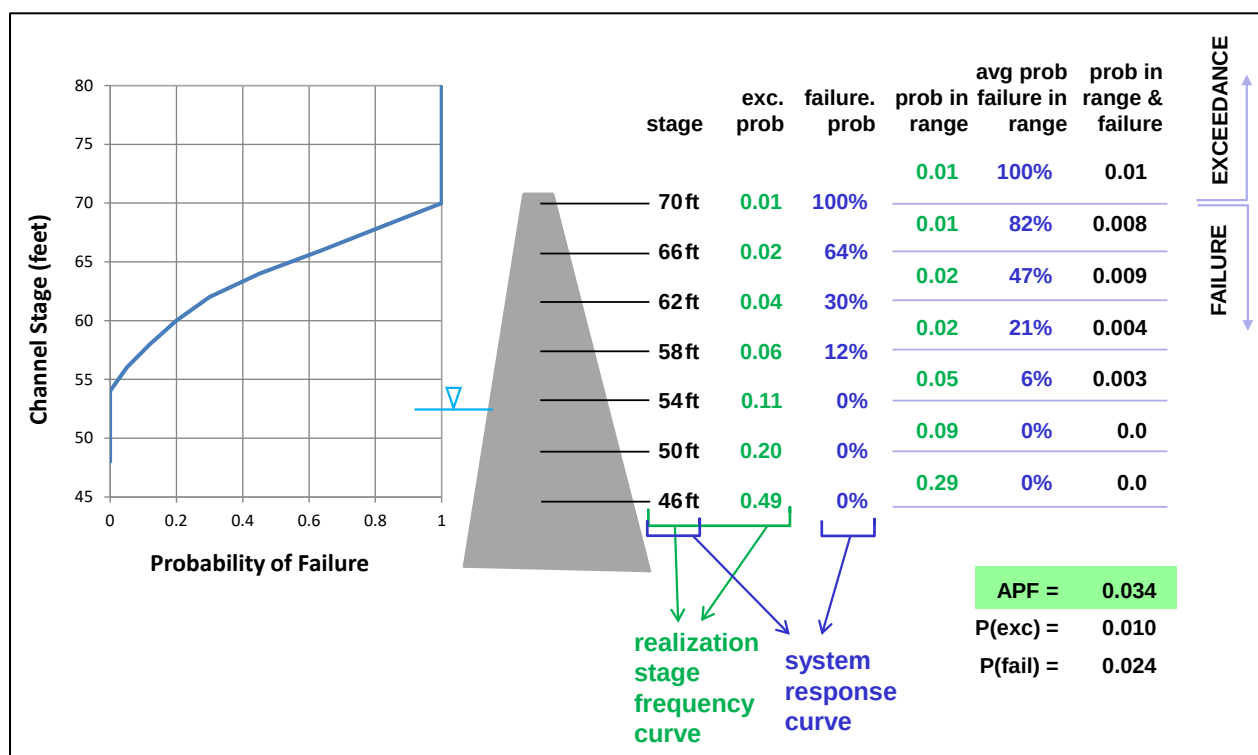


Figure 2-18. Computation of annual exceedance probability including exceedance and failure of a levee with system response curve

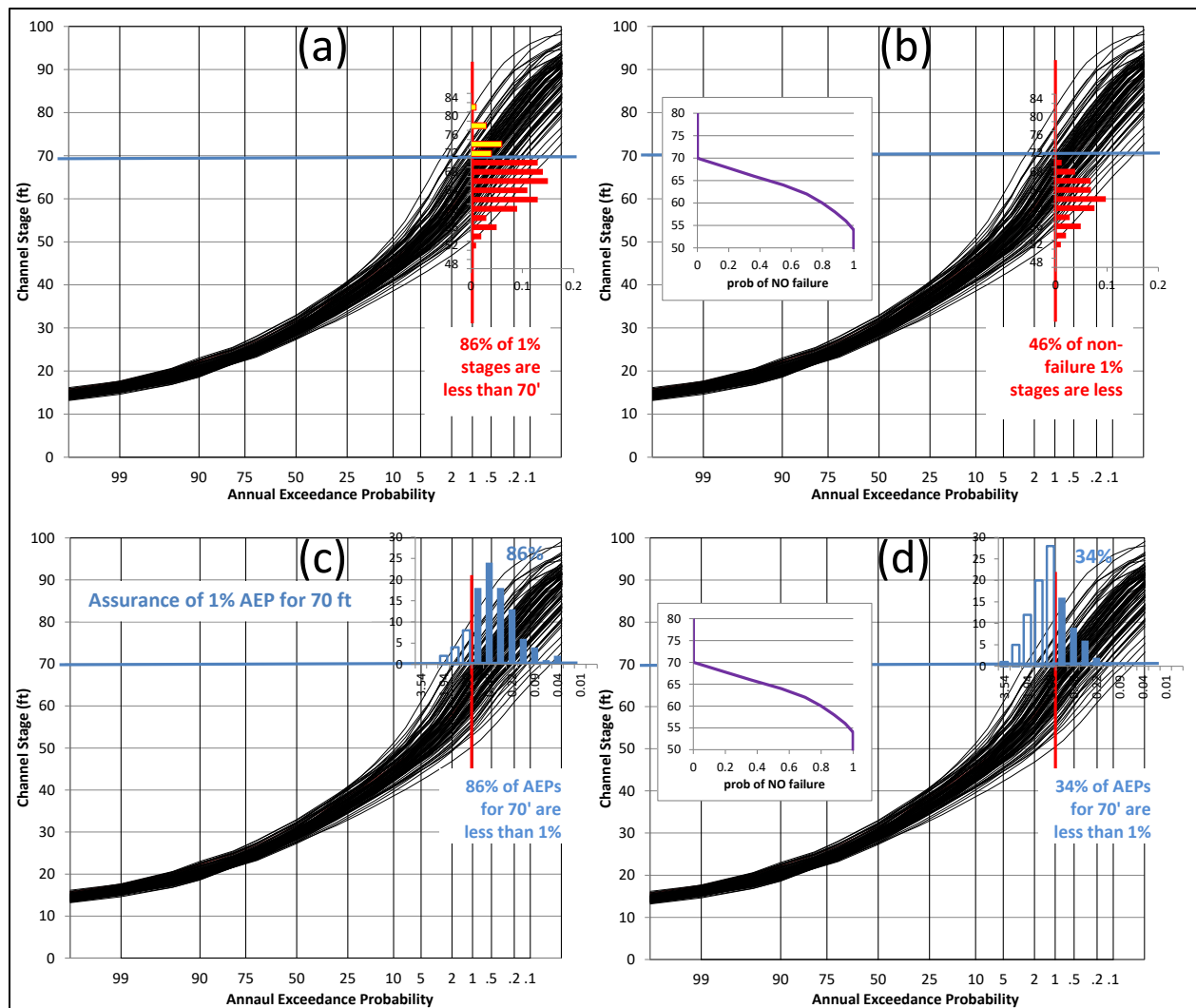


Figure 2-19. Inclusion of system response curve in assurance metrics

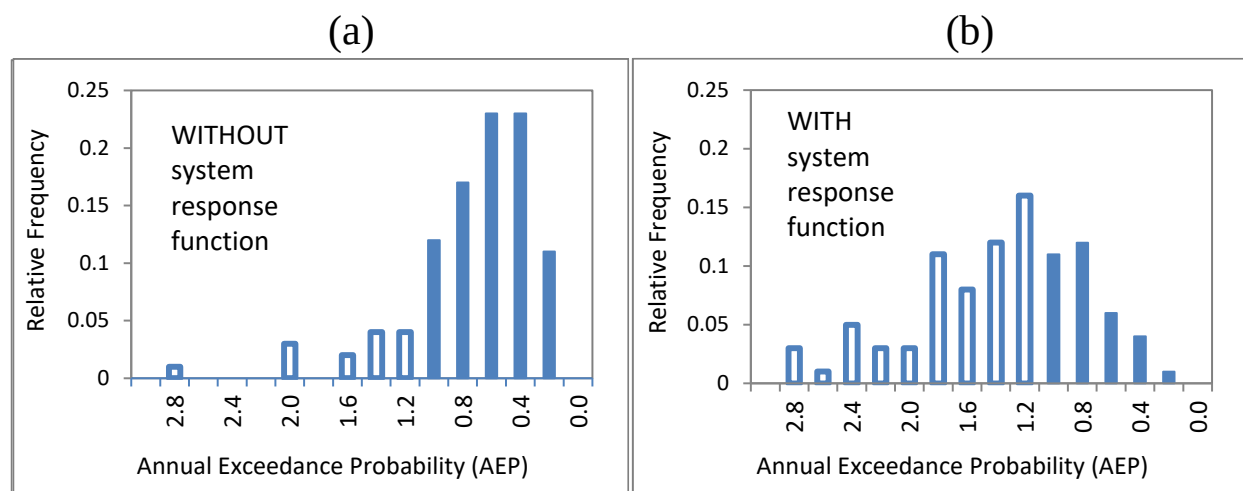


Figure 2–20. Assurance of annual exceedance probability, before and after including a system response curve

2–8. Event sampling

a. Using Monte Carlo simulation in an event-sampling approach is another common way to integrate the damage-frequency function to calculate its mean (expected) value, EAD. A Monte Carlo simulation replaces a probability distribution by a very large, random sample of values from that distribution. Using this approach for the risk computation, the discharge-frequency curve is replaced by a large, random sample of discharges, or flood events, which are each transformed to damage, resulting in a large random sample of damages. The average of the sample of damages provides the EAD, and the entire sample of damages forms the damage-frequency curve.

b. This event-sampling approach for integrating damage-frequency to compute EAD captures the natural variability in flow, as shown in Figure 2–21, using the layout of Figure 2–3. In the figure, pseudo-random $U[0,1]$ values shown in green are used as exceedance probabilities to sample discharges from the discharge-frequency function, which are then transformed to damage. A sample of 1,000 discharges is plotted with the discharge-frequency curve that the sample replaces in Figure 2–21(a), and the sample of 1,000 damages is plotted as an empirical cumulative distribution function (CDF) in Figure 2–21(d). Figure 2–22 shows the PDF form, or histograms, of the samples of discharge and damage, and the discharge-frequency curve.

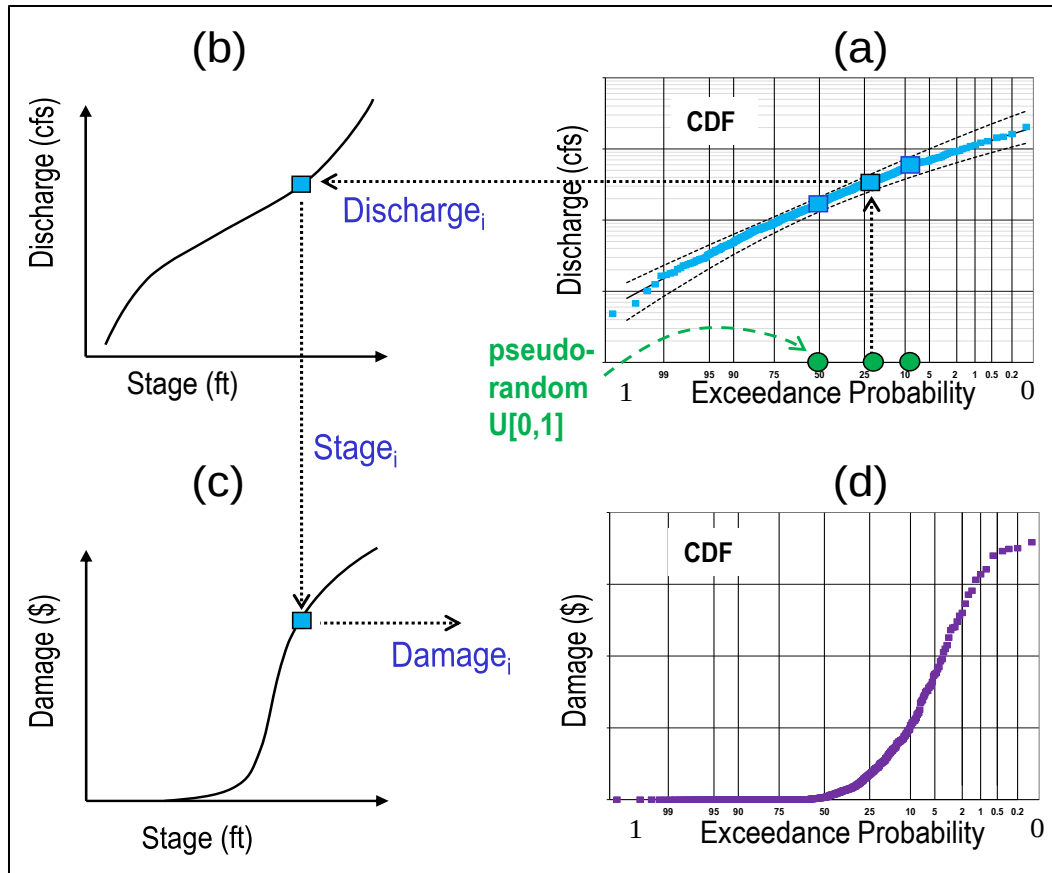


Figure 2-21. Use of Monte Carlo simulation with event sampling approach to integrate the damage frequency curve

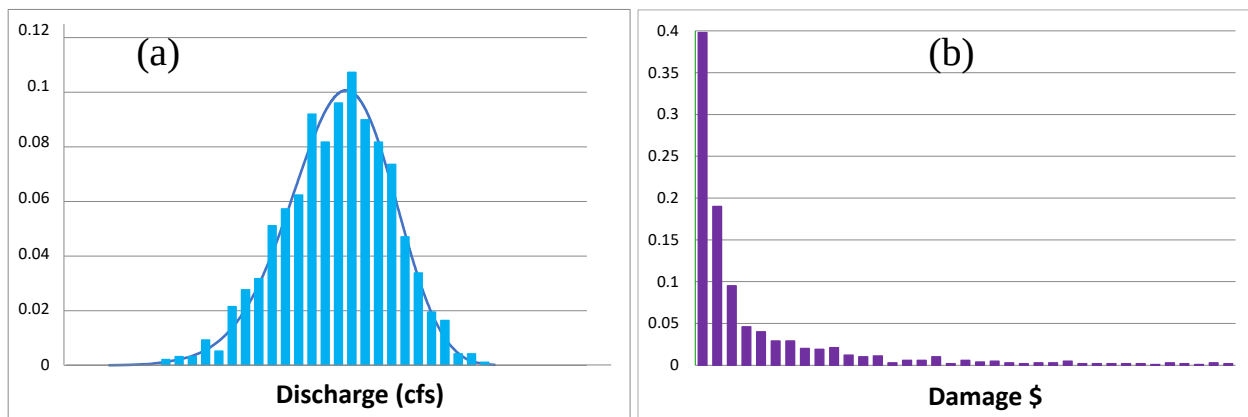


Figure 2-22. Samples from the use of Monte Carlo simulation to integrate the damage frequency curve

c. In the event-sampling approach, the transformation of discharge to damage can be performed by either using the simplified (one-to-one) relationships in Figure 2-3, as shown in Figure 2-21, or by forming a hydrograph for each sampled flood event and simulating it through a series of watershed models, as shown in Figure 2-23. The

objective of either approach to transformation is to produce a damage-frequency relationship. However, the ability to replace the simplified summary relationships with individual simulations of each flood event in detailed watershed simulation models is an important benefit of the event-sampling approach because those models can better capture the intricacies of reservoir operation, hydraulics, levee behavior, and inundation damage.

d. Further, simulating the watershed as a whole rather than evaluating separate index points allows the explicit representation of details, such as the impact of a levee breach in reducing downstream flow. Incorporation of a system response curve describing the likelihood of breach given channel stage is achieved more easily in the event-sampling approach, because each event in the analysis will either experience a failure of a given system feature or it will not. If a feature fails, the breach is captured in simulation of that event and the impact is carried downstream in the hydraulics and spreading models. Thus, the impacts are captured for other points in the watershed.

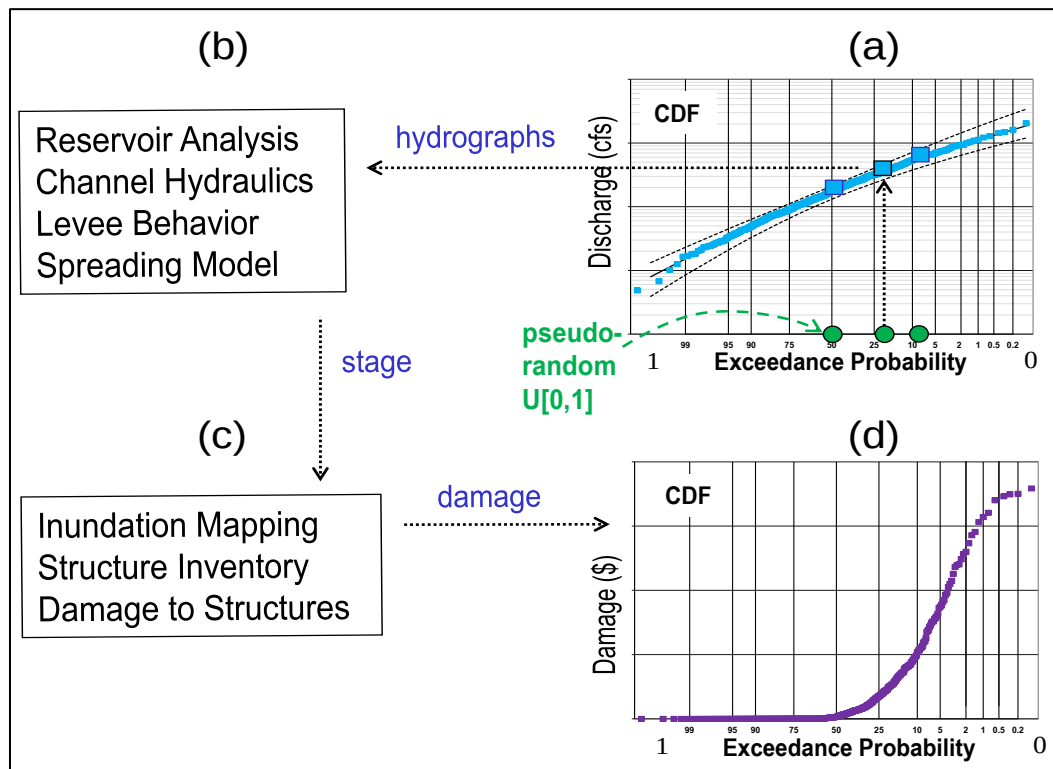


Figure 2-23. Risk computation with event sampling approach and watershed models

e. While the large, random sample of flood events that replaces the discharge-frequency function encompasses the natural variability in flood magnitude, this event sample can also incorporate other variabilities in flood events, such as the initial elevation of reservoir pools or the saturation and subsequent infiltration rate of a rainfall event in a watershed, when sampling and modeling begins with precipitation. These natural variabilities are also randomly sampled for each flood event, and the watershed models are initiated with their values. With this event-sampling process, the resulting

sample of damages incorporates the probability distributions of all natural variabilities in producing the estimate of the risk metrics that encompass natural variability (for example, EAD, AEP). A similar sample of an intermediate variable, such as channel stage at some system location, has the same property.

f. It is also necessary to incorporate the knowledge uncertainties in the risk computation. In the curve combination approach, the knowledge uncertainty in each of the summary relationships in Figure 2–3 was incorporated with curve sampling in Monte Carlo simulation, described in paragraph 2–6.c and Figure 2–9. With event sampling, when event magnitude is transformed from flow to stage or damage using watershed simulation models, knowledge uncertainties in model parameters can be captured explicitly, rather than condensed into the uncertainty around the summary relationships. However, since the event sampling approach uses Monte Carlo simulation for the task of integrating the damage-frequency function, it requires an additional level of Monte Carlo sampling to incorporate knowledge uncertainty.

g. This additional level of sampling occurs first and involves randomly sampling the knowledge uncertainties in the risk computation (the parameters of the discharge-frequency function, parameters in the watershed models, the failure stage of a levee, etc.) to initialize the watershed system models for a realization. Multiple-level Monte Carlo simulation is also called nested sampling, and in this case, the first level generates a realization capturing one instance of knowledge uncertainty, and the second level generates flood events in that realization that capture and represent natural variability.

h. Figure 2–24 shows an example of the first-level random sampling to generate a new discharge-frequency curve from its uncertainty (capturing one instance of uncertainty), producing the red frequency curve, and the second-level sampling of event magnitudes in that realization from the new red curve, producing the blue points. As the sample of events that form each realization are simulated in the watershed models to transform discharge to stage, inundation and damage, each realization thus spans (incorporates) natural variability and captures only a single instance of knowledge uncertainty in the discharge-frequency function. Figure 2–25 shows damage-frequency functions from several realizations of events, showing the result of several instances of uncertainty, with the EAD for a realization computed as the average of all event damages in that realization.

i. As with curve sampling, all realizations together capture the distributions of knowledge uncertainties, and thus the distributions of the risk metrics such as EAD and AEP, and provides values of the uncertainty-based metrics like assurance. Figure 2–26(a) shows the result of one realization as a histogram of a large sample of annual/event damages. A single realization also produces random samples of variables, such as regulated flows, peak reservoir pools, channel stages, and inundation levels at various watershed locations, from intermediate calculations, which are important for the metrics that describe system performance. Figure 2–26(b) shows a product of all realizations. With each realization producing an estimate of EAD, the result of all realizations is a large sample of EAD that demonstrates the impact of knowledge

uncertainty on the EAD metric (showing it uncertain as well). The two histograms in Figure 2–15 show how a similar sample of AEP or 0.01 stage would look with one value from each realization for a levee or an interior location.

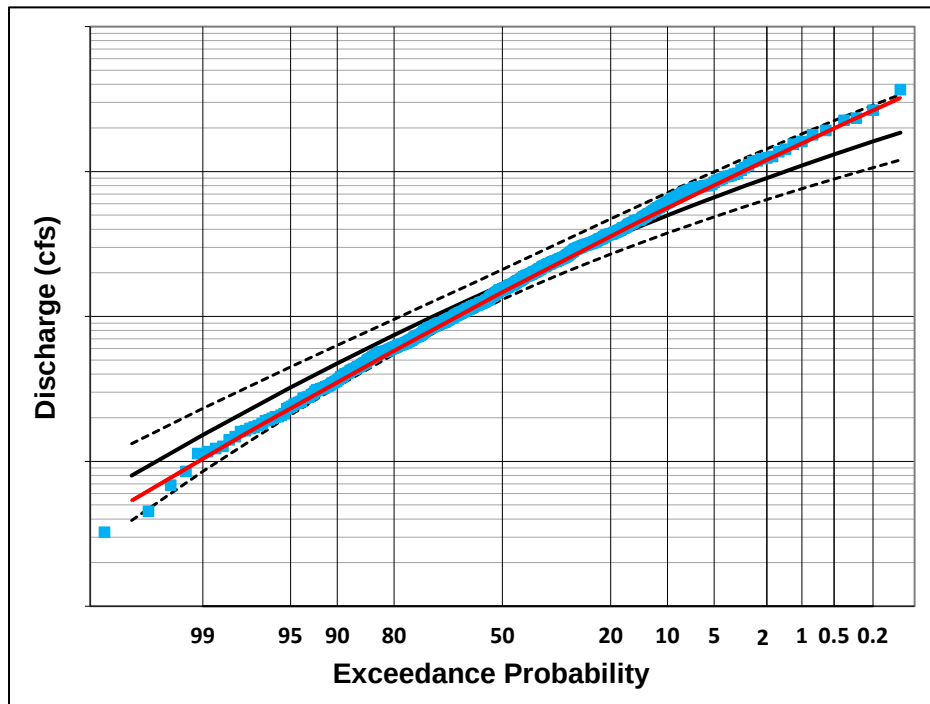


Figure 2–24. One realization of flood event discharges, incorporating one instance of knowledge uncertainty in the discharge frequency curve

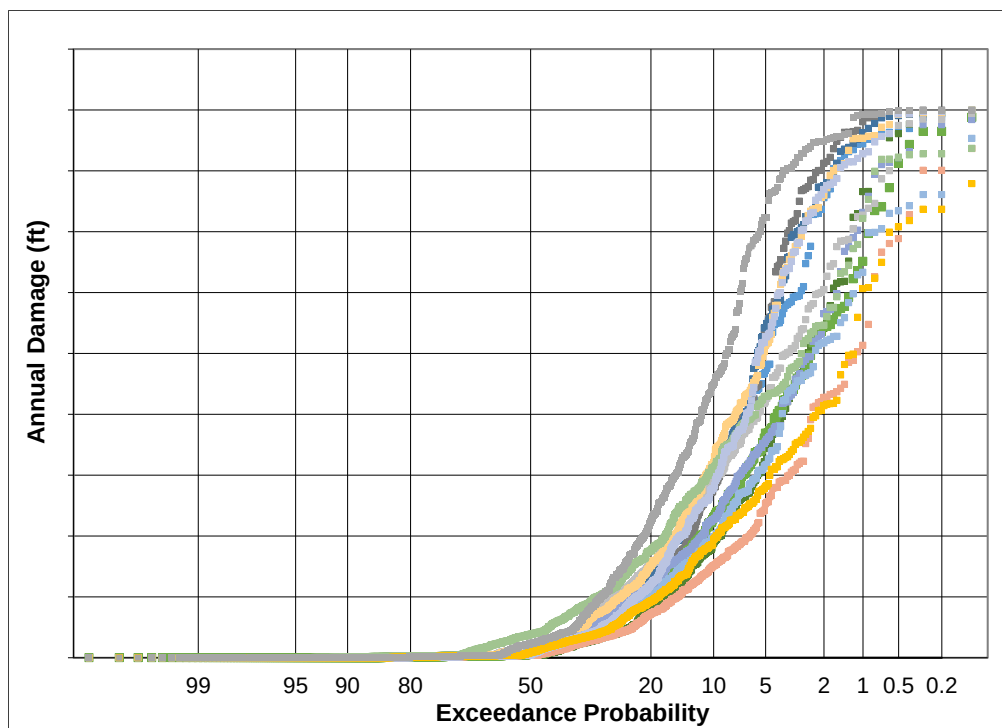


Figure 2–25. Fourteen realizations of the damage-frequency curve from two-level event sampling Monte Carlo analysis

j. The aspects of the risk computation that summarize realizations are the same with the event-sampling approach as with the curve-sampling approach. For example, metrics based on variability, such as EAD and AEP, are computed for each realization; while metrics that are based on and incorporate knowledge uncertainty, such as assurance of AEP and assurance of target stage (for example 1% CNP), are computed after all realizations. See paragraph 2–7.e for the discussion of assurance. As an example, assurance of 0.01 AEP is computed from the sample of AEP (one value of AEP from each realization) as the percentage of the sample that is less than 0.01.

k. When a system response curve is included in a risk assessment using the event-sampling approach, the chance of levee failure is realized in each event. This approach typically involves interpretation of the system response curve as uncertainty in the stage at which a levee would fail if that stage is reached. With this interpretation, a levee failure stage is randomly sampled from that uncertainty for each realization, and then, if stage in the event simulation reaches that failure stage, the simulation assumes a levee breach with all the subsequent impacts on damage area inundation and downstream flow. The outcome of levee failures as a percentage of total events that load the levee to a given stage thus equals the conditional likelihood of failure at that stage described by the response function.

l. There are trade-offs between the two Monte Carlo approaches described here. The curve-sampling approach is computationally efficient but represents a simplified depiction of risk, while the event-sampling approach allows use of detailed watershed

models in place of the simplified relationships in Figure 2–3, but suffers from a resulting increased computational burden. Often, the computational burden is justified when a higher level of fidelity is needed in modeling the watershed. For example, it may be desirable to use event sampling to capture the impact of upstream levee breaches on downstream stages. This type of hydraulic connectivity is generally not possible with the curve-sampling approach because the simplified relationships for one area in the watershed are treated as independent of the relationships for another area in the watershed.

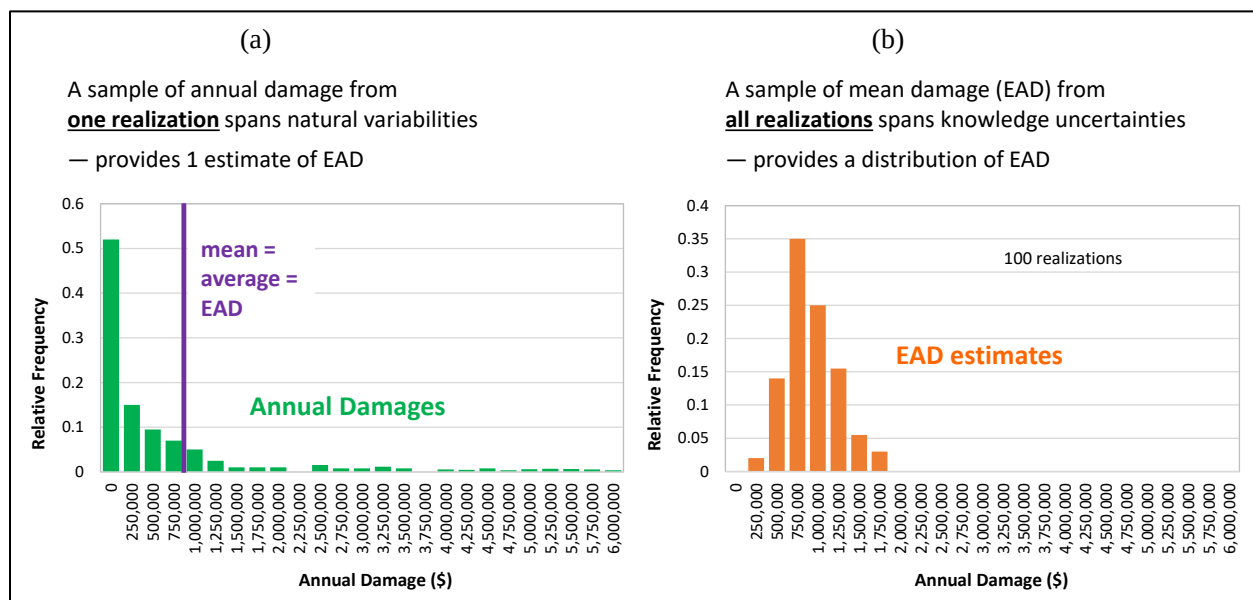


Figure 2–26. Depiction of the nested Monte Carlo simulation output, capturing natural variability as flood events in a realization, and knowledge uncertainties as expected annual damage estimates for all realizations

Chapter 3

Uncertainty of Discharge-Frequency Function

3–1. Overview

a. Chapter 2 describes how a risk assessment for USACE studies considers long-term performance of risk management systems against flood hazards, and the consequences of the floodwaters reaching people and property. As illustrated in Figure 2–3, computation of EAD includes a discharge-frequency function (a probability distribution of annual peak streamflow, also called a discharge-probability curve), a stage-discharge function, and a stage-damage function. In a more general sense, the risk computation starts with a frequency function for some random variable that describes flood magnitude, then proceeds with transformation of that variable to flood damage. The frequency function describes the relationship between annual peak discharge and exceedance probability (or precipitation or stage and exceedance probability) and represents the natural variability in annual peak discharge.

b. This chapter focuses on the discharge-frequency function, which describes the natural variability in flood magnitude. Since this function is not known with certainty, its use must also explicitly incorporate the knowledge uncertainty in estimating it. When the discharge-frequency function is developed from a precipitation-frequency function, the additional level of transformation from precipitation to discharge and that transformation's uncertainty must also be captured. Conversely, when a stage-frequency function is developed directly from gage data, less transformation is required, with less uncertainty.

c. Discharge-frequency functions may be developed as analytical or graphical probability functions (distributions). The choice of function type and method depends on the variable of interest and the type and availability of data that describes it. When observed or estimated data is available, an analytical probability distribution function may be estimated with that data using frequency analysis techniques. However, a graphical frequency function may be more appropriate in some cases, such as for a variable whose probability distribution cannot be represented with a smooth analytical function.

d. When little or when no gaged data is available, discharge-frequency function quantiles (discharge with a given exceedance probability) may be estimated individually from watershed characteristics or hydrologic modeling with frequency-based precipitation, producing individual discharge quantiles that are combined into a graphical function. However, when using this method due to lack of observed data, in some cases even such discrete discharge quantiles might nevertheless be consistent with a smooth analytical function.

(1) *Analytical.* An analytical frequency function is used when a sample of annual peak discharge values observed from a stream gage is available and this data can be reasonably fit with an analytical probability distribution. When stream gage data is available or streamflows can be estimated (such as reservoir inflows), the approach

used for USACE studies follows the recommendations in the federal Guidelines for Determining Flood Flow Frequency, Bulletin 17C (England et al. 2019) and in EM 1110-2-1415 and ER 1110-2-1450. An analytical function may also be used when the series of discharge quantiles developed from another approach is consistent with a smooth distribution. This consistency includes the credibility of extrapolation beyond the modeled values. Synthetic statistics as described in Bulletin 17B (IACWD 1982), or a least-squares fit can be used to estimate the parameters.

(2) *Graphical.* A graphical (nonanalytical) frequency function is used either when the sample of gaged or estimated streamflows is not reasonably represented by an analytical probability distribution (frequency function), or when no gage is present and the discharge quantiles are individually computed by other means, forming a resulting frequency curve that is not well-represented by an analytical probability distribution. When observed or estimated data is available, the approach to estimating a graphical frequency function from the sample of annual peak discharges uses Weibull plotting positions to estimate the exceedance probability of each ranked observed value, and drawing a function based on the plotted points. Examples of cases with data that are not well fit by analytical functions are regulated flows, mixed populations such as thunderstorm and tropical storm events, watersheds with development impacts, and stage-frequency functions.

3–2. Methods for development of frequency functions

The adopted function may be determined using one or more of the methods presented in Table 3–1 (adapted from Water Resource Council 1981).

a. At-site data for analytical function. For defining the discharge-frequency function, a frequency analysis is generally used when an annual maximum streamflow data sample is available and is consistent with a known probability distribution, such as Log-Pearson Type III (LPIII). The data sample may come from stream gage records or similar annual maximum discharges computed from reservoir mass balance or rainfall-runoff analysis. Procedures for developing discharge-frequency functions are presented in other USACE guidance, including, but not limited to, EM 1110-2-1415 and ER 1110-2-1450, and in federal guidance Bulletin 17C (England et al. 2019). Nothing in this manual is intended to replace that technical guidance; the intention here is only to summarize and provide context for using the probability functions for project risk evaluation.

b. At-site data for graphical function. When a frequency analysis approach is used for a variable that has an available sample of observed or estimated data, but the resulting relationship is not consistent with a smooth analytical probability distribution, a graphical frequency curve may be adopted. Rather than computing sample moments to estimate the parameters of a chosen probability distribution, this approach plots the ranked data against estimated exceedance probabilities, usually Weibull plotting positions, and defines the relationship with a graphical function consistent with the plotted points while respecting relevant hinge points and thresholds.

c. *Nearby data or regional parameters.* When data is available for a site near the location of interest, an analytical probability distribution for that site may be developed and transferred to provide an estimated frequency function. Results are best when the nearby site is on the same stream or in the same watershed. Further, in many areas in the United States, regional estimates of analytical probability distribution parameters are available from existing regression equations. The regional studies that produced these equations use frequency analysis results from gage sites to develop a statistical relationship (regression equations) between distribution parameters and watershed characteristics, such as drainage area, average slope, mean annual precipitation. In either case, an analytical distribution (generally LPIII) can thus be used, because distribution parameters are available.

d. *Regional quantiles.*

(1) Similar to regional parameters, regional regression equations that estimate annual maximum flow quantiles based on watershed characteristics have been developed throughout the United States (National Streamflow Statistics Application 2019). Though these regional estimation equations are based on frequency analyses that fit LPIII distributions for locations with available streamflow data, the resulting equations offer discharge quantiles (discharge with a specified exceedance probability) rather than distribution parameters.

(2) Together, these quantiles can be adopted as a graphical frequency curve. Or, if the resulting relationship is well represented by an analytical frequency function, synthetic statistics equations (IACWD 1982) or least-squares estimation can be used to estimate the analytical function's parameters (a mean, standard deviation and skew coefficient for an LPIII or other distribution). Discharge frequency quantile estimates can also be directly combined with at-site data in a frequency analysis using Bayesian fitting methods, or using methods described in Bulletin 17C (weighting of independent estimates).

e. *Watershed event modeling.* Given a calibrated and validated rainfall/runoff model that well represents the watershed of interest, discharge frequency quantiles can be estimated by rainfall events developed to each have specified exceedance probability, and assuming the resulting peak discharge has that same probability. Performing such an analysis produces discharge quantiles, and together these values form a graphical frequency function that can be adopted for the risk analysis. As with regional quantiles noted in paragraph 3–2.d, if the resulting function is smooth and consistent with an analytical distribution, synthetic statistics equations (IACWD 1982) or least-squares estimation can be used to estimate the analytical distribution's parameters (a mean, standard deviation and skew coefficient for use with an LPIII or other distribution).

f. *Stochastic event modeling.* Given a calibrated and validated rainfall/runoff model that well represents the watershed of interest, a discharge frequency function can be estimated by stochastically generating and simulating rainfall events along with other uncertain parameters in the hydrologic model (loss rates, unit hydrograph transform,

etc.) in an event-based Monte Carlo simulation. Performing such an analysis produces an empirical/graphical discharge frequency function. As with regional quantiles noted above, if the resulting function is smooth and consistent with an analytical distribution, a distribution can be fitted to the data by estimating the parameters with frequency analysis techniques.

g. Continuous watershed modeling. Given an observed record of precipitation in the watershed, and a calibrated and validated rainfall/runoff model that well represents the watershed of interest during long-term simulation, a continuous record of streamflow can be simulated and used in a discharge-frequency analysis. Simulated annual maximum streamflows are used in the same manner as an observed streamflow record, fitting an LPIII distribution using Bulletin 17C guidance as described in paragraph 3–2.a.

Table 3–1
Procedures for estimating discharge-frequency function without recorded events

Method	Summary of Procedure
Transfer	Discharge-frequency function is derived from discharge samples at nearby stream gages, or from discharge-frequency functions fit for those gages. Each discharge value, or each quantile (discharge value for specified probability), is estimated along a stream for the location of interest.
Regional estimation of individual quantiles or of function parameters	Discharge-frequency functions are derived from discharge samples at nearby gaged locations, and the function parameters or individual quantiles are related to measurable catchment, channel, or hydrometeorological characteristics via regression analysis. The resulting predictive equations are used to estimate function parameters or quantiles for the location of interest.
Hypothetical frequency events	Unique discharge hydrographs due to storms of specified probabilities and temporal and areal distributions are computed with a calibrated and validated rainfall-runoff model. Typically, the probability of discharge is assumed equal to the probability of precipitation, though this is not necessarily true if stochastic modeling techniques are appropriately employed (such as stochastic storm transposition).
Stochastic event modeling	Using event-based Monte Carlo simulation with a calibrated and validated rainfall-runoff model, probability distributions of precipitation and other factors are replaced by a large sample of storm events and simulated, producing a large sample of resulting peak flows from which to estimate an analytical or graphical frequency function.
Continuous simulation	Continuous record of discharge is computed from continuous record of precipitation with a calibrated and validated rainfall-runoff model, and annual discharge peaks are identified. The discharge-frequency function is fitted to series of annual hydrograph peaks, using statistical analysis procedures.

3–3. Sources of uncertainty in frequency functions

There are several sources for the uncertainty in discharge-frequency functions.

a. *Uncertainty of discharge-frequency function due to sampling error.* This source of error is generally greater than the others.

(1) It is assumed that in a sample of observed, annual maximum values (of discharge, stage, or precipitation) the data is random, independent, identically distributed (IID), and the data sample adequately represents the population of all possible values. However, a sample of limited size is probably not representative of the population, so parameter and quantile estimates will include sampling error. (Sampling error refers to the difference between the estimated value from a sample and the true value from the population.)

(2) For analytical frequency functions (probability distributions), the function parameters are generally estimated based on the moments of the data sample. The sample moment estimators generally used for distribution moments are consistent estimators, meaning that the estimates improve with sample size or a longer record, and the mean squared error (MSE) of these estimators thus decrease as sample size (record length) N increases. (For some distributions, including Log-Pearson Type III, the quantile variance is also asymptotically inversely proportional to the sample size.)

(3) Thus, the sampling error in the estimated discharge-probability function (including the parameters and the discharge quantiles, flow values with specified exceedance probability) can be effectively quantified and modeled by an effective record length (ERL), most simply equal to actual record length N when all data is systematic, but affected and often increased by other available information. While there are many other sources of error, the error due to limited sample size typically outweighs the others.

(4) Bulletin 17C (England et al. 2019) employs the expected moments algorithm to estimate the moments of the annual maximum discharge probability distribution, and also compute both confidence intervals (CI) and variance at each quantile to describe uncertainty in the resulting discharge-probability relationship. In addition to annual maximum values of observed streamflow, other data is often available to improve the probability distribution, such as a regional skew coefficient, historical or paleoflood estimates of large floods dating before the gage record, or evidence that large floods did not occur during a particular period.

(5) This information is included in the estimated probability distribution parameters provided by the expected moments algorithm procedure, and the improvement in the estimate is characterized in reduced CIs and quantile MSEs, and increased ERL. The ERL typically varies over the range of quantiles. An ERL based on the variance at the quantiles most important to the analysis should be used for risk analysis, with the 0.01 AEP quantile as a reasonable start.

b. Uncertainty due to unmet probability model assumptions and other adjustments.

Other uncertainties in estimating the frequency function can be harder to quantify explicitly but act to increase the parameter and quantile variances/MSEs while decreasing the ERL. The streamflow record used to estimate the discharge-frequency function must form a data sample of annual maximum discharge that is unregulated, random, and IID (from the same probability distribution, homogeneous).

(1) Annual maximum streamflows are assumed to be random and independent when instantaneous or short-duration, though longer-duration maximums can strain and sometimes violate the independence assumption. Multi-decadal trends toward wetness or dryness in the watershed may introduce some level of persistence in periods of consecutive years.

(2) If reservoirs, diversions, or other water control measures significantly alter the magnitude or timing of runoff, the data is considered regulated, and regulation is typically not consistent (and so not homogeneous or identically distributed) over time. The frequency analysis must first explicitly account for and remove the effects of the regulation in the streamflow record, adjusting discharge values to recreate the natural, unregulated discharges. To do so, the analyst must correctly identify and model the details of past regulation. These adjustments can be difficult, as operation during large floods cannot be represented with certainty.

(3) Naturally occurring phenomena may alter discharges. For example, an ice or debris dam may affect discharges similar to regulation. The outcome is a data set that is not identically distributed, so impacted values must be modified to remove those phenomena with appropriate adjustments to discharge before fitting a probabilistic/statistical model. EM 1110-2-1415, Bulletin 17B (IACWD 1982), and 17C (England et al. 2019) provide guidance regarding this type of situation. USACE has a web-based, statistical tool, the Time Series Toolbox, that should be applied to evaluate hydrologic and meteorologic datasets for evidence of nonstationarity. Any data adjustment for this purpose is uncertain in its accuracy.

(4) Watershed and meteorological conditions must be relatively constant, yielding a stationary record whose distribution does not change with time (identically distributed, homogenous data). When land use, hydrologic or meteorological properties change significantly over time in the watershed upstream of the location of interest, the data cannot be considered homogeneous (the identically distributed assumption is not true.) If the observed nonstationarity can be attributed to some cause, adjustments can be made to each value in the data set to derive a consistent, homogeneous (identically distributed) data set for frequency analysis, representing all data as either past, present, or future conditions. Any data adjustment for this purpose is uncertain in its accuracy, adding to the overall uncertainty in the discharge-frequency function.

(5) Annual maximum discharges used to estimate a discharge-frequency function should be from a common phenomenon to form a homogeneous dataset. For example, the sample of streamflows considered should not include both snowmelt runoff and thunderstorm runoff. In statistical analysis terminology, this combination is referred to as

a mixed population, and should not be treated as a sample from a single population for frequency analysis.

(6) Instead, to define a discharge-frequency function properly, the events should be identified and segregated by type and analyzed separately. The resulting probability functions can then be combined using an OR condition, following procedures described in EM 1110-2-1415 and standard textbooks on probability and statistics. In practice, if several types of common storms populate the record, yet form a smooth, consistent probability distribution, they are often reasonably and successfully analyzed together.

c. Uncertainty of discharge-frequency function due to model choice.

(1) Bulletin 17C specifies the LPIII distribution as the model for annual peak discharge, and standard practice has followed this recommendation. This probability distribution was selected as a compromise that can account for varied hydrologic conditions across the U.S., specifically when estimating the 100-year (AEP 0.01) discharge quantile, but it may not achieve the best fit for all cases. There are other probability distributions to model annual maximum discharges that could provide a better fit at a given location, and producing different quantile estimates than LPIII. Both the federal guidelines (England et al. 2019) and USACE policy permit use of different distributions and different methods if the selection is supported with adequate reasoning on a case-by-case basis.

(2) When estimating the LPIII distribution from annual maximum streamflow data, Bulletin 17C guidance estimates parameters using statistical moments of a sample (mean, standard deviation, and coefficient of skewness). The assumption of this method of moments parameter estimation procedure is that the sample moments are good estimates of the moments of the population of all possible annual maximum discharge values. However, this assumption does not always produce the best estimate.

d. Uncertainty due to transformation. This uncertainty may be from precipitation to discharge, or discharge to stage. A discharge-frequency function that was estimated from frequency-based precipitation events and rainfall/runoff modeling, or a stage-frequency function that was estimated with hydraulic modeling of discharge in a channel, contains modeling uncertainty from the rainfall/runoff and/or hydraulic models. This uncertainty must be included in the risk analysis, along with the uncertainty in the initial frequency function, whether it is precipitation or discharge frequency. The uncertainty in the various transforms can be explicitly modeled and quantified with stochastic methods or estimated using sensitivity analysis. Additional uncertainty is then captured with a decrease in the ERL.

3–4. Visual display of uncertainty in frequency functions

a. There are various ways to display the uncertainty in a discharge-frequency function (probability distribution of annual maximum discharge), with two ways being the most common. The probabilistic description of the discharge-frequency uncertainty can be displayed with a PDF at every quantile, visualized on a z-axis, or in summary with a

CI such as the 90% interval as shown in Figure 3–1. A 90% CI around a graphical frequency curve is shown in Figure 3–2.

b. Simply put, a CI is a range defined to have some probability of spanning the population parameter or quantile of interest. For an entire discharge-frequency curve, these CIs are curves that connect discharge intervals computed at each quantile (for each exceedance probability). For an analytical frequency analysis of at-site gage data using Bulletin 17C, these intervals are a product of the expected moments algorithm. Although such a plot is not required for the computations proposed herein, which instead characterize the uncertainty by quantile variance or ERL as shown in paragraph 3–5, it does illustrate the uncertainty in estimates of quantiles.

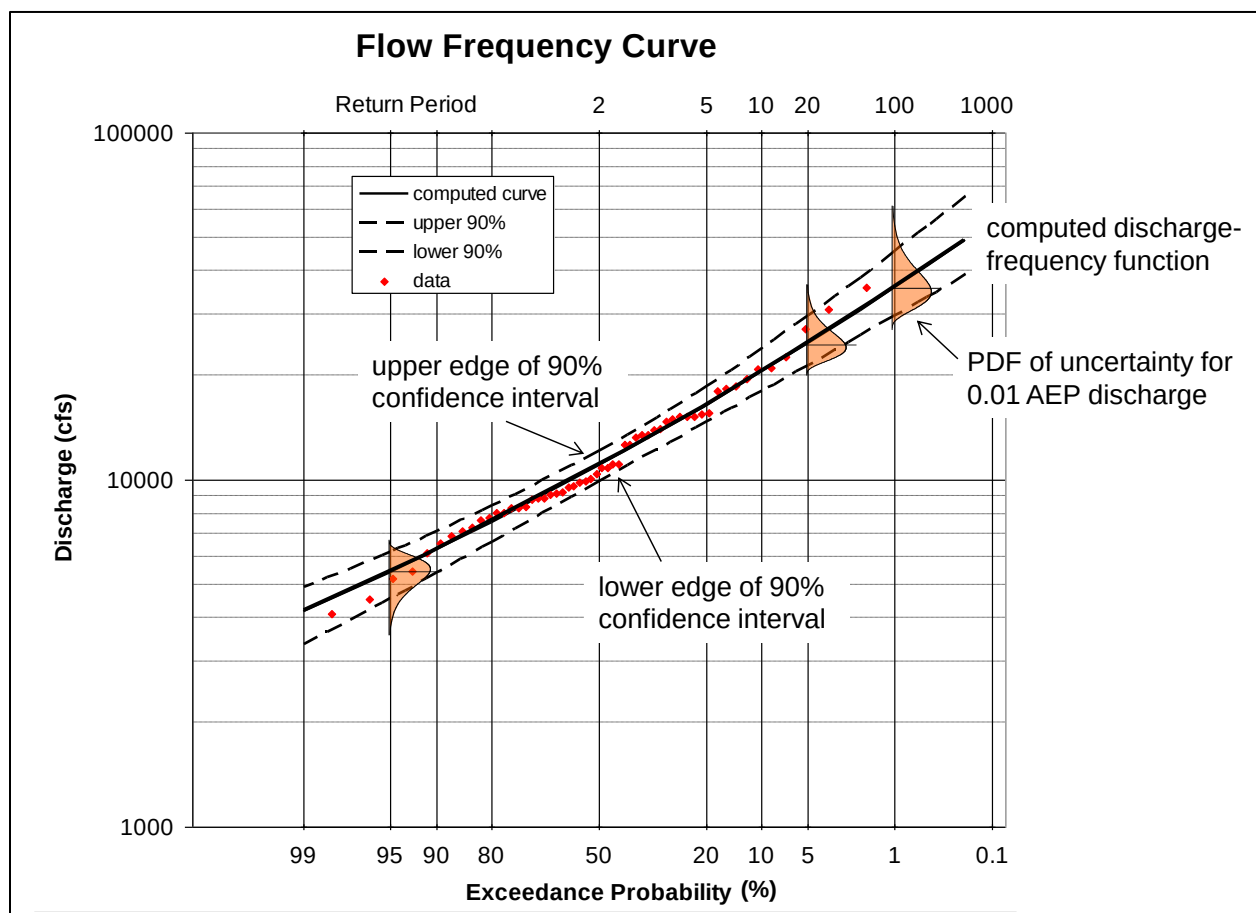


Figure 3–1. Uncertainty in analytical discharge-frequency function as probability density functions and confidence intervals

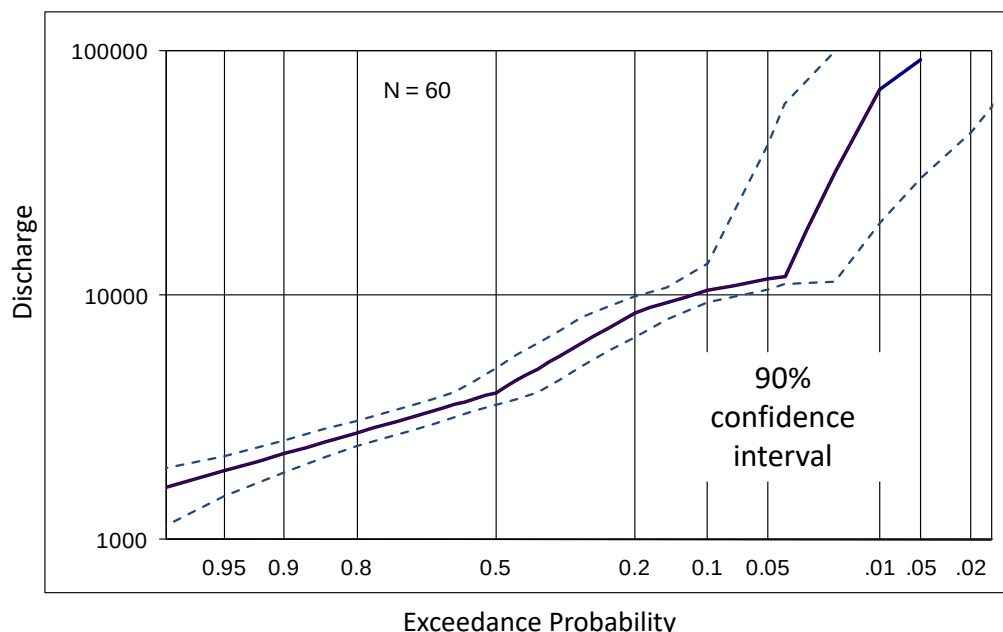


Figure 3-2. Uncertainty in graphical discharge-frequency function as confidence interval

3-5. Description of uncertainty in frequency functions

a. The data available and the approach used to estimate the frequency function impacts the uncertainty in that function, as does whether the function is analytical or graphical. For a frequency function developed by a frequency analysis of a systematic, gaged record of annual peak discharges, the record length (or, sample size, N) provides an understandable metric of the resulting uncertainty in the function due to sampling error (as described in paragraph 3-3.a(3)). The record length, when combined with the parameters of an analytical probability distribution (function), or the slopes of a graphical function, also defines the uncertainty at each quantile. The MSE of a parameter, quantile, or probability estimate is another common metric used to describe the uncertainty in the estimate and is similar to the variance.

b. MSE of probabilities, quantiles, and distribution parameters are inversely proportional to the size of data sample used to estimate the probability function. A larger sample size (longer record length), N , produces a frequency function with less uncertainty, so the relationship between N and MSE is inversely proportional. If a systematic (complete) data sample of size N is used to estimate a frequency function, producing a 1/100-year flow quantile ($Q_{0.01}$) estimate with a given MSE, then the same frequency function with the same MSE of $Q_{0.01}$, but estimated in some other way, is said to have an ERL equal to N . This relationship can be expressed mathematically, using variance in place of MSE, as shown in equation 3-1 and equation 3-2.

$$Var[\hat{X}_p|\theta, N] = Var[\hat{X}_p|\theta, ERL_p] \quad (3-1)$$

$$ERL_p \approx N \frac{Var[\hat{X}_p|\theta, N]}{Var[\hat{X}_p|\theta, ERL_p]} \quad (3-2)$$

where:

$\hat{X}_p$ = an estimate of the quantile with exceedance probability p

θ = a parameter set for the frequency function

N = the systematic sample size

$Var[\hat{X}_p|\theta, N]$ = variance of quantile $\hat{X}_p$, given parameters and sample size

ERL_p = the effective record length

Due to this relationship between sample size, N , and MSE or variance, there are two typical ways to state the uncertainty in a frequency function; ERL of the function as a whole, and variance or MSE of each quantile or parameter estimate. The methods used to randomly generate a frequency curve from within its uncertainty in a Monte Carlo risk analysis make the ERL the most commonly used uncertainty parameter.

c. One way to state uncertainty in a frequency function is as an ERL that describes the information content as the size of a systematic sample that would have provided the same amount of information. For example, the frequency function developed from regional parameters might contain the same amount of uncertainty as if it were estimated from a systematic, 37-year record of observed annual maximum streamflows. The $ERL = 37$ thus describes the uncertainty as if completely due to sampling error, or a limited sample size that, therefore, is not completely representative of the population of interest. A function based on frequency analysis of a dataset that includes some systematic data and some additional data, such as historical events, periods of historical nonexceedance, or regional skew estimates, generally has an ERL longer than the systematic record length. Further, depending on the type of information available for the estimation of the frequency function, the ERL might not be the same for all quantiles of the frequency function.

d. Thus, for flood risk analysis, the ERL for the quantiles that are most important to the analysis should be used, and without knowledge of those specific quantiles, 1/100 quantile may be used. An LPIII function estimated with Bulletin 17C methods using software referenced in that guidance should include the MSE (or the variance) at each quantile. Table 3–2 offers ERLs that are reasonable for discharge-frequency functions developed from at-site data, nearby data, or rainfall/runoff modeling using either historical precipitation or probabilistic precipitation and with or without adequate model calibration. Reductions in ERL to account for other uncertainty are also suggested in Table 3–2.

Table 3–2**Effective record length guidelines for methods of developing discharge frequency function**

Method of Frequency Function Estimation	ERL	Reasoning	Typical RL from study	ERL reduction	Resultant ERL (years)
Analytical distribution fit to observed data using B17C or Bayesian methods	ERL computed using Bulletin 17C methods	–	–	none	varies
Quantiles or analytical distribution estimated using regional regression analyses	Average record length used in regional study	Values are likely from StreamStats (USGS 2024a). Typical record lengths from the supporting reports are ~35 years.	35 ¹	none	35
Analytical distribution fit to 25+ year continuous simulation results using a calibrated/validated hydrologic model	90% of the ERL value computed using Bulletin 17C methods	This estimation technique should result in a distribution with less information content than one from observed discharge data.	44 ²	0.9	~ 40
Analytical distribution fit to 25+ year continuous simulation results using an uncalibrated/validated hydrologic model	50% of the ERL value computed using Bulletin 17C methods	Similar to above, but with uncalibrated/validated hydrologic model.	44 ²	0.5	22
Quantiles estimated from discrete event simulations (AEP of 1/2, 1/5, 1/10, 1/25, 1/50, 1/100, 1/200, and 1/500) using a calibrated/validated hydrologic model	75% of the average record length used in regional study	Typical average record lengths within Atlas 14 (NOAA 2024l) volumes are ~40–50 years for hourly stations.	45 ³	0.75	~ 34
Quantiles estimated from discrete event simulations using an uncalibrated/validated hydrologic model	50% of the average record length from Atlas 14/Atlas 2 (NWS 2023)	Similar to above, but with uncalibrated/validated hydrologic model	45 ³	0.5	~ 23

Notes:

¹ Typical average record lengths from reports used to drive StreamStats (USGS 2024a) are about 30–40 years.

² Hourly gridded Analysis of Record for Calibration precipitation/temperature is trustworthy and available throughout the continental United States (CONUS) from 1979 to current; thus, modelers could easily realize 44 years of continuous simulation output using hourly boundary conditions anywhere in CONUS. Furthermore, daily gridded Livneh (NOAA 2024d) precipitation/temperature boundary conditions are available from 1915 to 2011 and offer 96 years of possible continuous simulation output for larger modeling domains.

³ Typical average record lengths from Atlas 14 volumes (hourly stations) are about 40–50 years.

e. Another way to describe uncertainty in a frequency function is to specify the variance or MSE of each discharge quantile or frequency ordinate. For analytical frequency functions, variance or MSE can also or instead be stated for each distribution parameter. To use the variance or MSE of a parameter, a probability distribution, such as Normal, Chi-squared, or Beta, must also be specified for each parameter.

(1) Methods for fitting analytical frequency curves, such as the expected moments algorithm in Bulletin 17C (England et al. 2019) or Bayesian fitting techniques, provide estimates of these variance or MSE values for each quantile.

(2) For graphical frequency functions, an order statistics approach (Morgan and Henrion 1990) can be used for describing the uncertainty as an MSE or variance of each quantile. The order statistics logic, and computation of the variance of discharge quantiles, is described in CPD-72a (HEC-FDA 2014). Graphical frequency functions developed by estimating and then connecting discharge quantiles have different uncertainty than a function estimated from a data sample, but in practice, the uncertainty is described in the same way. The order statistics approach is theoretically based on an observed sample of data, and the likelihood that any ordered sample member is, in fact, greater than a given quantile (the likelihood that the third largest discharge sample member actually exceeds the 0.01 discharge quantile).

(3) The basic method is limited to describing uncertainty in the estimated function for the range of any observed data, or if developed by some other method, to a synthetic period of record that is equivalent in information content and range to the uncertainty in the simulation approach used to derive the frequency function. Beyond the range of this period of record, the method extrapolates the uncertainty MSE using asymptotic approximations of error distributions. A complete description of the order statistics approach, as implemented in in Hydrologic Engineering Center (HEC) software, can be found in CPD-72a (HEC-FDA 2014) (noted above). Figure 3–3 displays the quantile uncertainty around a graphical frequency curve as PDFs.

(4) However, most often a quantile variance (or MSE) is computed from each order statistics PDF, or from asymptotic approximation beyond the range of the sample data, and is paired with a Normal distribution to describe the quantile uncertainty for the frequency function.

(5) In flood risk analysis, the MSE or variance of the 1/100-year quantile (Q0.01) is often the chosen representation.

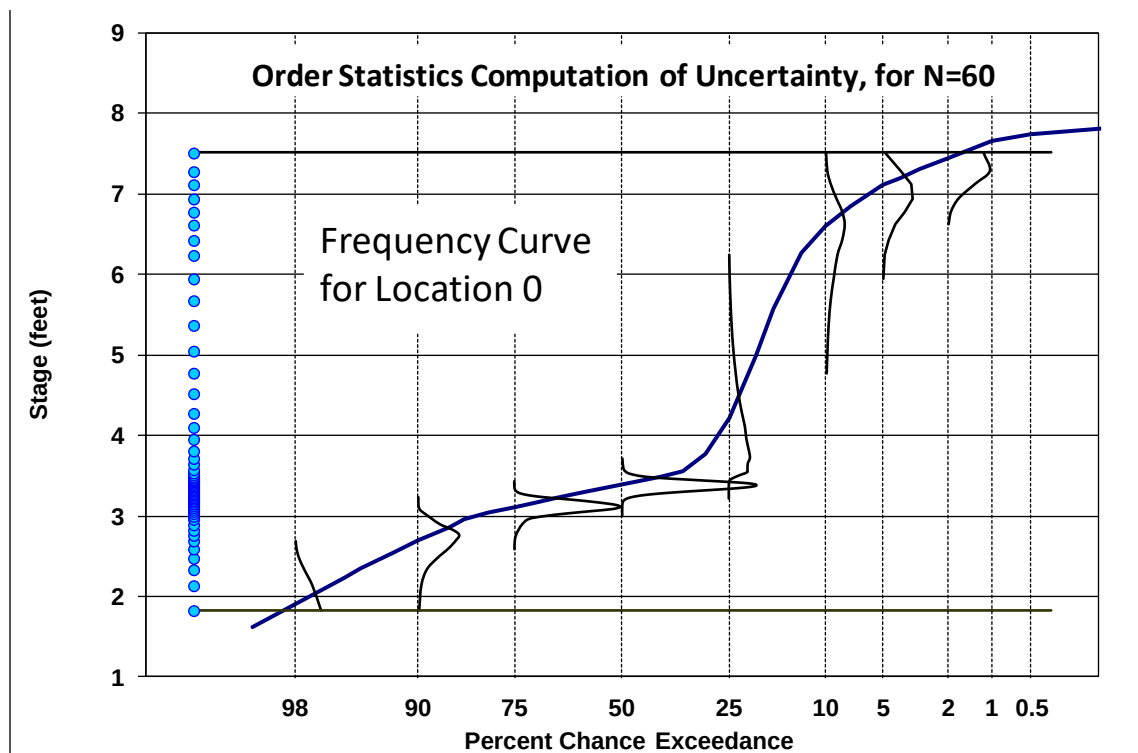


Figure 3-3. Quantile uncertainty from order statistics shown as probability density functions around a graphical frequency function

f. In some cases, when an ERL is required for parameterizing the uncertainty for Monte Carlo sampling, an ERL is estimated from the MSE of a parameter or quantile. For example, the MSE of the 1/100-year quantile ($Q_{0.01}$) may be of interest, and an ERL is thus estimated to capture the same MSE of a frequency function at that quantile. Sometimes, a procedure to estimate the frequency function offers an ERL for the resulting function. Other times, the procedure offers a variance/MSE or CI for the quantiles or parameters, and an ERL can be estimated. ERL is sometimes estimated by trial and error using a parametric bootstrap procedure (described in paragraph 3-6.c(1)), or as shown in equations 3-1 and 3-2.

g. Some Bayesian approaches to estimating a frequency function produce a collection of function parameter sets sampled from the posterior distributions of the parameters that, together, represent the uncertainty in those parameters (as large random samples). Depending on the approach used for risk analysis, these parameter sets could serve as the description of the uncertainty in the frequency function, and also the realizations of that uncertainty.

3-6. Ways to model uncertainty in frequency functions in Monte Carlo simulation

a. While the frequency function itself captures natural variability in annual peak streamflow (and ultimately the damage it could produce), the uncertainty in estimating the correct frequency function is a knowledge uncertainty. In general, natural

variabilities are captured within frequency-based risk analysis metrics, such as EAD and AEP, and knowledge uncertainties affect (decrease) how well we can estimate those metrics. Those knowledge uncertainties are also captured within metrics, such as assurance. Another distinction is that natural variabilities are irreducible, but knowledge uncertainty can potentially be decreased with an investment to collect more information.

b. In the Monte Carlo simulation approaches to risk analysis discussed in Chapter 2, a realization of knowledge uncertainty includes a frequency function that has been randomly generated from within its uncertainty. The collection of frequency functions resulting from all realizations should together capture the initial function's knowledge uncertainty. There are various methods for generating a frequency function for each realization, some that are used with analytical functions, and some that are used with graphical functions. To be used correctly, the statement or parameterization of the knowledge uncertainty must align with the needs of each Monte Carlo method.

c. Analytical frequency functions are defined by a probability distribution function and its parameters. Most commonly in flood risk analysis, the three-parameter LPIII is chosen, and the parameterization of the function uses the mean, standard deviation, and skew coefficient of the base 10 log of discharge. In Monte Carlo simulation that defines each realization with an instance of the function parameters from within their knowledge uncertainty, the sampling method must produce a set of parameters to define a probability distribution function for the realization.

(1) *Parametric bootstrap using effective record length.*

(a) Given an initial probability distribution, its parameters, and an ERL, the parametric bootstrap approach generates a pseudorandom sample of ERL number of values from the initial distribution and re-estimates the distribution parameters from that sample.

(b) Figure 3–4 shows an example of a single realization with $ERL = 30$. The initial LPIII distribution function is shown in pink, with a sample of 30 pseudorandom values from Uniform[0, 1] used as exceedance probabilities to generate 30 corresponding discharge values from the original function in light blue. A new LPIII distribution is then estimated from the sample of 30 discharge values, shown as the dark blue line (with the 30 values replotted against plotting positions in dark blue, as if a 30-year sample), and that LPIII distribution is used for the realization. The basic parametric bootstrap method is first-order accurate. Second-order accurate parametric bootstrap methods, such as the bias-corrected, accelerated bootstrap method, are also available.

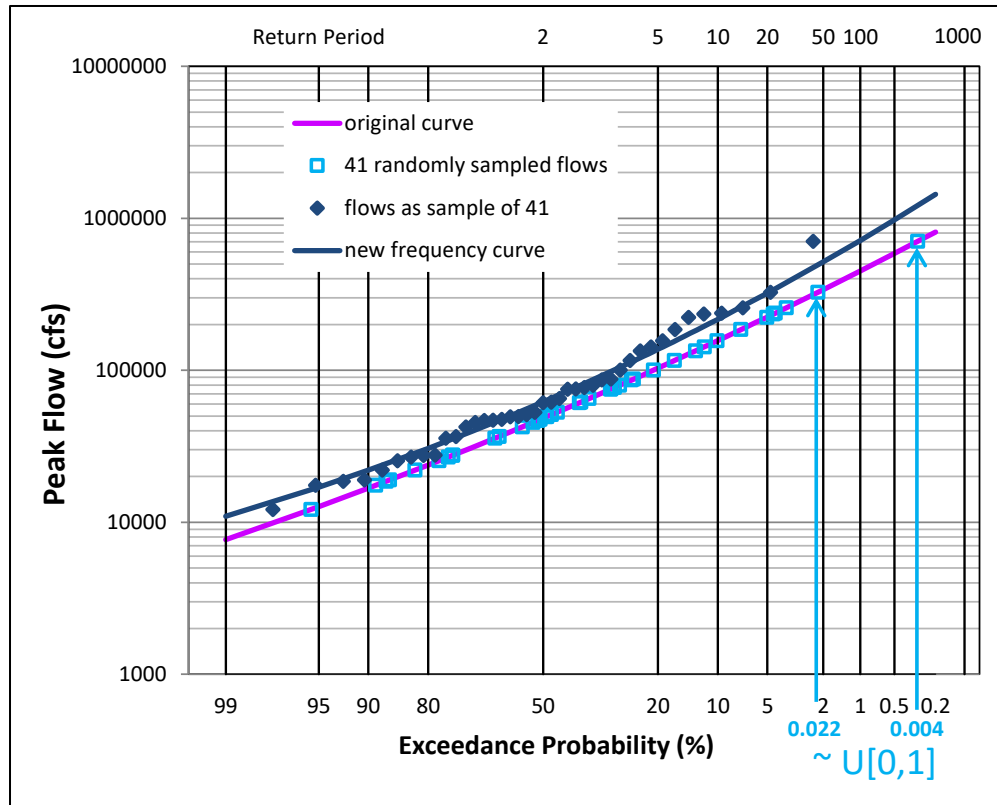


Figure 3–4. One realization of the parametric bootstrap approach to randomly generate a discharge-frequency function from within its uncertainty

(2) *Random sampling of parameters from uncertainty distributions and correlation matrix.* Another method to generate the parameters for an LPIII distribution is to randomly sample a value for each probability distribution parameter from its own uncertainty distribution. The description of uncertainty required for this approach is a probability distribution for the mean, for the standard deviation, and for the skew coefficient as well as a correlation matrix defining the linear relationship between those parameters. The simplest parameter uncertainty distribution is for the mean, which is Normally distributed around the assumed mean, with a variance equal to the variance of the initial LPIII divided by the ERL.

(3) *List of parameters sets.* Bayesian methods of estimating the discharge-frequency function produce a large random sample of parameter sets from their posterior distribution as output. Together these parameter sets fully describe the resulting function's uncertainty. Each of these parameter sets, and the frequency function it produces, provides a single realization of the frequency function's knowledge uncertainty. Therefore, a Monte Carlo simulation can simply take as input the list of parameter sets, and use one set for each realization of the discharge-frequency function.

(4) *Percentile sampling using quantile uncertainty.* Also called “parallel curves,” this approach samples one pseudorandom value from Uniform[0, 1] to select a percentile of the uncertainty distribution of each quantile. This approach is described more fully for graphical frequency curves in paragraph 3–6.d(3). This method is generally not recommended for analytical distributions because it can underestimate the variance, and more importantly, does not provide a parameterized analytical distribution.

d. Graphical frequency functions do not have parameters, but instead are defined by a list of exceedance probabilities versus quantile values (values with specified exceedance probability.) In Monte Carlo simulation that defines each realization with an instance of the graphical frequency function from within that function’s knowledge uncertainty, sampling methods must produce an entire frequency function for each realization.

(1) *Nonparametric bootstrap using effective record length.* While the parametric bootstrap method (described in paragraph 3–6.c(1)) samples pseudorandom values from an analytical frequency function to use in generating a “new” frequency function for the realization, a nonparametric bootstrap similarly samples pseudorandom values from a graphical frequency function, and re-estimates a new graphical function from those values versus plotting positions (estimates of exceedance probability). A pseudorandom sample of Uniform[0, 1] values of size ERL is generated, and used as exceedance probabilities to interpolate values from the graphical frequency function (generally using linear interpolation of log (discharge) and standard Normal deviate, Z_p). The sampled values are ordered, and Weibull plotting positions are assigned to each, producing a new graphical function.

(2) *Nonparametric bootstrap challenge.* A challenge of using the nonparametric bootstrap approach is how to define the function above the highest and below the lowest sampled values, because the most extreme value produced has return period 1/ERL. One reasonable option is to extend the function beyond the generated sample by maintaining the slope of the upper and lower end of the input frequency function, but this option could underestimate uncertainty.

(3) *Percentile sampling based on quantile uncertainty (also called parallel curves).*

(a) When an uncertainty distribution is defined for every quantile of a graphical frequency function, random realizations of the function can be generated based on the quantile uncertainty at each quantile described by a parametric probability distribution, such as Normal or Beta. For example, when the Normal distribution is chosen, each quantile of the graphical frequency function is defined by a Normal distribution with mean equal to the initial frequency function quantile and variance equal to the variance defined at that quantile.

(b) Typically, this approach to generating a random instance of the function uses a single pseudorandom Uniform[0, 1] value for the entire function, using it to specify a percentile of each uncertainty distribution along the entire curve. This sampling process, as well as the defined uncertainty PDFs at each quantile, is pictured in Figure 3–5. The

resulting frequency function (shown in blue) is referred to as “parallel” to the original because each quantile corresponds to the same percentile of the original function’s uncertainty.

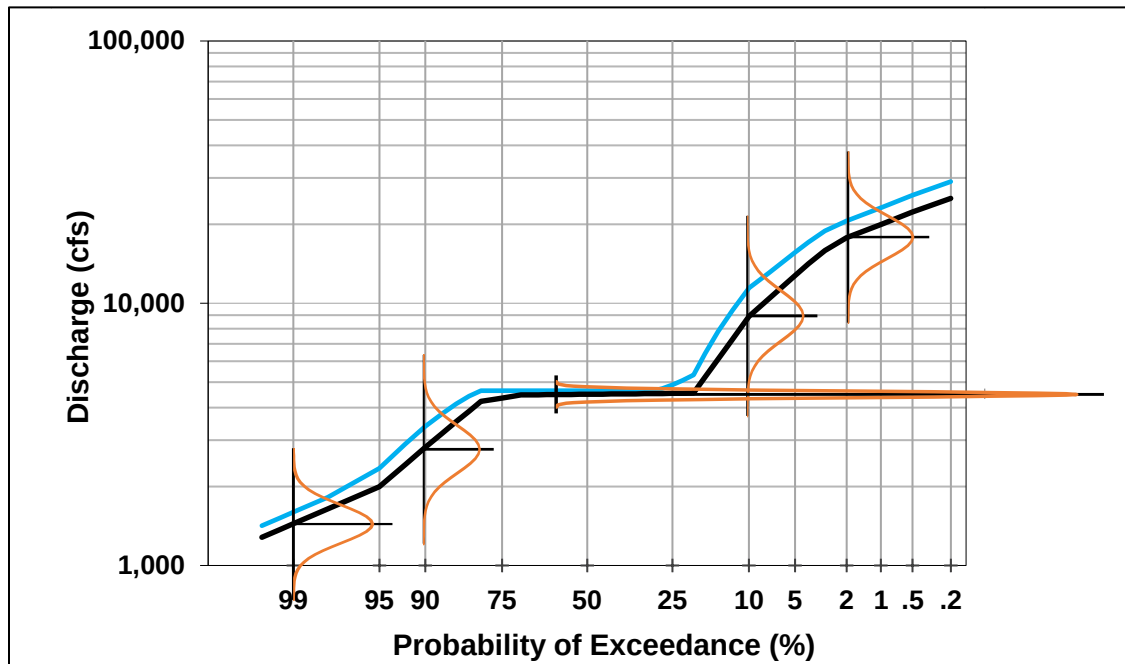


Figure 3–5. One realization of the parallel curve approach to randomly sample a discharge-frequency function from within its uncertainty

Chapter 4

Uncertainty in the Discharge-Stage Transformation

4–1. Overview

a. In a risk assessment as described in Chapter 2, the description of the system performance and the associated hazard includes a relationship between discharge and channel stage. The stage-discharge function predicts the maximum channel stage for a damage reach that results from any annual peak discharge from the discharge-frequency function. The channel stage predicted with the stage-discharge function should represent conditions at the most vulnerable point of the damage reach.

b. The Chapter 2 risk assessment includes the uncertainty in the relationships used, so the uncertainty in the stage-discharge function must be evaluated and described as defined in this chapter. The determination of stage-discharge uncertainty requires accounting for the uncertainty associated with factors affecting the stage-discharge relationship. For example, uncertainty might be introduced into the stage-discharge function due to variability in Manning's n as well as measurement errors from instrumentation or method of flow measurement.

c. Numerical hydraulic models of open-channel flow are commonly used in project studies to simulate channel stage given discharge to develop stage-discharge functions for both the without- and with-project conditions, and for both current and future conditions. Typical approaches range from one-dimensional, steady-state models to multi-dimensional unsteady models. Models are limited by the inherent inability of theory to exactly model the complex nature of the hydraulic processes. Data used in the models are also not exact, introducing errors in the model geometry and coefficients used to describe the physical setting.

d. Many of the factors that determine stage-discharge uncertainty, and that are estimated for modeling purposes, are time-dependent, both seasonally and during a flood event. Many of the factors are also spatially variable, both laterally and longitudinally in the channel and associated floodplain. In general, the more complex the flow conditions, the greater the need to use models that replicate the significant physical processes accurately, and the more challenging the task.

e. Stage-discharge uncertainty can be evaluated for all contributing factors, or for each factor individually. When the factors are analyzed separately, care must be taken to ensure that the resulting uncertainty from combining the factors is reasonable. An example is a stream where floods always occur significantly after ice melt, but where the ice creates significant stage increases when present. In this case, the uncertainty for ice should not be imposed in addition to the uncertainty due to increased resistance from early summer vegetation. In other words, correlation of separate factors should be considered in the analysis and accounted for in the combination of individual uncertainties, rather than simply assuming independence.

f. If flood inundation results from storm surges along a lake, wind-driven waves (run-up), a filling reservoir, or combinations of these events, then using a stage-frequency function is more appropriate for derivation of the damage-frequency function than a stage-discharge function combined with a discharge-frequency function. In these situations, a one-to-one relationship between stage and discharge is not feasible. EM 1110-2-1415 describes statistical analysis procedures for fitting a graphical frequency function with stage observations for a current, existing condition. The procedures are similar to those used for fitting a frequency function with a discharge sample when an analytical probability distribution is not appropriate, described in paragraph 3–2.b.

g. For future condition and other cases, the function must be defined based on model studies to capture the changed conditions. The model used depends on the condition to be analyzed. For example, if reservoir operation changes are proposed to reduce flood damage due to reservoir pool elevation rise, a reservoir-operation simulation model might be used to estimate the modified time series of lake levels. The stage-frequency function could then be fitted to this series with the methods in EM 1110-2-1415.

h. Channel bed, channel bank, and land surface erosion and deposition complicate evaluation of a stage-discharge function. Mobilization of bed and bank materials in alluvial channels alters the channel shape. If that happens, stage at a channel cross section is not a unique time-invariant function of discharge, channel geometry, and energy losses. Instead, the stage depends on material properties and the time history of discharge, and a movable-boundary hydraulics model is required to define the relationship for EAD computation. EM 1110-2-4000 provides guidance on analysis of erosion and deposition impacts. It identifies locations at which sedimentation problems are likely to occur and suggests design or maintenance solutions to those problems.

i. Ice accumulations can increase flood risk by restricting the flow in channels and conduits (causing higher channel stages) and by increasing pressure or forces on the physical structures. In cold regions, ice buildup and breakup must be anticipated, the impact must be evaluated, and project features must be adjusted to enable proper performance. With some measures, such as channel-lining improvements, this translates to an increase in project dimensions so the structures can withstand impacts of floating ice. Likewise, if ice is likely to form on a reservoir surface, the dam design must be altered to withstand the increased overturning moment due to the added force on the dam. EM 1110-2-1612 and the Cold Regions Research Engineering Laboratory can provide guidance.

j. The effect of debris is similar to that of ice; it can significantly reduce channel conveyance and constrict flows at obstacles. Examples are an increase in volume associated with runoff, constrictions at bridges, and accumulation of urban trash and waste in channels. If debris is mobilized and subsequently redeposited, it may adversely affect performance of pumps, gates, and other plan features. Proper maintenance should be included as a component of any plan to avoid these problems.

4–2. Methods for developing the stage-discharge function

The stage-discharge relationship can be an empirical curve that is based on observations of discharge and stage, or it may be based on a conceptual hydraulics model that represents the physics of flow in an open channel and the resulting relationship between discharge and stage. The choice of model depends on the data available and the particular needs of the study, although practically, the empirical approach is limited to cases in which measurements for the baseline without-project condition are available.

a. Developing stage-discharge functions using observed gage data.

(1) The most common and precise practice to relate channel stage with discharge is to plot observed discharge versus stage. The stage-discharge function for the baseline without-project conditions may be established from simultaneous measurements of velocity, cross-sectional area, and stage at a gaged stream cross section. Flow rate can be determined from the velocity and area measurements, yielding a sample of discharge and corresponding stage. The U.S. Geological Survey (USGS) provides a technical procedure for measuring stage and velocity at a given channel section and the development of stage-discharge rating curves (USGS 1977). The stage-discharge function is developed as the best-fit curve through the plotted observed stage measurements and computed discharges. In cases where stream gages are not located at the specific locations needed to support a study, it may be necessary to use numerical modeling to develop a stage-discharge function.

(2) Graphically, observations of stage can be plotted with corresponding indirect observations of discharge (computed from velocity measurements), and a best-fit line can be drawn by inspection or analytical fit. Decisions about best fit can be made considering the entire sample, or the sample can be separated into zones of flow for which connected curves are fitted subjectively. Figure 4–1 is a plot of stage discharge data for a stream with more than 70 years of record, where nonstationary effects have been removed from the record. Each point in the plot represents a simultaneous measurement of discharge and stage at a common location. As is typical with such sets of observations, many are for lower, more frequent (higher annual chance exceedance) discharge events, and fewer are available for rarer events.

(3) Rather than combining the set of observations to develop a single transform, the strategy here is to develop a smoothed, best-fit line for various discharge zones. The record is broken into sections to represent three zones of flow. The first zone is the within-bank flow zone; the second is the measured-out-of-bank flow zone (or bankfull to the highest measured flow), and the third is the rare event zone where occasionally an event may have been measured or estimated from a high-water mark. A minimum of 8 to 10 measurements out of bank is normally required for meaningful results. Unfortunately, it is uncommon to have measured events in the range of interest for flood risk management studies.

(4) Note that, similar to discharge observations for probability curve development, the observations of discharge and stage for transform development must be stationary; that is, they must be measurements of a common condition in the channel if a single transform is to be developed from them. So, for example, changes in gage position, stream geometry, and so forth must be identified, and observations adjusted or separated accordingly.

(5) Statistical outlier tests may be used to examine data anomalies. Engineering judgment is needed to identify and correctly handle extreme occurrences (coincidental effects of ice jams, debris blockages, etc.) in the development of the stage-discharge curve.

(6) A best-fit transform may be graphical or analytical. Analytically, a best-fit transform may be established with statistical fitting procedures, often using standard regression analysis. With those procedures, a model form for the transform curve is selected, and the parameters of that model are estimated with the sample of discharge and stage to fit the analytical model to the data. The method described in the National Handbook for Recommended Methods for Water-data Acquisition (USGS 1977) uses equation 4–1

$$Q = C(G - e)^b \quad (4-1)$$

to describe the stage-discharge relationship where:

Q = discharge

G = the stage reading

C , e , and b = coefficients used to match the curve to the data. It should be noted that the value of b is usually between 1.3 and 1.8

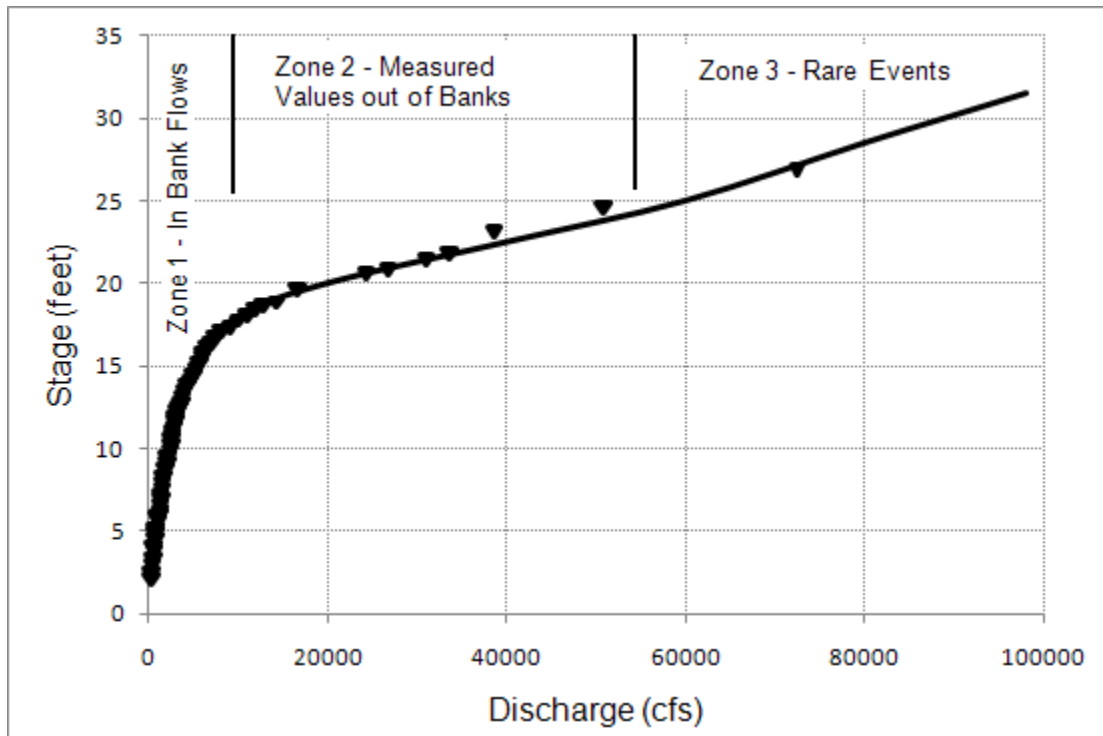


Figure 4–1. Stage-discharge plot showing uncertainty zones, observed data, and best-fit curve

(7) An alternative equation proposed by Freeman, Copeland, and Cowan (1996) is an exponential curve with decreasing exponents, shown in equation 4–2.

$$G = a + bQ^{1/2} + cQ^{1/3} + dQ^{1/4} + eQ^{1/5} + fQ^{1/6} \quad (4-2)$$

where:

G = stage in feet

Q = discharge in cfs

a through f = coefficients determined by a linear regression to fit the equation to the data

This equation yielded an R^2 better than 0.80 for 115 rivers and streams out of 116 analyzed. Additionally, for 75% of the streams the R^2 was better than 0.97.

Equation 4–2 does not accurately predict very low flows, but these are not generally of concern in flood risk management studies.

b. Developing stage-discharge functions using computed water surface profiles.

The stage-discharge function at the points of interest can be developed using a hydraulic model of the study area.

(1) A hydraulic model solves the governing equations of open-channel fluid flow to compute stage, given a specified discharge and parameters, such as channel geometry, Manning's roughness, bridges and culverts, and initial conditions. The discharge equations may be represented in steady or unsteady form, and in one or two

dimensions. Channel modeling methods are well known within USACE, and much guidance is available for developing, calibrating, and using such models. The USACE Hydrologic Engineer Center is a good resource for USACE hydraulic engineering tools, in addition to USACE guidance documents EM 1110-2-1416, EM 1110-2-1601, and ER 1110-2-1405.

(2) To derive the stage-discharge for baseline without-project conditions, a model is configured with a representation of those physical conditions and exercised with a wide range of discharges, computing the corresponding stages in the study area for each discharge. Those stage values may be plotted, tabulated, or even connected with a graphical function.

(3) Flood risk management projects may alter channel properties and corresponding model parameters. To derive the stage-discharge relationship associated with a proposed plan, model inputs are adjusted from the current without-project condition as appropriate, and the model is rerun for a range of discharges, yielding corresponding stages.

4–3. Sources of uncertainty in stage-discharge functions

The relationship of stage to discharge cannot be established with certainty, regardless of the method used to develop the transform function. It is, therefore, necessary to evaluate uncertainty in stage-discharge functions developed from both observed data and hydraulic models.

a. Stage-discharge functions from observed data. Sources of uncertainty about a transform derived from observations of discharge and stage include the following:

(1) Errors in measurements of stage and velocity (from which discharge is estimated). In many cases, these water surface elevations and velocities are measured under adverse conditions, perhaps from a boat or cableway. Although the USGS and other hydrographic agencies exercise diligence and care, the measurements are subject to errors.

(2) Uncertain knowledge of channel geometry from which cross-sectional area is computed to estimate discharge as the product of velocity and area. The topographic and bathymetric data from which geometry is determined may be surveyed with traditional methods or inferred from remotely sensed data, but even the most diligent field surveys are subject to measurement errors.

(3) Lack of stationarity of the series of discharge and stage observations. The presumption, as noted previously, is that conditions do not change from measurement to measurement across time. However, even seasonal variations in vegetation in the channel or debris content in the flowing water can violate this requirement. Adjustments may be made, but those adjustments are also uncertain.

(4) Uncertainty about the proper form of the mathematical function with which to represent the stage-discharge relationship when an analytical function approach is

chosen. Sensitivity analysis on the function and function parameters should be considered to understand the nature of the uncertainties.

(5) Uncertainty about the parameter values estimated for the function, particularly if the data set is small or if it is sparse in zones of highest flow.

(6) Uncertainty about if or how the relationship should be separated by flow zone.

b. Stage-discharge functions derived from a hydraulic model. Sources of uncertainty about a transform derived through application of a riverine hydraulics model include the following:

(1) Uncertainty about the choice of model type. As noted previously, models may be steady or unsteady and one- or two-dimensional. All are approximations of the true flow condition, with some simplifications of the governing equations. The optimal choice is never certain.

(2) Errors in the topographic and bathymetric data from which the model representation of the channel and floodplain geometry is inferred.

(3) Errors in measurements of dimensions and locations of structures in the channel and floodplain that impede flow. These man-made structures include culverts, bridges, piers, and other structures.

(4) Errors in measurements of dimensions and locations of water management features and structures that affect flow patterns. For example, if the stream is confined by levees, the elevations of the top-of-levee profiles are critical to predicting overflow locations, rates, and volumes. Errors in those levee profile elevation measurements have a significant impact on the accuracy of stage predicted for a specified flow rate exceeding those levees.

(5) Uncertainty about existence and impact of ice cover and ice jams in the stream.

(6) Uncertainty about downstream boundary conditions. This uncertainty is especially complex if stage at the index location is influenced by backwater at a downstream confluence, reservoir or lake, or sea. The most complex cases may require application of coincident frequency analysis methods, as described in EM 1110-2-1413 and EM 1110-2-1415.

(7) Uncertainty about the model parameters. For one-dimensional models, parameters include Manning's n and expansion and contraction coefficients. For two-dimensional models, additional parameters include eddy viscosity or the equivalent.

(8) Uncertainty about the impacts of erosion and deposition before or during a flood event.

(9) Uncertainty about the impacts of wind and wave run-up on elevation, given the flow.

4-4. Determination and description of stage-discharge uncertainty

Chapter 2 illustrates how a risk assessment for USACE studies explicitly considers the uncertainty about each of the functions included in that assessment. Therefore, the uncertainties identified must be quantified and represented mathematically for use in the assessment. This section addresses development of an uncertainty distribution for stage (including the form and the parameters of the distribution) for transforms developed using the two methods described above: (1) stage-discharge functions estimated using observed gaged data, and (2) functions developed using water surface profiles from hydraulic modeling.

a. Quantifying uncertainty in stage-discharge functions from observed data.

Developing a mathematical model of uncertainty about a graphically developed transform function is challenging. Many common statistical analyses are not possible because of the graphical rather than mathematical form of the function.

(1) The measure used to define the uncertainty of the stage-discharge relationship is the standard deviation of stage values (S) in a given discharge range. It is recommended that only data values for flows above bankfull be used in the computation of S (zones 2 and 3 in Figure 4-1), since low flows are generally not of interest in flood studies, and their uncertainty is of a different magnitude. Note that the objective is to calculate uncertainty in stage, not discharge. The stage residuals are the difference between the observed stage and the stage provided by the best-fit stage-discharge transform curve. The standard residual within a zone (or for the whole record), noted as S , can be estimated using equation 4-3.

$$S = \sqrt{\frac{\sum_{i=1}^N (X_i - M)^2}{N-1}} \quad (4-3)$$

where:

X_i = stage for observation i , which corresponds with discharge Q_i

M = best-fit curve estimation of stage corresponding with Q_i

N = number of stage-discharge observations in the range being analyzed

This computation may be completed by discharge zone, if the transform is separated as suggested in paragraph 4-2.a(3). Note that this approach describes only the error due to fitting the stage-discharge relationship to the data; it fails to account for other sources of uncertainty in stage identified in paragraph 4-3.a.

(2) The probability distribution of error around the best-fit, stage-discharge transform curves can vary significantly from stream to stream. The Gaussian (Normal) distribution can be used for the description of many rivers but not all. Freeman, Copeland, and Cowan (1996) found that for many streams, the data were much more concentrated near the mean value and the central portion of the distribution had much

shorter tails than is the case for a Normal distribution. On other streams, the distribution was markedly skewed, rather than symmetrical like the Normal. The gamma distribution can represent a wide range of stream conditions from normal to highly skewed and is another option for describing stage uncertainty. The triangular distribution is also an effective choice to capture asymmetry, as is the Log-Normal.

(3) The threshold for including data in the standard residual S described above is the bankfull discharge. Where explicit bankfull elevations and discharges are not known, the mean or median (2-year, 50% chance exceedance) annual maximum discharge may be used to define bankfull instead. Alternatively, Leopold (1994) recommends the 1.5-year recurrence interval (66.67% chance exceedance) in the annual flood series for the approximate magnitude of bankfull discharge.

(4) To account for additional uncertainties in paragraph 4–3.a beyond the standard residual of the best-fit stage-discharge relationship computed in paragraph 4–4.a(1), sensitivity of the graphical function to various errors may be considered through sensitivity analysis. For example, the impact of systematic errors in measurement of elevation can be addressed by adding or subtracting an increment of elevation to each observation, considering reasonable extreme values of those errors derived from expert judgment.

(5) With increments thus subtracted or added, an upper limit stage-discharge function and a lower limit stage-discharge function may be drawn, considering the range of observed stages for a given discharge. Then, if the spread is assumed to be Normally distributed (a common assumption for such errors), an uncertainty model's parameters can be inferred. With Normally distributed errors in stage, 95% of the error is within the range spanned by two standard deviations above and two standard deviations below the best estimate. Thus, the standard deviation of stage uncertainty could be estimated using equation 4–4.

$$S = \frac{\sum_{i=1}^N (X_{i,upper} - X_{i,lower})}{N} / 4 \quad (4-4)$$

where:

$X_{i,upper}$ = the upper limit stage corresponding to discharge Q_i

$X_{i,lower}$ = lower limit stage corresponding to discharge Q_i

N = the number of stage-discharge observations

Again, this equation can be applied by zone if the flow range is separated as described in paragraph 4–2.a(6).

b. Quantifying uncertainty in stage-discharge functions based on hydraulic modeling. In most cases, the hydraulic modeler is required to describe uncertainty for stage-discharge functions. The uncertainty in stage due to model and data limitations is best minimized by selecting the most appropriate model for the situation under study. After model selection, a model is adjusted and calibrated using observed flood data to

further minimize uncertainty in results from model applications for study conditions. A description of uncertainty about a stage-discharge transform developed using the model can thus be derived with sensitivity analyses, or via Monte Carlo simulations that account for likelihood of contributing factors.

(1) Sensitivity analysis and professional judgment can be used to estimate the uncertainty in stage-discharge relationships derived based on water surface profiles from a hydraulic model.

(a) Uncertainty in stage-discharge functions computed with hydraulic models can be described by developing reasonable upper and lower limits on stages based on the assumed uncertainties in those models and their parameters. These stage limits are defined with model parameters set at reasonable extremes, after estimating a realistic range of model parameter values. Figure 4–2 shows stage profiles computed for a stream reach, with channel n values and other parameters taken in combinations and fixed at reasonable extremes. An upper reasonable limit on stage is computed and plotted, as is a lower limit. Assuming that the spread of stage uncertainty is Normally distributed and that the resulting upper and lower stage limits define a 95% confidence range (± 2 standard deviations), the standard deviation of stage uncertainty can be estimated as one-fourth of this range, as with equation 4–5.

$$S_{Q_i} = (X_{i,upper} - X_{i,lower})/4 \quad (4-5)$$

where:

S_{Q_i} = a standard deviation of uncertainty in stage for a given discharge Q_i

$X_{i,upper}$ = the upper stage limit for that discharge

$X_{i,lower}$ = the lower stage limit for that discharge

Figure 4–3 can be used as a guide to estimate reasonable bounds of the Manning's n value model parameter in sensitivity studies, by suggesting a range to vary Manning's n in simulations.

(b) To the extent that the sensitivity analyses include all factors that contribute to uncertainty about the stage-discharge relationship, this approach implicitly captures all uncertainty, except for uncertainty in stages directly due to the accuracy of the terrain data used within the model. However, the separate impact of each contributing factor is obscured with this approach.

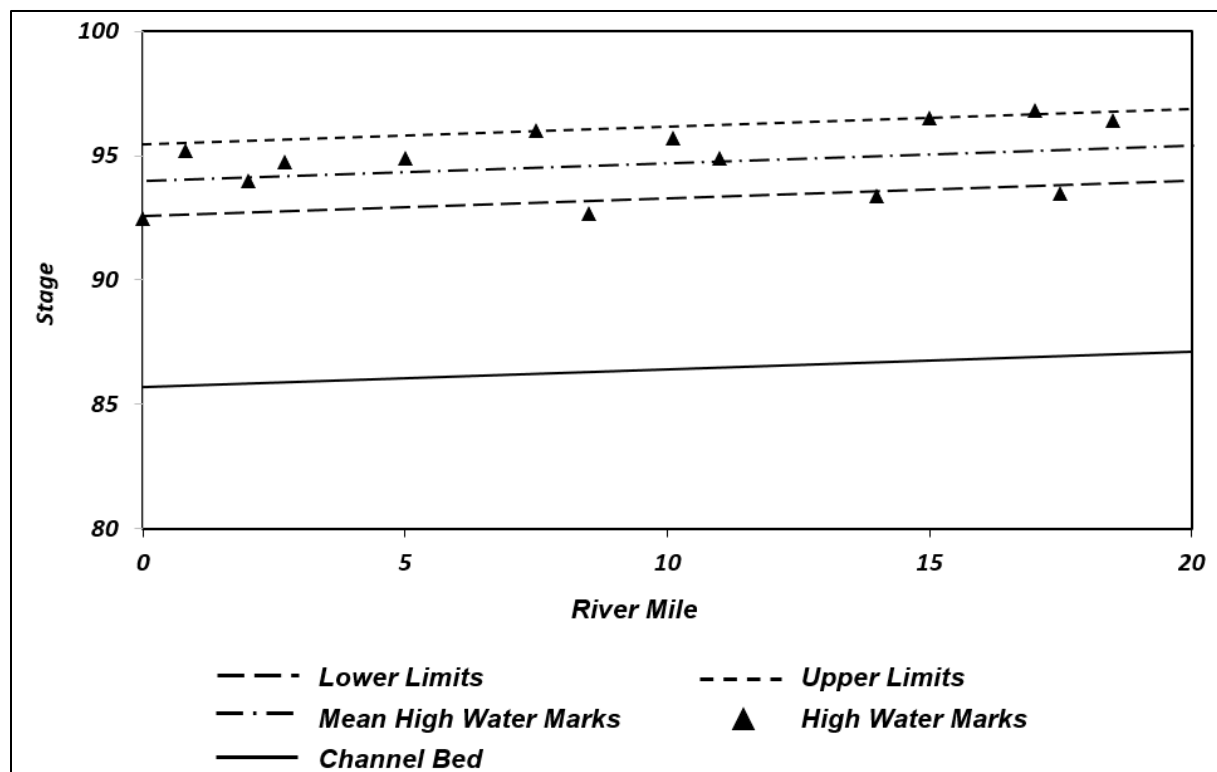


Figure 4-2. Water surface profiles from sensitivity analysis compared with high-water marks from field data

(2) Using Monte Carlo analysis to derive hydraulic model uncertainty can be described through detailed simulations that account for likelihood of contributing factors. This approach is similar to the sensitivity analysis-based approach described in paragraph 4-4.b(1), but involves more computes and random sampling.

(a) Monte Carlo analysis is a more systematic approach than sensitivity analysis to evaluating the impacts of the sources of uncertainty. Important parameters or other factors that contribute to the uncertainty of stage estimates are identified, along with reasonable ranges of those parameters or factors and measures of the likelihood of each value within the range (probability distributions). Then detailed simulation with random sampling of the uncertain model parameters and factors is completed, computing stages for specified discharges in each iteration. Subsequent statistical analysis of the stage results for each discharge explicitly describes the uncertainty in the associated stage. Practically, this is accomplished with Monte Carlo sampling integrated with the selected hydraulics model.

(b) This form of analysis (Monte Carlo) can be undertaken on a site-by-site basis, using data for and models of specific channel reaches. Studies completed at HEC (IWR 1996) and by Freeman, Copeland, and Cowan (1996) provided stage uncertainty model parameters that may be used if a site-specific analysis is not possible. These studies focused on uncertainty of stage due to uncertainty about the energy loss model

parameter (Manning's n). An uncertainty distribution for the n values can be estimated through professional judgment or with published study results.

(c) An example of published results is shown in Figure 4–3, which displays results from HEC and USACE Engineer Research and Development Center (ERDC) studies, indicating the variation of uncertainty (standard deviation) about n values versus average n values. This plot can be used to help derive a reasonable probability distribution for Manning's n values to be used in Monte Carlo analysis, and sensitivity analyses, in the absence of more detailed site-specific evaluations.

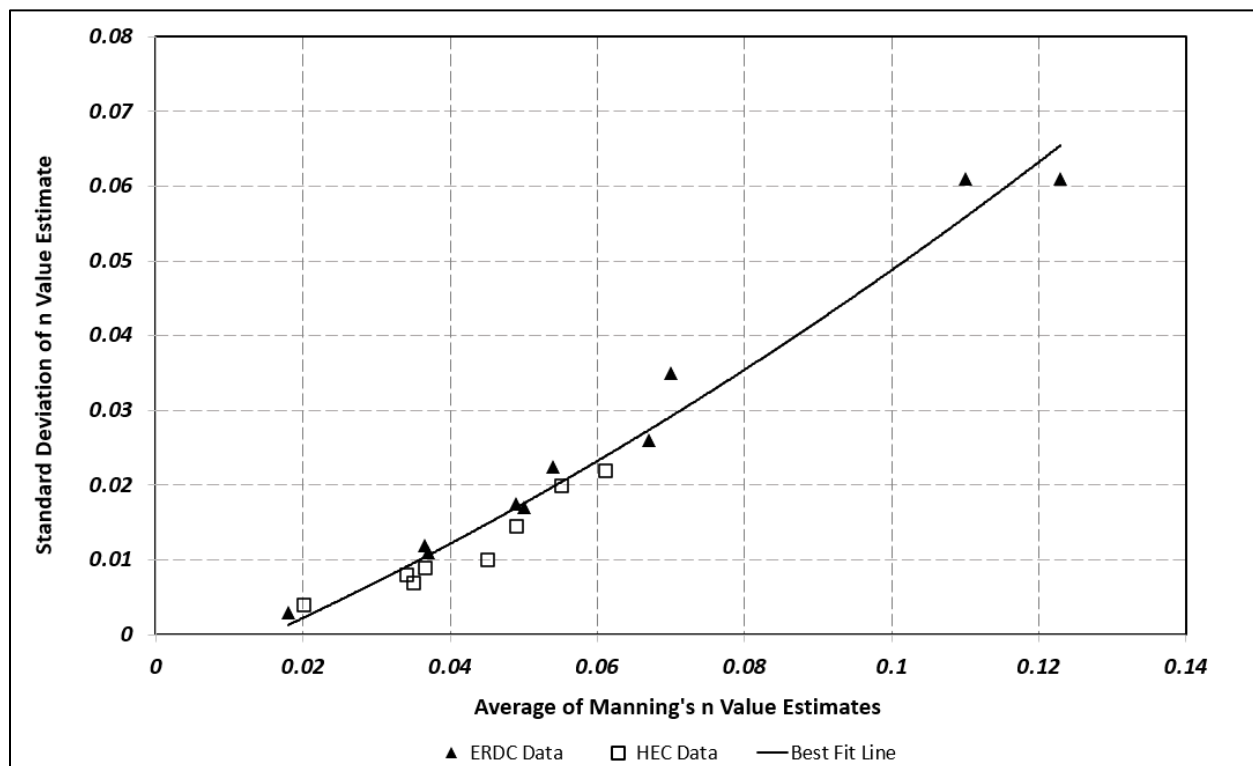


Figure 4–3. Uncertainty of Manning's n value estimates based on estimated mean values

(3) Combining uncertainties for computed water surface profiles (stages from hydraulic modeling) involves the following:

(a) If the impact of factors that contribute to uncertainty can be characterized separately through statistical analyses, and if those factors are independent, then the combined uncertainty model can be described by simply adding the uncertainty variances. To obtain the total uncertainty in computed water surface elevations, equation 4–6 can be used.

$$S_{total} = \sqrt{S_m^2 + \dots + S_n^2} \quad (4-6)$$

where:

- S_{total} = the total standard deviation of uncertainty in stage
- S_m = the standard deviation of uncertainty in stage due to the hydraulic model parameters
- S_n = the standard deviation associated with other independent uncertainty sources
- n = the number of sources of independent uncertainty included

(b) Clearly, as the uncertainty of each independent source increases and the number of independent sources increases, the overall uncertainty increases. Additional uncertainty sources, other than the hydraulic model parameters and the underlying terrain data, may include natural variability due to changes in bed forms, temperature, unsteady flow effects, seasonal roughness variations, changes in channel shape due to sedimentation, and many other factors. Any factor that contributes significantly to the uncertainty in the stage for a given discharge should be included in the uncertainty analysis. This may require separate analyses for that factor, then combining the uncertainties using equation 4–6.

(4) The preceding sections present a variety of methods for determining the standard deviation of uncertainty in stage derived using computed water surface profiles from hydraulic models. Whichever method is used, the standard deviation of uncertainty generated (S_{total}) should be compared to Table 4–1, which shows reasonable minimum standard deviation values. Provision of minimum standard deviation values is an effort to provide consistency in USACE risk analyses and should be adopted when values computed in the study are lower than those minimums.

(a) The USACE research documents “Accuracy of Computed Water Surface Profiles” (HEC 1986) and Freeman, Copeland, and Cowan (1996) provide information for estimating uncertainty in water surface profiles obtained when using a gradually varied flow model. The standard deviation of the Normally distributed errors in the estimated stages are based on topographic information and confidence in estimated Manning’s n value, with minimum values shown in Table 4–1. Thus, the values shown in Table 4–1 reflect uncertainty in computed stages due to terrain and reliability of Manning’s n values. Given that numerical models usually use fixed terrain data, the uncertainty due to the terrain is not included in sensitivity or Monte Carlo analysis.

(b) Additionally, without an adequate number of observed events with which to calibrate, accurate awareness of the uncertainty in water surface elevations due to roughness coefficients may not exist, and typical analysis can underestimate these uncertainties. To ensure these factors are included, the values of standard deviation in stage errors should not be less than the minimum values shown in Table 4–1, unless there is sufficient modeling analysis and justification.

(c) The values in Table 4–1 represent standard deviations of uncertainty in stage at design-level discharges. Standard deviations for events less than the design level decrease as a zero discharge is approached. Standard deviations for events greater than design levels may be equal to or higher than what is shown in Table 4–1.

Table 4–1
Minimum standard deviation of error in stage

Standard Deviation (in feet)		
Manning's n Value Reliability ¹	High-Resolution Geometry ²	Low-Resolution Geometry ³
Good	0.3	0.6
Fair	0.7	0.9
Poor	1.3	1.5

Notes:

¹ Where “good” reliability of Manning's n value equates to either excellent or very good model adjustment/ validation to a stream gage, a set of high-water marks in the watershed upstream of the project, and other data. “Fair” reliability relates to fair to good model adjustment/ validation for which some, but limited, high-water mark data are available. “Poor” reliability equates to poor model adjustment/validation or essentially no data for model adjustment/validation.

² High-resolution geometry equates to data collected in clear and unobscured areas with soundings of underwater cross sections and high channel width to digital elevation model (DEM) width ratio (ratio is greater than or equal to 10).

³ Low-resolution geometry equates to data collected in obscured areas with estimated underwater cross sections and low channel width-to-DEM width ratio (ratio is less than 10).

c. Probability density functions for describing stage uncertainty. The description herein suggests use of the Normal distribution to model uncertainty about stages in the stage-discharge transform curve. Using the Normal distribution is a common practice for models of errors and uncertainties when large samples are not available for comprehensive testing or when little published research is available to suggest an alternative. However, other distributions of the stage uncertainty may be used if the choice is supported and if parameters of the distribution can be estimated with the data and information available.

(1) Normal distribution: A two-parameter probability distribution defined by the mean and standard deviation. This distribution forms a symmetrical bell-shaped curve applicable to many kinds of data sets where values are equally likely to be greater than or less than the mean (symmetrical) and is also called a Gaussian distribution. The range created by the mean plus and minus two standard deviations spans 95% of the probability.

(2) Triangular distribution: A three-parameter probability distribution defined by a minimum stage, a maximum stage, and a most likely stage (mode). This distribution can be used for describing uncertainty that is not symmetric about the mean.

(3) Log-Normal distribution: A two-parameter probability distribution defined by the mean and standard deviation of the log of the variable. This is a nonsymmetrical distribution applicable to many kinds of data sets where the majority (more than half) of

values are less than the mean, but values greater than the mean can be extreme, such as with streamflow data.

d. Describing stage uncertainty for the full range of possible flows. A stage-discharge relationship must be derived for the full range of possible flows. Points on the curve are usually derived for the following probability-based events: 50%, 20%, 10%, 4%, 2%, 1%, 0.5% and 0.2%. Therefore, uncertainty about the mean stage estimates must also be described for the full range of events. Uncertainty is zero at the zero discharge stage. Uncertainty tends to increase as discharges increase toward the bank-full stage (while flows are contained in the main channel). Above bank-full stage, uncertainty in stage may increase, decrease, or stay the same. Specifically, when flows enter a floodplain, uncertainty may decrease because the change in stage for a given change in discharge becomes much smaller as the cross section widens. The full range of uncertainties can be computed from either the gaged stage-discharge data, or with a hydraulic model, as described in paragraph 4–2.

4–5. Analysis complexity

a. While the majority of water surface profile analyses are within the capabilities of one-dimensional, steady-state, hydraulic models, for streams that have rapidly varying flows or are subject to tides, an unsteady flow analysis may be needed. Sand bed streams may require mobile boundary modeling. Complex flow fields in unusual floodplains or estuaries may require multi-dimensional flow analysis. In such cases, when a single discharge-stage function is needed for a curve-combination risk analysis approach, a stage-discharge rating for the highest stages proportional to flow conditions of interest should be developed. The uncertainty associated with the rating is interpreted from the analysis results.

b. Often, sensitivity analysis as discussed below is an appropriate approach to such determination. If it is not possible to develop a rating from the results, analysis dealing directly with a stage-frequency relationship is likely to be necessary. Another option is use of the event-sampling approach to risk analysis, described in paragraph 2–8, which allows events to be modeled directly in hydraulic models during the analysis, and no one-to-one relationship between stage and discharge needs to be developed.

4–6. Sensitivity analysis and professional judgment

a. One approach to estimating stage uncertainty that can always be used, and was described for hydraulic modeling in paragraph 4–4.b(1)(a), is to estimate the upper and lower bounds on stage for a given discharge and convert that stage range to the needed uncertainty distribution. For example, if stage uncertainty follows a symmetrical Normal distribution, 95% of the possible stage range is encompassed by stages that are two standard deviations above and below the best estimate. Professional judgment could thus be applied to estimate the “reasonable” upper and lower bounds of stage, and the standard deviation estimated as the total range divided by four to determine a standard deviation.

b. Sensitivity analysis in which reasonable likely combinations of upper and lower bounds of model parameter values are also used to obtain a range of predicted stages for a given discharge could augment or serve as an alternative to the range determined from professional judgment.

4–7. Stage uncertainty for additional project conditions

The discussion in this chapter focused on estimating stage uncertainty for the existing without-project condition. The stage uncertainty for three additional conditions must also be evaluated: (1) the future without-project condition; (2) the initial with-project condition; and (3) the future with-project condition. If the flow and conveyance characteristics are expected to be markedly different from the existing without-project condition for any of those three additional conditions, further analysis is appropriate to estimate stage uncertainty specific to the differing conditions. If flow conditions and conveyance are expected to remain similar to the existing without-project condition, then stage uncertainty may be taken to be the same or similar to the existing without-project condition.

Chapter 5

Uncertainty in the Stage-Damage Transformation

5–1. Function development

a. Traditional method. A stage-damage function is a summary statement of the direct economic cost of floodwater inundation for a specified river reach. For residential structures, the function traditionally has been developed as shown in Table 5–1. Similar information on value, damage as a function of depth, and flood depth at a site is necessary to develop the stage-inundation damage functions for nonresidential structures and for other property.

b. Function development accounting for uncertainty. Much is uncertain in the procedure outlined in Table 5–1. Table 5–2 identifies some of the sources of uncertainty (IWR 1988). The rest of this chapter presents methods for describing uncertainty of these individual components. With these descriptions, an aggregated description of uncertainty of the stage-damage function can be developed with the procedure shown in Figure 5–1. In the figure, probabilistic descriptions are developed to describe uncertainty or errors in estimating (1) the first-floor elevation of the structure; (2) the percent damage to a structure for a given water depth; (3) the structure value; (4) percent damage to the contents for a given water depth; and (5) the structure-to-content value ratio. Each is sampled to develop a description of the overall uncertainty or error.

c. Uncertainty description. This uncertainty description then can be included in the sampling for EAD and AEP computations, as described in Chapter 2. This chapter addresses only the description of uncertainty in inundation flood damage to residential structures. However, the approach presented can be used to describe uncertainty of the stage-damage function for businesses, utilities, transportation and communication systems, and flood emergency costs.

Table 5–1
Traditional procedure for development of stage-damage function

Step	Task
1	Identify and categorize each structure in the study area based on its use and construction.
2	Establish the first-floor elevation of each structure using topographic maps, aerial photographs, surveys, and/or hand levels.
3	Estimate the value of each structure using real estate appraisals, recent sales prices, property tax assessments, replacement cost estimates, or surveys.
4	Estimate the value of the contents of each structure using an estimate of the ratio of contents value to structure value for each unique structure category.
5	Estimate damage to the contents of each structure due to flooding to various water depths using a depth-percent damage function for contents for the structure category along with the value from Step 4.
6	Transform each structure's depth-damage function to a stage-damage function at an index location for the floodplain using computed water-surface profiles for reference floods.
7	Aggregate the estimated damages for all structures by category for common stages.

Table 5–2
Components and sources of uncertainty in stage-damage function

Parameter/model	Source of uncertainty
Number of structures in each category	Errors in identifying structures; errors in classifying structures
First-floor elevation of structure	Survey errors; inaccuracies in topographic maps; errors in interpolation of contour lines
Depreciated replacement value of structure	Errors in real estate appraisal; errors in estimation of replacement cost estimation-effective age; errors in estimation of depreciation; errors in estimation of market value
Structure depth-damage function	Errors in post-flood damage survey; failure to account for other critical factors: flood water velocity, duration of flood; sediment load; building material; internal construction; condition; flood warning
Depreciated replacement value of contents	Errors in content-inventory survey; errors in estimates of ratio of content to structure value
Content depth-damage function	Errors in post-flood damage survey; failure to account for other critical factors: floodwater velocity, duration of flood; sediment load; content location, flood warning

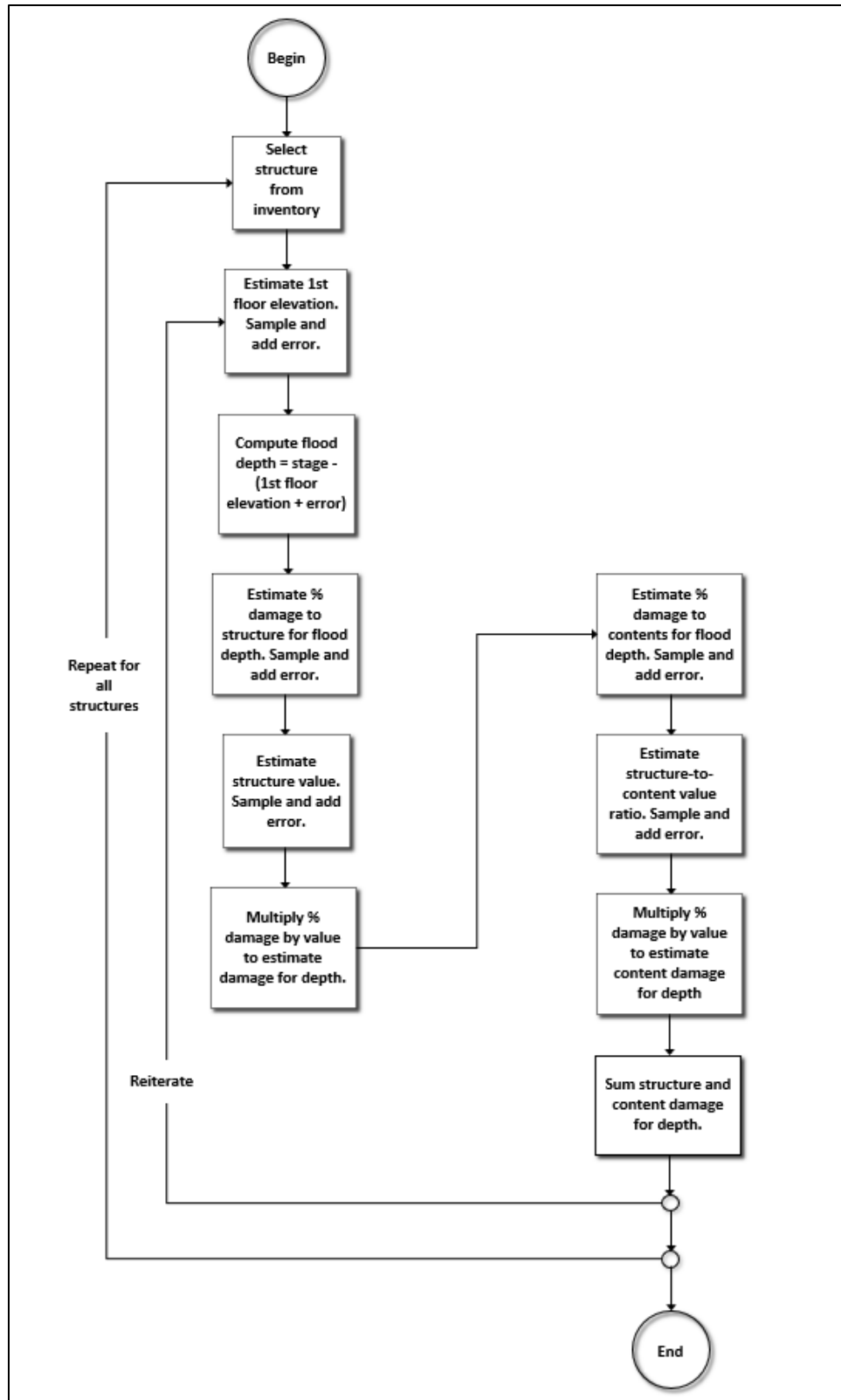


Figure 5–1. Development of stage-damage function with uncertainty description

5–2. Description of parameter uncertainty

a. *Structure value.*

(1) Structure value is a critical parameter of the stage-damage function, as it directly determines the structure damage and, indirectly, the content damage. Based on the “rational planner” model and the willingness-to-pay principle, depreciated replacement value is used as the appropriate measure of this value for USACE studies. Acceptable methods for estimating depreciated replacement value are summarized in Table 5–3.

(2) To develop a description of error or uncertainty in structure value, either professional judgement or sampling to fit a distribution may be used. A relative description of uncertainty is required to use that description for a group of structures (such as occupancy type) rather than a single structure alone. For instance, uncertainty about a Normally distributed, depreciated replacement value is often described in terms of the coefficient of variation (for example, the standard deviation of the replacement value divided by the mean replacement value), rather than simply the standard deviation. Coefficient of variation is defined by equation 5–1.

$$CV = \frac{\sigma}{\mu} \quad (5-1)$$

where:

μ = the mean of the parameter being examined

σ = the standard deviation of the parameter being examined

(3) Using the coefficient of variation allows two assets with different estimated replacement values to have the same relative amount of uncertainty, even though the value of each asset’s depreciated replacement uncertainties is not the same. For instance, if the average value of a vehicle in the floodplain is estimated to be worth \$10,000 with a Normally distributed standard deviation of \$1,000, then the coefficient of variation for this asset’s (the vehicle) depreciated replacement value is 10% (\$1,000/\$10,000). Thus, in this example, the uncertainty as measured by a \$500 standard deviation for a vehicle that is estimated to be worth \$5,000 is equivalent in relative terms to a standard deviation of \$2,000 for a vehicle that is estimated to be worth \$20,000.

b. *Professional judgment.* With this method, each structure’s value is estimated by an expert in real property valuation and is expressed as a range or as minimum and maximum values. From these, a uniform or triangular distribution can be fitted to describe the error. With a uniform distribution, all values between minimum and maximum are considered equally likely. With a triangular distribution such as that shown in Figure 5–2, the best estimate is considered most likely.

c. *Variable range.* Alternatively, a Normal distribution may be fitted. A common method for approximating the standard deviation in that case is to use the range of the variable. For instance, appraisers may attest that the depreciated replacement cost of a

structure is between \$60,000 and \$70,000. Dividing this range by four provides an estimate of the standard deviation of \$2,500. By implication, this method assumes that the error range is approximately equivalent to a 95% CI. One caution in using this method, as with any method using “expert judgment,” is that experts tend to be overly confident and provide an error range that is too narrow (Kahneman, Slovic, and Tversky 1982).

d. *Sampling to fit a distribution.* With this procedure, a sample of the structure values, stratified by structure category, is drawn from the real estate assessments and is used to estimate statistics that describe errors in each category. From these statistics, the parameters of a probability function are estimated. For example, from the mean and standard deviation of logarithms of values in each category, parameters of a Log-Normal distribution of values can be estimated. This approach uses natural variability as a proxy for knowledge uncertainty.

(1) Structure value uncertainty may be described as a function of the uncertainty about value per square foot and the square footage of a given structure.

(2) Suppose the value per square foot and square footage uncertainties for an occupancy type are independent, random variables (a potentially strong assumption, as there may be a positive correlation between square feet and value per square foot in typical real estate markets), and we wish to define the total structure value uncertainty as a coefficient of variation. Thus, equation 5–2 can be used to compute the total coefficient of variation in the structure value at the occupancy type level.

$$CV_{sv,sf} = \frac{\sqrt{Var(SV \times SF)}}{E[SV \times SF]} = \frac{\mu_{sf}^2 \mu_{sv}^2 \left[\left(\frac{\sigma_{sf}}{\mu_{sf}} \right)^2 \left(\frac{\sigma_{sv}}{\mu_{sv}} \right)^2 + \left(\frac{\sigma_{sf}}{\mu_{sf}} \right)^2 + \left(\frac{\sigma_{sv}}{\mu_{sv}} \right)^2 \right]}{\mu_{sf} \times \mu_{sv}} \quad (5-2)$$

where:

SV and SF = the random variables value per square foot and square footage, respectively

μ_{sv} and μ_{sf} = the occupancy type-level means of value per square foot and square footage, respectively

σ_{sv} and σ_{sf} = the occupancy-type-level standard deviations of value per square foot and square footage, respectively

e. *Uncertainty.* If the uncertainty about square footage can be reasonably described as negligible, then the relative uncertainty about structure value is equal to the relative uncertainty about value per square foot.

Table 5–3
Methods for estimating depreciated replacement value

Approach	Description	Comments
Replacement cost estimating using Marshall & Swift Foundational Property Valuations service (Marshall & Swift 2024) (both printed and computerized versions)	Develops a replacement construction cost estimate based on information on foundation, flooring, walls, roofing, heating system, plumbing, square footage, effective age, and built-in appliances. This estimate is then adjusted for depreciation.	No independent assessments of errors in resulting depreciated replacement value are available. Experienced building contractors and others may be useful in estimating error bounds.
Real estate assessment data	Involves adjusting real estate tax assessment values for deviations between assessed value and market value and subtracting land component of market value. Presumption is that remainder is depreciated replacement value of structure.	No general method for estimating error in resulting structure values. One approach is to compare results from a sample of individual structures to results using replacement cost estimating method. Random stratified sampling techniques should be used to assure that all structure categories and construction types are verified. Alternatively, verification should cover structure categories and construction types that are located in most flood-prone segment of study area. Sample size for verification should be sufficient to establish range of error, even if it is not large enough to empirically develop a frequency distribution of error in structure values. An easy, useful, approach to quantifying errors in structure values using real estate assessments is to query local real estate experts and appraisers.
Recent sales prices	Requires sufficient recent property sales in area for each type of structure and construction types for which structure value is to be estimated. As with real estate assessment data, adjustments must be made to subtract land value to yield structure component.	Theoretically, sale prices should be a more accurate basis for estimating depreciated replacement value than real estate assessments. Obvious source of error is estimating and subtracting land portion of sales price to yield structure value estimate. Methods for estimating error when using recent sales prices to estimate structure values are same as those when using real-estate assessment data.

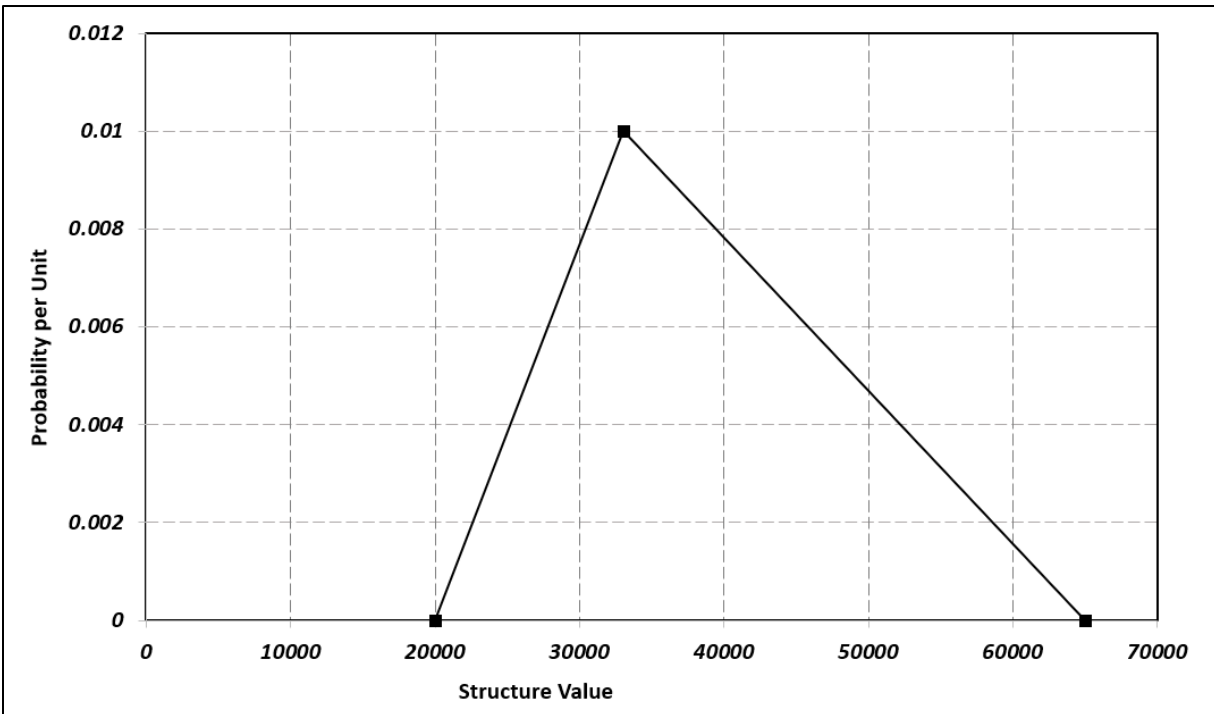


Figure 5–2. Example triangular distribution of structure value

f. Content-to-structure value ratio.

(1) A common approach to estimating residential content value is to estimate that value as a fraction of the structure value. This approach mimics that typically employed by residential casualty insurers in setting rates and content coverage for homeowners' insurance; however, the value of contents found in any structure is highly variable. It may reflect the wealth and income of the occupants, their personal tastes and lifestyles, and a variety of other factors.

(2) Table 5–4 shows computed means and standard deviations of content-to-structure value ratios based on large samples of the Federal Emergency Management Agency's (FEMA) Flood Insurance Administration claims records. These nationwide averages are not appropriate for all cases, but in lieu of better site-specific information, the values in this table can yield estimates of parameters of a probability distribution of errors. The best use of the table is to specify a triangular distribution using the minimum and maximum values directly and using the mean as the most likely value.

(3) In some instances, content values may have been developed by using survey or inventory methods. Recognize that where content values are directly measured there is still uncertainty in the actual content value due to errors in inventories, pricing, and age. It is difficult to judge the overall effect of these potential sources of uncertainty on content values. One easily implemented method is to request that the individual completing the survey or the inventory provide an estimate of the accuracy of the provided information.

Table 5–4
Content-to-structure value ratios ^{1,2}

Structure Category	Number of Cases	Mean	Standard Deviation	Minimum	Maximum
1 story – no basement	71,629	0.434	0.250	0.100	2.497
1 story – basement	8,094	0.435	0.217	0.100	2.457
2 story – no basement	16,056	0.402	0.259	0.100	2.492
2 story – basement	21,753	0.441	0.248	0.100	2.500
Split level – no basement	1,005	0.421	0.286	0.105	2.493
Split level – basement	1,807	0.435	0.230	0.102	2.463
Mobile home	2,283	0.636	0.378	0.102	2.474
All categories	122,597	0.435	0.253	0.100	2.500

Notes:

¹ The means and deviations in the table are less than ratios commonly used by casualty insurance companies, but they reflect replacement costs rather than depreciated replacement costs.

² Research by the Institute for Water Resources (USACE 1992a, USACE 1992b) suggests that errors may be described best with an asymmetrical distribution, such as a Log-Normal distribution. In that case, the parameters of the error distribution cannot be estimated simply from the values shown in this table.

g. First-floor elevation. Estimation of flood damage using depth-percent damage relationships requires specification of the first-floor elevation of the structure. This elevation may be established via field or aerial surveys or by reference to topographic maps. Table 5–5 describes the elevation errors for each of these methods (HEC 1986). This description of errors can be used to estimate parameters of a probability distribution of errors. If a Gaussian Normal distribution is assumed to model the errors, the indicated standard deviation can be used. Alternatively, professional judgment can be used to determine the most likely, minimum, and maximum values, and a triangular distribution can be fitted.

h. Foundation height and ground elevation. If foundation height and ground elevation is used in the risk assessment, then first floor elevation uncertainty should be described as the combination of foundation height uncertainty and topographic uncertainty. The variance of error about the first-floor elevation is then the sum of the variance of error about the ground elevation with the variance of error about the foundation height. It may be safe to assume that the error about the ground elevation is independent of the foundation height if the two measurements are collected through different methods (for example, ground elevation through light detection and ranging (LiDAR) or foundation heights by counting steps). If the sources of error are assumed to be independent, then the overall standard deviation of error may be calculated as in equation 5–3.

$$\sigma_{x,y} = \sqrt{\sigma_x^2 + \sigma_y^2 + 2(\sigma_{xy})} \quad (5-3)$$

where:

σ_x^2 and σ_y^2 = the variances in the two sources of uncertainty

σ_{xy} (not to be confused with $\sigma_{x,y}$) = the covariance in the two sources of uncertainty

Table 5–5
First-floor elevation error and standard deviation calculated from results in accuracy of computed water surface profiles

Method of Elevation Estimation	Error, ¹ in ft	Standard Deviation, ² in ft
Field survey, hand level	0.2 @ 50'	0.1
Field survey, stadia	0.4 @ 500'	0.2
Field survey, conventional level	0.05 @ 800'	0.03
Field survey, automatic level	0.03 @ 800'	0.02
Aerial survey, 2-ft contour interval	0.59	0.3
Aerial survey, 5-ft contour interval	1.18	0.6
Aerial survey, 10-ft contour interval	2.94	1.5
Topographic map, 2-ft contour interval	1.18	0.6
Topographic map, 5-ft contour interval	2.94	1.5
Topographic map, 10-ft contour interval	5.88	3.0

Notes:

¹ Errors for aerial survey and topographic maps are calculated at the 99% confidence level, assuming the deviations from the true elevation are Normally distributed with zero mean and indicated standard deviations.

² Standard deviation for field survey assumes that error represents a 99% CI and assuming Normal distribution.

5–3. Description of uncertainty in form of depth-damage functions

a. The final elements required to develop the stage-damage function are the structure and content depth-damage functions. These are models of the relationship of depth of flooding at a structure to the damage incurred. As with other models used in plan evaluation, these models are not known with certainty. For example, Table 5–6 shows factors that should be included in, but that are commonly omitted from, a damage prediction model.

b. The impact of including or excluding these factors may be explored through sensitivity analysis, with the factors shown in Table 5–6 incorporated to develop a more complex relationship (IWR 1988). For example, if duration is incorporated, a depth-duration-damage function might be developed. This function can be used in the EAD and AEP computations. In display of plan performance, the computed EAD and AEP values is then identified as those computed with the alternative models.

c. An alternative approach suggested by the Institute for Water Resources is to treat model uncertainty directly as parameter uncertainty. In that case, the error in percent damage for each depth is described with a Gaussian Normal probability distribution.

Table 5–6
Factors, other than depth, that influence damage

Factor	Effect
Velocity	Major factor aggravating structure and content damage. Limits time for emergency floodproofing and evacuation. Additional force creates greater danger of foundation collapse and forceful destruction of contents.
Duration	May be the most significant factor in the destruction of building fabric. Continued saturation causes wood to warp and rot, tile to buckle, and metal objects and mechanical equipment to rust.
Sediment	Can be particularly damaging to the workings of mechanical equipment and can create cleanup problems.
Frequency	Repeated saturation can have a cumulative effect on the deterioration of building fabric and the workings of mechanical equipment.
Building material	Steel frame and brick buildings tend to be more durable in withstanding inundation and less susceptible to collapse than other material.
Inside construction	Styrofoam and similar types of insulation are less susceptible to damage than fiberglass and wool fiber insulation. Most drywall and any plaster will crumble under prolonged inundation. Waterproof drywall will hold up for long periods of inundation. Paneling may be salvageable when other wall coverings are not.
Condition	Even the best building materials can collapse under stress if the construction is poor or is in deteriorated condition.
Age	May serve as an indicator of condition and building material.
Content location	Important factor, as small variations in interior location of contents can result in wide variation in damage.
Flood warning	Major reduction in both content and structural loss can be made through flood fighting and evacuation activities when there is adequate warning.

5–4. Stage-damage function using the opinions of experts

a. The approach illustrated in Figure 5–1 does not reflect the methodology typically employed to estimate damages for nonresidential property. For these unique properties, the stage-damage function may be developed as a consequence of post-flood surveys or through personal interviews with plant managers, plant engineers, or other experts. Then, instead of employing dimensionless depth-percent damage functions, damages incurred at various water-surface elevations are approximated directly.

b. To describe uncertainty in these cases, the experts should be asked to estimate the most likely damage for a range of depths, to provide a range of damages for each depth, and provide their confidence that the range contains the actual damage value

that would occur. These opinions on the range and confidence can be used to estimate the parameters of a probability distribution that describes error for each depth. If the respondent cannot or will not provide information other than an estimated range, the analyst can use the mid-point of the range as the mean and one-fourth of the range as the standard deviation; this assumes a Normal distribution of errors and inclusion of 95% of all damages in the stated range.

5–5. Approach with limited data

In some flood risk management planning studies, data in the detail or format for proper analysis of uncertainty is not available, and the cost to enhance existing data to conduct an uncertainty analysis is not justified. In those cases, the planning team must take care to acknowledge likely sources of uncertainty and their impact.

- a. The mean stage-damage function is likely most sensitive to error in the first-floor elevation, other things being equal.
- b. The error in damage at any stage is not symmetrically distributed around the mean damage. This is particularly true at the lower stages because damage cannot be negative. Thus, the probability of overestimating damage is greater.
- c. Although the dispersion of damages about the mean, as measured by the standard deviation, increases with increasing stage, the coefficient of variation (standard deviation divided by mean) decreases with increases in stage. Thus, the error in damage, expressed as a fraction of the mean damage decreases as the stage (and hence, mean damage) increases. This is due in part to the truncation of damage at zero. It is also a consequence of lessening sensitivity of error to error in first-floor elevation as stage increases.

5–6. Event-based stage-damage transformation

The development of a stage-damage function is a required step for the curve combination approach to calculating EAD. However, watershed modeling approaches as described in paragraph 2–8 can be paired with event-based, stage-damage transformations using the same approaches described above on an event-by-event basis instead of developing and storing a function between stage and damage. The event-based, stage-damage transformation allows the risk assessor to model time dependence in flood damage by comparing reconstruction rates to the time between damaging events of a given realization, where the realization reflects the entire period of analysis.

Chapter 6

Evaluating Flood Risk with Levee or Floodwall Systems

6–1. Overview

A levee or floodwall reduces flood risk by separating the channel from the floodplain and reducing the frequency with which the flood hazard inundates the floodplain. Flood risk is reduced but not eliminated because there is always a probability that water can access the floodplain as a result of breach failure at stages below the crest or from overtopping. The procedures in this chapter demonstrate how to assess flood risk in areas defended by levees or floodwalls. The procedures apply equally to levees or floodwalls, though typically only one measure at a time is referred to at a time in this chapter. A simplified cross section of a channel with an adjacent floodwall and example channel water surface (exterior) and floodplain water surface (interior) are illustrated in Figure 6–1. In the illustrations, water surface elevations for a range of peak flood stages are depicted as (a), (b), and (c).

a. The water surface elevation of the channel indicated by the solid line (a) in Figure 6–1(a), is below the toe of the levee and has no access to the floodplain. The floodwall is above the water surface, and therefore, has no influence on inundation of the floodplain.

b. The water surface elevation of the channel indicated by the solid line (b) in Figure 6–1(a) is loading the floodwall but has no access to the floodplain. The solid line illustrates a scenario where the floodwall successfully keeps the water in the channel and out of the floodplain. Figure 6–1(b) shows that stage (b) in the channel has no stage in the floodplain if the floodwall does not breach. The water surface elevation of the channel indicated by the dashed line (b) in Figure 6–1(a) represents a scenario where the floodwall failed (breached), and water accessed the floodplain. For illustrative purposes, Figure 6–1(b) depicts a hypothetical scenario where floodplain stage in the failed scenario is equal to the stage in the channel. However, the stage in the floodplain can be different than the channel depending on hydraulic considerations.

c. If the water surface elevation in the channel were to exceed the crest of the floodwall, as illustrated by line (c) in Figure 6–1(a), water in the channel can access the floodplain. This scenario reflects a scenario where the floodwall fails and breaches when the water surface elevation in the channel exceeds the floodwall crest. This failure mode is referred to as overtopping. For illustrative purposes, Figure 6–1(b) depicts a hypothetical scenario where floodplain stage in the breach scenario is equal to the stage in the channel. However, the stage in the floodplain can be different than the channel depending on hydraulic considerations.

d. The relationship of breach and no-breach scenarios over a range of water surface elevation in the channel and in the floodplain is illustrated in Figure 6–2(b). The solid blue line reflects the relationship has the presence of a floodwall that did not fail. There is no water in the floodplain until the channel stage reaches the crest of the floodwall, at which point the water surface elevation in the floodplain reflects the inability

of the floodwall to contain the flood. This figure illustrates the overtopping scenario. This example depicts a hypothetical scenario where floodplain stage in the failed scenario is equal to the stage in the channel. Paragraph 6–4 describes situations where the channel and floodplain water surface elevations are not equal.

e. Flood risk assessments account for the probability of failure of the levee or floodwall using a system response curve—a relationship between the probability of failure and the hazard loading. The system response curve describes the probability the levee or floodwall will fail to contain the water to the channel, given the water surface elevation in the channel. The remaining sections of this chapter describe how to incorporate the system response curve into the methodology presented in Chapter 2.

f. Information on the design of levees and floodwalls can be found in EM 1110-2-1913 and EM 1110-2-2502. Information on the development of geotechnical system response curves to estimate the probability of levee breach can be found in ETL 1110-2-588.

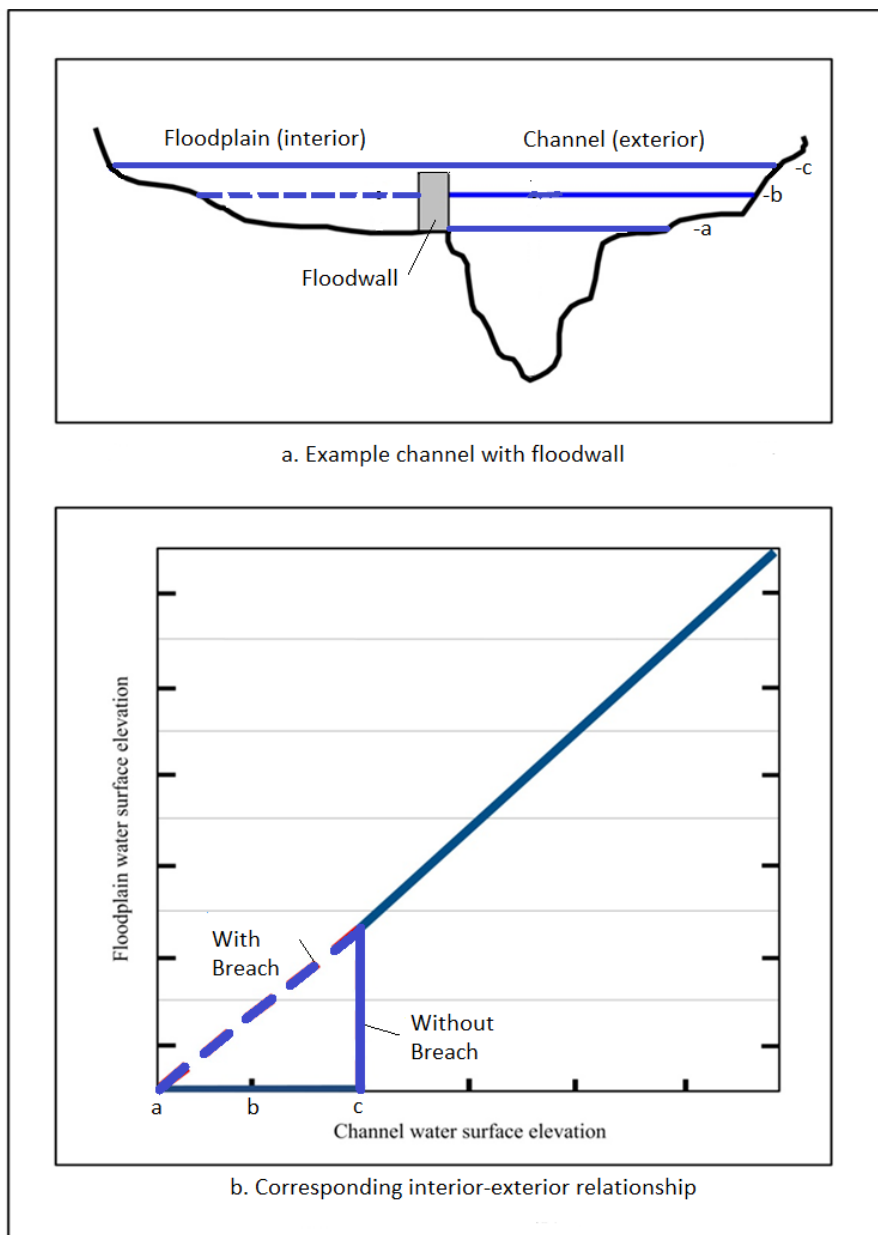


Figure 6-1. Example of cross section and simple interior-exterior function

6-2. Incorporation of system response in risk assessment procedures

Since the frequency of floodplain inundation depends on the ability of a levee or floodwall to contain floodwater, flood risk assessments should account for the probability of failure of the levee or floodwall. Uncertainty about levee or floodwall structural performance is described with a system response curve.

a. The combined system response curve in Figure 6-2 (black dashed line) displays, for any selected channel water surface elevation (river stage), the probability that the levee or floodwall will fail to prevent the external flood from entering the interior

floodplain. The combined system response curve represents the combined effects of all failure modes that may contribute to levee or floodwall failure: overtopping, stability, underseepage, erosion, and through seepage. The level of analysis and scope applied in developing the system response curve should be appropriately scaled to the specific circumstances of a study. At a minimum, the overtopping failure mode must be explicitly included in the risk assessment.

b. The analyst should select the additional failure modes to be reflected in the combined system response curve. If the other failure modes are unimportant for economic risk, then overtopping may be the only failure mode represented in the system response curve used in the economic risk calculation. Detailed USACE guidance on evaluating failure modes and developing the system response curve is given in EM 1110-2-1913.

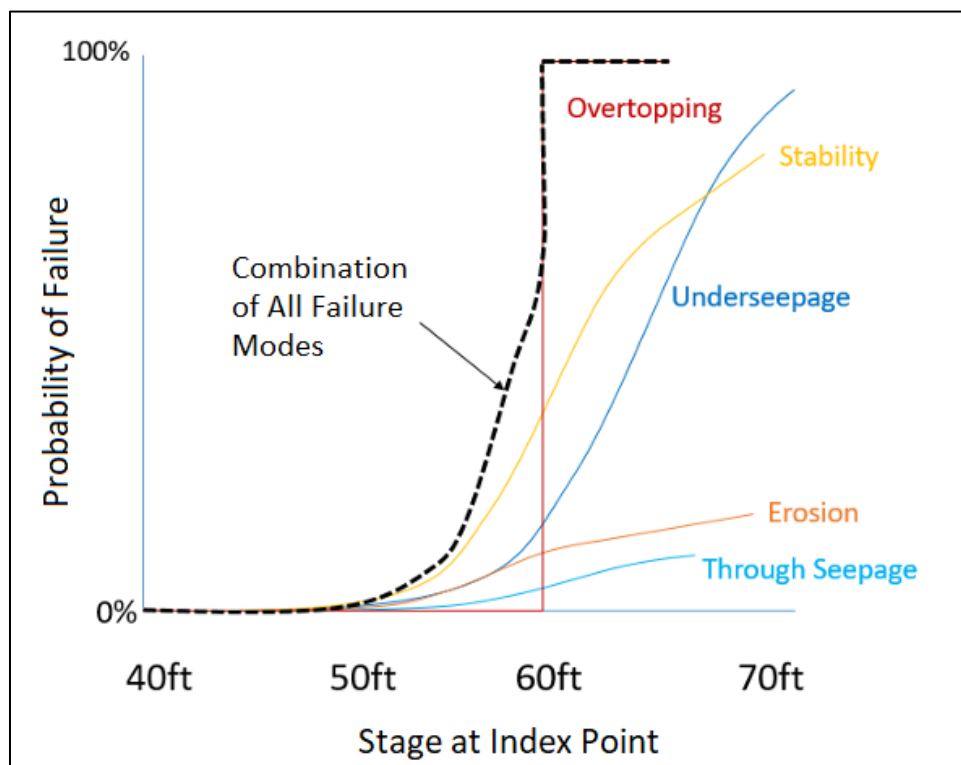


Figure 6-2. Example levee system response curve (modified from ETL 1110-2-588)

c. Errors in defining the levee or floodwall structural performance may arise from the following uncertainties:

- (1) The true dimensions of the levee or floodwall.
- (2) The behavior of unseen properties of the levee, including core materials and materials beneath the levee. These properties have a significant impact on seepage through and under the levee.

(3) Levee slope stability. Slope failure may lead to catastrophic failure of the levee, with a resulting breach. This failure could lead to interior flooding even when exterior stage is far below the levee crown elevation.

(4) Levee condition, including existence of animal burrows, impact of vegetation, and other levee conditions.

(5) Impacts of surface erosion and wave attack on levee reliability, which could lead to catastrophic failure.

(6) The status of the closure of openings in the line of defense provided by the levee.

(7) Flood fighting, emergency repairs, and other actions that may be taken to avert overtopping or failure of the levee.

(8) The impact of seismicity on levee or floodwall structural performance.

d. With a stage-damage function calculated and a system response curve identified, risk can be calculated as a function of the levee. From the two functions, damage and the probability of levee failure are matched according to stage. Damage is multiplied by the probability of levee failure to produce a relationship between probability-weighted damage and the stage of the channel. If damage is expected in a nonfailure scenario before the levee is overtopped, then the probability of nonfailure times the nonfailure damage is included in the probability-weighted average. The resulting probability-weighted, stage-damage function is integrated to calculate EAD. This procedure is described in paragraph 2–3.f.

6–3. Uncertainty in flood risk

a. The risk computation strategy shown described in Chapter 2 is implemented in USACE software with a Monte Carlo sampling framework. In that framework, the annual maximum discharge probability function is sampled randomly, considering the range of uncertainty about the function. Various transformations are made, as discussed in Chapter 2, with each function also sampled randomly, adding or subtracting a component of error to discharge or stage values, depending on the uncertainty about the function.

b. In the performance computation, the system response curve is used to assess the likelihood that levee or floodwall breaches for stages below the crest or due to overtopping. If the levee or floodwall is considered to breach, a stage in the floodplain representing the breach is used to compute the consequences. If the levee does not breach, no consequences are assumed. This computation is repeated for each sample.

6–4. Relationship between channel stage and floodplain stage

a. The breach of a levee or floodplain can result in floodplain water surface elevations that are different than the channel. Hydraulic modeling of a range of flood magnitudes should be used to determine if it is appropriate to assume they are equal. If the assumption of water surface equalization is inappropriate, then hydraulic modeling of assumed no breach and assumed breach scenarios should be conducted over the range of flood magnitudes. The no-breach assumption is used to represent the stage-discharge relationship in the channel relative to the system response curve at the index point. The floodplain stage from the breach assumption is then used to estimate consequences in the floodplain if the levee were to breach.

b. Uncertainties in the assumption of equalization are typically not quantified in a risk analysis. However, understanding the sources of uncertainty is necessary to develop a hydraulic model that is reasonable. Uncertainties in developing the model may arise from the following:

(1) Topographic and bathymetric data from which the model's representation of the channel and floodplain geometry is inferred.

(2) Hydraulic properties of the levee when it functions as a weir. For example, TOL elevation likely varies along the stream because of construction methods, differential settling, erosion, and other hydraulic properties. This variation leads to uncertainty about the discharge computed as a function of overflow head and length of the overtopped levee or floodwall.

(3) Hydraulic modeling methods. Flow over a levee and across a floodplain may be simulated with a steady or unsteady model, in one or two dimensions.

(4) Dimensions and locations of structures that impede or control flow in the floodplain. These structures include roadway embankments, culverts, bridges, piers, and even buildings in the floodplain.

(5) Modeling assumptions including the roughness coefficient (Manning's n values), overflow weir coefficient, and expansion and contraction coefficients.

(6) Impacts of erosion and deposition before or even during a flood event.

(7) Wind and wave run-up and overtopping discharge.

(8) Levee breach assumptions. In that case, the interior-exterior function should reflect flow through the breach, not just flow over the levee. A hydraulic model, with the opening described, can provide information necessary to construct the appropriate interior-exterior water surface elevation relationship in that case.

(9) Performance of interior drainage facilities. Water may rise to damaging levels in the interior floodplain, even as the levee contains the exterior channel flow.

6–5. Interior drainage facility and closure system performance and uncertainty

a. A levee or floodwall reduces the hazard of the interior area to flooding from the exterior channel. However, the interior area is exposed to another hazard: flooding due to precipitation falling behind the line of defense provided by the levee. The levee, when performing as designed, blocks flow of water from the channel onto the floodplain. But that same levee also blocks the natural runoff of excess water into the channel from the floodplain. Interior drainage facilities accommodate this runoff, reducing the hazard.

b. Interior drainage facilities may include the following:

(1) Channels or pipes and associated facilities to collect interior area runoff and carry the runoff to the levee.

(2) A detention pond at the line of defense to collect and store interior runoff until it can be conveyed through or over the levee to the channel.

(3) A culvert through which the interior drainage is conveyed to the channel. Inflow to the culvert may be from a pond or directly from the interior channels or pipes. The culvert is a gravity-flow structure that functions when the head differential between the interior stage and the exterior stage is sufficient to cause flow into the channel. Commonly, such a culvert includes a valve or gate that may be closed to ensure that floodwaters do not enter the interior area if the channel stage is high.

(4) A pump to remove water from the interior area when gravity flow is not possible because of higher heads in the exterior channel. Inflow to the pump may be from the pond or directly from the interior channels or pipes.

c. EM 1110-2-1413 provides guidance for planning, analyzing, and designing interior drainage facilities that include all these facilities. EM 1110-2-1413 also identifies procedures for quantifying the interior hazard due to precipitation and runoff in the interior floodplain and the coincident hazard performance due to water within the channel.

d. Risk assessment procedures described herein generally are consistent with methods described in EM 1110-2-1413, so an independent interior risk assessment should follow the guidance given there. However, to accurately characterize the total risk in an interior area, the hazard from both interior and exterior flood sources must be combined for the EAD computation. When evaluating effectiveness of measures, it is inappropriate to compute EAD separately for each flood source and add those values together; this results in double-counting some amount of damage.

e. In some cases, the line of defense provided by a levee or floodwall must have openings for roadways or railways. Gates or other structures are included to block flow through the openings. HEC (1996) provides an in-depth description of closure types, with assessment of system performance, function, and workability of each closure gate.

f. Closure structures must be operated to prevent flow. Flow prevention requires full functionality and operation of all components of the structure, and action to close the structure. The success or failure of the components and actions has a significant impact on the risk computed. If gates are not closed or function improperly, risk is significantly greater, as water from the channel will flow through the opening and onto the floodplain. Uncertainty associated with gate closure must be included in the risk assessment.

g. No direct capability for uncertainty analysis associated with closure of openings is included in commonly available risk assessment software. Nevertheless, this uncertainty must be considered in USACE risk analyses. The following are reasonable options for accounting for the uncertainty:

(1) Adjusting the system response curve to reflect uncertainty associated with necessary actions and functionality. In this case, the probability of failure for a given exterior stage increases to reflect the uncertainty.

(2) Undertaking a sensitivity analysis to determine risk without closing and risk with closing. The results may be weighted by the likelihood of successful closure of openings. HEC (1996) offers weighting suggestions.

6–6. Wind waves and uncertainty

a. Wind waves can be a significant factor in evaluating flood risk associated with levees in areas with long, open-water fetch lengths. Wind waves are short-period waves that are not included in typical procedures used to estimate flood stage assuming still-water conditions. The two primary wind wave considerations for evaluating flood risk associated with levee measures are the probability of breach due to wind wave erosion and wind wave overtopping.

b. Specific guidance for wave computations is provided in EM 1110-2-1100, with requisite calculations differing somewhat for riverine, large lake, estuarine, and coastal situations. Common model parameters include fetches, wind or coastal wave directions, and wind speeds. From these parameters, wave heights, periods, and directions are computed. If uncertainty associated with various parameters is described with a probability distribution, the resulting increment in stage due to waves, given the still-water elevation, can also be described with a probability distribution.

c. The occurrence of high wind speeds may not occur at the same time as high flood stage. Therefore, coincident frequency analysis should be included in the assessment.

d. One potential method of accounting for the probability of wind wave related failure for levees is to include the failure mechanisms in the development of system response curves using event tree analysis. For example, for a flood event with a given peak still-water elevation, estimate the probability of wind wave erosion and wind wave overtopping causing a breach of the embankment.

Chapter 7

Evaluating Flood Risk for Reservoirs or Diversions

This chapter describes how flood risk involving a reservoir or diversion can be estimated. Sources and methods to evaluate the uncertainty in reservoir and diversion system performance are identified. EM 1110-2-1417, EM 1110-2-1420, and EM 1110-2-3600 provide guidance for planning, designing, and evaluating system performance of flood management (flood control) reservoirs or diversions.

7–1. Evaluating flood risk for a reservoir

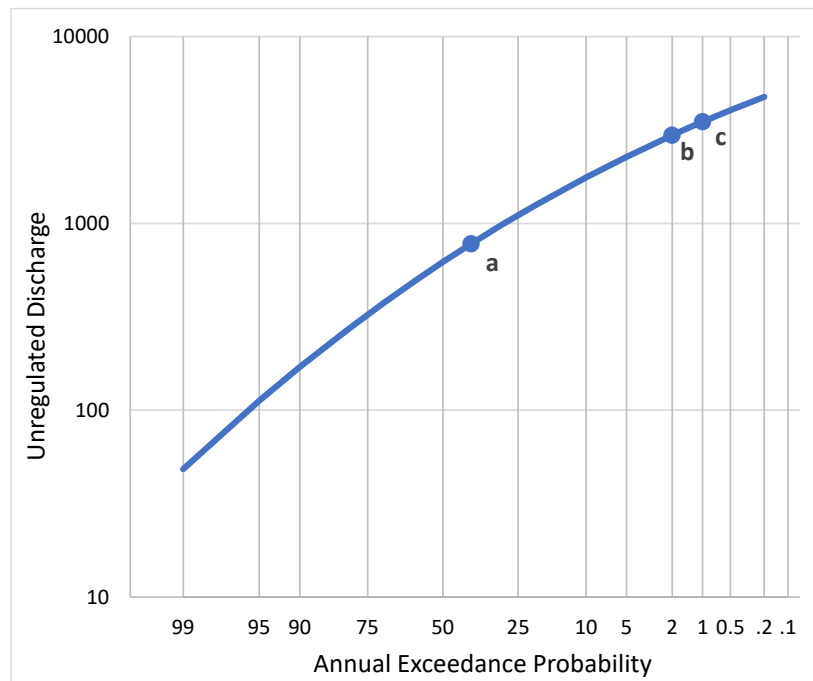
a. A reservoir reduces flood risk by reducing peak downstream flows. This is accomplished by maintaining empty storage space in the reservoir during the flood season, and then temporarily storing upstream runoff during a flood and releasing that water at a lower rate over a longer duration. Those releases are managed to flow limits (called objective flows) at downstream locations (control points) that are defined based on flood consequences. When the magnitude of an upstream flood exceeds the capacity of the reservoir to manage the release to the objective flow limit (when flood volume exceeds storage space), priority is given to reducing the chance of dam failure by discharging necessary flows over a spillway, which may exceed downstream flow limits.

b. A reservoir's flood management operation changes the shape and form of the discharge-frequency function. While the unregulated inflow frequency function may look like Figure 7–1(a), the regulated outflow frequency function has a shape similar to that shown in Figure 7–1(b). The shape of the outflow frequency curve is a function of the inflow flood frequency. For inflow floods smaller than point (a), the outflow is less than the downstream objective flow rate and the inflow is released with minor regulation to maximize storage space available for later inflows. For inflow floods larger than point (a) but less than point (b), the reservoir's available (empty) storage capacity is sufficient to manage the outflow to the objective flow rate. Thus, for some range of exceedance probability, the peak regulated discharge immediately below the reservoir is the same.

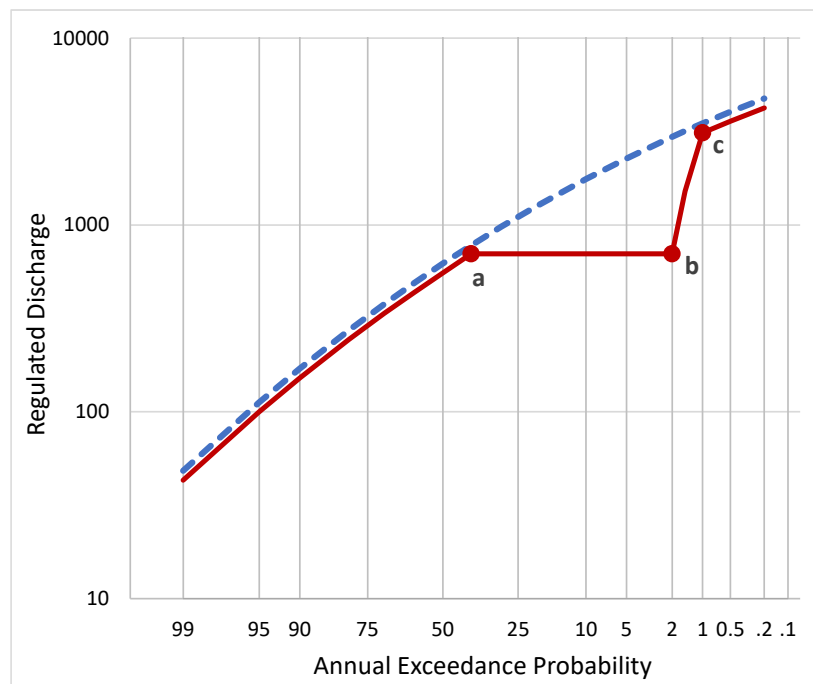
c. For rarer inflow events, the reservoir does not have adequate capacity to store the volume of water needed to manage peak release to less-than-objective release rate. At point (b) in Figure 7–1(b), the regulated function again changes slope to show the inability to maintain objective release. For flood magnitudes greater than point (b) but less than point (c), regulated discharges increase, eventually to the level at which they equal or nearly equal unregulated discharges. This increase reflects the inflow flood volume, becoming large enough to fill the available (empty) reservoir storage space before the flood peak, causing outflows to match inflows at inflows greater than point (c). The inflow values at which the outflow changes occur (points (a), (b) and (c) in Figure 7–1(b)) depend on the volume of the reservoir, release capacity, the manner of operation, and the downstream channel capacity.

d. The probabilities at which those changes occur depend on the inflow frequency functions and the flood event volume for a given peak inflow. Note that Figure 7–1(b) depicts regulated discharge immediately below a regulation structure, such as a

reservoir. Some locations are impacted by discharge regulation by several upstream reservoirs and diversions, and thus have a regulated discharge frequency curve with less obvious features than this figure.



a. Unregulated Discharge Frequency Function



b. Regulated Discharge Frequency Function

Figure 7–1. Unregulated inflow discharge probability and regulated outflow discharge probability functions

7-2. Unregulated-to-regulated transform

a. In the risk computation procedure shown in Figure 2-5, based on representing the system with summary relationships or curves, the unregulated discharge-frequency function is transformed to a regulated discharge-frequency function, and thus to a model of flood frequency in a system that includes reservoir(s) with flood storage. The regulated discharge-frequency function is then used in the risk computation depicted in Figure 2-3, as the discharge-frequency function in the upper right. The unregulated-to-regulated discharge transform may be developed using methods identified in EM 1110-2-1417 and summarized below. None of the methods require an estimate of the AEP of inflow or outflow peaks to define the transform function.

(1) *Continuous observations of unregulated and regulated discharge.* In this case, the annual maximum reservoir inflow peaks and the corresponding outflow peaks are determined from inspection of a historical time series. Use of this approach depends on having a long record of regulated flows that resulted from a consistent operation infrastructure and strategy over the entire record. Both inflow peaks and outflow peaks are plotted, as illustrated in Figure 7-2, and a function is fitted graphically. This approach has the advantage of using observed flow data, thus implicitly including actual operation decisions and features of the physical system. The selected period of record used in this method should include only years that have consistent water control operations, and the same reservoirs must have existed within the system for entire time. This requirement often limits the length of record that is available.

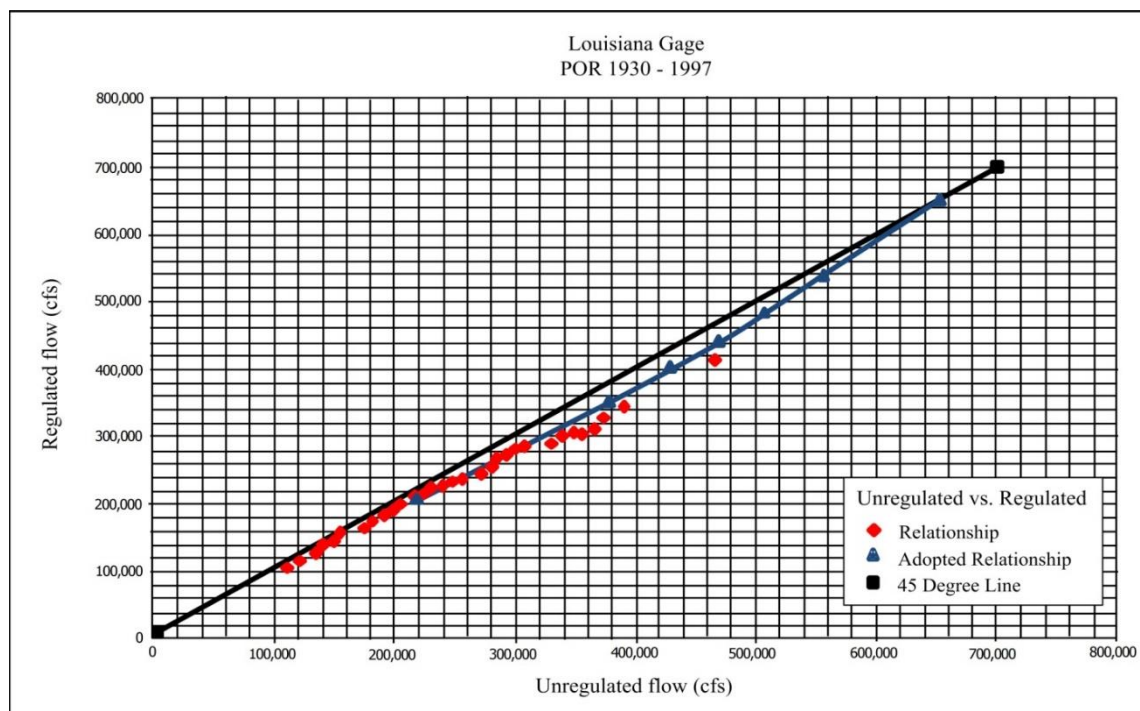


Figure 7-2. Example of unregulated-regulated discharge transform, fitted with historical data

(2) *Continuous simulation of unregulated and regulated discharge.* This continuous simulation approach may be used when adequate observed regulated flow data is not available due to lack of gaged data, a reservoir still in the planning phase, or because the historical record does not reflect consistent regulation or the proposed water control plan. With this procedure, a simulation model of the reservoir (or reservoir system) operated with a specified water control plan is developed. The model is used to simulate a continuous record of historical unregulated flows to compute the resulting regulated flows at the site of interest. Unregulated peak inflows and simulated peak outflows are then used in place of observed flows to develop a relationship as shown in Figure 7–2. If unregulated time-series data are not available, they may be generated by hydrologic simulation models.

(3) *Calibrated model to compute inflows.* For example, a calibrated rainfall-runoff-routing model with observed rainfall data as the forcing function may be used to generate streamflow. When a reservoir already exists, unregulated inflows can also be estimated from computed inflow, based on observed regulated outflows and changes in storage. EM 1110-2-1415 suggests that simulations of scaled-up historical events be used to extend the transform to include more rare events, with a maximum scaling factor of three.

(4) *Simulation of representative flood events.* With this approach, a smaller set of unregulated inflow floods is selected for simulation in a reservoir system model, as described in paragraph 7–2.a(2). This set of floods may be historical or hypothetical flood events, scaled or not. Each event is routed through the reservoir system model, simulating operation of the reservoir(s). It is important to include flood events with various different spatial and temporal patterns, or hydrograph shapes, to capture the variability in the volume associated with a given peak inflow. Inflow and corresponding outflow peaks are determined and plotted. The required unregulated-to-regulated transform is then fitted graphically or analytically. If this method is used, the simulated flood duration should exceed the reservoir’s critical duration as described in EM 1110-2-1420.

b. When performing risk analysis using the Monte Carlo event-sampling approach described in paragraph 2–8, rather than the summary-curve combination approach that uses an unregulated- to regulated-discharge transform function, the transformation process occurs directly in watershed simulation models, rather than with a transform function. A reservoir operation model is thus included in the sequence of watershed models that ultimately transform an unregulated flood event hydrograph through regulated discharge, channel stage, floodplain inundation (if a large enough flood event), and subsequent damage. Several reservoir simulation model alternatives might be developed, capturing both current operation and proposed future operation of the reservoir or reservoir system.

7–3. Sources of uncertainty

Uncertainties in flood risk assessments involving reservoirs include the following:

- a. Variability in the initial state of the reservoir. The reservoir storage at the onset of a flood event affects the volume available to attenuate the peak of the inflow event.
- b. Uncertainty about the physical properties of the reservoir pool, including the relationship between stored volume and pool elevation, or in outlet release capacity. This uncertainty, in turn, translates into an uncertain discharge if a flood event is large enough to produce uncontrolled outflow.
- c. Uncertainty about the physical properties of the dam and outlet works, including (but not limited to) elevations of the spillway(s), inlets on the reservoir side of the dam, and outlets on the downstream channel side; size, shape, and condition of the spillway and outlet works; and operability of gates, valves, and other physical properties. This operability may vary during a flood event, as debris buildup on gates or other unpredictable environmental factors affects operation capabilities.
- d. Uncertainty about properties of the channel between the reservoir and a control point location. If the control point location is not immediately downstream of the reservoir, releases from the reservoir may be attenuated as the discharge is translated through the stream channel. This attenuation can be simulated with a hydraulics model, and doing so helps reservoir operators make releases with that attenuation in mind. Physical properties of the channel must be known to accomplish this awareness and this simulation as well as parameters of models of the hydraulic processes. None are known with certainty.
- e. Variability in the actual operation of controllable gates, valves, and outlet features. Water control manuals dictate the desired operation for flood management. However, most manuals permit some flexibility based on current watershed conditions and foresight. Further, with imprecise knowledge of reservoir inflows, precise operation is impossible, even if the releases to be made are accurately prescribed and operational features are instantaneous.
- f. Variability in performance of other related water management facilities that may affect reservoir flood operation. For example, if levees along a stream reach downstream of a reservoir fail, a deviation from the normal operational procedures may occur. Similarly, if the reservoir shows signs of failure, deviation from the normal operational procedures may occur.

7–4. Resulting uncertainty in unregulated-to-regulated transform

- a. Variabilities and uncertainties described in paragraph 7–3 can be reflected as uncertainties in the unregulated-to-regulated transform. A distribution of errors about the best estimate is illustrated in Figure 7–3. In Figure 7–3, the dashed diagonal line represents the case of no regulation, with regulated discharge equal to unregulated discharge. The solid line represents a best estimate of the peak regulated discharge,

given a peak unregulated discharge. The distribution shown—in this case a Normal distribution—displays the likelihood that the actual regulated discharge is the value shown. The uncertainty PDFs should be viewed as on a Z-axis coming out of the page.

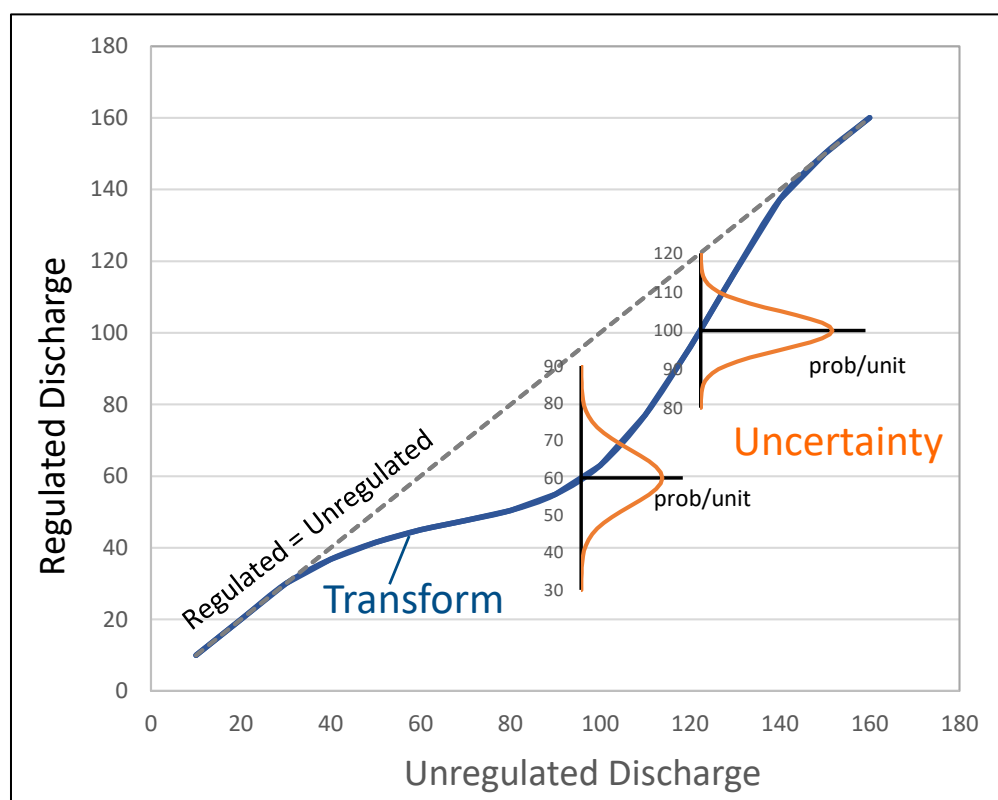


Figure 7-3. Illustration of uncertainty distribution about unregulated-to-regulated discharge transform

b. Any appropriate distribution can be used to describe the spread of uncertainty of regulated discharge about the best estimate, if that selection is supported with appropriate research. Current USACE software for a risk assessment includes the Normal, Log-Normal, and triangular distributions.

c. The parameter(s) for the selected distribution of uncertainty may be estimated as follows:

(1) If historical data are used, separate the observed unregulated peak discharges into categories or ranges, and analyze the frequency of resulting regulated discharges for each range. Make a histogram, and compute sample statistics of the regulated discharges. Use this simplified statistical analysis to infer the shape and appropriate parameters of a probability distribution of uncertainty about the best estimate of the regulated discharge transform for a given peak unregulated discharge. For example, if a Normal distribution is appropriate, use the chosen transform function value as the mean, and the standard deviation of the regulated discharges in the range as the standard deviation.

(2) Select critical, uncertain factors identified in paragraph 7–3. Identify combinations of the factors that define the best case (lowest), the most likely case, and the worst case (highest) operation scenario. Compute regulated peaks for various events under these scenarios, then select a probability distribution that is consistent with the location and spread of values computed, estimating parameters of that distribution from the computed peaks. For example, a triangular distribution might be selected if outflow is extremely unlikely to be outside the range of the lowest and highest values computed; that is, greater than the predicted worst case or less than the predicted best case. The parameters of that distribution (minimum, maximum, most likely) can be inferred directly from the computed discharges. Or, if the best case and worst case values are symmetrically distributed around the most likely case, a Normal distribution might be selected.

(3) Complete a probability tree or fault tree analysis in which a limited number of critical, uncertain factors are selected, and alternative values are postulated for each. This procedure is conveniently represented with a tree diagram as shown in Figure 7–4. In Figure 7–4, only two sources of uncertainty are considered: initial storage in the reservoir when a given flood occurs, and manner of operation. The uncertainty associated with each value is quantified by assigning a probability value to each “branch” of the tree, conditioned on the preceding branches. Simulating operation for a historical or hypothetical event with the various combinations yields a set of regulated discharges, the probability of which is the product of the probabilities of the branches along the path. The results can be examined, and a distribution fitted if desired.

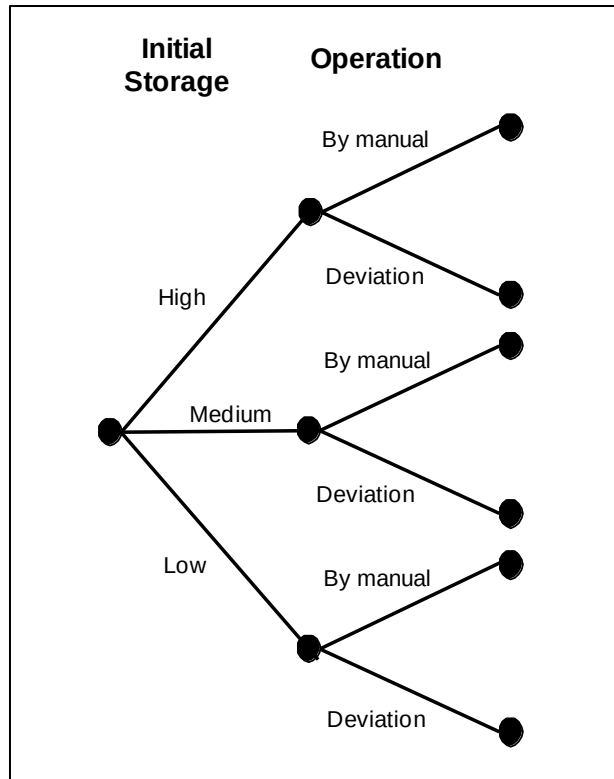


Figure 7–4. Example of probability tree

(4) As an alternative to the previously described methods, a probability distribution of uncertainty of each selected factor can be identified, and Monte Carlo sampling, coupled with reservoir simulation, can be used to generate a large sample of regulated discharges for a given an unregulated inflow. A distribution of errors can be fitted to the resulting Monte Carlo regulated flow sample, thus describing the total uncertainty.

7–5. Dam safety evaluation

Dam safety is the art and science of ensuring the integrity and viability of dams such that they do not present unacceptable risks to the public, property, and the environment. It requires the collective application of engineering principles and experience, and a philosophy of risk management that recognizes that a dam is a structure whose safe functioning is not explicitly determined by its original design and construction. It also includes all actions taken to routinely monitor, evaluate, identify, or predict dam safety issues and consequences related to failure, including ensuring all reservoir regulation activities are performed according to established water control plans. These actions are to be performed in concert with activities to document, publicize, and reduce, eliminate, or remediate, to the extent reasonably practicable, any unacceptable risks. Dam safety evaluations must follow regulations described in ER 1110-2-1156.

7-6. Evaluation of flood risk for a flood diversion

a. A flood diversion reduces flood risk by decreasing downstream discharges and so reducing the flood hazard. A diversion reduces discharge in the main channel, removing and directing a portion of the discharge elsewhere. The diversion facility may be a simple bifurcation in the channel, a culvert into which discharge is directed, or a side-channel spillway. The amount of water diverted may be controlled with gates, valves, or stop logs; or the diversion may be hydraulically controlled, depending on the physical properties of the facilities and the energy of the flowing water. EM 1110-2-1417 provides guidance for planning, designing, and evaluating system performance of a diversion.

b. A diversion affects the discharge-frequency function in a manner similar to changes caused by a reservoir. With a diversion, the unregulated-to-regulated discharge transform could have a shape similar to that shown in Figure 7-1. For lesser flow rates, little flow is typically diverted from the main channel into the diversion channel. Thus, the transform predicts that regulated peak flow equals unregulated peak flow in that range. When the unregulated flow reaches a threshold rate, the diversion redirects a portion of the flow away from the main channel and into the diversion channel. In that case, the regulated peak flow in the main channel is less than the unregulated peak, as indicated.

c. For very high, unregulated flow rates, the diversion has less impact; while it does reduce the main channel peak from the unregulated condition, that reduction is proportionally smaller. The corresponding regulated peak discharge-frequency function, therefore, is similar to that shown in Figure 7-1(b). Discharges associated with a given probability (quantiles) for the unregulated and regulated functions are the same when no discharge is diverted. As discharge is diverted from the main channel, the regulated discharge-frequency function falls below the unregulated function. For rarer events, the diversion does not have adequate capacity to convey all excess water. As unregulated discharges increase, regulated discharges also increase to the point that they are nearly equal to unregulated values (at Point c in Figure 7-1(b)). A difference from the reservoir transform is that a diversion always reduces the peak regulated discharge by some amount, while a reservoir sees regulated discharge equaling peak discharge when all storage is filled volume prior to the event peak.

d. In the risk computation procedure illustrated in Figure 2-5, flood risk for a diversion is characterized by the unregulated-to-regulated discharge relationship. With that function, the unregulated discharge-frequency function is transformed to a regulated discharge-frequency function. Methods for developing the function are similar to the ones described in paragraph 7-2 for development of the transform for a reservoir. The primary difference is that a simulation model of the channel network with any diversion structures is used, instead of a simulation model of a reservoir (though often the tool for simulation can do either or both tasks). Otherwise, regulated and unregulated peak discharges are determined and the relationship determined graphically, as described in paragraph 7-2.

e. Similar procedures can be used for assessment of the impact on risk of an unintentional or accidental diversion, for example, when a levee is overtopped or breaches. In that case, discharge is diverted from the channel, thereby altering downstream peak and volume. Again, the relationship of regulated-to-unregulated discharge can be determined with simulation to define the regulated peaks. In this case, those peaks are associated with discharge that remains in the channel at the index location, downstream of the unintentional diversion. For the risk computation, the transform is used in a manner identical to the transform developed for an engineered diversion.

f. When performing risk analysis using the Monte Carlo event-sampling approach described in paragraph 2–8, rather than the summary-curve combination approach that uses an unregulated- to regulated-discharge transform function, the transformation process occurs directly in watershed models. A regulation model is thus included in the sequence of watershed models that ultimately transform an unregulated flood event hydrograph through regulated discharge, channel stage, floodplain inundation (if a large enough flood event) and subsequent damage.

7–7. Sources of uncertainty about diversion system performance

Risk management attributable to a diversion is uncertain because the operation and system performance of the diversion are uncertain. Sources of this uncertainty include the following:

a. Uncertainty about the physical properties of the diversion inlet structure. Inflow to the diversion channel is controlled by hydraulic properties of the inlet structure, which may be a side-channel weir, in-stream barrage, or other engineered structure. The size, shape, and other properties of such structures affect the volume of water diverted. However, these properties are not known with certainty. Even if the properties can be measured as built, the properties change with time caused by erosion, deposition, or debris buildup, which changes the manner of a diversion.

b. Variability in operability of gates, valves, and components of a diversion. In some cases, inflow to the diversion channel is controlled with gates, valves, pumps, stop logs, or some combination of these features. The manner of, or even the ability to operate, these features varies during a flood event and from one event to another. This variability may be based on design, or due to lack of proper maintenance, debris buildup on gates, or other environmental factors that adversely affect operation capabilities; all are unpredictable.

c. Uncertainty about properties of the diversion channel. Rate of flow through the diversion channel, and hence the rate of diversion of excess water from the main channel, is a function of the channel dimensions and resistance to flow. Dimensions can be measured, but not without error. Resistance to flow may change seasonally or with deposition or erosion in the channel, so it is not reliably predictable.

d. Variability in the downstream boundary condition of the diversion channel. In some water control systems (such as those in the Central Valley of California) short connector channels divert excess discharge from main rivers into parallel, engineered flood-bypass channels. For those systems, the water surface in the channel receiving the diverted flow forms the downstream boundary condition for the diversion channel; with high downstream head, the rate of diversion may be reduced. This downstream condition may vary during a flood event, and varies from event to event.

e. Uncertainty about properties of the channel between the point of diversion and the downstream index location. If the index location is not immediately downstream of the point of diversion, discharge that remains in the main channel (after some portion is diverted) is attenuated as it is translated through the stream channel. This attenuation can be simulated with a hydraulics model. Physical properties of the channel must be known to accomplish this simulation, along with parameters of models of the hydraulic processes. Neither is known with certainty.

f. Variability in system performance of other related water management facilities that may affect diversion performance or operation. For example, if the diversion channel itself is flanked by levees, the stability of those levees affect the performance of the diversion. For example, a levee breach might increase the rate of discharge into the diversion channel, as less energy is required for discharge into the otherwise protected floodplain.

7–8. Quantitative description of uncertainty about diversion system performance

The manner in which a diversion can regulate discharges in a system is never known with certainty, owing to factors identified in paragraph 7–7. Accordingly, properties of the unregulated-to-regulated discharge transform function for a diversion are not certain. That uncertainty is described explicitly with a probability distribution and incorporated into risk computations. Probability distributions used, and the manner in which parameters for those distributions may be estimated, are consistent with the distributions described for reservoir operations in paragraph 7–4.

Chapter 8

Evaluating Flood Risk for Channel Modifications

8–1. Overview

Channel modifications for flood risk management include modifying the geometry of the channel to increase the effective flow area at one or more locations; reducing the boundary roughness; and reducing the energy loss due to obstructions, curvature, or impediments to flow. Any one or a combination of these actions reduces the stage at a cross section in the channel for a specified flow rate, as it makes the channel a more efficient conveyance. EM 1110-2-1416 provides more details about the hydraulic engineering principles that apply and methods for analysis of the effects.

8–2. Evaluating flood risk for a channel modification

a. Per the discussion in paragraph 2–3.g(1), the impact of a channel modification is represented with a modified discharge-stage transform, as shown in Figure 2–3(b). Figure 8–1 illustrates such a transform, showing both the without-project function (dashed line), and that with the modification in place (solid line). The stage is lower for a given discharge if the modification successfully reduces risk at the index location. In Figure 8–1, the modification is more effective for lower discharge rates, reducing the channel stage significantly there, while achieving smaller reductions in elevation for higher flow rates. The actual shape and properties of the modified function depend on the details of the modification.

b. The with-project discharge-stage transform is commonly developed with a numerical model of the channel hydraulics at the points of interest in the system. As described in paragraph 4–5, such a model solves the governing equations of flood discharge to compute stage, given the channel properties and model parameters. As channel conditions are modified, the properties and parameters of the model are adjusted accordingly, the model rerun for a wide range of discharge values, and the corresponding best estimates of the stages computed. The results may be plotted, tabulated, or fitted with an empirical function.

8–3. Sources of uncertainty

The following sources of uncertainty about the with-project discharge-stage transform should be considered:

- a. Uncertainty about the choice of model.
- b. Errors in the topographic and bathymetric data.
- c. Errors in measurements of dimensions and locations of structures.
- d. Errors in measurements of dimensions and locations of water control features.
- e. Uncertainty about existence of and impact of ice cover and jams.

- f. Uncertainty about downstream boundary conditions.
- g. Uncertainty about model parameters.
- h. Uncertainty about impact of erosion and deposition.
- i. Uncertainty about impacts of wind and waves.

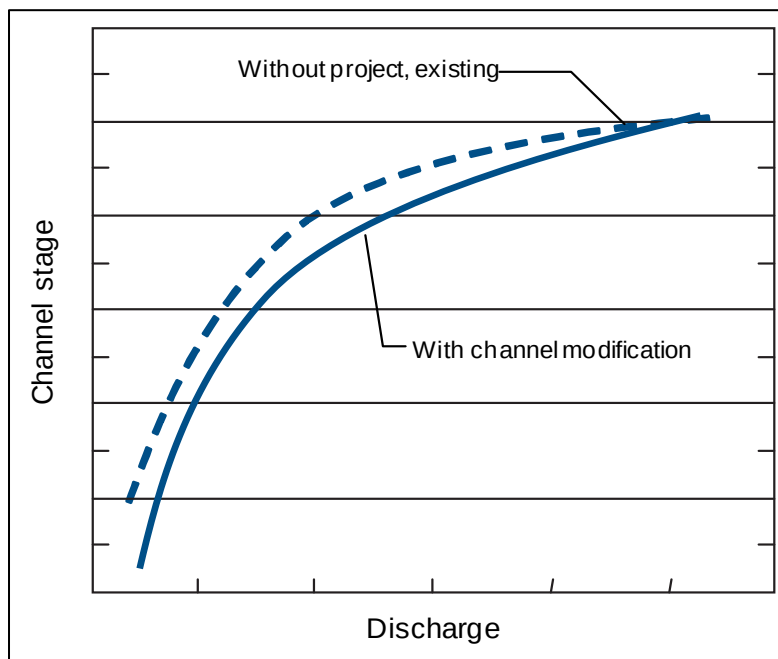


Figure 8–1. Example of discharge-stage transform, without and with-channel modification

8–4. Risk assessments

Risk assessments for USACE studies require explicit accounting for uncertainty for both without- and with-project conditions. Therefore, the uncertainty associated with the discharge-stage transform must be described for the modified channel case generally following the procedures described in Chapter 4.

a. For the with-project, modified channel condition, some error sources will be less critical and may contribute less error than for the without-project case. For example, in the natural condition, surveying and establishing the geometry of a channel cross section are difficult. However, the modified-condition cross section is established as a consequence of project construction, with some control over the engineered geometry. While this control is also subject to error, it is more certain than, for example, channel elevations determined from a USGS topographic map. On the other hand, the with-project condition in future years may have greater uncertainty as channel erosion, deposition, and maintenance cannot be forecast with certainty.

b. Likewise, if the modified condition includes relining the channel to reduce resistance, the roughness can be controlled with greater precision, and the roughness model parameter should be less uncertain (although never certain).

c. Adjustments to the uncertainty model parameters, such as reducing the standard deviation if the Normal distribution is used to model uncertainty, may be appropriate. Any adjustment must be explained and supported.

Chapter 9

Evaluating Flood Risk for Nonstructural Measures

9–1. Overview

a. *Nonstructural floodproofing.* This measure reduces risk by reducing the consequences of flooding. This contrasts with structural measures, such as levees and dams, which reduce risk by instead reducing the probability of flooding reaching a specific location. There are three primary types of nonstructural floodproofing measures: dry floodproofing, wet floodproofing, and structure elevation. With each of these measures, the depth of water at the floodplain location of the damageable structure does not change, but the damage incurred is reduced.

b. *Dry floodproofing.* This measure involves sealing building walls with waterproof compounds, impermeable sheeting, or other materials to prevent the entry of floodwater into a damageable structure. Any openings in the building perimeter, such as doors or windows, must be sealed with a closure structure.

c. *Wet floodproofing.* This measure allows floodwater to enter a structure, and vulnerable items such as utilities, appliances, and furnaces are waterproofed or relocated to higher locations.

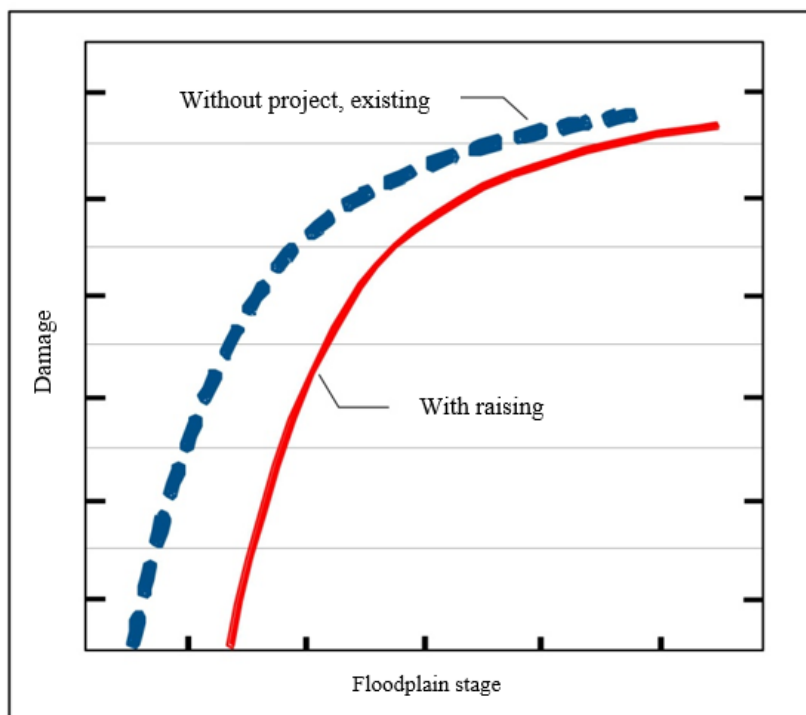
d. *Structure elevation.* This measure involves raising a structure in place so that the damageable portion of the structure (typically the first floor) experiences a reduction in frequency of flooding during high-water events.

e. *Relocation.* This measure involves moving one or more structures away from a flood hazard. If all structures and their contents are removed from the flood-prone area, the stage-damage function is reduced to zero damages for all stages in the range of practical interest.

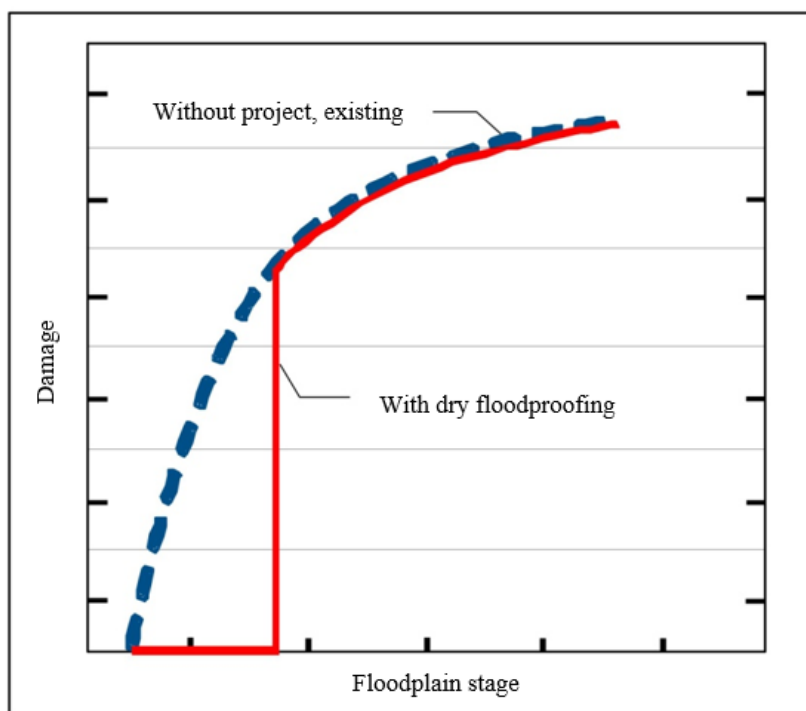
9–2. Risk computations for floodproofing

a. For risk computations, the impact of floodproofing is represented with a modified stage-damage relationship, as shown in Figure 9–1. Figure 9–1(a) shows, for structure elevation, the without-project, stage-damage function (dashed line) and the with-project function (solid line). The shape and other characteristics of the stage-damage function depend on the details of the property represented, but the without- and with-project functions have the same shape in this case. The impact of structure elevation for floodproofing is that the with-project function is shifted by an elevation increment equal to the distance by which the structure is raised. Consequently, the damage incurred is less for a given floodplain stage.

b. Figure 9–1(b) shows the without-project, stage-damage function (dashed line) and the with-project function (solid line) for dry floodproofing a structure. With dry floodproofing, no damage is incurred until the dry floodproofing is overtopped. Then the damage is equal to the damage incurred without the dry floodproofing, as the stage within the structure reaches equilibrium with the water outside.



a. Without project stage-damage function (dashed line) and with-project function (solid line)



b. Without project stage-damage function (dashed line) and with-project function (solid line) for dry floodproofing

Figure 9–1. Stage-damage functions modified by floodproofing

9–3. Modified functions

a. The modified functions are developed in a manner similar to the without-project, stage-damage relationship, as described in Chapter 5, taking into account the shifted location in the case of raising or the limited flood risk management provided with a corresponding change in damage due to dry floodproofing.

b. Modifications to the stage-damage function cannot be predicted without error, due to the following:

(1) The with-project function depends on the without-project function, and that is not known with certainty. Sources of error in the without-project function are identified in paragraph 5–3.

(2) Details of construction of floodproofing features, while planned exactly, cannot be executed exactly. Thus, the constructed height of a floodwall or the distance by which a structure is raised may be slightly more or slightly less than what the design calls for, and the resulting stage-damage relationship will be different than anticipated.

9–4. Risk assessment with uncertainty

For a risk assessment with uncertainty, as required for USACE studies, the uncertainty about the best-estimate, stage-damage function must be described with a probability distribution. For modified conditions, this assessment is done in the same manner as described in paragraph 5–2. However, certain error model parameters may be adjusted to account for increases or decreases in uncertainty. For example, if dry floodproofing is included as a risk management measure, the uncertainty about the modified stage-damage relationship may be increased slightly from the without-project condition to reflect the uncertainty about structural performance of the dry floodproofing.

Chapter 10

Assessing System-Wide Risk

10–1. Situation

a. USACE flood-risk management projects seek to reduce flood risk, as it is defined in this manual. Strategies shown in Table 10–1, if properly planned, designed, constructed, and operated, can meet this goal. In the simplest case, these strategies can be implemented for a single impact area, reducing risk for that area. However, if the impact area is hydraulically connected to other impact areas, these strategies must be planned and evaluated in a system-wide context.

Table 10–1
Strategies to reduce flood risk

Strategy	Example
Decrease hazard by decreasing discharge for a selected exceedance probability or design event.	Improved reservoir operation: storing more water earlier in a flood, then releasing it at a lesser rate later.
Decrease hazard by decreasing the channel stage corresponding to a given discharge.	Channel improvements, such as deepening or widening.
Decrease hazard by decreasing the floodplain stage corresponding to a given river stage.	Levee improvements: levee raising, levee strengthening, and levee realigning or relocating.
Decrease hazard by mitigating for increased flow durations or velocities at any location in the system, including project site.	Levee strengthening to mitigate for increased flow durations, or bank protection to mitigate for increased velocities.
Decrease vulnerability and thus consequence by reducing the damage incurred by water reaching a given floodplain elevation.	Enhanced flood response and emergency planning.

(1) Changes implemented to reduce risk at one location may increase risk elsewhere in the system. In that case, the risk is transferred rather than reduced, and may be unacceptable depending on the nature of the transferred risk. Conversely, transfer of risk may be an acceptable component of risk management via risk communication. Shared responsibility and shared accountability for implementing measures beyond USACE authority and budget leads to shared decision making. This expectation means that a USACE decision on selecting the recommended plan cannot be made in isolation from the preferences of other entities.

(2) Ideally, USACE should lead a deliberate and collaborative process to reach agreement on a recommended plan informed by the evaluation and comparison of the alternative plans. Up-front engagement and participation in the formulation, evaluation, and plan recommendation is required of all cost-sharing sponsors. USACE should include all other stakeholders who have financial obligations or other implementation obligations (such as building code changes or land-use restrictions) for plan implementation to the extent that they are willing to participate. Also, other federal agencies and nonfederal government agencies may have projects and programs that

are part of an alternative plan in the final array of plans, or may have responsibility to approve a plan before it may be implemented. USACE will share the evaluation and comparison information and engage with all cost-share partners and stakeholders and engage them in selecting a recommended plan.

(3) As an example, to illustrate the need for and methods of system-wide analysis, this chapter focuses on improvements to levees. Some of the material that follows was adapted from the publication, “Options for Measuring, Preventing, and Mitigating Impacts due to Improvements to the Sacramento and San Joaquin Flood Control Projects” (Ford 2007).

b. As described in Chapter 6, a levee manages flood risk for people and property by blocking flow from the channel onto the adjacent floodplain. If water level in the river channel rises above the top of the levee as illustrated in Figure 10–1, the levee will be overtopped, with flow over the levee onto the floodplain.

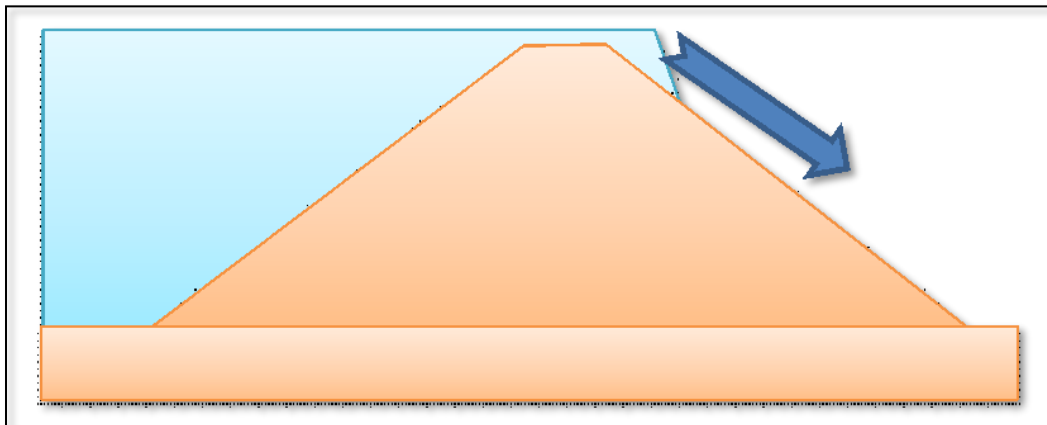


Figure 10–1. Levee overtopping

(1) Raising a levee increases the elevation of the top of the levee, thus reducing the likelihood of overtopping. When a levee is raised, material is added to the top of the levee. To ensure stability of the levee as it is raised, the base of the levee must also be increased by the addition of material, typically on the landside of the levee. The width of the top of the levee also must be increased to maintain a stable side slope on the levee.

(2) Quantitative evaluation of the impacts of raising a levee requires hydrologic, hydraulic, geotechnical, and economic information. This information can be developed and presented as a set of functions, as shown in Figure 2–3. The discharge-stage transform (Figure 2–3(b)) predicts water level in the channel, given discharge. As described in paragraph 4–5, this transform is developed by application of principles of hydraulics, with a mathematical model that represents the energy and momentum of flowing water, along with the conservation of volume of that water. Any modification to the channel geometry or channel roughness affects the energy and conservation, yielding a change in the discharge-stage transform. That change can be predicted with the hydraulics model.

(3) Changes to the channel at one location may have an impact elsewhere in an interconnected system. For example, if an improvement increases the velocity of flow in one reach of the system, the impact of that increase is felt downstream. If an improvement raises the channel stage in one reach of the system, the impact of that rise is felt upstream. Consequently, the model used must represent the flow regime upstream and downstream of the location of proposed improvements to permit adequate assessment of changes to the discharge-stage function, and hence, to the risk elsewhere in the system.

c. The purpose of raising a levee is to reduce the probability of overflow from the river onto the protected floodplain. Until a levee is overtopped, the stage in the floodplain is zero (putting aside, for the moment, the uncertainty associated with occurrence of a levee breach that would permit flow into the interior area). With zero stage in the floodplain, the damage incurred is zero. As the levee is raised, the channel stage at which water overtops the levee and flows into the interior floodplain increases, and the discharge that the channel can convey without overtopping increases. This increased channel capacity corresponds to a lower probability of exceedance and less risk for the impact area.

d. Raising a levee may have unintended impacts outside the area for which the raise reduces flood risk. The change in TOL elevation alters the channel geometry. This change may, in turn, affect discharge and stage upstream and/or downstream of the improvement.

(1) For example, in the leveed system illustrated in Figure 10–2, suppose that the levee that manages flood risk for Impact Area 3 from River F is raised from elevation 60 to 62 feet. This levee raise reduces the probability of overtopping and hence reduces the expected damage for Impact Area 3.

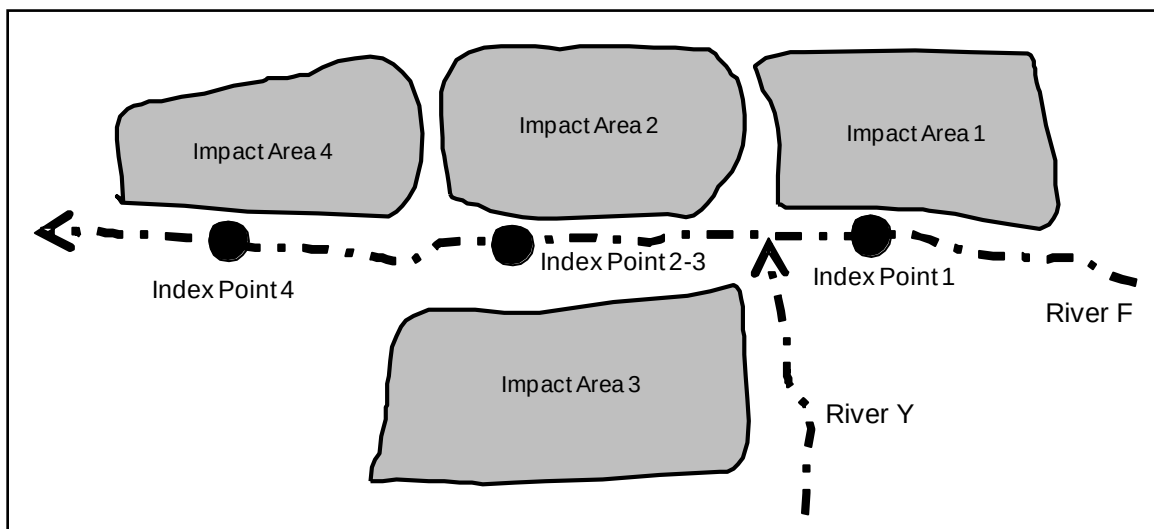


Figure 10–2. Example of interconnections in system

(2) Now the flow is constrained somewhat, contained in the channel at Index Point 2-3 in Figure 10–2 between elevation 60 to 62 feet, moving downstream. The storage that was provided incidentally and unintentionally by overflow into Impact Area 3 at elevation 60 feet has been eliminated. As a result, downstream discharge at Index Point 4 is greater for a given probability (between elevations 60 to 62 feet) and the hazard there has increased. The greater discharge at Index Point 4 potentially yields greater stage in the channel. If the levee that manages flood risk for Impact Area 4 is not sufficiently high, it will be overtopped. The interior floodplain elevation will increase, and risk will be greater. In that case, risk has been transferred.

e. In addition to overtopping, flooding in the interior floodplain may be a consequence of instability of the levee embankment due to seepage flow through or beneath the embankment; erosion of the levee; or flow around or through pump stations, gravity outlets, or other structures that pass through the levee.

(1) Levee strengthening (also called hardening) includes actions that reduce the risk of a levee breach due to seepage, erosion, or failure due to flow around river structures. These actions are illustrated in Figure 10–3, which shows a slurry wall at the core of the levee. This impervious wall limits seepage flow through or beneath the embankment.

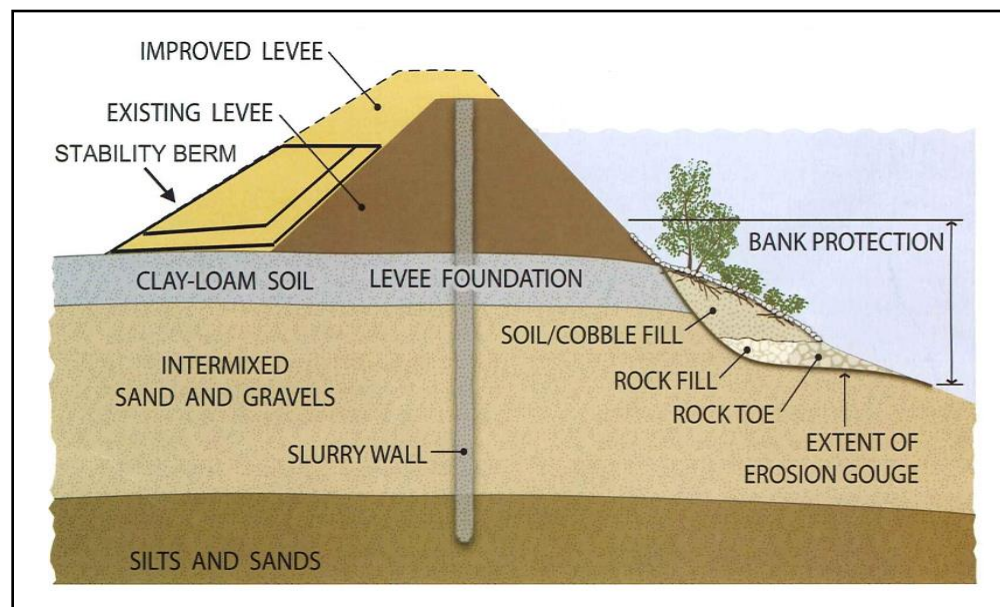


Figure 10–3. Illustration of levee raising and strengthening (SAFCA 2006)

(2) Seepage through the levee may lead to sliding of the interior face of the levee. The stability berm shown in Figure 10–3 is designed to reduce the risk of this kind of sliding. Figure 10–3 also shows how erosion of the channel side of the levee may be prevented with bank protection and fill.

f. The intended purposeful impact of levee strengthening is to reduce seepage, erosion, or flow through or around river structures, preventing levee failure except by overtopping. This intended impact is reflected in the system response curve that describes the uncertainty associated with levee structural performance and the interior-exterior function.

(1) The indirect, unintended, incidental consequences of levee strengthening may include increasing discharge and stage elsewhere in the interconnected river system, and increasing the likelihood of levee failure elsewhere in the interconnected system caused by migration of erosive forces.

(2) The first incidental consequence is an outcome of removal of the incidental storage that exists in the without-improvement condition. If a levee breaches or is overtopped, water from the channel enters and is stored temporarily in the interior floodplain. Availability of this storage, which comes at the expense of flooding of the interior area, reduces downstream discharge. Removal of the storage increases downstream discharge, and perhaps the water surface elevation, increasing the likelihood of levee failure there.

(3) Similarly, the risk across the channel may increase as an incidental consequence of strengthening. In Figure 10–2, for example, if the levee for Impact Area 2 is strengthened, but the levee for Impact Area 3 is not, the strengthening may increase the probability of failure of the Impact Area 3 levee. The hardening may shift the distribution of erosive forces in the channel.

g. The intended, direct, purposeful impact of levee relocation or realignment is reduction of channel water surface elevation at the location of the improvement. That reduction yields a reduction in floodplain stage, which, in turn, reduces risk.

(1) Repositioning a levee away from the channel, also known as levee setback, is illustrated in Figure 10–4 (Ford 2007). Levee setback is an improvement that provides a wider channel cross section, which, in turn, permits a given discharge to pass at a lower water surface elevation. Lower elevation in the channel reduces the likelihood of floodplain inundation, as the probability of levee failure—either by overtopping or another failure mechanism—is reduced.

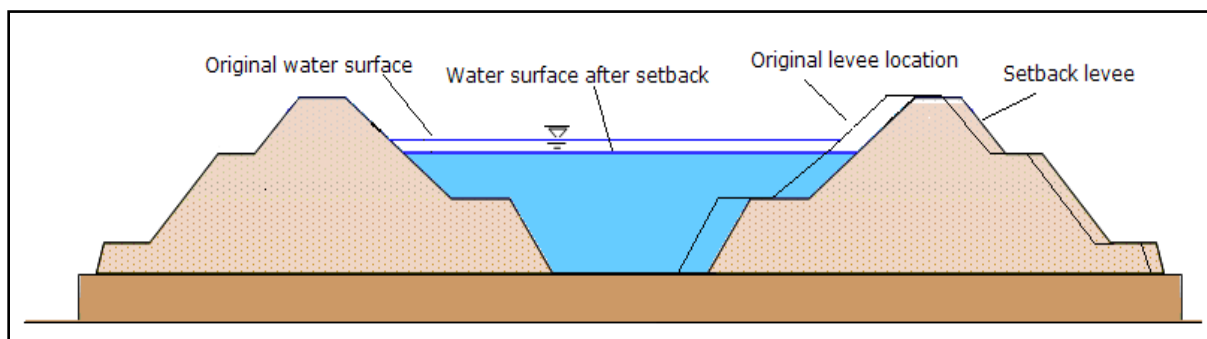


Figure 10–4. Levee setback

(2) Levee relocation may affect conditions outside the site of the relocation as a consequence of the hydraulic connectivity of the channels. For example, if the Impact Area 2 levee in Figure 10–2 is moved back, the channel width increases there. The increased width yields a greater channel area and greater potential for storage of water in the reach. This storage may decrease flow downstream, also reducing stage there for a specified AEP. On the other hand, the increased width and greater cross-sectional area may increase the channel conveyance and thus increase the downstream discharge and stage, ultimately increasing the risk downstream.

(3) Levee realignment consists of straightening the channel by removing bends and oxbows. Water flows in a straightened channel with less energy loss, thus permitting a lower stage. As with the setback, this lower stage reduces the probability of overtopping or breaching and lowers the likelihood of levee failure.

(4) Levee realignment may have incidental impacts elsewhere in the interconnected hydraulic system. For example, if the Impact Area 2 levee in Figure 10–2 is realigned, the stage for a given discharge will decrease. However, velocities may increase there and downstream. These greater velocities increase the likelihood of erosion, thus increasing the probability of levee failure. Thus, this example of levee realignment transfers the risk from one site to another by increasing the hazard.

10–2. Strategy

Risk analyses for USACE studies—whether planning studies, operational studies, or permitting studies—must assess risk and risk management at the project site and at all locations reasonably considered to be affected by risk management actions at the project site. The procedures described in this manual are, in general, appropriate for those studies, with some adaptation to reflect the effects of hydraulic connectivity.

10–3. System-wide risk assessment

A system-wide risk assessment for USACE studies includes the following steps:

a. Define the spatial extent of the system for which risk (and risk transfer) must be assessed, and select index locations within that extent for the risk assessment.

(1) The extent of the hydraulically interconnected system must be defined as the first step in a risk assessment. This extent must be broad enough to include channel reaches and floodplains downstream and upstream of a project site that a reasonable analyst expects to be influenced by changes in discharge rate, volume, or corresponding water surface elevation at the site. Within that extent, impact areas should be identified and index locations selected to permit fair assessment of risk transfer. Guidance for identifying impact areas and selecting index locations is included elsewhere in this manual.

(2) Review of hydraulic model results aids in determining the appropriate extent. For example, examination of computed water surface profiles identifies locations upstream or downstream of a project site at which changes in channel geometry at the

site will impact water surface elevations. Care must be exercised and results scrutinized to judge if changes in computed elevations logically are related to the changes in channel geometry or if changes seen in the model results are an artifact of computational imprecision. In some cases, downstream flows at a confluence will increase for an alternative, but the increase is due to a change in timing between contributing hydrographs. Consideration should be given to whether the change in timing is expected to be reflected in historical events, or whether the change in timing is an artifact of the synthetic hydrology developed for the planning study.

b. Identify the without-project condition for all components of that system.

(1) The basis for assessing impacts of alternative plans is the without-project condition, so this condition must be described system wide for a risk assessment. That description must include the current without-project condition and future conditions, as risk computations must span the expected life of a project.

(2) Paragraph 2-4b of ER 1105-2-100 requires that the without-project condition be defined as the “most likely condition expected to exist in the future in the absence of a proposed water resources project.” Therefore, defining this condition demands a forecast of future conditions. ER 1105-2-100 requires that the forecasts “consider all other actions, plans, and programs that would be implemented in the future to address the problems and opportunities in the study area in the absence of a USACE project.” Forecasts should consider national and regional socio-economic conditions; and within that context, plans that have been adopted and other current planning efforts with high potential for implementation or adoption are to be considered as part of the future without-project condition.

(3) For example, in Figure 10–2, consider evaluation of risk for a project proposed for Impact Area 2. The without-project, unregulated, discharge-frequency function is required there, along with a transform that will permit development of the unregulated to regulated discharge transform there. If the without-project condition is judged to include the levees of Impact Area 1 in their current deficient condition, those levees may fail and reduce the discharge at Impact Area 1 and subsequently at Impact Area 4, as water is stored in the floodplain. But if the levees for Impact Area 1 are presented in the without-project conditions as raised or strengthened, that action will eliminate or reduce storage for certain events, thus increasing the discharge downstream at the project site. That increased discharge must be considered when evaluating without-project risk throughout the system.

(4) The analyst must consider carefully whether such actions have a high potential for implementation, including the potential for funding from tentative sources. Likewise, the forecast of future without-project conditions must weigh the possibility that future modifications themselves will transfer risk, and thus be limited in their design.

c. Collect or develop the necessary functions and transforms to compute without-project risk at all index locations within the system.

(1) Risk computations are completed on a location by index location basis following the procedure described in Chapter 2. Each of the functions must be developed for each index location. The unregulated discharge-frequency function (Figure 2–5(a)) must include all flows that accumulate at the index location, including tributary inflows upstream. The unregulated-regulated flow transform (Figure 2–5(c)) must represent, in the aggregate, the impact of all regulation upstream of the index location. This impact includes those of intentional regulation by upstream reservoirs and diversions, and the incidental impact of regulation if water overtops or breaches upstream levees and flows onto an adjacent floodplain. The discharge-stage transform (Figure 2–3(b)) is a localized function representing conditions at each index location, unaffected by upstream conditions but including perhaps the impact of downstream conditions if backwater influences stage.

(2) When uncertainty is considered, a levee system response curve is likewise required at each index location, and that is a localized function (Figure 2–6). Finally, the stage-damage relationship (Figure 2–3(c)) is required to assess the economic risk. If life safety is to be considered, a relationship between potential loss of life and stage is required for each location.

(3) In addition to the various functions required for a risk assessment, the uncertainty about each must be described. This task is completed following the general guidance presented in this manual.

(4) Assess without-project risk at all index locations. Risk and other measures of performance identified in this manual are computed location by location within the extent of the system. Results of without-project risk assessment may be reported as shown in ER 1105-2-101.

(5) Simulate system behavior and system performance with the features of the proposed project in place as necessary to revise and modify all functions and transforms throughout the system to reflect changes due to the project.

(a) Actions taken in this step depend on measures included in the proposed plan. For example, if the plan includes addition of flood storage or modification of the way available storage is operated, a reservoir system simulation model such as HEC-ResSim may be developed and exercised with a period of record or selected hypothetical events. Through this model, a new unregulated-to-regulated discharge transform can be developed.

(b) Similarly, if the proposed project includes modifications to the channel through levee setbacks and similar features, these features must be simulated to derive new transforms for downstream locations. Those transforms may change as a consequence of the channel alteration, as described in paragraph 10–1.g.

(c) The with-project assessment must include a forecast of future conditions with proposed project features in place. ER 1105-2-100 requires that such forecasts be consistent with the forecasts in the without-project assessment. Thus, as described in

paragraph 10–3.b, an assessment must consider the effects of future flood risk management measures throughout the system in the absence of the proposed project.

(6) Compute with-project risk and indices of performance at index locations system wide. As described in paragraph 10–3.c, risk and other measures of performance identified in this manual are computed point by point within the extent of the system.

(7) Determine if risk is transferred by comparing risk and performance indices system wide.

(a) Once EAD and various indices of performance are computed and reported, system-wide impact of a proposed alteration can be assessed. Of course, at the index location that represents the project site, a reasonable, feasible project reduces risk. In that case, EAD is less, AEP is less, and reliability of flood risk management is greater. But these outcomes may not be true for all index locations within the system.

(b) As described in paragraph 10–1.g, levee modifications may lead to decreases in indices of reliability of flood risk management. If these adverse changes are determined to be significant, then the proposed modification must be rejected, as it transfers, rather than reduces, risk.

(c) From a purely economic risk perspective at the national level, risk transfer may be acceptable if the net risk for the nation is reduced. However, this transfer may be inequitable, fails to account for intangible benefits and values, and ignores regional economic development. Currently no firm guidance exists for determining significance of adverse impacts. Analysts should consult with HQUSACE staff and experts at ERDC and HEC for assistance with determining significance of adverse impacts.

Chapter 11

Understanding and Estimating Potential Life Loss

11–1. Need for evaluation of life safety risk

a. Flood water can be one of the most destructive forces on earth, especially if caused by an event that unexpectedly overwhelms an existing flood defense or by catastrophic breach of a dam or levee. Flooding can have many types of severe consequences, including economic, social, cultural, and environmental. Events, such as flooding caused by Hurricane Katrina and the tsunami in Japan, have caused thousands of people to lose their life and unknown billions of dollars in damages. By the same token, dozens of floods (some from similarly unexpected events like a dam or levee breach) occur every year with no resulting loss of life and relatively minimal property damage.

b. A primary goal of the USACE flood risk management program is to reduce the risk of loss of life from flooding. As such, potential loss of life is assessed and considered separately from economic and other considerations. Evaluation of alternatives should focus on the costs, flood damage reduction benefits, and the effectiveness of these measures while meeting essential agency guidelines and identified societal concerns. The preferred plan should be the one that provides the best tradeoff between the full range of comprehensive benefits and associated cost of achieving those benefits.

c. Interagency engagement is essential for plan formulation and for securing an understanding of the capabilities and evaluation procedures of the various agencies. Interagency engagement is most important for realizing the goal of shared responsibility for life safety risk management actions. This chapter focuses on approaches for estimating loss of life within the context of a risk assessment to support flood risk management.

d. Estimation of the magnitude of potential life loss resulting from a flood requires consideration of the following essential elements:

(1) The initial spatial distribution of people, or population at risk (PAR), in the threatened area at the onset of the flood event. This distribution should consider all important elements, including how people are distributed in the area based on time of day, and likely locations of transient population.

(2) The redistribution of people within and outside of the inundated area due to the effectiveness of warning, the response of people to those warnings, and the opportunity for and effectiveness of evacuation. Those that remain in the impacted area when the flood arrives are identified as exposed population.

(3) The flood event, including specific characteristics of the initiating event (for flooding caused by dam or levee breach, the following factors are important: the breach location, geometry, and rate of breach development); the water level and volume available to flow into the inundation area; the time of day, detection time of the event

relative to initiation of flooding; and the extent, velocity, depth, and arrival times throughout the downstream inundation area. Many of the same factors are important in nonbreach scenarios, although there is often less uncertainty surrounding the characteristics of the event.

(4) The degree of shelter provided by the setting where people are located (structure, vehicle, on foot, etc.) at the time of arrival of the flood water and how that shelter withstands the flood characteristics at that location.

(5) Fatality rates for the exposed population who remain in the inundation area at the time of arrival of the flood water. Fatality rates at a specific location consider the physical character of the flood event and the degree of shelter provided by the setting where people are located at the time of arrival and after the flood wave has passed for those who survive it.

11–2. Considerations when estimating potential loss of life

a. Clearly, the full consideration of all the elements listed above is a very complex problem that requires understanding of the physical processes and how they impact the flood characteristics (flood event characteristics and flood routing), human responses, and the performance of technological systems and the built environment (such as warning and evacuation systems, transportation systems, and buildings under flood loading).

b. Historically, during a flood event, not all people who lose their lives do so by drowning; for example, the stress can trigger heart attacks or other fatal ailments. Generally, methodologies used for estimating loss of life focus on direct loss of life from flooding (drowning). Indirect loss of life can be addressed qualitatively or quantitatively, through other means as necessary.

c. Methods for estimating potential loss of life from flooding should be applied within a scalable framework, meaning the level of detail for a given assessment is proportional to the life safety risks and decision that is required. For instance, the presence of existing levees or dams in the study area and their potential for failure may be an important consideration in scaling the analysis. This scalable framework can be divided into two broad application categories—screening and detailed. Generally, screening assessments generate results that are appropriate for informing initial risk classification and prioritization and scoping of additional studies. More detailed or higher level assessments can be used to verify findings from screening assessments and to inform the decision-making process for selection of the appropriate risk management approach.

(1) Regardless of the level of study, the method used to estimate potential loss of life should follow the USACE state-of-practice described herein.

(2) Screening-level assessments should generate reasonably conservative results. Conservative, in this case, means that in the face of significant uncertainty, results from a screening assessment tend toward risk estimates that overestimate risk rather than

underestimate it. This is “erring on the side of safety,” such that if an assessment is considered high risk after a screening assessment, it will be prioritized for more detailed assessments to support development of risk reduction measures. There is an art to this conservative bias. If the bias is too large, then the screening assessment loses credibility. If the bias is too small, some projects may be considered low risk that are actually high risk, which reduces the efficiency and effectiveness of the overall risk management program.

(3) More detailed assessments either reinforce the findings from the screening assessment or provide a more accurate (less uncertain) estimate that results in the understanding of risk being lower than it was from a screening assessment.

(4) USACE applies simulation-based methodologies to estimate potential loss of life by modeling each essential element listed in paragraph 11–1.d, including how the PAR is distributed throughout the threatened area, when warning is issued, how that warning spreads among the PAR, how the PAR reacts to the warning, how the PAR evacuates over the available road network, how quickly the flood waters arrive, and how those flood waters affect the survivability of people trapped in structures and vehicles or caught out in the open. These simulation-based methods are generally applicable to all inundation scenarios. The simulation-based methods supported by USACE include LifeSim (Fields et al. 2012), which was developed at Utah State University but is now undergoing further development by USACE, and the HEC-FIA (Flood Impact Analysis) software, which was developed by USACE (HEC-FIA 2012) and contains a simplified version of LifeSim.

11–3. Guidance for estimating potential loss of life

a. Details on how essential elements for estimating loss of life are included in USACE simulation methods are described below:

(1) *Initial distribution of people.* PAR is the number of people located within the maximum inundation boundary for a given scenario upon initiation of that event. For example, the PAR for an evacuation warning that goes out on 0800 on Sunday morning due to impending breach will include everyone who is located within the inundation boundary extents for that flood scenario at 0800. LifeSim and HEC-FIA require a structure inventory with population as input to define the initial distribution of people in the potential inundation area. The structure inventory should consist of both permanent and transient population.

(a) Since shelter provided at location when the flood arrives is significant in the determination of whether someone can survive flood conditions, the structure inventory should include information such as occupancy type (single family residential, commercial, etc.), height (one story, two story, etc.) and construction type. The National Structure Inventory (NSI) is the industry standard for initial structure inventories, but analysts should refine the base NSI as necessary to meet the needs of the study.

(b) The number of people in a structure often varies between day and night and weekday and weekends in residential, commercial, and industrial areas. Population in a structure or area can also vary significantly on a seasonal basis for campgrounds and other types of recreational facilities. Consider a range of different scenarios to capture the temporal variations in the numbers of people in a structure. The number of people estimated in each structure should apply to the time that an official public warning to evacuate is issued for the event being considered.

(c) It is important to consider the fact that often floods, or flood-initiated failure events for a dam or levee, occur only during a specific season of the year. This means the potential for large flood events is commonly highly correlated with season of the year. Day and night populations estimates should consider the shift of population in an area due to working in or out of the area during the day and returning home during the evening, and other similar considerations. The NSI contains population estimates for day and night for each structure, but seasonal and weekend population distributions are not included.

(d) Transient population (campground and other recreation) should be considered explicitly and included as appropriate when developing structure inventories. Recreational PAR for a campground should be based on visitation information for impacted downstream recreation areas (not the ones around the lake) and the duration of average visit when available. The NSI does not contain structures or population counts for most transient PAR categories, so it is up to the analyst to include these in the structure inventory if deemed significant.

(2) *Redistribution of people.* For a given flood scenario, LifeSim and HEC-FIA simulate the warning and evacuation processes (Figure 11–1) to determine the number of people exposed to flood waters. The exposed PAR is the subset of the initial PAR that does not evacuate to safety before the arrival of flood water at their location. The timeline identifies points in time and the time periods between those points in time (delays) required to warn and evacuate the PAR.

(a) The points in time include hazard occurrence time, imminent hazard identified, Emergency Management Agency (EMA) notified of hazard, evacuation warning issued, first alert received by public, evacuation begins, and destinations reached. The periods between the points in time are referred to as delays. They include imminent hazard identification period, hazard communication delay, warning issuance delay, first-alert diffusion delay, protective action initiation (PAI) delay, and evacuation delay.

(b) The warning timeline starts just prior to the hazard initiating. The period between when the hazard is identified and when the hazard occurs is the imminent hazard identification period. Note that the imminent hazard identification can happen before or after the hazard occurs. Next, the hazard communication delay is the period it takes for whoever identified the hazard to alert the EMA. Once the EMA is notified of the hazard, information regarding the hazard (a warning) must be communicated to the public. The time it takes for EMAs to compile the message and issue a warning is the warning issuance delay.

(c) At this point the warning is issued to the PAR. It takes time for the warning or first alert to be disseminated to the public. This period is referred to as the first-alert diffusion delay. Once the population receives the first alert, they will decide to take protective action or not. This decision process is the PAI delay. If they decide to act (evacuate), the time it takes for them to reach their destination is the evacuation delay. For the evacuation to be successful, the protective action must begin in time for the PAR to reach safety before the hazard arrives. Each delay should be represented with uncertainty in life estimation process.

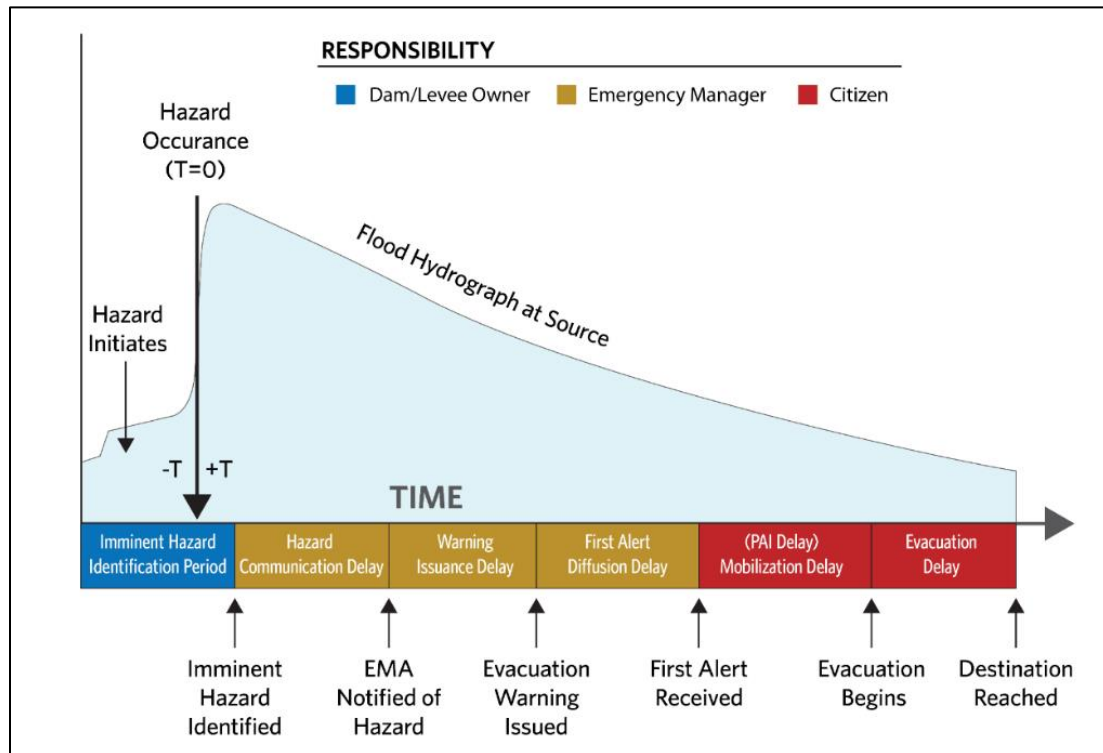


Figure 11–1. Warning and evacuation timeline

(d) Three entities are involved in the warning and evacuation timeline: (1) the dam/levee owner/operator or flood forecast agency, (2) the EMA, and (3) the PAR. The entities, their roles, and their impacts on the warning-evacuation timeline are shown in Table 11–1.

Table 11–1
Entities, roles, and impacts on the warning and evacuation timeline

Entity	Role	Impact on Timeline
Dam or levee owner operator or flood forecast agency	• Detects the potential or actual failure • Weighs decision to notify EMAs • Notifies EMAs • Carries out standard operating procedures related to the dam failure to the extent possible	The actions of the dam or levee owner/ operator impact the warning delay time. In some cases, a bystander or other observer detects the failure mode. In that case, the incipient or actual failure may be reported directly to an EMA.
Emergency Management Agency (EMA)	• Receives notification from dam owner/operator • Weighs decision to warn the public • Warns the public	The actions of the EMAs affect the warning delay time, the first alert diffusion time, and the PAI.
Population/sub- population (there may be more than one sub- population)	• Receives first alert • Mobilizes (starts to evacuate) • Travels the fastest path to an accessible safe destination • Reaches a safe destination or is caught in transit	The characteristics and actions of the population (or sub-populations) affect the first alert diffusion time and the PAI.

(e) Life loss estimates are highly sensitive to when a warning is issued relative to onset of flooding, and there can be significant uncertainty about when a warning should be issued. This is especially true when estimating the time of warning issuance for a given dam or levee breach scenario. Experience indicates that there is sometimes a reluctance to issue dam or levee failure warnings. The operating procedures or emergency plans that may be available for a dam or levee system should provide some guidance regarding when a warning should be issued; however, there is no assurance that a warning will be initiated as directed in a plan.

(f) Life loss modeling for dam and levee safety studies can be used to highlight weaknesses in the warning process and provide some guidance on how improvements in the process could potentially reduce the loss of life if a breach or overtopping scenario were to occur. Uncertainty and sensitivity analysis should be used to provide information on how significant warning issuance delay is in a life-loss estimate.

(g) Factors that influence each delay time from Table 11–1 are briefly described below. For more information, see EP 1110-2-17. These factors can be measures, and corresponding delays can be predicted by discussions with the operations personnel and emergency managers involved. When obtaining input from operations personnel and emergency management personnel, it is important to carefully describe the flooding scenario, including the key assumptions that define the development and detection of the event, so that all associated factors in estimating warning issuance times for structures are considered.

(h) It is useful to have more than one responsible person involved in this expert elicitation process since different individuals often think of different important factors and their judgments may vary resulting in a range of estimates of warning issuance times. If a potential failure mode analysis is being performed, the warning issuance times should be estimated by the group during discussion relevant to each failure mode.

1. Imminent hazard identification and hazard communication delay are primarily the responsibility of the owner/operator of flood defense infrastructure, or the regional flood forecast agency. The imminent hazard identification and communication delay rely on effective emergency action planning (EAP) to efficiently identify the hazard and communicate the emergency to downstream EMAs. An effective EAP should include a detailed monitoring and intervention plan, which increases the likelihood of an early hazard identification. The EAP should also include message templates and notification flowcharts, and identify the availability of redundant communication channels, which reduces the hazard communication delay.

a) For most breach mechanisms where the breach progression is observable prior to catastrophic failure of the dam or levee, the time when a warning is issued should be determined by first estimating the time when imminent hazard is acknowledged relative to the time of breach (collapse of embankment). The imminent hazard time for these failure modes is the time at which a dam/levee owner determines that a failure is likely to occur, and they decide to recommend that the responsible EMA issue an evacuation warning.

b) Nonbreach scenarios follow a similar process. The warning time is determined by first estimating the time at which an event is expected to induce flooding or a particular threshold condition (such as river stage) is surpassed. At that time, those with the forecast knowledge should notify the authorities responsible for issuing an evacuation order.

2. The warning or evacuation issuance time is defined as the point in time when an official alert or warning is first sent out by the responsible EMA to the PAR. Prior to warning initiation, there is a decision-making process that occurs within the EMA. The amount of time that decision takes to make is called the warning issuance delay and covers the time from when the agency learned about the impending hazard and when they flip the switch on whatever warning systems they have at their disposal.

a) Factors that most influence the warning initiation delay are whether the EMA has a warning plan and procedures written down, and whether they have a warning decision matrix in place.

b) A warning decision matrix describes all the possible issues (or evidence of potential issues such as increased seepage under levee), what that means in terms of a potential flood threat, and what message should be sent to the PAR under that scenario. If all these items have been thought through ahead of time, the process will be much more efficient during an actual event.

3. Warning issuance delay is modeled in LifeSim and HEC-FIA using warning issuance delay curves. A warning issuance delay curve defines, with uncertainty, an estimate of how long it will take for the EMA to issue a warning after receiving notification of a threat. Warning issuance delays can vary greatly and should be estimated based on discussions with local emergency managers. It is usually appropriate to define different warning issuance delays for different EMAs. USACE uses a consequence elicitation process to determine warning issuance delay curves. That elicitation process follows guidance from social sciences on the questions that should be asked of the emergency managers and methods for using their answers to select appropriate diffusion curves. Contact the USACE Risk Management Center for up-to-date consequence elicitation guidance.

4. The amount of time it takes for warning to spread through a community is known as the first alert or warning diffusion time. Warning diffusion varies between communities and events.

a) Some dissemination channels reach more people more quickly than others, and some types of people are easier to reach than others. Warnings can come from a formal emergency management system, through informal communication processes, or directly from cues in a person's environment. It can be an alert (signal) or a notification (message). Formal alerts and warnings can come via several different communication channels involving both new (mobile phone or internet) and established (TV, radio, siren, route alert) technologies. Each type of channel has strengths and weaknesses, such as the speed of dissemination, ability to convey information, and susceptibility to failure.

b) The receipt of a warning is also influenced by the characteristics of the people for whom the message is intended. These characteristics include the activities that people are engaged in, where they are located, the time of day, reception impediments (visual or hearing impairment), and the personal resources available.

c) The warning dissemination process is modeled in LifeSim and HEC-FIA using warning diffusion curves. A warning diffusion curve defines the relationship between time from warning issuance and the percentage of the PAR that has received that warning. Warning diffusion speed can vary greatly and should be estimated based on discussions with local emergency managers. In many cases, it is necessary to define different warning diffusion curves for different sub-populations. USACE uses a consequence elicitation process to determine diffusion curves. That elicitation process follows guidance from social sciences on the questions that should be asked of the emergency managers and methods for using their answers to select appropriate diffusion curves. Contact the USACE Risk Management Center for up-to-date consequence elicitation guidance.

5. PAI delay time, or mobilization time, is defined as the amount of time between when a warning is received by a person and when that warned person takes the recommended protective action (leaving their structure).

a) Like warning issuance delay and warning diffusion times, PAI delay times are modeled with uncertainty in LifeSim and HEC-FIA using PAI delay curves. These PAI delay curves contain two important pieces of information for determining the number of people that have evacuated their structures when the flood arrives: (1) the percentage of warned people that mobilize over time; and (2) the maximum mobilization percentage.

b) The maximum mobilization percentage defines the highest percentage of people expected to mobilize, given the characteristics of the potential inundation scenario, the warning message, and many other factors, including demographic and cultural considerations and, in some cases, the effects of past evacuation experiences. History has shown that 100% compliance is rare when a population is ordered to evacuate an area in response to an imminent flood threat, whether it is due to a person's inability or unwillingness to leave. Like the warning dissemination functions, it is important that the PAI delay functions be defined based on the characteristics of the PAR and flood scenario for a given study.

c) USACE uses a consequence elicitation process to determine PAI delay curves. That elicitation process follows guidance from social sciences on the questions that should be asked of the emergency managers and methods for using their answers to select appropriate diffusion curves. Contact the USACE Risk Management Center for up-to-date consequence elicitation guidance.

d) Life loss estimates are highly dependent on the PAI delay curves, especially when long warning times exist, as the percentage of people that do not evacuate become the primary driver of potential life loss.

6. The combination of warning and mobilization functions defines the rate at which people leave their structures after an evacuation warning is issued. LifeSim contains a detailed evacuation simulation component that considers the location and timing of vehicles entering the road network, road type, density of cars on the road within the affected area, percentage of people with live traffic updates, user-defined evacuation routes and destinations, depths and velocities of water over time, and many other factors to determine the efficiency, and ultimately the success, of the evacuation process for an individual scenario.

a) Traffic flow is modeled using a dual-regime modified Greenshields' approach (Greenshields 1935), which determines traffic speed based on road type, number of lanes, vehicle density, minimum stop-and-go speed, and multiple other factors. If jammed, the percentage of people identified as having live traffic updates are allowed to determine new routes and adjust accordingly. As with any aspect of the life loss estimation process, the amount of time spent on analyzing the evacuation process should be appropriate for the scope and characteristics of the study.

b) For highly urbanized areas, or for scenarios with short warning times where traffic congestion or traffic routing could make a significant difference in the ultimate loss of life estimate, detailed dynamic traffic routing over existing road networks should be

considered using LifeSim. For other studies, simplified approaches to estimating evacuation effectiveness may be appropriate, such as those in HEC-FIA.

b. Anyone who occupies (lives, works, or recreates in) potential inundation areas downstream of a dam or behind a levee is at risk of losing their life. The magnitude of that loss of life is based on the extents of the flooding, how quickly the flooding spreads, and the depths and velocities where people are located when the flooding arrives. Any estimate or report of loss of life from flooding must be accompanied by a reasonable description of how these factors influenced the overall estimate. These factors can be estimated through application of hydraulic models. A variety of one- and two-dimensional hydraulic models are available that can be used to model various dam and levee failure and nonfailure scenarios. For example, USACE studies typically use Hydrologic Engineering Center's River Analysis System (HEC-RAS) 2D to model dam and levee break scenarios (USACE 2023).

(1) At a minimum, development and post processing of the hydraulic model for consequence assessment scenarios requires terrain data, aerial photography, and physical characteristics of the dam or levee and potential breach scenarios being modeled (if breach scenarios are being modeled). Additional data requirements may include downstream hydraulic structure information (such as bridges, levees, dams), storage areas, stream centerlines, and other features important to the hydraulic analysis. Many of these items are detailed in previous chapters of this document. The level of data needed and processing time required for model development should be consistent with the type of risk assessment being performed. The best available data that can be acquired and processed within the resources scoped for a particular risk assessment should be used.

(2) Inputs to a hydraulic model that most influence the flood characteristics for a given flood scenario are listed below. Like any modeling effort, results of hydraulic models can be highly uncertain. These uncertainties are reflected in the estimation of potential loss of life. The modeler must understand how the factors can affect the accuracy of the flood characteristics so they can improve the quality of the results and communicate significant uncertainties. Estimates of potential loss of life should include a description of how each of the factors identified in paragraphs 11–3.b(3) through 11–3.b(8) were accounted for in the modeling process.

(3) The development of any hydraulic model in support of a life loss assessment requires a reasonably accurate representation of the terrain data. The level of effort to acquire and process terrain data should be consistent with the type of risk assessment being performed. The best available data that can be acquired and processed within the scope and available resources of a risk assessment should be used. Generally, lower resolution (30 meter or 10 meter) USGS DEMs do not include necessary details about many important features that have a significant influence on the inundation characteristics for a given scenario (levees, highways, etc.). These features should be added to the available terrain data set, especially in areas where they exist primarily to reduce flooding in urban areas, or higher resolution terrain data should be acquired.

(4) The level of the water in a reservoir or river that leads to the flood event (breach or overtopping of the flood defenses) typically influences the consequence estimate. Also, inflow conditions into the reservoir and coincident tributary flow into the modeled reaches downstream of a dam may have a significant impact on consequence determination.

(a) For example, if a dam breaches with very little water against it and no inflow, then that breach is less likely to cause loss of life and catastrophic damage than if it breaches under full conditions with a probable maximum flood-type inflow. When a dam or levee breach model is run to estimate the inundation characteristics at downstream locations, care should be taken to properly account for not only the elevation of the water against the structure when the breach occurs, but also the volume of water that is available to flow through that breach (reservoir volume and inflow if significant, or river hydrograph) and at downstream locations.

(b) From a hydrologic or loading perspective, a proper risk assessment requires an evaluation of the full range of hydrologic loading conditions and possible breach mechanisms that could lead to appreciable consequences.

(5) The estimation of an overtopping or breach location, dimensions, and development time are often crucial, especially when estimating loss of life due to breach of a dam or levee. Breach parameters directly affect the estimates of flow coming out a breach and how quickly flood waves reach downstream locations. Unfortunately, the breach location, size, and formation time are often the most uncertain pieces of information in a dam or levee safety risk assessment.

(6) Breach characteristics can be estimated in several ways, including comparative analysis (comparing existing structure to historical failures of dams or levees of similar size, materials, and water volume); regression equations (equations developed from historical dam failures to predict peak outflow or breach size and development time); and physically based computer models (computer programs that attempt to model the physical breaching process by using sediment transport/erosion equations, soil mechanics, and principles of hydraulics).

(a) All of these methods are viable techniques for estimating breach characteristics. However, each of these methods has strengths and weaknesses and should be considered as a way of estimating the parameters and not used as absolute values. Historical breach information, regression equations, and physically based computer models all have limitations that must be well understood when they are applied.

(b) For detailed life loss assessments, it is important to estimate a range of parameter estimates for the breach size and development time for each failure scenario/event, and then perform a sensitivity analysis of the breach parameters to identify their effect on the outflow hydrograph, downstream stages and flows, and warning time to any PAR.

(7) Each set of breach parameters produce a different outflow hydrograph. However, once these hydrographs are routed downstream, they tend to converge toward each other. Hydrographs from different assumed breach parameters can converge to produce similar peak flow and stage in a surprisingly short distance. The rate at which the hydrographs converge depends on many factors: steepness in the rise of the breach (outflow) hydrograph, volume of the breach hydrograph, slope of the downstream reach, roughness of the downstream reach, available storage in the downstream floodplain, etc.

(a) The modeler should route all of the breach outflow hydrographs downstream through the entire study area to fully evaluate the effect of the breach parameters on the resulting flood hydrographs and inundation levels.

(b) The modeler should use engineering judgment to select the set of breach parameters that are considered to be most appropriate for the dam and for each event/pool elevation, and show how the range of breach parameters can impact the range of potential loss of life.

(8) To complete dam breach (or nonbreach) consequence assessments, coincident inflow hydrographs downstream of the dam are required to accurately predict inundation extents and the associated estimate for loss of life. When modeling dam failure scenarios, inflow hydrographs to the reservoir upstream of the dam site and areas downstream of the dam establish antecedent conditions for the water bodies and river systems in the study watershed. Downstream inflow hydrographs become even more important when determining incremental consequences (the difference in consequences between with- and without-failure scenarios). Methods for estimating coincidental inflows include the Hydrologic Modeling Method, the Exceedance Flow Rate Method, and the Residual Rainfall Method (USACE 2024h). These should be applied based on engineering judgments following the accepted USACE guidance.

(a) The degree of shelter provided by the setting where people are located can have a large impact on whether someone loses their life during a flood event. After the warning, mobilization, and evacuation processes have been simulated and flood scenarios have been developed, LifeSim and HEC-FIA categorize those people that are exposed to flooding into two flood lethality zones. The classification is a function of shelter effectiveness and the depths and velocity of flooding at that location.

(b) Shelter effectiveness is determined based on the type of shelter and associated stability criteria. For example, if located in a wood building, the shelter effectiveness is determined based on the depths and velocity of flooding at that location, the stability criteria for wooden structures, and the height of the structure (if the structure survives but depth exceeds the height of the structure, the shelter effectiveness is negligible). Another example is if someone is caught out in the open while evacuating on foot. In that scenario, the shelter effectiveness should be based on the relationship between the depths and velocities at that location and predefined human stability criteria.

c. Fatality rates are applied to the exposed population, which is the subset of the PAR that are flooded before they could leave the flooded area, whether they are still in their house or overcome during the evacuation process. HEC-FIA and LifeSim categorize exposed population into flood (lethality) zones, which distinguish physical flood environments where historical rates of life loss have distinctly differed. Each flood zone is physically defined by the interplay between available shelter and local flood depths and velocities, as summarized below.

(1) High hazard conditions in which flood victims are typically swept downstream or trapped underwater and survival depends largely on chance; that is, the apparently random occurrence of floating debris that can be clung to, being washed to shore, or otherwise finding refuge safely. Historical fatality rates range from about 50% to 100%, with an average rate of about 70%.

(2) Low-hazard conditions where people are exposed to relatively calm floodwaters and their stability, or the stability of their shelter, is not at risk. However, a hazard exists because people are encountering water in locations not meant for such interaction. Historical fatality rates are virtually zero in low-hazard conditions.

11–4. Sources of uncertainty in estimating loss of life

a. *Estimate loss of life.* Any attempt to estimate loss of life is highly uncertain. Therefore, the effort should focus on identifying the factors that contribute the most to the overall estimate and should contain reasonable uncertainty and sensitivity analysis about those factors and their contribution to the overall life loss assessment.

b. *Initial population distribution.* Sources of uncertainty in the initial distribution of people include the following:

(1) Lack of knowledge about the actual location and number of structures in the potential inundated area. This uncertainty is a function of how the structure inventory was developed.

(2) Lack of knowledge about the actual distribution of people in each structure at the onset of a flood event.

c. *Redistribution of population.* Sources of uncertainty in the redistribution of people during the warning, mobilization, and evacuation process include:

(1) Uncertainty about identifying a potential problem (breach and/or forecasted event).

(2) Uncertainty about if people responsible for issuing an evacuation warning do so based on available evidence.

(3) Uncertainty in how quickly the warning spreads through the PAR, which is a function of not only the warning system but also the activities that the PAR is engaged in at the time of the warning.

(4) Uncertainty in how people respond to a given warning; specifically, how long they take to mobilize or if they will mobilize at all.

(5) Uncertainty in how the evacuation process will proceed, including how people choose to evacuate (public transportation, private vehicle, on foot) and how they choose to move through the available evacuation network.

d. Uncertainties in characterization of the flood event. Sources of uncertainty in the flood event, which is defined by arrival time, depth, and velocity at all locations throughout the inundated area include:

(1) Uncertainty in breach parameters, including initial breach location, breach growth rate, breach formation time, and final breach dimensions.

(2) Uncertainty caused by terrain data inaccuracy including whether the terrain model used includes those features (levees, raised roadways, etc.) that could have a significant impact on how flood waters travel through the floodplain.

(3) Uncertainty in downstream coincident inflows.

e. Uncertainties in shelter provided by location when flood water arrives. Specifically, how well the structure will withstand the combined depths and velocities at that location, including:

(1) Uncertainty in structure height, including structure foundation height and height of ceilings and roof.

(2) Uncertainty in structure construction type, which is important when determining if a structure can withstand the flood characteristics and avoiding collapsing on potential occupants.

(3) Uncertainty in stability criteria of structures, vehicles, and people.

f. Uncertainties in fatality rates. Specifically, if using average fatality rates for lethality zones rather than the full fatality rate distribution for each zone as show in Figure 11–2.

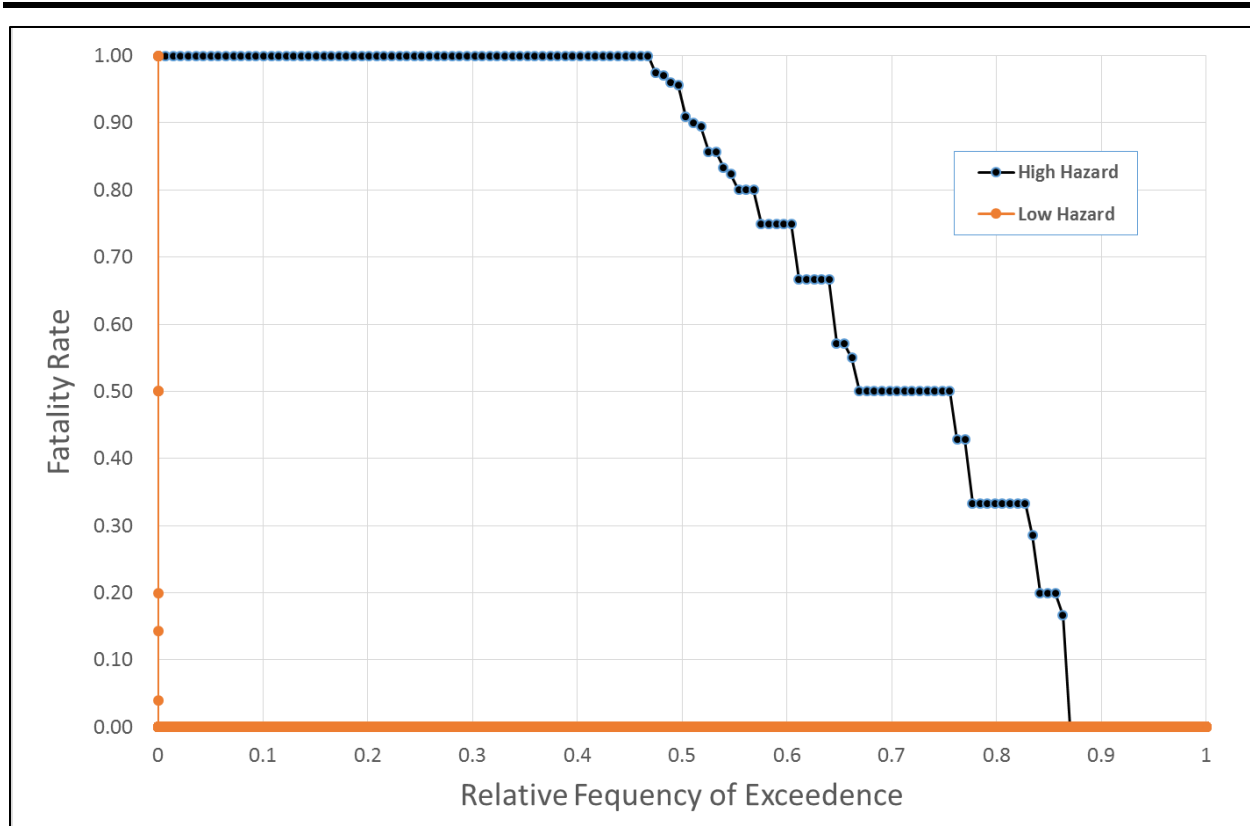


Figure 11-2. LifeSim fatality rate distributions

11-5. Quantitative description of uncertainty in estimating loss of life

a. Any estimate of potential loss of life from flooding contains a significant amount of uncertainty, and therefore, describing that uncertainty is necessary to provide the appropriate information to decision makers on how to use those estimates.

b. Quantitative descriptions of uncertainty in life loss estimates are preferable to qualitative as most people are unable to grasp how uncertainties in individual factors influence the overall life loss estimate. Therefore, estimates should include, at a minimum, a sensitivity analysis that shows the most likely estimate as well as the reasonable minimum and maximum values. Preferably, the quantitative estimate should be developed using a Monte Carlo approach where distribution of uncertainty in life loss estimate could be fully described.

c. Estimates for those parameters or inputs that are highly uncertain should be determined by informal or formal team opinion or expert opinion elicitation as appropriate, depending on the nature of the study (if the results are likely to be highly contentious, then formal expert-opinion elicitation may be more appropriate).

(1) This process involves selecting the appropriate team or experts for the issue, review and discussion of pertinent background information, documenting the factors that make influence the parameter in a positive or negative manner (more or less likely,

more or less efficient, etc.), conducting initial anonymous elicitation, review and discussion of initial results, and conducting additional rounds of elicitation if necessary. Elicited values should include the “best estimate” and “reasonable best case” and “reasonable worst case” scenarios.

(2) This process not only helps build the case for each variable that factors into the life loss estimate, but it also provides a range of values for each variable that can be used to define the uncertainty about them.

d. There are many ways to portray loss of life estimates with uncertainty. Tables displaying mean estimates along with median, maximum, minimum, and standard deviation from the mean are useful and should be provided at a minimum. However, graphical representation of sensitivity and uncertainty analysis should be used to better portray certain concepts. Some examples are provided in the following paragraphs.

(1) Figure 11–3 illustrates results from a sensitivity analysis that shows how an earlier mandatory evacuation warning can potentially reduce loss of life for a dam breach. The x-axis is the time of day when breach occurs (0 is midnight, 2 is 0200, etc.) and the y-axis is the estimated loss of life. The chart shows not only how loss of life varies significantly depending on when the breach occurs, it also shows that loss of life during the night is much more sensitive to when warning goes out than loss of life during the day.

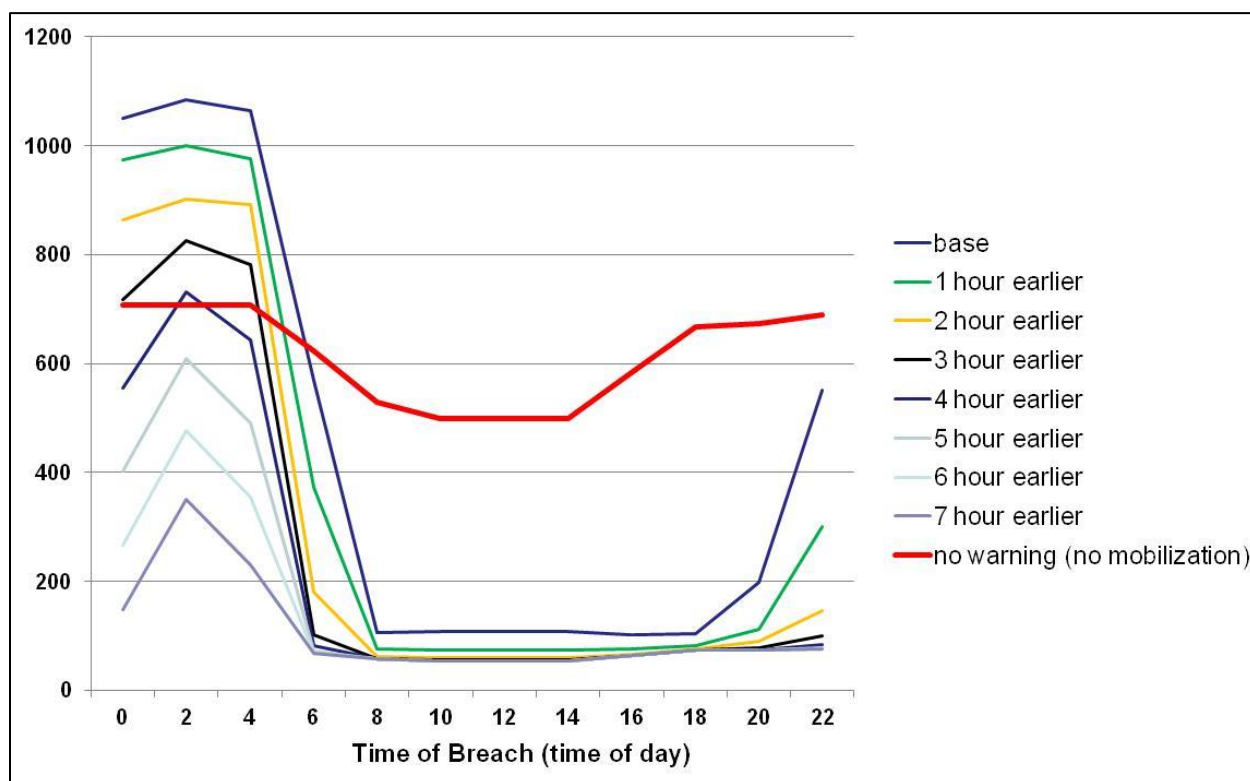


Figure 11–3. Displaying loss of life results from sensitivity analysis

(2) Figure 11–4 illustrates the range of results from an uncertainty analysis on the potential impacts of an advanced warning system risk management measure through whisker plots, where the line in the middle of the two boxes is the median value, the lower and upper boxes represent the middle 50% of results, and the lower and upper lines are the first and fourth quartiles. The whisker plot in this case shows that while the overall loss of life is reduced with an improved warning system, the uncertainty in the estimate is also larger than the results for the existing warning system.

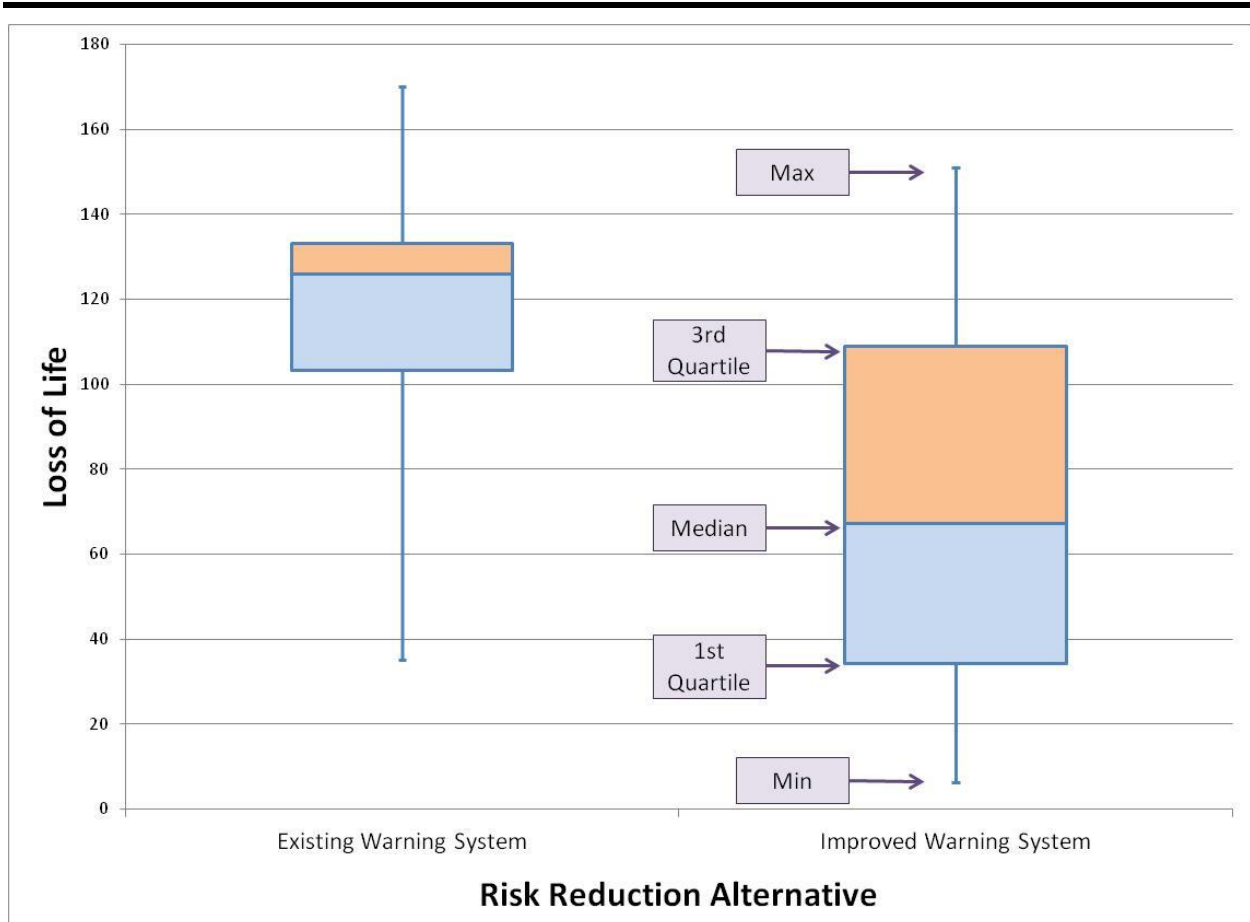


Figure 11–4. Displaying loss of life results from uncertainty analysis

11–6. Estimating expected annual life loss

Estimating EALL generally follows the same approach as outlined in Chapter 2 of this document for estimating EAD.

Chapter 12

Computing Flood Hazard and Exposure in Coastal Systems

12–1. Overview

a. Chapter 2 of this manual describes, in concept, how risk is computed for USACE studies. For the computation, exposure to the flood hazard must be described. This chapter describes the methods for computing the flood hazard and wave-forcing exposure for coastal systems.

b. Flood risk for an impact area subject to coastal storm-related flooding from large water bodies, such as oceans or lakes, is primarily a result of storm water levels (SWLs) that are a function of astronomical tide; storm surge; wave setup; wave run-up; and long-term, sea level change; and can be combined with heavy rainfall runoff and riverine flooding. Local winds and currents can contribute to SWLs. Overtopping of levees, floodwalls, revetments, and beach/dune systems and consequential flooding can occur from steady overtopping from extreme flood levels, but flooding can also result from direct wave attack. Erosion on or around these systems can lead to breaching and exacerbated flooding. Coastal storms also produce high winds. The winds contribute to surge and waves but also can independently produce significant damage to infrastructure.

(1) Coastal flood damage risk computation can be quite complicated primarily because of the multiple correlated and complex nonlinear processes, and because extreme coastal storms are relatively rare at any specific location. Waves and water levels are usually correlated, being caused by a local storm. However, very large storm waves can travel thousands of miles, so wave hazard can occur from multiple events simultaneously. Waves are random and are characterized by directional wave spectra. While water level is a single variable, wave spectra are usually parameterized with a characteristic wave height (H_s or H_{m0} = significant wave height), period (T_p = spectral peak period), direction (θ = mean wave direction). In shallow water, waves and water levels are strongly influenced by one another. Waves contribute to water levels in the form of wave-setup, while wave height is limited by wave breaking as water depth decreases.

(2) Wave energy is usually conceptualized in terms of regions of the energy density spectrum with rule of thumb separation by frequency range as follows: local wind waves ($f = 0.1 - 2$), swell ($f = 0.05 - 0.1$), infragravity ($f = 0.005 - 0.05$), where f is frequency (1/s). The wave energy density spectral characteristics evolve in the nearshore and inland with energy dissipation through wave breaking and bottom friction, and a dramatic shift to lower frequencies. The infragravity frequencies in inland areas may be low enough to be considered as increases to the water levels in some cases. In inland areas, there may be local wave generation as well as waves propagating inland from offshore. Even though water level is a single variable, it is the result of the interaction of its components. Surge and tide interact nonlinearly and the phasing and duration can be important.

(3) Surge, waves, wind speed, and rainfall for a specific storm are typically correlated while tide, seasonal water level variations, decadal water level variations, and long-term relative sea level change (RSLC) are usually independent variables. The types and seasonality of atmospheric disturbances that produce coastal hazards also need to be considered. Coastal areas along the Atlantic mainland are routinely exposed to both tropical and extratropical storms, which peak at different times of the year and have different statistical characteristics. Time of year can be important because storm intensity and frequency tend to peak at specific times of the year.

(4) As an example, in some cases, the highest spring tides may roughly coincide with the peak of annual storm intensity. Also, the peak of hurricane season coincides with the peak of water temperature in the Gulf of Mexico and so corresponds to the highest static sea surface level of the year. Coastal flood risk assessment requires computing the joint probability of water level and wave responses and sometimes local hydrology and riverine stage for the forcing hazard. Several approaches are presented in this chapter depending on the storm hazards.

c. In relatively rare cases, coastal risk assessment can proceed using an approach somewhat similar to the riverine approach. That is, a flood stage probability function can be defined for a reach of flood protection, then transformed to a damage-probability function and subsequently integrated to compute EAD. This is a frequency-based approach founded primarily on water level. However, wave damage or erosion is commonly an important factor in coastal risk computation. In coastal areas with no flood protection or hardened structures (levees and floodwalls), the analysis is usually based on peak values of forcing parameters. HEC-FDA is based on a stochastic analysis computation approach termed “event-based,” where a single AEP forcing condition is defined (for example, SWL and H_{m0} defined at the 1% AEP) and associated responses determined.

(1) Melby et al. (2021b) illustrated that this approach can produce large errors for coastal applications because the multi-dimensional, hyper-surface that represents the 1% AEP joint probability has an infinite number of possible combinations of the wave and water level forcing parameters. A single value or small number of samples is not likely to characterize the risk. For that reason, coastal studies usually require a stochastic simulation approach termed “response-based,” where storm-forcing parameters are sampled from one or more populations, by-storm responses computed (for example, overtopping) and the damage-probability function computed from those by-storm responses.

(2) For some coastal systems, like beaches, the time-dependent characteristics are critical because beach-dune systems continuously evolve. For these systems, risk assessment should consider time-dependent characteristics. An example of a beach-dune system risk assessment is given in Appendix B. Time dependency, combined with the strong nonlinearity of the processes and the larger number of correlated processes, represent the key differences between inland and coastal flood hazard risk assessments. This complexity in coastal systems occurs in the physical processes (wind, surge and tide, and wave contributions to water level interact nonlinearly in

shallow water and are correlated), time dependency (tide, waves, and surge occur with different time scales and with different phasing and can be the result of multiple simultaneous storms), and sparseness in the long-term storm record.

(3) In addition, coastal extreme water levels have high local spatial correlation but lack significant spatial correlation regionally. Therefore, accurate coastal risk assessment often demands complex numerical hydrodynamic modeling and complex statistical analysis. Fortunately, new data resources and statistical simulation software that quantify coastal storm hazards are available that can dramatically simplify the coastal flood hazard analysis, and these are described in this manual.

d. This chapter covers the primary requirements for coastal flood hazard quantification to support a risk assessment. The probabilistic coastal hazard analysis (PCHA) can be complex and computationally intensive to do accurately, assuming the state-of-practice multivariate probabilistic methods are followed. Figure 12–1 suggests a strategy that takes advantage of the significant amount of coastal hazard information available. This approach is used for coastal planning projects. In particular, probabilistic coastal hazard studies have been conducted for all hurricane-exposed regions of the U.S. and these high-fidelity, statistical and process modeling simulations are available primarily from the Coastal Hazards System (CHS) (USACE 2024a) and, in some cases, from FEMA.

(1) If the appropriate CHS study is incomplete or inadequate for a particular study, the CHS regional probabilistic hazard study can be used as a basis and it can be augmented with additional process simulations. In addition, the Wave Information Study (WIS) (USACE 2024b), National Centers for Environmental Prediction/National Center for Atmospheric Research (NCEP/NCAR) Reanalysis (NOAA 2024i), and other resources as described in paragraph 12–3, provide hindcasts that, when combined with measurement data, may provide sufficient storm response data in many cases. Figure 12–2 summarizes the steps within a probabilistic coastal hazard study as discussed in detail in this chapter.

(2) In Figure 12–2, tropical storm analysis is on the left and extratropical storm analysis is on the right (green boxes). The blue boxes identify historical storm analysis for defining storm suite. The yellow boxes indicate optional modeling to define base or augmented storm suite. The orange boxes indicate responses for storm suite and associated epistemic uncertainty. Finally, the gray boxes represent storm response probabilistic analysis to define coastal hazard curves. This process is followed for studies that comprise the CHS (USACE 2024a). In Figure 12–1 and Figure 12–2, the general approaches are provided separately according to whether the storm is characterized as tropical or extratropical storm.

(3) While this chapter describes the various steps in Figure 12–1 and Figure 12–2 in detail, it is expected that much of the complex effort required to accurately quantify the hazard has been completed and can be accessed from the online data and product sources described herein.

e. This chapter starts with a description of the coastal flood hazard drivers and associated data and hindcasts. Next, general univariate and multivariate extremal analysis methods are summarized. Coastal flood risk planning studies should consider the univariate and multivariate probability distributions of wave and water level parameters, at a minimum.

(1) Once the flood hazard is quantified, coastal flood risk assessment follows stochastic simulation approaches available in USACE categorized by: (1) event-based simulation where the risk is based on a single statistic of forcing parameters, for example, the SWL and H_{m0} associated with the 1% AEP; and (2) response-based simulation where the stochastic simulation computes response for individual storm responses and then computes the final response distribution.

(2) Methods to consider uncertainty, both aleatory and epistemic, are summarized within this context. The methods vary by the coastal region because the coastal storm characteristics vary by region.

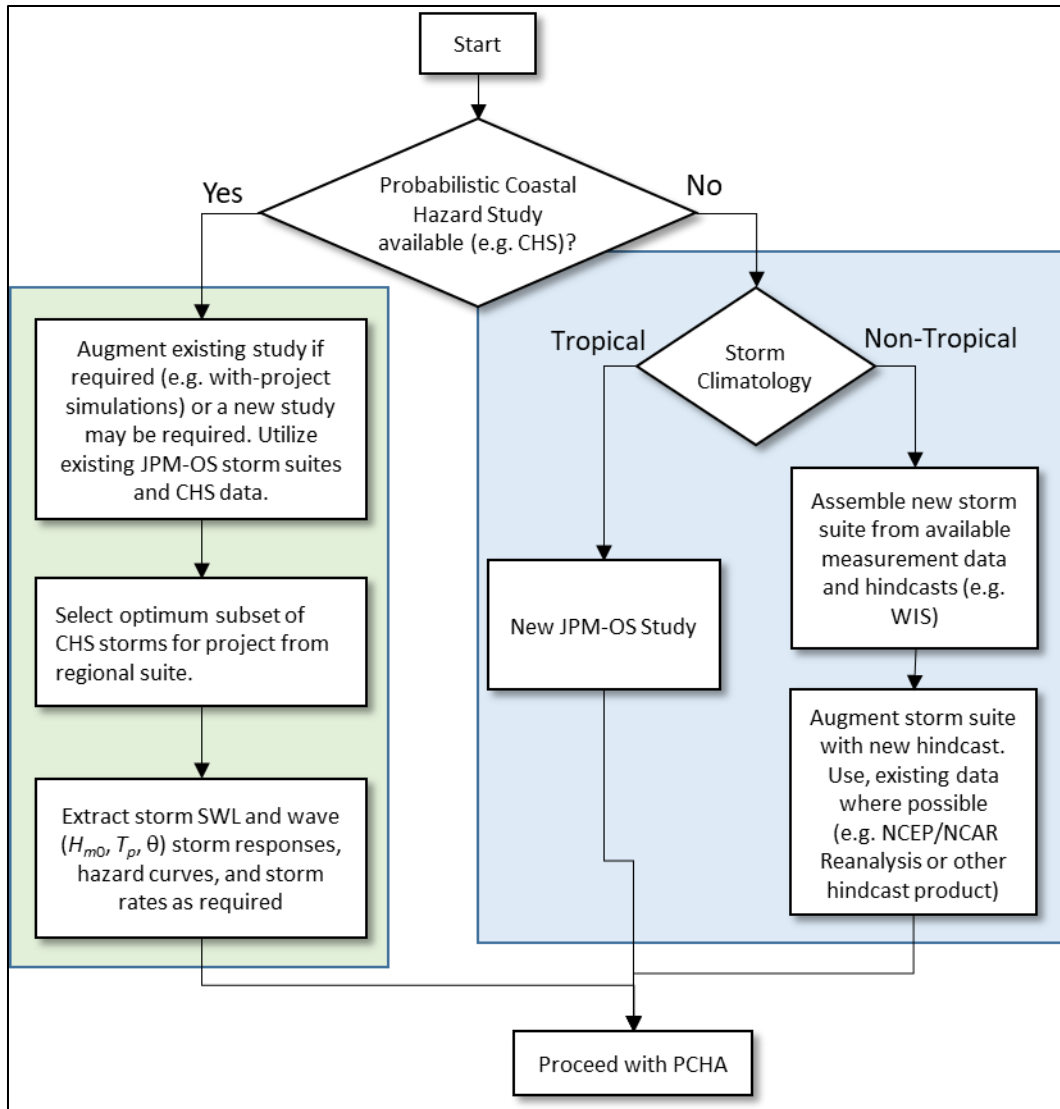


Figure 12–1. Probabilistic coastal hazard study outline that takes advantage of precomputed hazards

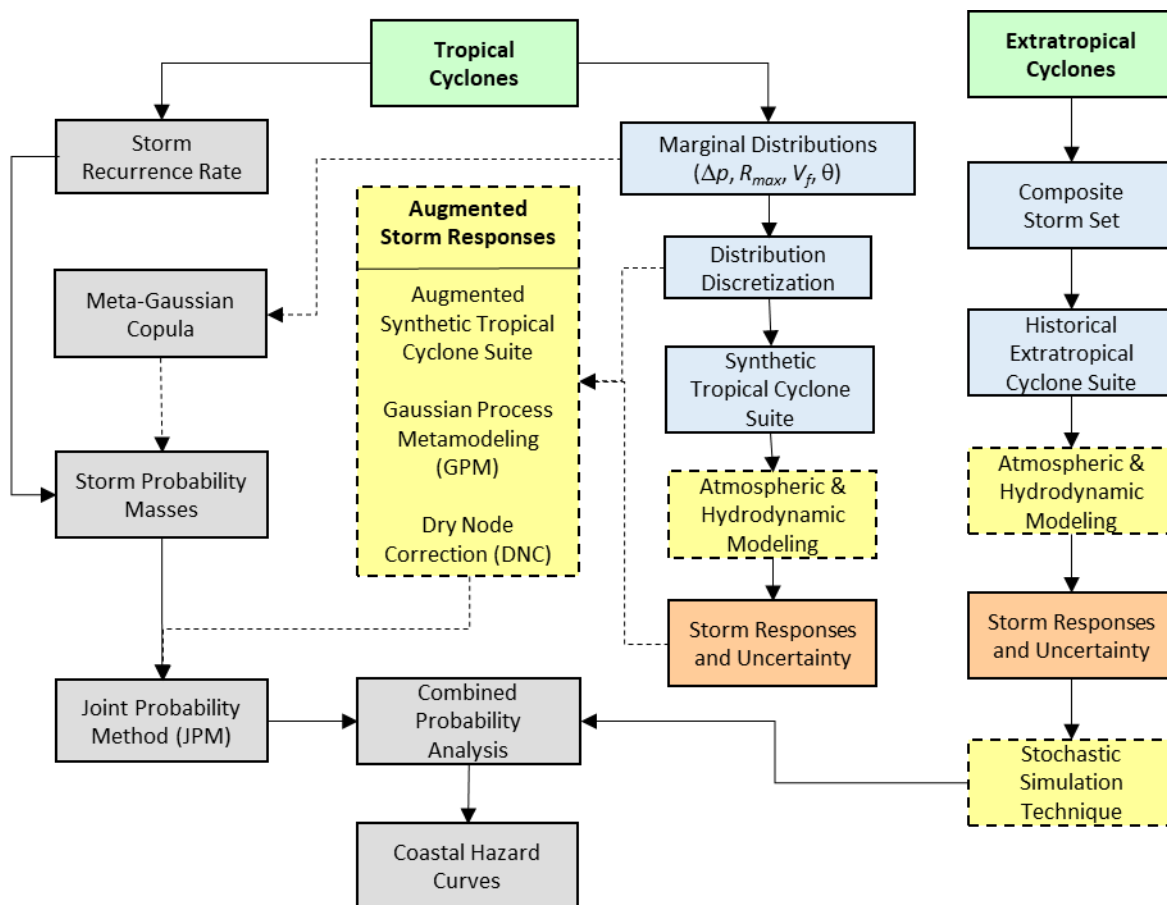


Figure 12–2. Schematic of probabilistic coastal hazard study for a mixed-storm population (based on Nadal-Caraballo et al. (2020))

12–2. Background for extreme water level analysis

a. Extreme coastal total water levels (TWLs, Figure 12–3) are the result of the complex interactions between multiple oceanographic, hydrological, geological, and meteorological forcings that act over a wide range of scales (astronomical tide, large-scale storm surge, wave setup, wind setup, precipitation, fluvial discharges, monthly mean sea level, subsidence, etc.). Magnitudes of extreme TWLs, as compared to elevations of important backshore features, such as the crest of coastal protection structures or sand dunes, are particularly germane to coastal risk.

b. TWLs are also critical in terms of computing depth-limited wave heights. Under certain conditions, some of the complex processes contributing to total water level can be simplified. For example, linear superposition of water level components (surge, tide, waves, and sea level change) is often appropriate. However, the contribution of each component of a TWL has the potential to significantly alter spatially and temporally varying flood hazards, resulting in a complex and nonlinear problem where linear superposition results in large errors.

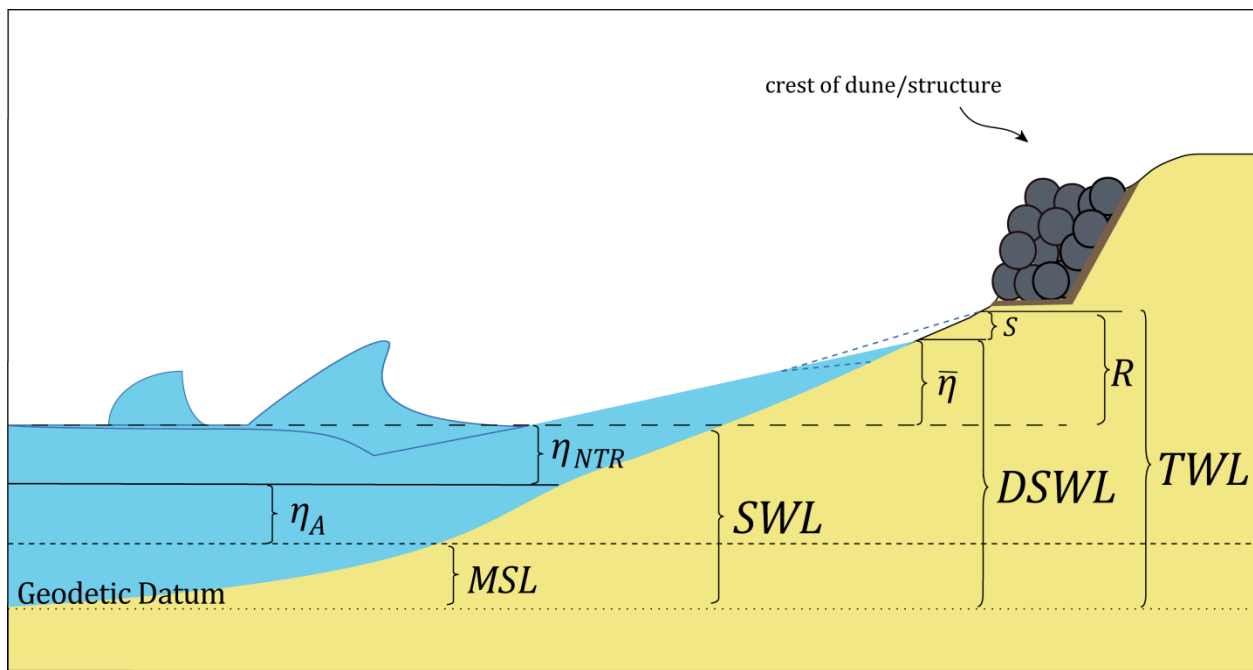


Figure 12-3. Generic profile view of the components of coastal water levels

c. At any given time, the elevation of the SWL relative to a fixed datum is composed of at least three components as defined in equation 12-1.

$$SWL = MSL + \eta_A + \eta_{NTR} \quad (12-1)$$

where:

MSL = the mean sea level,

η_A = the deterministic astronomical tide, and

η_{NTR} = the nontidal residual (NTR). The η_{NTR} is given as:

$$\eta_{NTR} = \text{seasonal cycle} + MMSLA + SS + \eta_Q \quad (12-2)$$

and is defined as any elevation change in the still-water level not related to the astronomical tide. The seasonal cycle can be significant (order 1 foot) due to cyclical changes in water temperature and other forcings. The η_{NTR} also includes low-frequency monthly mean sea level anomalies ($MMSLA$) associated with water level variability due to water temperature and the geostrophic effects of large-scale currents (processes associated with the El Niño Southern Oscillation). The combination of seasonal cycle and $MMSLA$ is often referred to as *steric water level* η_{NTR} and also includes high-frequency components due to the presence of winds and low atmospheric pressure (storm surge, SS). In sheltered environments, water level fluctuations from precipitation and river discharge, η_Q , also can contribute significantly to η_{NTR} .

d. The dynamic still-water level, DSWL, combines the SWL as defined in equations 12–1 and 12–2 with wave-induced changes to the mean sea surface and wave-induced water level fluctuations on the order of minutes, and is defined in equation 12–3.

$$DSWL = SWL + \bar{\eta} + \eta' \quad (12-3)$$

where:

$\bar{\eta}$ denotes the mean water level in the presence of waves, including wave setup (a super elevation of the water level due to wave breaking that reaches its maximum at the shoreline, Figure 12–3) and setdown

η' denotes additional water level fluctuations due to waves caused by processes such as bound long waves and wave groups. In many situations the DSWL should be used to compute depth-limited breaking wave heights for project applications. In situations where swash processes are negligible (unbroken waves impinging directly on a breakwater), the TWL is equal to the DSWL. When waves run up on coastal protection structures or beaches, swash processes are non-negligible, and the total water level is defined in equation 12–4.

$$TWL = DSWL + S \quad (12-4)$$

where:

swash, S , consists of contributions from both the incident band, S_{inc} , and the infragravity band, S_{ig} , as defined in equation 12–5.

$$S = \sqrt{S_{inc}^2 + S_{ig}^2} \quad (12-5)$$

e. The time-varying location where the water meets the land is termed wave run-up, R , and includes both the wave setup and swash oscillations around the wave setup. Equations 12–1 through 12–5 suggest that water level contributions can be computed through linear superposition of its components. However, the various processes interact nonlinearly, so there is an error in linear superposition termed the nonlinear residual (NLR). NLR is usually small on the open coast but can be large in inland flooded areas (Nadal-Caraballo, Melby, and Gonzales 2015; Melby et al. 2021b). For the case where NLR is suspected to be large, the various processes should be modeled together in a single, coupled numerical model, like CSTORM (Melby, Nadal-Caraballo, and Ebersole 2012), that couples a circulation and storm surge model (such as ADvanced CIRCulation, ADCIRC) with a wave transformation model (such as STWAVE).

f. Flooding in coastal systems is typically a result of storm-generated surge and waves. Storms are atmospheric disturbances with low atmospheric pressure and strong winds. The main contributors to increased water levels during a coastal storm are the wind stress, wave stress, and low atmospheric pressure at the storm center. The largest contributor to storm surge is typically the wind blowing over the water. The magnitude of the wind-driven surge is a function of the wind speed (higher surge with stronger winds), wind direction, water depth (higher surge for shallower basins), and the distance over

which the wind blows. The wind-driven vertical increase in water level from a strong storm can be a few feet to a few tens of feet.

g. The atmospheric pressure component of storm surge is called the barometric tide. This component is a bulge of water due to the lower atmospheric pressure in the center of the storm. The barometric tide is a few feet or less. Storm surge increases the water level locally for periods of hours and, in rare cases, for a day. The wave stress component of storm surge is called wave setup. Wave setup results from the transfer of momentum from breaking waves to the water column and results in increases in water level of approximately 15% to 20% of the breaking wave height. Wave setup can contribute several feet to the total storm surge in areas directly exposed to large waves. Wave run-up is typically the same order of magnitude as the wave height. While run-up is not usually explicitly used for flood risk computation, it is an indicator of wave overtopping if the run-up exceeds the crest elevation of the CSRM structure.

h. Astronomical tides are the periodic rising and falling of Earth's oceans in response to the gravitational attraction of the Moon, Sun, and all other celestial bodies and the rotation of the Earth. The moon has the strongest effect on the tides because it is nearest the earth. The timing and elevation of the tides vary by location and are influenced by the relative alignment of the Sun and Moon, the shape of the coastline/basin, and the near-coast water depths. High tides may occur approximately twice a day (semidiurnal with a period of approximately 12 hours 25 minutes), once a day (diurnal), or a mixture of the two (mixed tide). The tidal range (the difference in height between high and low water) varies on a 2-week cycle, with the maximum range (spring tide) occurring when the Moon and Sun are aligned relative to the Earth (at the time of a new or full moon) and the minimum range (neap tide) occurring when Moon and Sun are separated by 90° when viewed from the Earth.

i. Smaller amplitude variations occur at various other frequencies. Tides alone are seldom a source of flood damage, but high tides in combination with storm surge and wave setup often impact flooding. For example, in 2012, Superstorm Sandy made landfall at high tide near The Battery in Manhattan resulting in significantly higher TWLs than the same storm landfalling at low tide. Although Superstorm Sandy extreme SWL resulted in the storm of record for this area, the surge was similar to other extreme storms that had occurred in the 20th century in this area. However, the surge coupled with high tide produced water levels that were much more extreme than had occurred in modern history in the area of Manhattan, northern New Jersey, and southern Long Island (Nadal-Caraballo and Melby 2014). Fair-weather or sunny-day flooding, where high tides inundate developed areas, is becoming more common for low-elevation areas as a result of increased coastal development and sea level rise and, in some cases, subsidence.

j. Storms and tides are independent (uncorrelated) in the analysis of water levels at the time scale of a storm (hours to days), but storm surge and waves are generally dependent, primarily generated by wind from the same storm event and peaking at approximately the same time. Water levels in interior regions may or may not be correlated with coastal storms. There is the potential for coastal storms and hydrologic

impacts to be related in multiple ways: (1) antecedent rainfall and runoff combined with SWLs from a separate hydrometeorological event; (2) coincident rainfall and runoff with SWLs from a separate hydrometeorological event, and (3) coincident rainfall and runoff from the coastal storm or rain bands associated with the storm combined with SWLs from that same storm.

k. For example, rainfall and runoff associated with a slow-moving hurricane or extratropical storm may be correlated to the storm surge, but flooding from a large river within the interior basin is likely independent of coastal storms. Tropical storm frequency peaks during the period August to October, when river flow is generally low, thus these exterior/interior flooding events are usually physically independent but statistically (negatively) correlated. Tropical storms with extraordinary rainfall, such as Hurricane Harvey, can have increased flood risk due to the combination of SWLs and runoff, particularly near river outlets in estuaries.

l. Coastal water levels also vary in the long term due to local ground motion, local sea level changes and global sea level changes or time-varying lake levels. Local ground movement can, for example, be caused by post-glacial uplift or subsidence. Local sea level changes occur due to seasonal and decadal water temperature and current variations. Global sea level changes occur from long-term changes primarily driven by ice melt and thermal expansion. The National Oceanic and Atmospheric Administration (NOAA) present estimate of global sea level change is +3.6 mm/year (for the period 1993–2018) (NOAA 2024k). Usually, local sea level change is on the order of a few mm/year but some areas with strong subsidence have higher levels.

m. Finally, lake levels vary due to variations in precipitation and evaporation as well as varying anthropogenic influences, like lake level management. Sea level change is treated deterministically within USACE according to ER 1100-2-8162 and EP 1100-2-1. For the Great Lakes, lake levels can be treated as per Melby, Nadal-Caraballo, and Ebersole (2012) and Nadal-Caraballo, Melby and Ebersole (2012), where the anthropogenic contributions are eliminated and the resulting long-term variations treated stochastically.

n. Coastal storms usually develop over days and can continue to strengthen, transform, or weaken as they traverse the body of water they are over or near. Depending on the size of the water body and the path of the storm, landfall and coastal impacts can occur rapidly for a matter of hours within an area of tens of miles or result in longer duration impacts that occur over multiple days and tide cycles that can cover large regions of the coast. Even then the most acute impacts are often constrained to multiple hours for any one location. Typically, tropical systems are more focused and result in shorter duration and more localized impacts, while extratropical systems result in longer duration regional impacts. Once again, even for the longer duration events, the most acute coastal impacts for any one location are typically on the order of hours.

o. The coastal storm hazard consists primarily of waves and surge with tidal phase interaction. This can be contrasted with riverine flooding that occurs over a much longer time of days to weeks, particularly for a large drainage basin. In other words, damaging

coastal storms have relatively short durations, produce damaging impacts over relatively small spatial areas, and the most intense events are relatively rare. These are, of course, generalities and there are notable exceptions through the coastal storm history. In the future, due to sea level rise, the longer duration, wide-scale, extratropical events that may not have been particularly damaging in the past are likely to become more damaging since the relatively small surge will be on top of higher overall sea levels. Extreme storms, particularly hurricanes, are relatively rare, so measured coastal storm hazards due to extreme events at any given location are rare. Measured extreme coastal storm hazards are also limited by our spatially sparse measurements along the U.S. coast.

p. The result is that historical extremal analysis using water-level measurements does not effectively characterize storm flood risk. As a result, coastal storm response extremal statistical analysis methods have been evolving in recent years to take advantage of new technologies and more diverse data resources. This is described more in this chapter.

12–3. Data availability

Water level, wave, and hydrometeorological data requirements for coastal flood damage are summarized in EM 1110-2-1413 and EM 1110-2-1100. The following list summarizes the existing measurement sources:

a. NOAA’s National Ocean Service provides observations about water levels, tides, surge, and sea level change through the Tides and Currents web page (NOAA 2024k) (<http://tidesandcurrents.noaa.gov/>) that covers the mainland U.S. and territories. The USGS also provides some coastal water-level data (USGS 2024b) (<http://waterdata.usgs.gov/nwis/rt>). Specific USACE Districts maintain water-level gages as well.

b. Tropical storm parameters and other historical storm characteristics are well summarized in the Hurricane Database 2 (HURDAT2) (NOAA 2024a), available from the NOAA National Hurricane Center (<https://www.nhc.noaa.gov/data/#hurdat>) (NOAA 2024h). An additional source for these types of data is the NOAA National Climatic Data Center (NOAA 2024e), which distributes IBTrACS (International Best Tracks Archive for Climate Stewardship, NOAA 2024c), a global tropical cyclone parameter dataset.

c. Following are some coastal wind, wave, and water-level probabilistic hazard products and hindcasts.

(1) A comprehensive resource for probabilistic coastal storm hazards is the USACE CHS (<https://chs.erdcdren.mil/>) (USACE 2024a). This data resource provides high-fidelity, coastal, regional wave and surge water-level modeling results by storm and in a statistical context, including aleatory and epistemic uncertainties. For extratropical storms, modeled historical events are provided. For tropical storms, mostly synthetic events sampled from a joint probability have been modeled and provided within CHS to better characterize the stochastic response.

(2) For regions populated within CHS (USACE 2024a), the storm suites span practical probability space. AEPs for SWL and wave parameters from joint probability models are provided based on PCHA following the process shown in Figure 12–2. Overall storm probabilities are provided for tropical events. Epistemic uncertainties are provided as confidence levels associated with AEPs for wave and water-level parameters. CHS also provides storm data screened from continuous measurements along with associated extremal analysis.

(3) A resource for continuous wave hindcasts is provided by the USACE WIS (<https://wis.erdc.dren.mil/>) (USACE 2024b). WIS provides consistent and continuous simulated historical winds and waves for U.S. coastlines for a 20 to 60-year record, depending on location. A continuous hindcast of meteorological conditions is provided by NOAA called NCEP/NCAR Reanalysis at <https://www.esrl.noaa.gov/psd/data/gridded/data.ncep.reanalysis.html> (NOAA 2024i).

d. Measured meteorological data, including winds and pressures, are archived by the NOAA National Climatic Data Center (NOAA 2024e) (<https://www.ncei.noaa.gov/cdo-web/>).

e. Wave and over-water meteorological data can be acquired from the NOAA National Data Buoy Center (NOAA 2024f) (<http://www.ndbc.noaa.gov/>).

f. Ice modifies the storm responses in the Great Lakes and Arctic areas. Ice can increase the wind drag, and thereby increase waves and surge, or can inhibit waves and surge when solid ice forms on the surface. Ice data for the Great Lakes can be acquired from NOAA Great Lakes Environmental Research Laboratory (<http://www.glerl.noaa.gov/data/ice/>) (NOAA 2024j). For arctic sea ice data, the National Snow and Ice Data Center at the University of Colorado (UCB 2024) provides data on percent ice coverage (<http://nsidc.org/data>).

g. Modeling results and measurement data are available from NOAA's regional Integrated Ocean Observing System (NOAA 2024b) web site <http://www.ioos.noaa.gov>.

12–4. Extreme value statistical models

a. Statistical models of extreme values are rarely directly fitted to the continuous raw measured data primarily because the nonextreme events contaminate the extremal data and yield erroneous results. Instead, it is most common to extract extreme events from the historical record and fit only these using extreme value theory (Coles 2001). For the statistical analysis of historical extreme events, one of two main types of series is produced: block maxima series (BMS) or partial duration series (PDS). For the following univariate and multivariate statistical analyses, there are several USACE tools available. One tool, called StormSim, is specifically for coastal engineering and the StormSim routines can be used to automate most of these analyses (USACE 2024d) (<https://github.com/Coastal-Hazards-System/StormSim>).

b. The most well-known approach to develop a BMS consists of recording or sampling the maximum event for each year or month over the duration of the data. The

resulting series are known as either annual maxima series (AMS) or monthly maxima series (MMS). The construction of a BMS is a simple and straightforward approach that can be applied with few restrictions. A frequent objection to using AMS is that secondary extreme events in any one year are discarded even if they exceed the annual maximum of other years (Madsen, Pearson, and Rosbjerg 1997) and minor events that are not extreme are included from years with little storm energy.

c. Since coastal gage record lengths using modern accurate measurement techniques are virtually always too short for accurate estimation of extremal distributions, all extreme data are very valuable and any loss of data through poor screening techniques should be avoided. Therefore, the use of MMS is strongly preferred over AMS.

d. The most common PDS is developed using the peaks-over-threshold (POT) approach (Pickands 1975; Davison and Smith 1990). Using POT, the PDS is constructed by sampling all peaks over a certain threshold. The POT approach requires that only IID peaks are sampled to prevent counting multiple peaks from a single storm as unique extreme events. Also, the extreme values that compose a PDS must meet the conditions of a Poisson-distributed process: independence, homogeneity, and stationarity (Lang, Ouarda, and Bobée 1999; Luceño, Menéndez, and Méndez 2006).

e. As a rule, based on mathematical considerations, the selected threshold value should be high enough to satisfy the Poisson process hypothesis. In the POT technique, the most significant parameters are the time lag required for the extreme events to be considered as IID, often referred to as inter-event time, τ ; and the number of individual storms per year, which is usually referred to as mean rate or sample intensity, λ . For water level and wave gage records, it is common to sample an average of roughly two events per year (Nadal-Caraballo, Melby, and Ebersole 2012).

f. The extremal distribution is characterized using POT data above a predefined threshold. Although return-level estimates produced from this model are independent of threshold, the value determines which data are used in the model, and therefore, influences the resulting estimates. Plots of threshold-independent parameter estimates are often used in combination with the mean residual life plot to select an optimal threshold that is high enough for the limit to hold but low enough to provide robust estimates. Nadal-Caraballo and Melby (2014) provides an example of selecting the threshold for coastal waves and water levels that limits the high-frequency tail to only extreme storms.

g. Historical extreme events are typically fit to statistical models under the assumption that the statistical population is stationary (for example, no trend). It is, therefore, important that any trends or other nonstationarities are removed from the dataset prior to fitting. These can then be added back at the end of the statistical analysis, if necessary, often with the use of Monte Carlo sampling. It is, however, well recognized that there is a need to provide extreme sea levels that are specific to a point in time and these estimates vary in time. There are a number of approaches that can be applied to address this. A common approach is to de-trend the data relative to a

reference (base) year. For measured water levels, for example, the de-trending is routinely done by rotating the record about the mid-year of the National Tidal Datum Epoch and then de-meaning (Nadal-Caraballo and Melby 2014).

h. For future scenarios, the common practice is to perform analysis for different instances in the future. For example, for sea levels, the common practice is to perform the extremal analysis at start of project (SOP) and, across the project's service life for example, SOP+50 years and SOP+100 years in the future using sea levels corresponding to SOP, SOP+50 years, and SOP+100 years, respectively. RSLC is computed according to USACE guidance previously discussed using the Sea Level Analysis Tool (SLAT) (USACE 2024c).

i. As discussed in prior chapters, it is common to choose the marginal (univariate) statistical model that best fits the data, often from a selection of flexible distributions such as the Weibull or Log-Pearson Type III. While this approach can be useful to study the behavior of the bulk of the data, care must be taken when the aim is to extrapolate outside the range of the data since a good-fitting model can give erroneous extrapolations. This is a particular problem, for example, when the high-frequency tail is fit to data that include nonextreme values. For example, for coastal water levels, it is crucial to separate storms from tidal variations and other nonextreme data. When extrapolating to extreme values, it is more important that the chosen statistical model has a strong theoretical basis to justify its use in extrapolation.

j. According to extreme value theory (Coles 2001), data from BMS have a corresponding distribution in the generalized extreme value family. When data are sampled by means of POT, the peak values above a fixed threshold should conform to the generalized Pareto distribution (GPD) so long as the threshold is large enough for the asymptotic limit to hold. These two distributions are now increasingly being widely applied for extreme value modeling in many different fields and are the state-of-practice for coastal hydrodynamic response extremal analysis (Nadal-Caraballo and Melby 2014).

k. There are many methods for estimating the best parameter values of a distribution. These include the least square method, the method of moments, and the maximum likelihood method (MLM). The primary method used is MLM as this has a strong theoretical basis. The goal of this method is to determine the distribution parameters that maximize the likelihood of the given sample. In other words, the resulting best-fit parameters correspond to the distribution from which the data were most likely to correspond to. Fourth-generation scientific computer programming systems contain tools that facilitate the application of these methods (MATLAB®, R, Python). StormSim (USACE 2024d) uses MATLAB to facilitate application of the methods described in this Chapter.

l. By definition, the CI is a measure of the sampling uncertainty associated with statistical estimates. The CI, which is represented by a percentage, indicates the chance that the true value of interest lies within the range of values bounded by the interval. The bounds of the CI are known as confidence limits (CL). A CI of 80%

indicates that there is a 20% chance ($1 - CI$) that the true value is outside the CI. In the case of $CI = 80\%$, the upper and lower CLs are 90% and 10%, respectively. There is a 10% chance that the true value lies above the upper CL = 90% and a 10% chance that the true value lies below the lower CL = 10%. CIs may be produced from distributions fitted by MLM, both for parameter and return level estimates.

m. The most commonly used method for doing this is based on the asymptotic Normal distribution of MLM estimates and always produces an interval that is symmetrical about the best estimate. CI can also be estimated through the application of a bootstrap resampling technique. Bootstrapping is a type of Monte Carlo simulation that consists of drawing new samples a specified number of times from the original sample distribution to create multiple resamples from which the distribution of the expected value can be estimated.

n. Parametric fitting methods, such as MLM, often require numerical optimization algorithms to apply in practice so were not always available before the widespread use of personal computing. Historically, this led to the use of ad hoc graphical methods for fitting certain distributions via plotting the data on special graph paper and drawing a line of best fit. To do this, each data point must be assigned a cumulative probability for which there are many alternative formulations (plotting positions formulas), each of which each gives a slightly different fit. However, the use of graphical fitting methods is no longer recommended given that computers can easily be used to apply more robust approaches such as MLM, which do not depend on plotting positions.

o. Though modern fitting methods do not use plotting positions, the goodness-of-fit of a distribution is often judged by plotting the fitted data against model estimates; for example, on an AEP scale. This, again, requires cumulative probabilities to be assigned to each data point from which corresponding AEPs may be calculated. Most authors assign the i^{th} smallest value the probability using equation 12–6.

$$p_i = \frac{i-a}{n+1-2a} \quad (12-6)$$

for different values of a , where n is the number of values in the sample. In practice, the value of a has negligible effect on the AEPs assigned to all but the very largest data values. When $a = 0$, equation 12–6 has the form of the Weibull (1939) plotting position, equation 12–7.

$$p_i = \frac{i}{n+1} \quad (12-7)$$

p. In extreme value analysis, the Weibull plotting position is widely accepted as estimate of the empirical distribution. Coles (2001) states that for an ordered sample of IID observations, the empirical distribution is defined by equation 12–7.

12–5. Coastal flood hazard probability functions

a. *Overview of extremal statistical analysis methods for coastal studies.*

(1) Coastal storm populations are defined by the cyclone type as primarily tropical, extratropical, or hybrid/combined. Other storm classifications exist, such as mesoscale convective systems, and should not be ignored. However, convective systems do not typically alter the hazard extremal distributions. Hybrid or transitional storms, where, for example, a tropical cyclone transitions to an extratropical storm as it transits from southern to northern latitudes, are most often included in the primary classifications (Nadal-Caraballo, Melby, and Gonzales 2015). Tropical and extratropical storms are fundamentally different. Tropical storms consist of warm core, well defined, mostly uniform cyclones. Extratropical storms are cold core, poorly defined cyclones or disorganized combinations of cyclones and frontal systems.

(2) Tropical storms originate in tropical latitudes (between the Tropic of Cancer in the northern hemisphere at 23° 26' 16" N and the Tropic of Capricorn in the southern hemisphere at 23° 26' 16" S), while extratropical storms form in the middle latitudes (roughly between 30° and 60° from the equator). Tropical cyclones usually transition to extratropical storms as they progress up into higher latitudes. However, storms that originated as tropical cyclones are usually treated as tropical cyclones in extremal statistical analysis studies (Nadal-Caraballo, Melby, and Gonzales 2015).

(3) For the coastal regions within the Gulf of Mexico, the storm population causing flooding is commonly considered to be composed entirely of tropical cyclones for coastal hazard extremes. However, Melby et al. (2021a) noted that both nontropical and tropical storm events influenced beach morphology in north central Texas. For the Atlantic mainland coast and the Pacific Ocean islands, the storm population is a mixture of tropical and extratropical storms. The further north, the more influence extratropical storms have on extremes. For the Great Lakes, continental Pacific coast, and Alaska, storm populations are virtually all extratropical, although tropical storms off the coast of Mexico can impact southern California.

(4) Unlike tropical cyclones, extratropical cyclones are usually not well-defined weather systems. So, for extratropical storm exposure, the typical approach is to identify storms from continuous measurements using a POT technique. Extratropical storms are usually sufficiently spatially consistent and sufficiently frequent that long-term, historical gage records reasonably define the statistical model that can be used to define more extreme events than have occurred in the past. Wave and water level observations are the primary resource for discovering storm events. However, historically, wave and water level measurements have been spatially sparse because of relatively few gages and because of intermittent gage failure, so wind measurements should be queried as well to assure that all significant storms are captured.

(5) The CHS (USACE 2024a) has identified significant extratropical storms for the Great Lakes and Atlantic coastlines and provides hydrodynamic modeling and hazard curves for these regions. It is becoming more common to model the coastal flood

hazard through accurate, high-resolution, numerical hydrodynamic models so this data resource can be very useful.

(6) In tropical storm-exposed areas, tropical cyclones are almost always well defined, relatively uniform cyclones that can be parameterized. Hurricane parameter data, such as central pressure deficit, radius to maximum wind speed, forward speed of storm, heading and landfall location, are used to construct a joint probability model that is then optimally sampled to define a minimum set of synthetic storms that spans and adequately discretizes the probability response surface (Resio et al. 2013; Nadal-Caraballo, Melby, and Gonzales 2015). This approach is often termed JPM-OS, which stands for joint probability method with optimal sampling. JPM-OS-defined storms are called synthetic because they are not historical events.

(7) The optimal sample set of cyclone parameters is used to define synthetic tropical cyclones that are subsequently modeled using physics-based, high-resolution, numerical meteorological and hydrodynamic models. In this case, the relative probability of the event is primarily defined by the JPM-OS model. The primary benefits of this approach are high-accuracy joint probability of responses and accurate storm hazard computation where nonlinear coupled physics are maintained.

(8) The probability distributions of responses due to coastal extratropical and tropical cyclones are computed independently, as shown in Figure 12–2. The response distributions associated with extratropical events are determined from extreme value analysis of measured or modeled historical storm responses, while the response distributions due to tropical cyclones are computed from modeled synthetic storm responses based on the joint probability analysis of meteorological parameters. Combined water level distributions are estimated by summing probabilities associated with both sets of results for a range of water levels (Nadal-Caraballo, Melby, and Gonzales 2015).

b. Joint probability methods.

(1) *Overview.* Coastal flood protection risk evaluation invariably requires joint probability models for accurate depiction of flood hazard risk. While complex and time consuming to construct, large errors or misleading results are likely if an overly simplistic statistical model is used. Joint probability models usually consist of quantifying marginal extremes for each parameter, quantifying the dependence structure between parameters, and integration of the joint probability density of the response of interest. The methods are normally applied with liberal simplifications to bound the uncertainty and make the problem tractable under the typical time and funding constraints of a USACE planning or engineering study.

(2) *Bounding assumptions.* For example, complete independence and complete dependence bounding assumptions are often made. Figure 12–2 provides a schematic of a joint probability analysis for a mixed storm population consisting of both tropical and extratropical storms. The steps illustrated are described in the following sections. Note

the separate analysis of tropical storms on the left and extratropical on the right, with combined probabilities computed near the end.

(3) *Joint probability for extratropical storms.* As stated above, for extratropical storms, the joint probability model is usually constructed based on response parameters like water level, wave height, and storm duration.

(a) *Storm sampling.* As seen on the right side of Figure 12–2, the first step is screening the measurements to determine a storm list for modeling. For a specific point, a single gage may suffice to determine storms, and so a POT analysis can be conducted. However, the results are neither an optimal list nor comprehensive if the area of interest spans miles of coastline. Nadal-Caraballo, Melby, and Ebersole (2012) provides an approach for optimal sampling of extratropical storms for a specific region using observations from a number of gages. The methodology is called generation of composite storm set (CSS) and is summarized as follows:

1. Use the POT technique to sample storms at each applicable gage. Multiple types of measurements might be used here to define storms (water levels, waves, winds).
2. Isolate the extratropical storm population by eliminating tropical cyclones and combine all storms into one list.
3. Develop full storm set (FSS) probability distributions by fitting GPD to, for example, NTR values estimated from each of the gages.
4. Establish various CSSs with a range of numbers of storms (for example, 250, 225,...100) by rank ordering and eliminating storms producing low responses and fit GPD to the CSS extreme values.
5. Minimize the number of storms in the CSS by selecting the CSS with the fewest number of storms that matches the FSS distributions.
6. Validate the CSS by comparing to the FSS for the full range of practical AEPs (0.2, 0.1, 0.04, 0.02, 0.01, 0.004, 0.002, 0.001).
7. The above methodology can be applied to, for example, water level or wave height, if that is the dominant forcing parameter for the project.

(b) *Joint probability model.*

1. For general joint probability analysis, copula methods are becoming routine. Meta-elliptical copulas are frequently used since this family often outperforms other multivariate copulas in capturing the full range of dependence. They also excel due to simplicity of construction and ease of parameter estimation, particularly in the case of the meta-Gaussian copula (MGC). The MGC consists of a multivariate CDF constructed by linking a set of marginal probability distributions with a multivariate Gaussian copula as the dependence structure.

2. To apply the copula methods, marginal distributions are computed for the parameters of interest using POT and fitting a GPD. Then the dependence between the parameters is fit with a joint distribution and the fitted distribution is integrated. Fourth-generation computer programming languages, like MATLAB and R, are typically used because they contain routines that support these types of computations. StormSim (USACE 2024d) also includes routines for fitting these distributions.

(c) *Selecting values from joint probability method.* For most flood-risk studies, the dominant parameter is water level, and the correlation of other parameters with water level is determined. Since the water level and wave height are correlated, for a specific AEP SWL, the solution is multi-valued. In some cases, the response is entirely dependent on a single parameter and selection of response is relatively simple. However, for many cases in coastal engineering, the relative contributions of the various parameters are not clear. This is especially true for flooding associated with surge and waves where overall flood levels are a strong function of both water level and wave height.

1. As an example, Figure 12–4 gives joint probability results for two AEPs, 1% and 0.2%, for Milwaukee gage measurements on Lake Michigan. It is clear that a range of water levels and wave heights define both of these AEPs. For a frequency-based analysis, various methods are possible for sampling the joint wave height with water-level AEP. The MGC can be sampled using the Monte Carlo sampling.

2. A simpler method that can be used as a crude approximation is to pick three values: (1) the highest water level and associated wave height, (2) highest wave height and associated water level, and (3) the mid-point on the exceedance curve. Finally, each storm can be sampled in a Monte Carlo simulation and responses computed by storm.

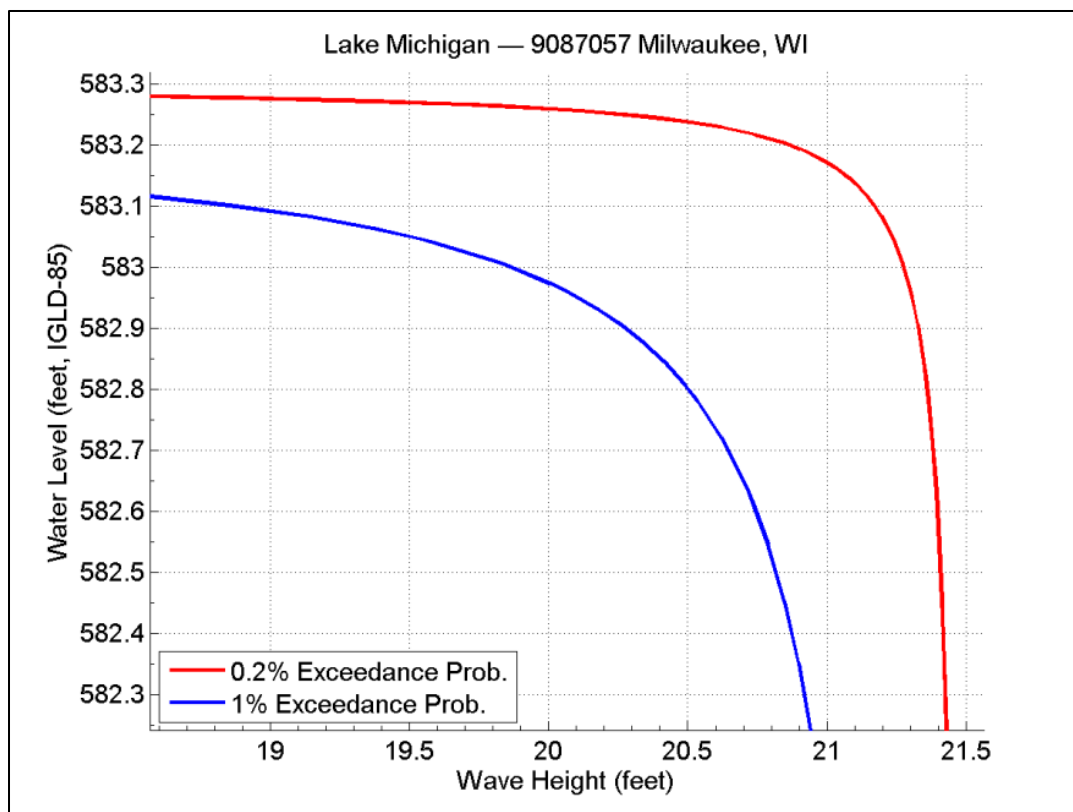


Figure 12-4. Lines of constant exceedance probability from a joint probability model for Milwaukee gages located on Lake Michigan

(4) *Joint probability for tropical storms.*

(a) The joint probability approach for tropical storms uses the JPM-OS methodology as illustrated on the left side of Figure 12-2. This is the process followed for PCHA in CHS (USACE 2024a). JPM studies have been conducted for all tropical storm-exposed coastlines of the U.S. and the data are available in the CHS, so it is expected that new JPM studies will not require a complete analysis as described in this section. The following steps of this approach were extracted from Nadal-Caraballo, Melby, and Gonzales (2015). More detail on application of these methods is contained in FEMA 2012.

1. Assemble storm parameter data. Parameters of interest are landfall location, heading direction, central pressure deficit, radius of maximum winds, and forward speed. The HURDAT2 (NOAA 2024) database of historical tropical cyclone parameters covers the period from 1851 to present. The typical project study uses only tropical cyclone parameter data since roughly the 1940 coinciding with the period of U.S. Air Force hurricane hunter overflight data collection because data prior to this are much less reliable. However, data prior to this should inform the selection of possible storm tracks and storm parameters.

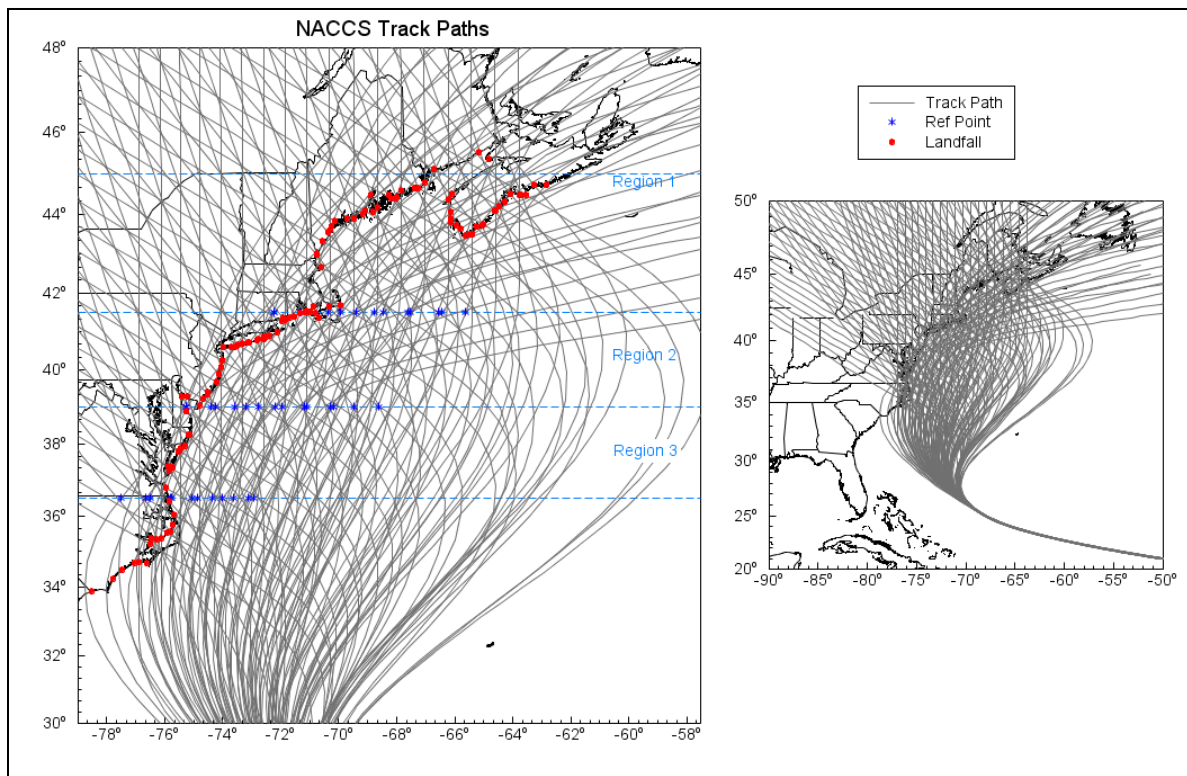
2. Construct marginal distributions for each storm parameter for each subregion.

3. The marginal distributions are discretized using techniques, such as Bayesian Quadrature, to arrive at an optimal sample of storms that well represents the probability response surface.

4. From the landfall location, approach angle, discretized marginal distributions, and historical storm tracks, families of storm tracks are generated that are representative of the historical record. Figure 12–5 shows an example of synthetic storm tracks generated for the North Atlantic Coastal Comprehensive Study (USACE 2024e).

5. Along-track parameter variations are defined by historical storm averages for various ranges of central pressure deficit, forward speed, and storm size. Generally, for synthetic tropical cyclones, storm parameters are constant over the majority of the storm life, but the storm intensity begins to fall and storm infilling occurs just prior to landfall. This is represented by the along-track variations of storm parameters.

6. The storm recurrence rate (SRR) along the coast is quantified based on the Gaussian kernel function where distance-adjusted mean weights are computed using a Gaussian PDF with an optimal kernel size. An example is given in Figure 12–6. The SRR is a measure of the storm frequency and has units of storms/yr/km, or $2.78\text{E-}4$ storms/yr/km. The SRR is multiplied by a radial distance (r) in km to compute the project-specific storm rate. The SRR should correspond to a coastal reference location at or near the mid-point of the study area. The radial distance used should be equivalent to at least the distance along the coastline from the mid-point of the study area to one of its boundaries, such as 200 km. In this example, the SRR in storms/yr is then computed as $RR[\text{storms/yr}] = SRR[\text{storms/yr/km}] \times 2r$. For example, $SRR = 0.00278 \times 2(200) = 0.1112 \text{ storms/yr}$, which is equivalent to 1 storm every 9 years ($1/0.1112 \text{ storms/yr} = 8.99$). This value is region-specific.



**Figure 12–5. Example of JPM-OS synthetic tracks from
Nadal-Caraballo, Melby, and Gonzales (2015)**

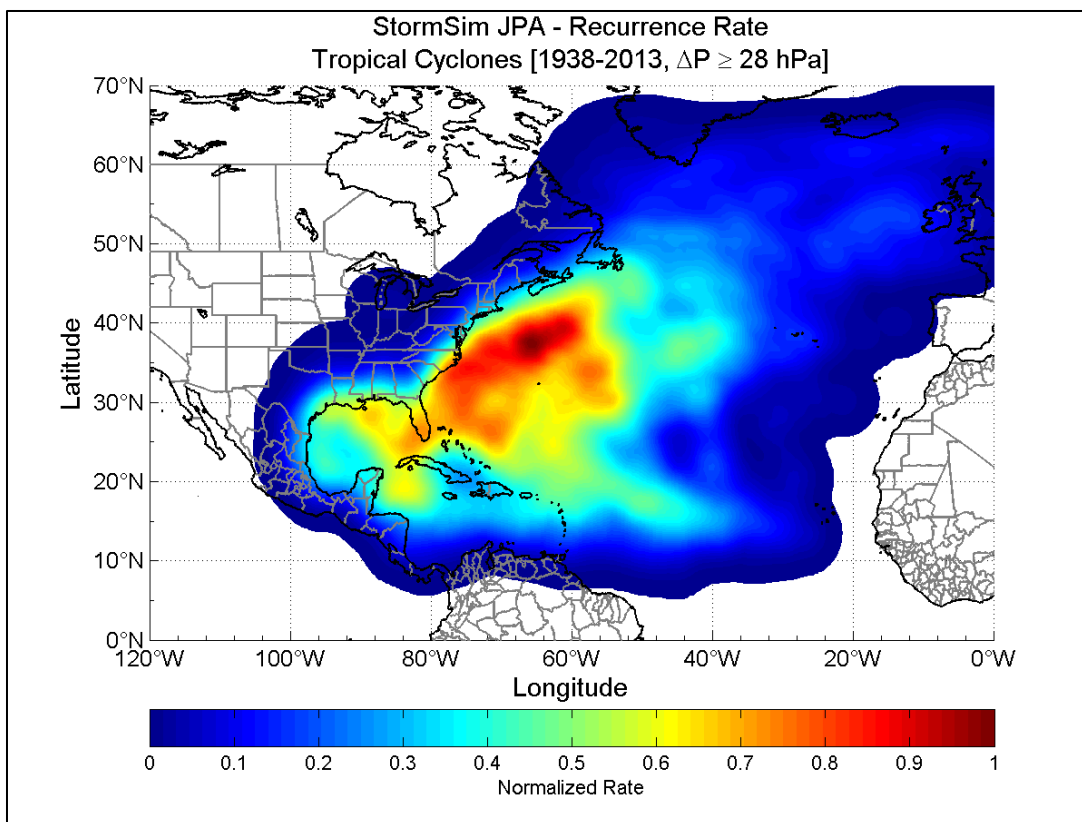


Figure 12–6. Example of storm recurrence rate for tropical cyclones in the Atlantic and Gulf of Mexico basins (Nadal-Caraballo, Melby, and Gonzales 2015)

7. The usual approach in a PCHA study is to model the synthetic storms using high-fidelity numerical models. A climate model, such as planetary boundary layer (PBL) (Thompson and Cardone 1996), is used to develop wind and pressure fields. Then these fields force wave transformation or circulation models or a coupled circulation and wave transformation model, such as CSTORM (Massey, Wamsley, and Cialone 2012). CSTORM includes ADCIRC (for circulation, storm surge, and tides) coupled with STWAVE (for wave transformation). These modeled storm responses are stored in CHS (USACE 2024a) and can be downloaded. Note that the output from the STWAVE spectral wave model within CSTORM does not include wave diffraction or reflection and other nonlinear wave effects such as infragravity waves. A separate phase-resolving wave model, such as FUNWAVE, should be run to model these phenomena.

8. A complication of this storm modeling is representing tidal fluctuations in an accurate way because there are many possible realizations of tides, but modeling many realizations of tides requires many expensive numerical simulations. For the Gulf of Mexico studies, tide range is small and is included as an epistemic uncertainty within the PCHA (Melby et al. 2021b). For the North Atlantic region, tide range is relatively large and requires explicit treatment where it is added to the SWL (Nadal-Caraballo, Melby, and Gonzales 2015).

9. As discussed in paragraph 12–2, NLR error is usually small along the open coast, allowing linear superposition of water level components. However, it can be significant in shallow waters and inland locations. For North Atlantic Coast Comprehensive Study Report (NACCS), NLR was found to be small over the vast majority of the region but high at specific locations. These locations were very sparse and located in confined areas like isolated back-bay channels and small embayments. So linear superposition of random tides and RSLC on the SWLs was valid over most of the domain, and the NLR was accounted for as a small adjustment for bias and an additional epistemic uncertainty term for variability. The spatially varying NLR was included in the NACCS data within CHS (USACE 2024a). In addition, hydrodynamic simulations included CSTORM runs for each storm with a single random tide so the nonlinearities could be resolved, if required.

10. In contrast, the NLR errors were determined to be very large for coastal flooding in northern Texas, so all storms were modeled using CSTORM and the only linear superposition was relatively small epistemic uncertainty added within the stochastic simulation (Melby et al. 2021b).

11. The results of step 12–5.b(3)(a)7 are high-fidelity waves, winds, and water levels for all storms across the region of interest. The final step is to compute probabilities of the events and of the responses. The response average annual exceedance frequency is computed at each spatial point using the joint probability integral, which was expressed by Nadal-Caraballo, Melby, and Gonzales (2015) in continuous and discrete forms as in equation 12–8.

$$\begin{aligned}\lambda_{T(\hat{x}) > \tau} &= \lambda \int P[T(\hat{x}) + \sigma > \tau | \hat{x}, \sigma] f_{\hat{x}}(\hat{x}) f_{\sigma}(\sigma) d\hat{x} d\sigma \\ &\approx \sum_i^n \hat{\lambda}_i P[T(\hat{x}_i + \sigma) > \tau | \hat{x}_i, \sigma]\end{aligned}\tag{12–8}$$

where:

$\lambda_{T(\hat{x}) > \tau}$ = annual exceedance frequency of storm response T due to forcing vector $\hat{x} = f(x_o, \theta, \Delta p, R_{max}, V_t)$

λ = SRR at site

$P[T(\hat{x}) + \sigma > \tau | \hat{x}, \sigma]$ = conditional probability that storm i with parameters $\hat{x}$ generates a response larger than τ

n = the number of events

$\hat{\lambda}_i$ = probability mass of synthetic storm i

σ = unbiased error or epistemic uncertainty

The tropical storm parameters are:

x_o = landfall location

θ = heading at landfall

Δp = pressure deficit

R_{max} = radius to maximum wind speed

V_t = translational speed

In the discrete integral of equation 12–8, the probability mass of a given storm ($\hat{\lambda}_i$) is defined as $\hat{\lambda}_i = Ap_i$, where p_i is the product of the storm's discrete joint probability and the tropical cyclone track spacing (in kilometers). In other words, each storm's probability mass is computed as the product of its SRR, the discrete probability (or weight) determined from the joint probability model of tropical cyclone parameters, and the tropical cyclone track spacing.

(b) Equation 12–8 provides the tropical cyclone response that is based on a conditional joint probability distribution. Response T could be, for example, SWL, H_{m0} , or overtopping rate. Generally, peak values for each parameter are used. The approach for SWL is as follows: Each synthetic storm, represented by a combination of parameters, produces a peak water-level response at a particular site that is estimated from the numerical coupled hydrodynamic modeling. To estimate the water level frequency, bins of elevation are created and a probability mass function is constructed for a range of elevations that encompass all the storm results. The rate for each storm is associated with the bin that corresponds to its peak water level response to estimate the response PDF. The width of the bins should be as small as practicable.

(c) For example, Nadal-Caraballo, Melby, and Gonzales (2015) used a 0.01 m bin with a range of 0 to 10 meters for SWL. The cumulative distribution is calculated by adding at each water level elevation all the rates accumulated in the bins with higher response. A StormSim (USACE 2024d) routine performs this computation using data directly from the CHS (USACE 2024a).

(5) *Combined probabilities.* The probabilities from the separate storm populations (tropical and extratropical) are computed as the sum of probabilities for specific responses. So, if we say that P_{TC} (SWL = 9 ft) = 0.012 and P_{XC} (SWL = 9 ft) = 0.023 are the tropical and extratropical probabilities for a SWL of 9 ft, respectively, then the combined probability is P_C (SWL = 9) = 0.012 + 0.023 = 0.035. The entire combined hazard curve can be constructed by summing many discrete values of the two populations. Again, A StormSim (USACE 2024d) routine performs this computation using data directly from the CHS.

c. *Simulation methods to generate storm probabilities for life-cycle analysis.*

(1) Storm sampling as part of a Monte Carlo life-cycle (MCLC) simulation can be done from historical storms or from a JPM-OS synthetic suite or in combination. If the storm suite is composed of historical storms, and these storms are represented in the same frequency as occurred historically, then the sampling probabilities for all storms are the same and the annual sampling rate is dictated by the Poisson distribution with mean rate from historical storms. When performing the MCLC simulation, the number of events to be sampled per year is dictated by the Poisson distribution (equation 12–9). The Poisson PDF is given by:

$$f(x|\lambda) = \frac{\lambda^x}{x!} e^{-\lambda} ; x = 0, 1, 2, \dots, \infty \quad (12-9)$$

where:

λ is the mean of the distribution and, in the example given previously, corresponds to $\lambda = SRR[storms/yr] = 0.1112\ storms/yr$. Thus, on average, one storm should be sampled every 9 years for this case. This number is site specific and varies dramatically.

(2) For storm suites composed of JPM-OS synthetic events, the probability of each storm must be defined a priori. The usual approach for conducting the MCLC simulation using JPM-OS storms is as follows: Iteratively proceed year by year, sample the number of storms in a year from the Poisson distribution with mean of SRR, then sample each storm based on its probability from an appropriate CDF. This requires computation of the SRR at the site of interest, which yields the mean of the Poisson distribution as well as the relative frequencies of the storms. In some cases, it is necessary to further subsample events by their relative frequencies.

(3) For example, Melby et al. (2021a) sampled extratropical storms according to the Poisson distribution defined for the extratropical cyclone population, sampled tropical storms according the Poisson distribution defined for the tropical cyclone population, and then subsampled the tropical cyclones according to three intensity groups to assure that life cycles contained the representation from the various populations. To compute the CDF of the storms, the storms are ranked in decreasing frequency order, the cumulative sum is computed, and the probability values are normalized to unity (from 0 to 1). The sampling should be performed using random numbers from 0 to 1. The original storm indexes are retained when doing the ranked order for identification of the storms. Both the SRR and the storm probabilities are populated in the CHS (USACE 2024a) regionally. StormSim (USACE 2024d) MCLC simulation software follow the general approach given above.

(4) For a particular project study, a subset of storms may be desired due to the computational requirements of using all storms. For example, a typical region has roughly 250–300 storms. Computing levee overtopping for 250 storms on varied tides for many different structure configurations and RSLC scenarios may be too computationally costly. So, a subset of storms is usually acquired for alternative analysis in planning studies. In that case, the storm CDF changes and the resulting relative probabilities need to be recomputed from the CHS values. Paragraph 12–8.c describes the CHS/StormSim (USACE 2024a, d) capabilities for storm subsampling.

12–6. Epistemic uncertainty in coastal risk assessment

a. It has become routine in tropical storm coastal flood risk studies to quantify aleatory and epistemic uncertainty. Epistemic uncertainties can be accounted for in sampling from general Monte Carlo simulations or within JPM-OS studies, as described in equation 12–9. Commonly, measurement or hydrodynamic modeling uncertainties are included. The typical JPM-OS approach is given in Resio et al. (2013) and Nadal-Caraballo, Melby, and Gonzales (2015). The primary uncertainties that have been addressed are:

- (1) Random astronomical tide phase if tide is not explicitly included.
- (2) Storm track variations not captured in synthetic storm set.
- (3) Errors in hydrodynamic modeling and grids associated with epistemic uncertainty.
- (4) Errors in meteorological modeling associated with simplified PBL winds.
- (5) Nonlinear interaction of tide and surge.

b. The normal approach is to assume that the error terms are independent, their effects can be combined by addition, and the combined error can be represented as a Gaussian distribution with zero mean. The standard deviation can be computed as (FEMA 2012) as shown in equation 12–10.

$$\sigma_{\varepsilon} = \sqrt{\sigma_{\varepsilon 1}^2 + \sigma_{\varepsilon 2}^2 + \sigma_{\varepsilon 3}^2 + \cdots + \sigma_{\varepsilon n}^2} \quad (12-10)$$

c. The overall flood hazard uncertainty has been populated in the CHS (USACE 2024a) as CIs under the assumption of Normally distributed errors. In addition, Nadal-Caraballo, Melby, and Gonzales (2015) discuss methods for incorporating the above uncertainty into risk evaluations.

12–7. Stochastic simulation (flood hazard assessment approaches)

a. The FEMA nomenclature for computational approaches to flood risk is adopted where frequency-based approaches are referred to as event based and storm-based approaches are referred to as response based. An example of an event-based approach is a design based on a specific AEP or average return interval. HEC-FDA planning software uses this approach as does the overtopping tool used for South Louisiana tropical storm flood risk studies following Hurricane Katrina. For an event-based approach, an example is to use the 1% AEP flood level (SWL), wave height (H_{m0}), associated wave period (T_p), and mean wave direction from a joint probability model to define acceptable overtopping discharge rate. CHS provides SWL, H_{m0} hazard curves, associated (T_p), and wave direction that provides these inputs.

b. Similarly, a typical approach as implemented in the overtopping tool uses the same inputs, but additionally performs a Monte Carlo simulation to quantify the effect of epistemic uncertainties summarized in paragraph 12–6 as well as the uncertainty of the empirical equations used to compute overtopping. StormSim (USACE 2024d) contains a similar capability for stochastic simulation using the event-based approach. The primary advantage of event-based simulation over response based is that it is simpler. The primary disadvantage is that it is inaccurate. The methodology produces significant errors that are unpredictable. It constrains the forcing to a specific AEP, but the response AEP varies spatially, producing varying levels risk.

c. A storm-based analysis approach, including time-dependent approaches, is referred to as response based. Response-based, stochastic simulation samples each storm based on its relative probability, adds uncertainty to all of the parameters, computes the response for the storm, then proceeds through all storms. Once complete, response distributions are computed for all of the response parameters and specific statistics of responses are selected. Melby et al. (2021b) provides an example of response-based approach for a CSRM project containing levees and floodwalls using StormSim (USACE 2024d). Beach-fx and G2CRM planning software systems are based on response-based, time-dependent, life-cycle approaches. An example using Beach-fx is given in Appendix B.

d. The advantage of a response-based simulation approach is that response statistics are more accurately represented and response statistics are spatially consistent, producing more spatially consistent risk estimates. The primary disadvantage of response-based approaches is that they are more computationally demanding. However, with modern computers and available software to support these approaches, this is not a significant deterrent.

e. For a time-dependent approach, the modeled storms can be assembled into a life cycle. These storms can be rearranged in any random order and can be simulated on random tides to provide sufficient realizations to achieve a stable variance. One key benefit of the life-cycle approach is that varied risk management approaches during a project life that depend on events within the life cycle can be modeled. In addition, time-dependent processes, like beach morphology and RSLC, can be modeled as they would occur in a project. However, take care when linearly superimposing water level components to avoid significant NLR errors that are often present for inland levees and canals. In those cases, hydrodynamic numerical modeling is required with surge, tide, and sea level change scenarios.

12–8. Application considerations for coastal risk assessments and hydrodynamic modeling considerations for stochastic simulations

a. For coastal, it is common to model storm responses for alternatives using CSTORM or other hydrodynamic models because of the complex nonlinear, nearshore response. Often, the without (existing)-project alternative can be obtained from the CHS (USACE 2024a) data resource. For with-project alternatives, it is common to model all alternatives with a subset of storms on several sea level change scenarios. As shown in Melby et al. (2021b), using a relatively simplistic linear superposition of the water level components can yield reasonable estimates of response but can also produce very large NLR errors for shallow or inland flooding areas. If linear superposition is reasonably accurate, surge and waves can be modeled together, and then tides and RSLC can be linearly superimposed during stochastic simulation. This approach facilitates life-cycle analysis and other time-dependent analysis where thousands to tens of thousands of combinations of conditions are required.

b. In cases where linear superposition is not sufficiently accurate, modeling of all scenarios may be required. In this case, usually the storms are modeled at specific times during the project life, such as start of project, end of project life, and, for example, using historical and intermediate sea level change curves. It may be difficult to perform accurate life-cycle simulations because a wide variety of combinations of parameters is not modeled. One approach that overcomes this limitation is to incorporate the added uncertainty from the NLR error into the life-cycle simulation and account for the added error in levels of confidence. Another approach that is gaining wider application is to use a relatively small set of modeled events and scenarios to train an artificial intelligence, machine learning, or surrogate model of the hazards and then use this model in the life-cycle simulation.

c. A typical risk assessment usually requires a without-project scenario and multiple with-project scenario alternatives, all modeled on multiple sea level change scenarios. The number of permutations can quickly overwhelm a reasonable project budget and schedule. StormSim (USACE 2024d) includes an artificial intelligence tool for efficiently subsampling the full CHS storm suite (USACE 2024a). The iterative storm selection tool uses a genetic algorithm in an optimization scheme to select the subset of storms that best matches the hazard curve. USACE projects have used this tool to develop suites of 20–25 storms each for alternative evaluation. For each storm suite, different probability masses for the storms are required.

Appendix A References

Section I

A-1. Required Publications

These documents define policy and basic methods directly related to hydrologic engineering for flood risk management planning by USACE. All are promulgated by the HQUSACE in Washington, DC.

Unless otherwise indicated, Army and USACE publications are available at <https://armypubs.army.mil/> and <https://www.publications.usace.army.mil>. Coastal Science & Engineering (CSE) publications are available at <http://coastalscience.com/>. Hydrologic Engineering Center (HEC) publications are available at <https://www.hec.usace.army.mil/>.

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ECB 2019-08

Managed Overtopping of Levee Systems

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A-3. Prescribed Forms

This section contains no entries.

Appendix B

Coastal Storm Risk Assessment Example

B–1. Executive summary

a. Edisto Beach is a managed beach and dune system at Edisto Island, South Carolina. Edisto Island is approximately 42 miles southwest of Charleston, SC, and straddles the Charleston County and Colleton County border. The Town of Edisto Beach, the largest incorporated township on the island, is on the Colleton County side of the island and located behind 31,500 feet, or just under 6 miles, of managed coastline.

b. Study area vulnerability assessment and economic risk analysis was performed using Beach-*fx*, an ERDC-developed Monte Carlo life-cycle simulation model. Coastal morphology inputs were developed using observed historical erosion rates and modeling results from ADCIRC, Beach Morphology Analysis Package (BMAP 2024), and SBEACH. Beach-*fx* interfaces the study area economic data with coastal morphology inputs to evaluate the overall coastal storm risk. Coastal storm risk was evaluated at three rates of RSLC (Low, Intermediate, and High) according to ER 1100-2-8162. In addition to uncertainties accounted for in the coastal morphology modeling, uncertainties were added to key economic variables, including depreciated replacement value, first floor elevation, content-to-structure value ratio, damage curves, and time to rebuild.

c. Future without-project condition damages for Edisto Beach total \$64.1 million in mean present value, with \$44.4 million from structure and content damages, \$17.5 million from emergency nourishment costs, and \$2.2 million in local armoring costs. The EAD were calculated over 300 iterations and reached equilibrium near the 85th iteration (variance approached 0.0%).

d. Four proposed alternatives were modeled in the future with-project condition, and included varying dune heights, berm widths, and groin extensions. Each alternative was simulated with 300 iterations and Alternative 4, with a maximum dune height of 15 feet and a maximum berm width of 75 feet, was selected as the National Economic Development (NED) plan. The NED plan reduced EAD in Edisto Beach by \$2.7 million, with an average annual cost of \$1.1 million, to produce \$1.6 million in average annual net benefits (AANB).

e. Risk assessment showed that the selected NED plan maintained economic justification across all three RSLC functions and produced a positive AANB calculation in 272 of 300 iterations. In other words, given the economic and coastal morphology in the model, the proposed NED plan has a 90.7% probability of producing a positive return with a mean result of \$1,619,000 AANB.

B–2. Introduction

a. Overview.

(1) This risk assessment appendix provides an example of coastal flood hazard risk analysis. The example is related to a managed beach and dune system at Edisto Island, South Carolina, which undergoes morphology change and erosion when exposed to repeated coastal storm hazards, which results in consequential flooding and damage to infrastructure shoreward of the dune. The risk analysis study implements a life-cycle analysis approach with Monte Carlo sampling from historical storm events that are superimposed on varied tides. The primary risk analysis software employed was *Beach-fx* (Gravens 2007). The beach/dune response and flood elevations were precomputed and stored in a database. The wave and water level conditions for historical storms were acquired from an existing database of storms that had been previously modeled.

(2) The Monte Carlo life-cycle simulation, as described in Chapter 12, consists of marching through time, randomly selecting storms, looking up the response from the response database, determining damage from each storm, summing the damage over the life cycle, and computing the net benefits. Although historical storms are used herein, this study could have sampled synthetic JPM-OS storms in a similar manner, although the storm probabilities would have been more variable. In addition, for beach projects, extreme storms do not drive the decision process. This can be contrasted with levee- and floodwall-focused projects where extreme storms are much more influential in the decision process. So, for this study, quantifying the entire practical probability space was not essential to define the optimal alternative. However, for coastal levee or floodwall risk analysis, the entire range of practically possible storms is likely to be necessary.

b. Project history.

(1) Historical erosion of Edisto Island, South Carolina, has led to a long history of shoreline management activities. The first two groins were constructed in 1948 and construction of additional groins continued over the next decade, bringing the total number of groins on the Edisto Beach shoreline to 17 by 1958. The groins were built from north to south and, as the erosion continued to move south, additional groins were constructed. By 1975, a total of 34 groins had been constructed. Table B–1 provides details on the groin construction timeline.

(2) Despite the construction of groins, erosion continued to threaten Palmetto Boulevard near the pavilion. As a result, in 1954, the South Carolina Highway Department undertook the first nourishment of Edisto Beach. Approximately 830,000 cubic yards of material, consisting of a mixture of sand, shells, and mud, was dredged from the marsh behind the island and placed between Groins 1 and 12. Unfortunately, much of the material was not suitable for beach fill and the fine portions washed away quickly. The coarser sand and shell fractions remained on the beach and added to those transported to Edisto from Edingsville Beach (CSE 2006).

Table B-1
Groin construction chronology

Groin Number	Construction Completion
1	(1948)
2	1948
3-4	1949
5-8	1954
9-12	1953
13-17	1958
18-19	1962
20-21	1964
22-25	1969
26	1970
27-29	1972
30-33	1974
34	1975

(3) While localized erosion problems have persisted along Edisto Beach, the groin field significantly reduces the rate of sand loss along the oceanfront. More specifically, periodic surveys have shown that erosion rates in Groin Cells 1–27 have been less than 1 cubic yard per foot per year (cy/ft/yr) in recent years (CSE 2003).

(4) The next beach nourishment project on Edisto Island took place in 1995, when approximately 155,000 cubic yards of fill was placed between Groins 1 and 17 (Pavilion to Chancellor Street) and Groins 24 and 28 (Laroche Street to Billow Street). This beach-fill project was accompanied by major improvements to the groins in those areas.

(5) Shoreline erosion continued along Edisto Beach, and in 2001, two houses in the 700 block of Palmetto Boulevard (near Groin 12) were lost. Despite the groin improvement and beach-fill project in 1995, the shoreline continued to erode. The most recent beach nourishment project on Edisto Island was constructed between March and May 2006. The project added approximately 850,000 cubic yards (192,100 cubic yards in the state park area) of beach-compatible material along 18,258 feet (3,200 feet in the state park) of shoreline from the state park to Groin 27. The Great Lakes Dredge and Dock completed the project at a cost of \$7,697,500 (CSE 2006). The locations of the 1954, 1995, and 2006 beach nourishment projects are illustrated in Figure B-1.

B-3. Coastal processes

This section provides a summary of the geological framework, key environmental conditions, and active coastal processes that characterize the vulnerability of Edisto Beach to economic losses through coastal storm-induced damages to existing infrastructure.

a. *Coastal conditions.* Existing coastal processes at Edisto Beach are driven by high-energy waves and water levels generated by both tropical and extratropical storms. Significant tropical cyclones impacted the Edisto Beach shoreline approximately once every four years over the past 100 years. Tropical storms occur between June and November, with more than 65% of them occurring in the months of August and September. Extratropical storms are more frequent, occurring on average once every year and a half. Extratropical storms typically occur in the late summer and early fall (September and October) and again in late winter and early spring (January through March), with most occurring in February.

(1) Tropical storm events are typically fast-moving storms associated with elevated water levels and large waves; whereas, extratropical storms are slower moving with comparatively lower water-level elevations and large wave conditions. Both storm types can produce beach erosion and morphology change as well as coastal inundation, leading to economic losses to improved property backing the ocean and inlet shorelines of Edisto Beach. Although economic losses are most often realized in the wake of major storm events, it is long-term chronic erosion that creates the vulnerability to major economic losses through volumetric depletion of beach material in the active profile, reduction in beach berm width, and reduction in dune crest elevation and dune volume.

(2) Not all storms produce measurable economic damages, but all contribute erosional vulnerability and so are important for risk evaluation. The long-term, chronic erosion is driven by gradients in the longshore sand transport rate and depends on sediment supply from updrift beaches.

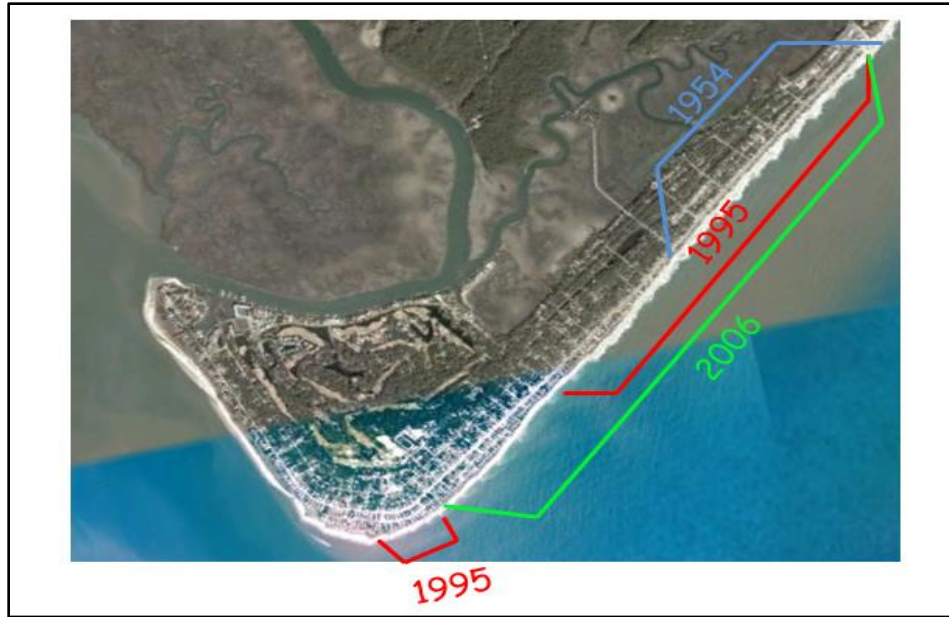


Figure B–1. Historical beach nourishment placement areas

b. Longshore sediment transport regime. Net longshore sand transport along Edisto Beach is from north to south, and the magnitude of the longshore sand transport rate tends to increase moving from north to south. Intra-annual reversals in the longshore transport direction at Edisto Beach can be significant and are readily observed by shoreline position changes within groin compartments. These intra-annual transport direction reversals are driven by seasonal changes in the incident wave direction. During the stormier late fall/winter/early spring seasons, net transport direction is to the south; whereas, during the milder late spring and summer seasons the net transport direction is often directed to the north. Within the groin compartments, an accretional berm fillet develops on the updrift side of the groin and an associated erosional scarp develops on the downdrift side of the groin.

(1) Consequently, when sand transport is in the long-term net direction (north to south), a wider beach berm is observed at southern end of the groin compartment; whereas, narrower beach berms are present at the northern end of the groin compartment. The opposite occurs during periods of net longshore transport reversals, with accretion and wider beaches at the northern end of the groin compartment and narrower beaches at the southern end of the groin compartment. For this reason, characteristic representative beach profiles developed for modeling purposes were generated using an average of all surveyed profiles within a given groin compartment.

(2) Taking a wider view of the regional coastal setting improves the overall understanding of the coastal processes that have led to the increasing vulnerability of Edisto Beach to storm-induced damages. Figure B–2 provides an aerial view of the South Carolina coastal region between North Edisto Inlet (upper right) and South Edisto Inlet (lower left). Net longshore transport is directed north along Botany Bay Island, located to the south of North Edisto Inlet, a reversal in the net transport direction to the

south occurs near Edingsville Beach, which is centrally located between the two large inlets. The barrier island at Edingsville Beach is low lying and frequently over washed during storm conditions. Sand is sequestered in the extensive inlet shoals and is washed over the low-lying barriers into the coastal marshes. As a result, the sand supply delivered to Edisto Beach by the prevailing coastal processes has diminished considerably over the past century and is expected to continue to diminish into the future.

(3) The presence of the groin field at Edisto Beach has significantly reduced shoreline erosion within the project study area and has been essential to the stabilization of the Edisto Beach shoreline. Nevertheless, sand supply to Edisto Beach from the north is insufficient to maintain natural storm protection in the form of significant beach berm widths and a protective dune feature. In the absence of beach nourishment, erosion within the groin cells will continue and, ultimately, exchange of sediment between groin cells will be cut off. When this occurs, it is expected that shoreline erosion will increase, first in the vicinity of the point and along the South Edisto Inlet shoreline and then to the north along the ocean-facing groin cells.



Figure B-2. Longshore sand transport regime between North Edisto Inlet and South Edisto Inlet

(4) Gross longshore sand transport rates in the vicinity of Edisto Beach have been estimated at approximately 210,000 cubic yards/year, with about 44,000 cubic yards/year directed to the north and about 167,000 cubic yards/year directed to the south. The net longshore sand transport rate is estimated at approximately 123,000 cubic yards/year and directed to the south (CSE 1993).

c. *Geomorphology of Edisto Beach.* Edisto Beach is at the southern end of what was once a classical, pro-grading, drumstick-shaped barrier island common in South Carolina. However, over time, erosion in the central portion (Edingsville Beach area) of the larger barrier island system due to a net longshore transport divergence has resulted in opening of new tidal inlets (Frampton Inlet, Jeremy Inlet, and an un-named inlet north of Frampton Inlet) and loss of littoral sediments to developing shoal features at those inlets. Continued erosion has reduced the central barriers to little more than swash shoals that allow littoral material to wash over the barriers and become trapped in the coastal marshes. As a consequence, the barrier island at Edisto Beach is transitioning to a landward-migrating, transgressive barrier island.

(1) The geomorphology of Edisto Beach is unique among South Carolina beaches in that the sediment composition of the beach is coarser grained than most South Carolina beaches, with a median grain size of approximately 0.4 mm (CSE 2006) and significant shell content. The relatively coarse median sediment grain size results in comparatively steep foreshore slopes. Within the oceanfront groin compartments, the foreshore slope is approximately 1 on 10; within the state park north of Edisto Beach, the foreshore slope is slightly milder at 1 on 15; and the foreshore slope along inlet shoreline is milder still, at approximately 1 on 25. These steep foreshore slopes, together with a fairly high tidal range (average spring tide range is 6.3 feet), reduce the beach area between the low-tide terrace and the fore dunes compared to other South Carolina beaches.

(2) Due to these geomorphic conditions, wave energy associated with storm conditions is not dissipated to any large degree before it reaches the relatively low fore dunes present on the barrier island. The overall average dune crest elevation within the project study area is about 10.5 feet (NAVD88), although dune crest elevations vary between a minimum of 8.5 feet (NAVD88) and 12 feet (NAVD88) in different reaches within the study area. In the state park area and in the vicinity of the point, the average dune crest elevation is 10 feet (NAVD88). Along the Edisto Beach, ocean-fronting shoreline, the average dune crest elevation is 11.5 feet (NAVD88), and on the inlet-fronting shoreline, the average dune crest elevation is approximately 9 feet (NAVD88).

(3) Dune volume above the berm elevation can be used as an indicator of storm vulnerability, and within the project study area, the overall average dune volume above the berm elevation was estimated at approximately 4.3 cubic yard/feet based on representative, idealized beach profiles developed for modeling purposes. However, as with the dune crest elevation, this quantity varies between a minimum of approximately 1.4 cubic yard/feet north of the camping area in the state park and the inner inlet sub-reach to a maximum of 7.9 cubic yard/feet in Groin Cell 6.

(4) In the state park area, the average dune volume above the berm elevation is 3.3 cubic yard/feet. Along the Edisto Beach, ocean-fronting shoreline, the average dune volume above the berm elevation is 5.5 cubic yard/feet; whereas, in the vicinity of the point the average dune volume above the berm elevation drops to 3.7 cubic yard/feet, and along the inlet shoreline the average dune volume above the berm elevation drops to just 2.8 cubic yard/feet. A detailed description of the analysis performed to develop the representative idealized profiles for modeling purposes is provided in paragraph B-4.

d. Sea level change.

(1) The closest NOAA tidal gauge to Edisto is the Charleston, South Carolina (NOAA 8665530) gauge. Edisto is approximately 31.5 miles from Charleston. The mean sea level trend at Charleston, is 3.51 millimeters/year (1.15 feet/century), with a standard error of 0.19 mm/year based on monthly mean sea level data from 1901 to 2024 (Figure B-3). Tidal datums for the Charleston gauge relative to NAVD88 are displayed in (Figure B-4). The SLC curve historic rates presented in this document have been updated to reflect present SLAT output and USACE guidance for the station closest to Edisto Beach. The original analysis relied on outputs generated based on an interpolation between SLC outputs at both Charleston, South Carolina (NOAA 8665530) and Fort Pulaski, Georgia (NOAA 8670870). Best practice is to use information collected at the tidal gauge located closest to the study site with greater than 40 years of historic data available. This change in approach does not result in a significant change in the approximation of SLC between 2009 and 2069 (a difference of 0.1 ft +/- 0.03 ft).

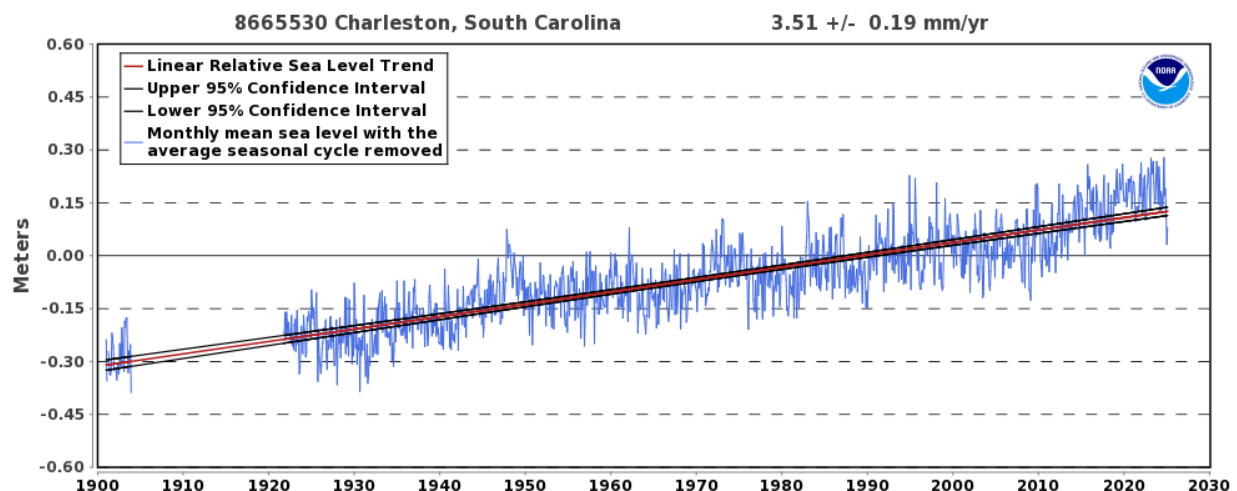


Figure B-3. Mean sea level change trend at Charleston, South Carolina

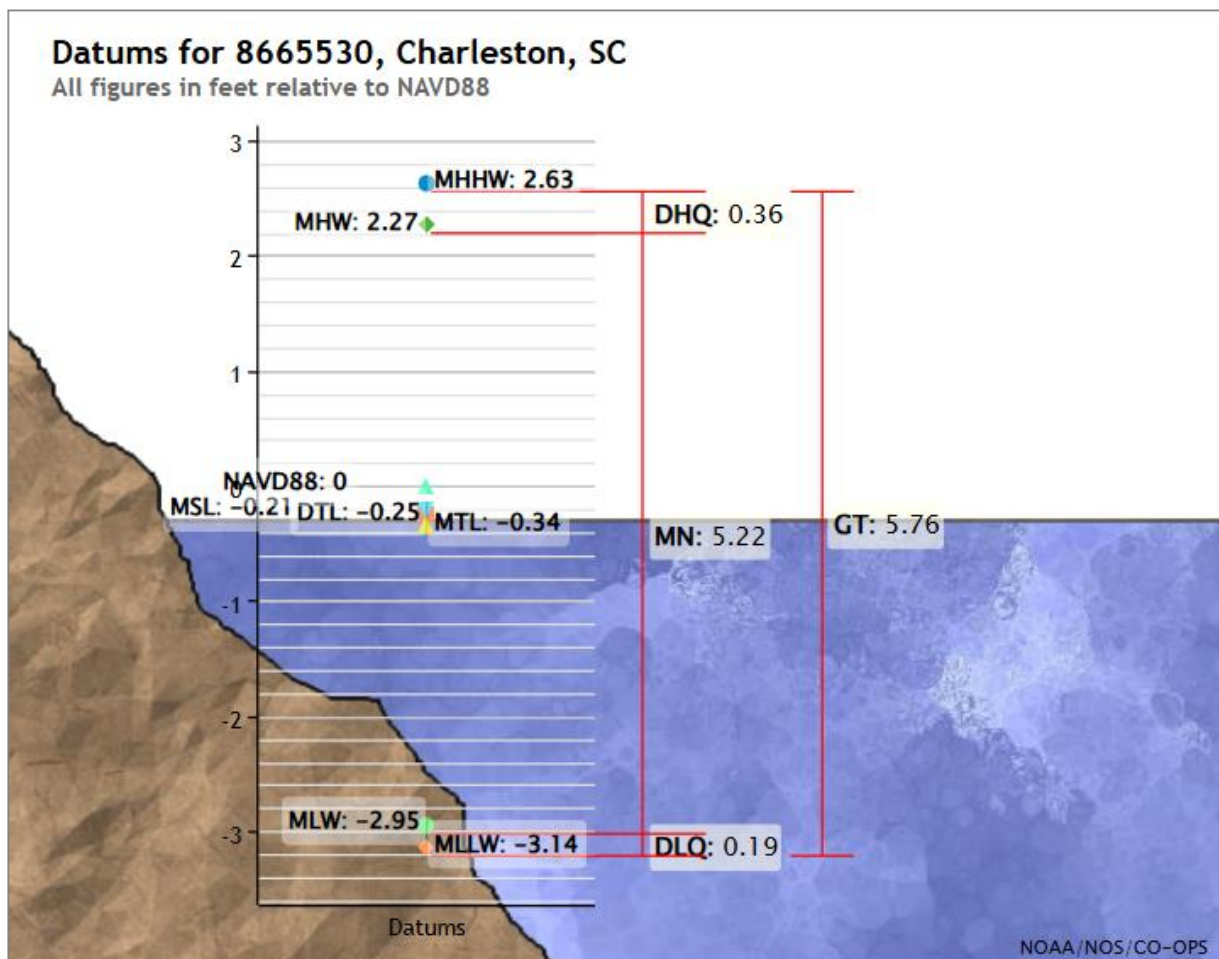


Figure B-4. Tidal Datums for National Oceanic and Atmospheric Administration gauge 866530 at Charleston, South Carolina

(2) The USACE “Low” rate represents a linear extrapolation of the historical rate of mean sea level change. The “Intermediate” rate of future sea level change was computed using a modified National Research Council (NRC) Curve I (equation 3 in ER 1100-2-8162). The “High” rate of future sea level change was computed using a modified NRC Curve III (equation 3 in ER 1100-2-8162). The relationships for future sea level change as outlined in that ER are coded within Beach-*fx* and sea level change is internally computed continuously throughout the simulated project life cycle.

B-4. Development of storm suite

a. *Water levels.* Storm-generated water levels along the open coast and up the major tributaries of South Carolina have previously been investigated by Dr. Norman W. Scheffner of the USACE Engineer Research and Development Center (ERDC) Coastal and Hydraulics Laboratory of the Waterways Experiment Station in Vicksburg, Mississippi for the USACE District, Charleston (CESAC). The findings of that investigation are presented in the report titled, “Coast of South Carolina Storm Surge

Study,” dated May 2000 (Scheffner 2000). Tropical and extratropical storm events that have historically impacted South Carolina were simulated with the ADCIRC (ADvanced CIRCulation) long-wave hydrodynamic model.

b. Advanced circulation modeling. The large-domain, long-wave hydrodynamic ADCIRC model (Luettich 1992) was used to provide water-surface elevations for this study. The ADCIRC model is an unstructured, grid-finite element, long-wave hydrodynamic model developed under the USACE Dredging Research Program (DRP).

(1) The model was developed as a family of two- and three-dimensional codes with the following capabilities:

(a) Simulating tidal circulation and storm surge propagation over large computational domains while simultaneously providing high resolution in areas of complex shoreline and bathymetry. The targeted areas of interest include continental shelves, nearshore areas, and estuaries.

(b) Representing all pertinent physics of the three-dimensional equations of motion. These include tidal potential, Coriolis, and all nonlinear terms of the governing equations.

(c) Providing accurate and efficient computations over time periods ranging from months to years.

(2) The ADCIRC model solves the depth-averaged, generalized wave continuity equation formulation of the governing equations and has been extensively applied to projects requiring frequency analysis of storm events. The general methodology developed for these previous studies was applied to the South Carolina storm surge investigation (Scheffner 2000). Tidal and storm surge-water surface elevation data were archived at 38 stations, including one station immediately north of Edisto Island (-80.18945 32.56440) and one station immediately south of Edisto Island (-80.35598 32.48759).

(3) The 20,000-node computational domain (Figure B-5) used in the generation of the DRP east coast, Gulf of Mexico, and Caribbean Sea tidal database formed the initial grid for this study because the tidal boundary conditions along the eastern boundary (60° east Longitude) had already been determined. Additionally, proper flow connectivity between the Gulf of Mexico, Caribbean Sea, and the South Atlantic Bight was assured by accurately defining the bathymetry of all basins.

(4) For example, by modeling the entire domain, the flow and surge distribution resulting from hurricanes moving toward the study area from the Caribbean or Gulf of Mexico is properly simulated. Minimum node-to-node spacing of this initial grid was on the order of 5 kilometers. Minimum resolution along the open coast and up the major tributaries of the study area was decreased to approximately 700 meters to provide sufficient detail of the local bathymetry and topography. The increased resolution of the study area is shown in Figure B-6.

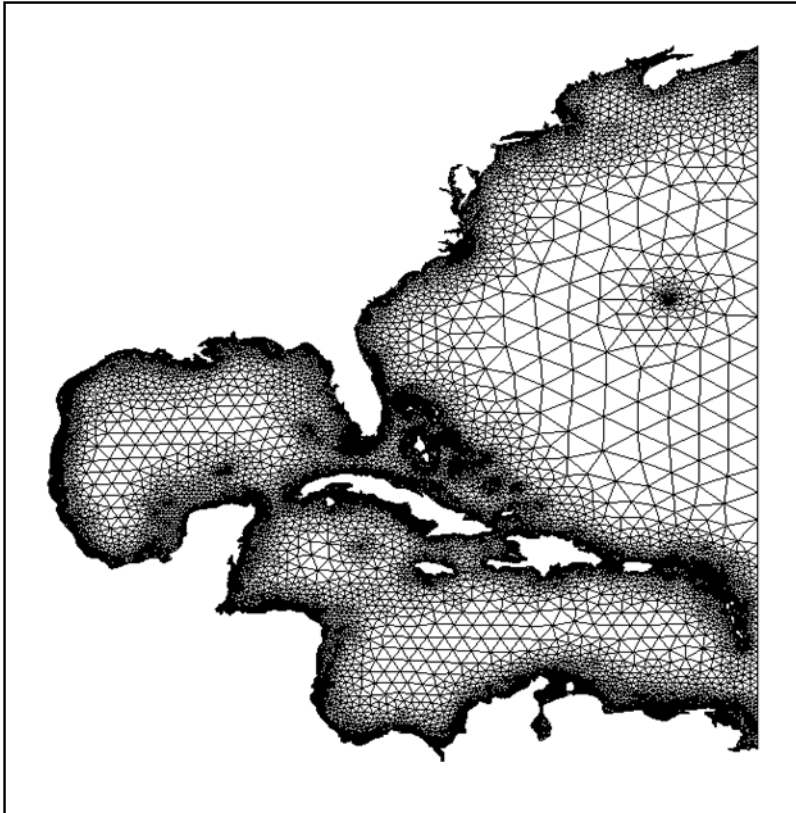


Figure B-5. The dredging research program grid of the East Coast, Gulf of Mexico, and Caribbean Sea

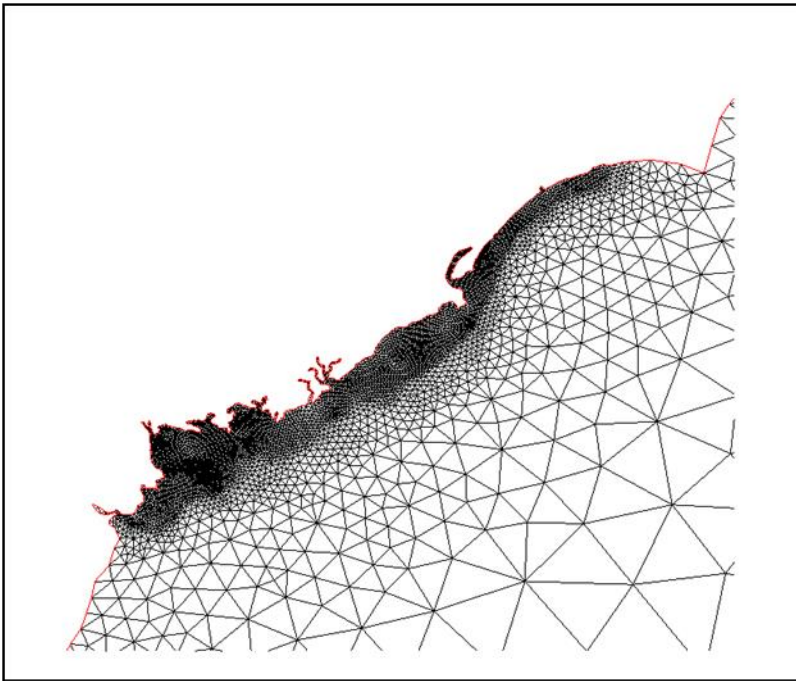


Figure B-6. Grid resolution in the South Carolina study domain

c. *Tropical storms.*

(1) A tropical storm database (Scheffner et al. 1994) was generated during the DRP through simulation of 134 historically based storm events along the east coast, Gulf of Mexico, and Caribbean Sea. For 486 discrete locations along the United States coast, storm events that produced a storm surge of at least 1 foot were archived and indexed according to event, location, and surge. This indexed database was used to define an initial training set for the present study (Scheffner 2000).

(2) Ideally, historical events represent the full range of possible event intensities. If this occurs, the historical events can be used directly to develop the full training set of storms. For extratropical events, this is generally the case because extratropical events occur often, cover extremely large areas, and persist for long periods of time (for example, days). However, with tropical events, this is often not the case. At many locations, the worst-case tropical event scenario may not yet have occurred in the historical record but may be represented by a historical event with a slightly shifted path or larger/smaller radius to maximum wind. Due to this, some augmentation of the historical events is often necessary.

(3) This was found to be necessary for the South Carolina study because station locations of interest span over 150 miles. For example, Hurricane Hugo made a near-perpendicular landfall near Charleston on 21 September 1989. Hurricane Hugo produced severe surges for areas north and east of Charleston; however, areas south and west of landfall were not significantly impacted (Scheffner 2000).

(4) To supplement the training set so that all stations within the study experience a maximum intensity event and thereby fill the vector space with events ranging from nominal to intense, six additional storm events were added to the initial training set. Four of these events were developed as perturbations of Hurricane Hugo and two were developed as perturbations of the unnamed hurricane of 1910, as Hugo and the 1910 hurricane were the two most intense events in the historical record. As a result, maximum surge elevations were experienced just east of landfall; however, negligible surge elevations were experienced southwest of landfall and the surges were minimal further northeast of landfall.

(5) Four hypothetical events were created by assuming the historical path of Hurricane Hugo was shifted 1° and 2° west of landfall and 1° and 2° east of landfall. These four combinations, along with the historical event, produced maximum surges throughout the study area. For the 1910 hurricane, two hypothetical events were created by assuming the historical path was shifted 1° both east and west of landfall. The final training set consisted of 30 events: 24 historical and 6 hypothetical (4 representing perturbations of Hurricane Hugo and 2 representing perturbations of the hurricane of 1910). The final training set is shown in Table B–2 (Scheffner 2000).

d. *Extratropical storms.*

(1) In a similar manner to the tropical event database described above, an extratropical storm event database was generated within the DRP. This database was constructed by driving the ADCIRC model with wind fields extracted from the United States Navy Fleet Numerical Meteorology and Oceanography Center's database of winds for the 16-year winter storm period (defined as September through March) of 1977 through 1993 (1977–1978, 1978–1979, etc.). Data are provided at 6-hour intervals on a 2.5° latitude and longitude grid. The extratropical storm database consists of surface elevation and current hydrographs at each of the 486 stations described above.

Table B–2
Tropical storm training set inventory

#	HURDAT* Storm Number	Given name	Date* (mm/dd/yyyy)
1	194	NOT NAMED	10/09/1910
2	194A	1910-A	–
3	194B	1910-B	–
4	196	NOT NAMED	08/23/1911
5	217	NOT NAMED	07/11/1916
6	292	NOT NAMED	09/06/1928
7	296	NOT NAMED	09/22/1029
8	299	NOT NAMED	08/31/1930
9	353	NOT NAMED	08/29/1935
10	398	NOT NAMED	08/05/1940
11	440	NOT NAMED	10/12/1944
12	449	NOT NAMED	09/12/1945
13	463	NOT NAMED	09/20/1947
14	465	NOT NAMED	10/09/1947
15	521	NOT NAMED	08/28/1953
16	526	FLORENCE	09/23/1953
17	541	HAZEL	10/05/1954
18	562	FLOSSY	09/21/1956
19	589	GRACIE	09/20/1959
20	597	DONNA	08/29/1960
21	643	ALMA	06/04/1966
22	669	GLADYS	10/13/1968
23	777	DAVID	08/25/1979
24	797	DENNIS	08/07/1981
25	839	KATE	11/15/1985
26	872	HUGO	09/10/1989
27	872A	HUGO-A	–

#	HURDAT* Storm Number	Given name	Date* (mm/dd/yyyy)
28	872B	HUGO-B	–
29	872C	HUGO-C	–
30	872D	HUGO-D	–

Note:

*The HURDAT (NOAA 2024) storm number designation refers to the storm identification number of the events in the National Hurricane Center (NOAA 2024h) data base of historical tropical events and the time signifies the first time of storm on record.

(2) These data contain severe events occurring during the 16-year sequence of winter months; however, unlike tropical events that are clearly distinguishable, identification of individual extratropical events within the records requires additional analysis (Scheffner 2000).

(3) Time series surface elevation plots corresponding to an archived station near the center of the study area were analyzed. Each time series represented surge with no tide. The time series of water surface elevations for 16-year seasons were plotted and 9 extratropical events were identified and extracted from the time series to populate the extratropical storm event database. The approximate starting time for each of the nine events is shown in Table B–3 (Scheffner 2000).

Table B–3
Extratropical storm training set inventory

Storm Number	Starting Date (mm/dd/yyyy)
1	02/13/1979
2	09/01/1979
3	02/08/1983
4	02/24/1983
5	03/13/1983
6	09/05/1984
7	10/24/1985
8	01/01/1987
9	02/13/1987

e. Development of plausible storm suite.

(1) The historical (and hypothetical) tropical and extratropical storm events identified as discussed in paragraph B–4 were expanded to form what is known as a plausible storm suite that was used to drive beach evolution within Beach-*fx*. The procedure followed to generate a site-specific plausible storm suite for Edisto Beach for use in the numerical estimation of storm-induced beach morphology response during this feasibility study involved five broad steps:

- (a) Identification of project-specific significant storm events (discussed previously).
- (b) Extraction of the storm surge hydrographs corresponding to the identified significant storm events.
- (c) Estimation of wind-wave conditions corresponding to the identified significant storm events.
- (d) Statistical characterization of project specific astronomical tides and estimation of representative high, mean, and low tidal ranges.
- (e) Development of 12 plausible total water level hydrographs for each of the identified significant storm events by linearly adding surge and tide.

(2) The first two steps were completed by reviewing and utilizing the data from the South Carolina Storm Surge study detailed above. Final storm surge hydrographs were constructed by averaging the hydrographs from the two Edisto stations for each tropical and extratropical storm in the database.

(3) Preparation of the storm surge hydrographs for their use in SBEACH (Larson 1990) beach morphology simulations involved identifying the portion of the ADCIRC simulation relevant to morphology response modeling and processing the storm surge hydrographs, which involved clipping the storm surge hydrographs and applying a mild smoothing of the hydrographs as necessary. The storm surge hydrographs were analyzed to identify which portions of the modeled event were essential for capturing the beach response in SBEACH. For example, if the averaged hydrograph from the South Carolina Storm Surge Study was 150 hours long, and only the 100 hours in the middle of the data were appropriate and necessary, the 25 hours at the beginning and the 25 hours at the end were clipped from the record. The next step was to smooth portions of the time series that required some degree of smoothing.

(4) The contribution of astronomical tides to the total water elevation hydrograph was developed by first performing a statistical analysis of tides at Edisto Beach. The aim of the analysis was to estimate three statistically significant tidal ranges by quantifying:

- (a) A high tidal range representing the mean of the highest 25% of all tidal ranges occurring at Edisto Beach.
- (b) A mean tidal range representing the mean of the central 50% of all tidal ranges occurring at Edisto Beach.
- (c) A low tidal range representing the mean of the lowest 25% of all tidal ranges occurring at Edisto Beach.

(5) Next, semi-diurnal cosine tide signals were generated with ranges corresponding to the computed high, mean, and low tidal ranges. Each of the historical

and hypothetical storm surge hydrographs were combined with the idealized cosine astronomical tide hydrographs to generate the suite of plausible total water elevation time series. For each tide range (high, mean, and low), the storm surge hydrograph was added to the cosine astronomical tide hydrograph at four phases of the tide signal, aligning peak storm surge with high tide, mean tide falling, low tide, and mean tide rising. This procedure produces 12 plausible total water elevation representations of each of the historical storm events. Those events associated with the mean tide range are weighted double the weight of those events associated with the high and low tide ranges.

(6) For the two historical storms that involved hypothetical representations, the hurricane of 1910 and Hurricane Hugo, the combined weight of all representations of the historical is equal to the combined weight of the other historical events. The plausible storm suite includes five representations of Hurricane Hugo, four hypothetical storm tracks and the historical storm track; whereas, the other hurricanes (with the exception of the hurricane of 1910) involved just the historical storm track. For example, the combined total weight of the 12 representations of Hurricane Dennis is 16 (4 with a weight of 1 associated with the high tide range, 4 with a weight of 2 associated with the mean tide range, and 4 with a weight of 1 associated with the low tide range).

(7) However, for Hurricane Hugo, which involves five representations of the storm (one historical and four hypothetical), the weighting is as follows: 20 with a weight of 0.2 associated with the high tide range; 20 with a weight of 0.4 associated with the mean tide range; and 20 with a weight of 0.2 associated with the low tide range; which results in a combined weight of 16, the same as Hurricane Dennis. The relative frequencies of storms could have been generated by a JPM-OS approach as well, if JPM-OS had been used to define the storms prior to modeling.

(8) Waves for the extratropical events were available from the WIS database . All of the wave height time series (tropical and extratropical) were reduced, where applicable and necessary, by limiting wave heights according to the depth-limited, breaking wave criteria based on water depth during the event at the SBEACH computational boundary. When the wave record length was shorter than the surge record length, a standard minimum wave height and period were added to the time series. The minimum wave height was selected as a weighted average of the mean wave heights of the first two bins in the WIS analysis. Bin1 (0.0–0.5 m) was assigned 0.25 meters and represented 11.67% of the 20-year record, while Bin2 (0.5–1.0 m) was assigned 0.75 meters and represented 49.49% of the 20-year record (equation B–2).

$$0.25(0.1167/0.6116) + 0.75(0.4949/0.6116) = 0.65 \text{ meters} = 2.1 \text{ feet} \quad (\text{B-1})$$

(9) Peak wave heights were adjusted regarding the timing of the peak storm surge so that the peaks were more or less aligned. Wind-waves for the tropical storms were obtained from the WIS hindcast for those storms occurring between 1980 and the present. For those tropical storms occurring prior to 1980, a parametric prediction technique for waves in hurricanes was employed as described in Part II-2-2-c of EM 1110-2-1100.

B–5. Representative beach profiles

a. EM 1110-2-1100 provides some guidance on how to determine baseline damages by including the existing or without-project condition of the project study domain. Morphologic features of the existing beach, such as dune height, berm width, and offshore profile shape, typically vary along the project study domain. To accurately estimate storm erosion response for the existing condition, EM 1110-2-1100 suggests developing a set of representative morphologic reaches to describe variations in profile shape along the project domain.

b. Morphology analysis software applications such as BMAP (2024) or RMAP (Regional Morphology Analysis Package) (2024), can be used to define morphologic reaches by analyzing profiles, grouping similar profiles, and calculating an average representative profile for each reach. According to EM 1110-2-1100, the profile characteristics that should be considered when developing morphologic reaches include dune height and width, berm width, nearshore and offshore profile slopes, sand grain size, presence of seawalls or other structures, and proximity to inlets.

c. The Edisto Beach Hurricane and Storm Damage Reduction feasibility study employed Beach-*fx*, the USACE Monte Carlo life-cycle simulation model, for estimating shore protection project evolution and cost-benefit analyses (Gravens 2007). An overview of the general hierarchical data structure employed in Beach-*fx* is provided in Figure B–7 taken from Beach-*fx* User’s Manual, Version 1.0 (Rogers et al. 2009). Within Beach-*fx* the overall unit of analysis is the “project,” a shoreline area for which the analysis is to be performed. The project is divided for purposes of analysis into “reaches,” which are contiguous, morphologically homogeneous areas.

d. The structures within a reach are referred to as damage elements (DEs) and are located within lots. All locations are geospatially referenced using a cartographic coordinate system, such as state plane coordinates. This project definition scheme is shown schematically in Figure B–8, in which the shoreline is linearized into reaches. Each reach is associated with a representative beach profile that describes the shape of the cross-shore profile and beach composition.

e. The profile is the basic unit of beach response. Natural beach profiles are complex; for the modeling, a simplified or idealized beach profile, representing key morphological features defined by points, is used as shown in Figure B–9. The idealized profile represents a single trapezoidal dune with a horizontal berm and a horizontal upland landward of the dune feature.

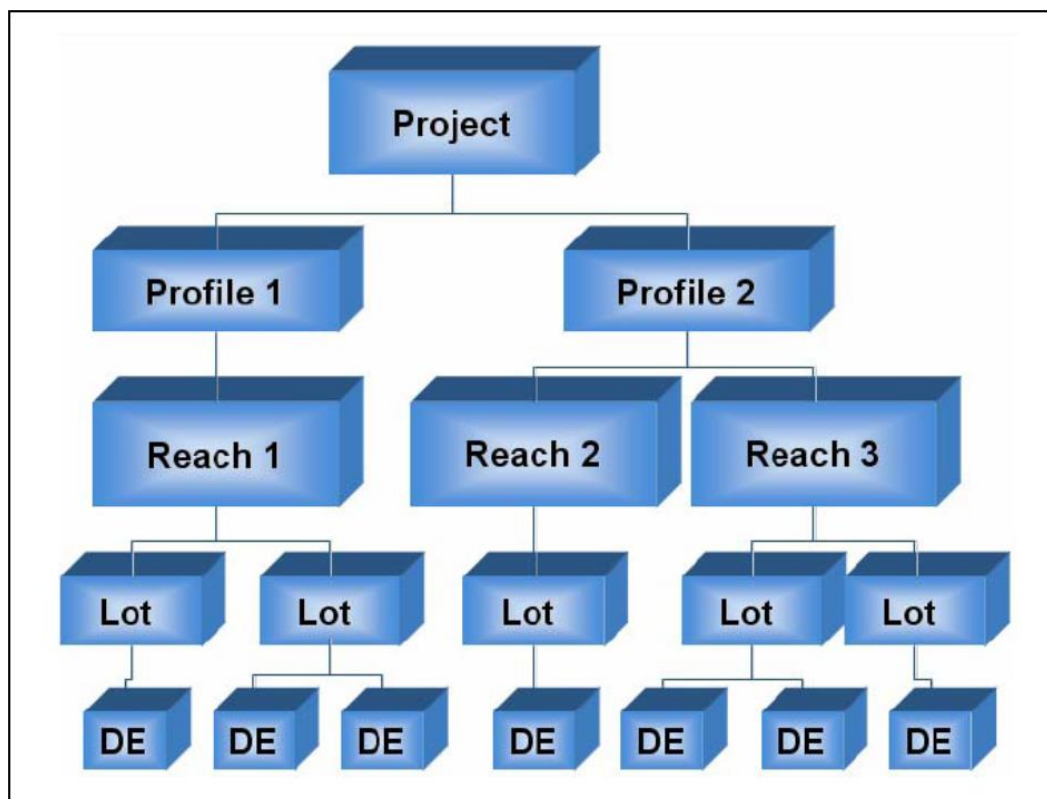


Figure B-7. Hierarchical representation of Beach-fx data elements

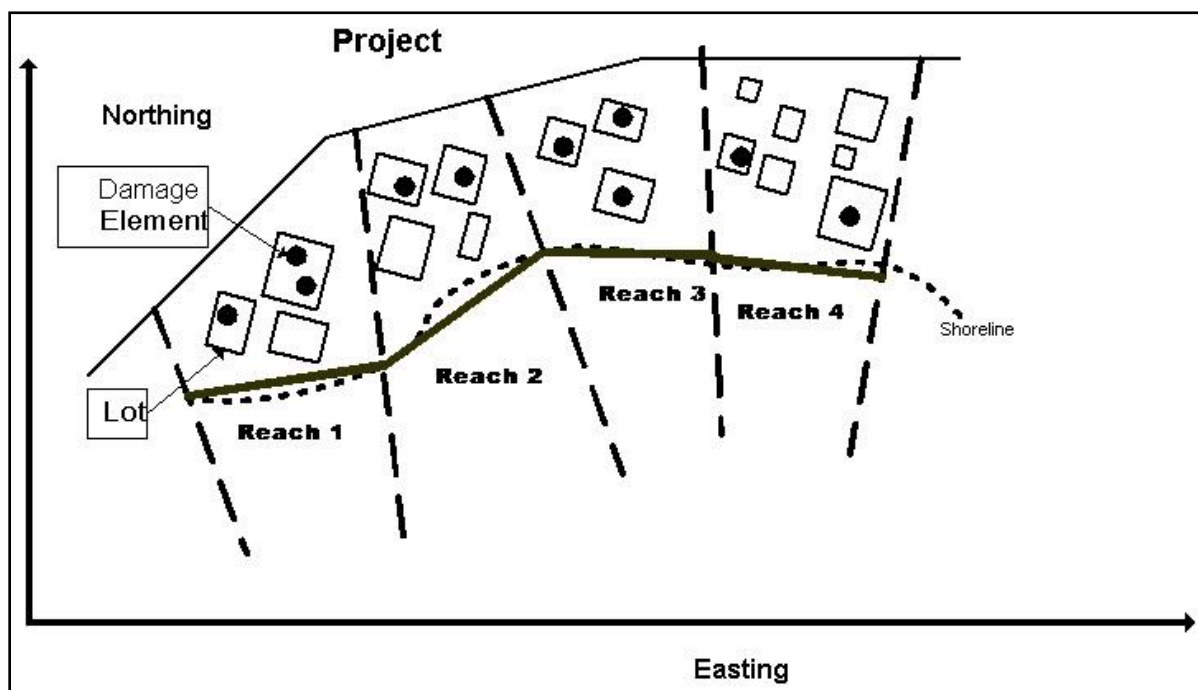


Figure B-8. Beach-fx schematization of the project study area

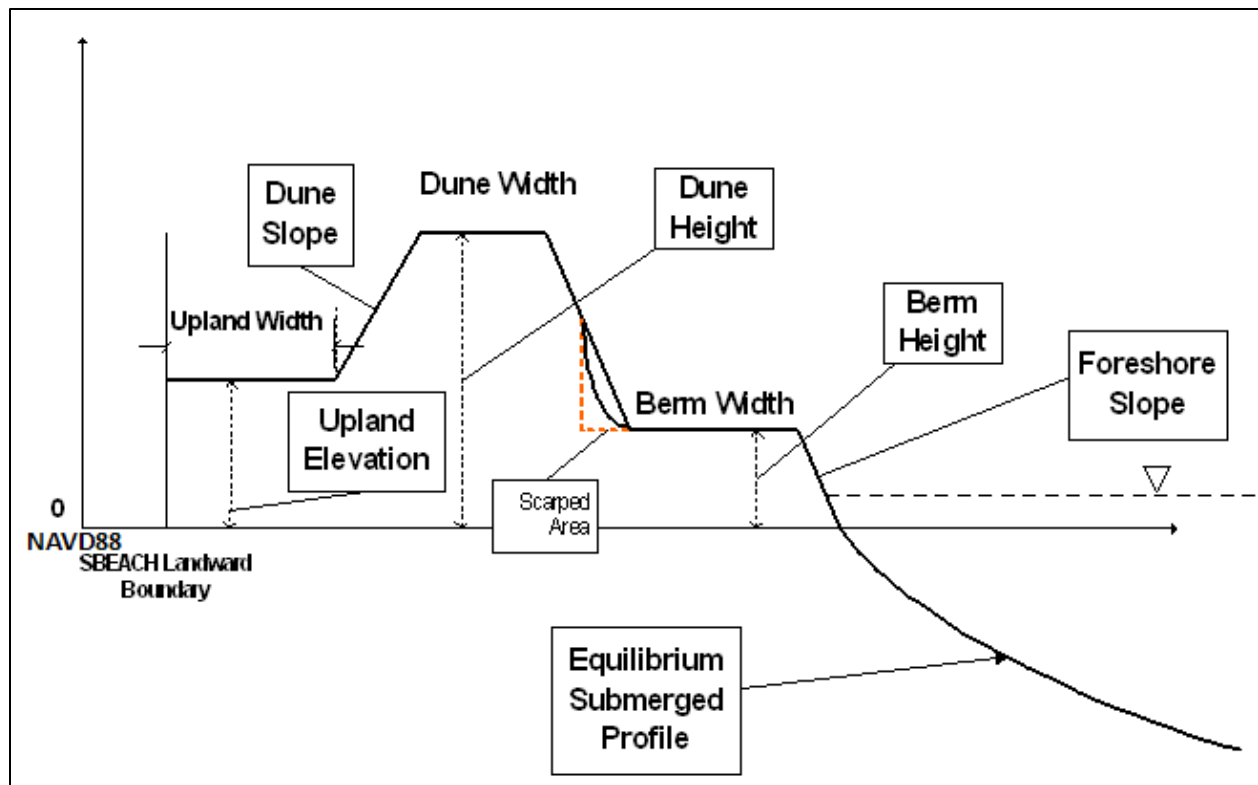


Figure B-9. Beach-fx idealized beach profile

f. The submerged portion of the profile is represented by a detailed series of distance-elevation points that are determined through an analysis of available beach profile information. For the Edisto Beach project, the detailed submerged beach profile was developed by averaging across multiple surveyed beach transects containing similar offshore slopes.

g. The beach morphology of Edisto Island, particularly the Town of Edisto Beach, is heavily influenced by the presence of the 34 groins, which are spaced an average of every 600 feet along the Atlantic shoreline of the Town of Edisto Beach. Profiles have been monitored at Edisto Island at 90 locations yearly since 2004. Figure B-10 shows the distribution of these 90 locations and clearly shows that the primary area of emphasis is the Atlantic shoreline of the Town of Edisto Beach. Figure B-11 provides a more detailed view of some of the monitoring locations and their relationship to the groins.

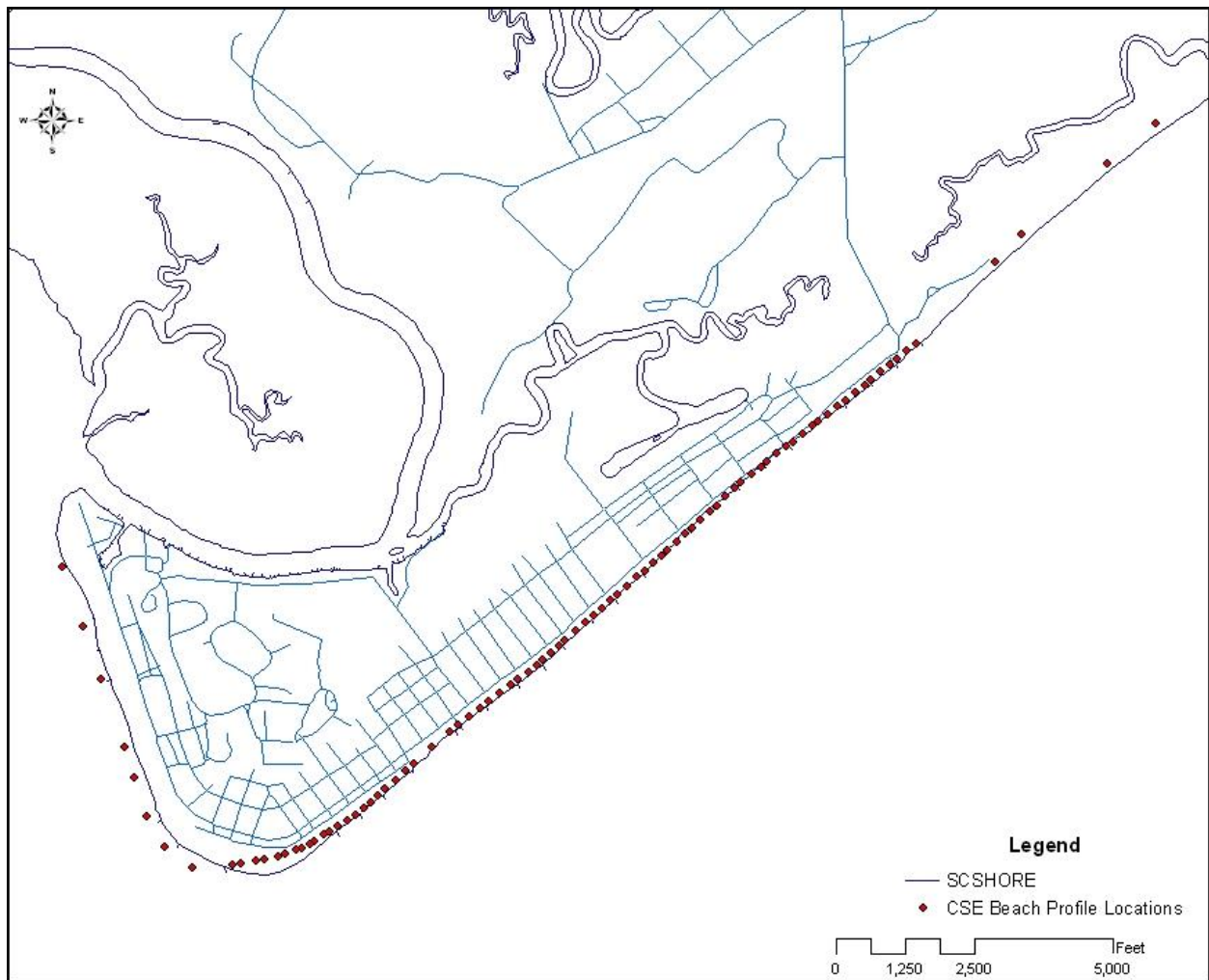


Figure B-10. Location of beach profile monitoring stations

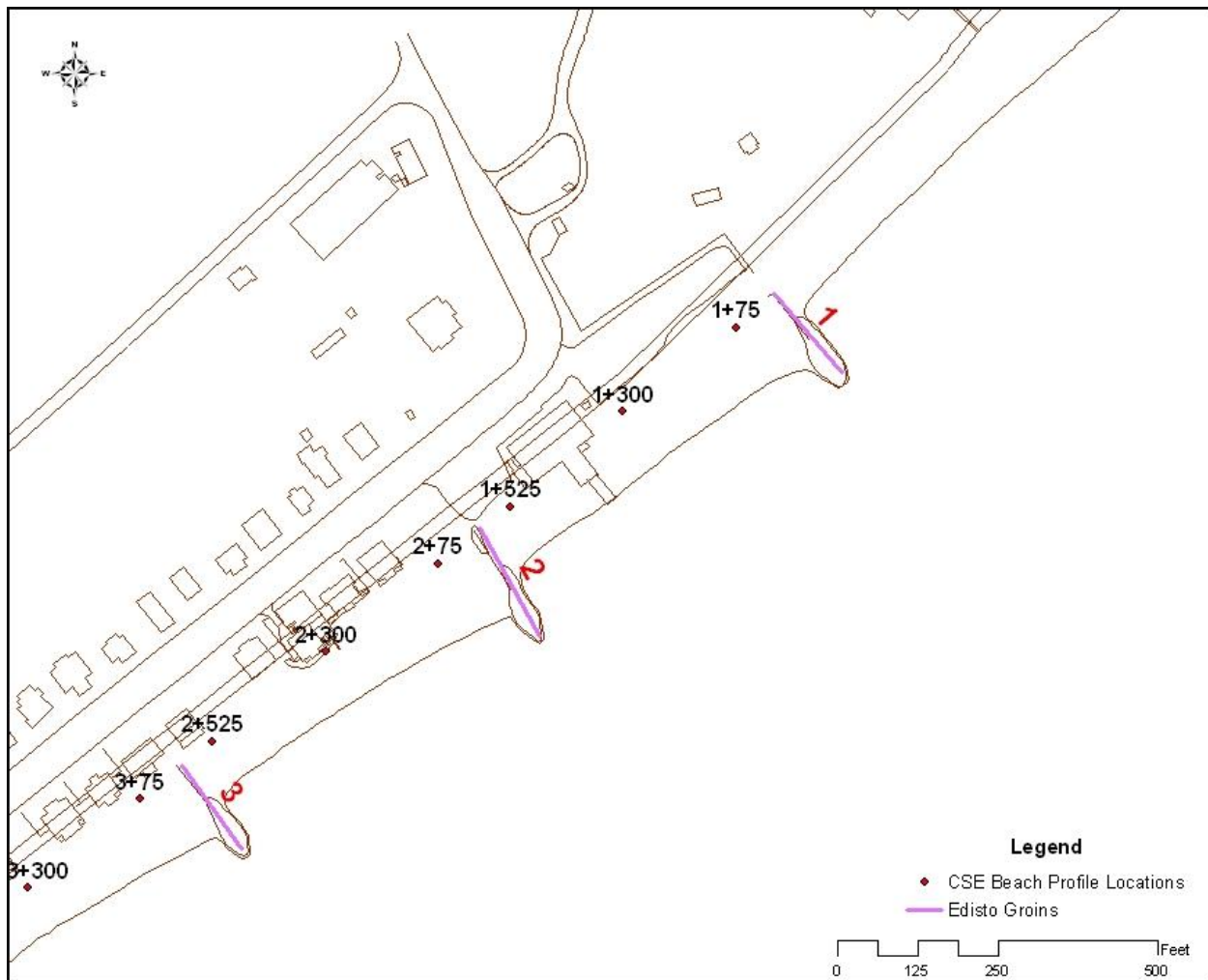


Figure B–11. Location of beach profile monitoring stations with respect to groins

h. The beach profile analysis that led to the development of the idealized representative beach profile began by first computing the average beach profiles within each of the groin compartments. The profile surveys employed in this analysis were those surveyed in August 2004, November 2005, July 2007, and July 2008. After computing the average submerged profile within each of the groin compartments, the shape of the offshore profile was compared across all the groin compartments and similar profiles were combined and an average submerged profile was computed for similar shape offshore profiles across multiple groin cells.

i. In the end, a total of 14 representative submerged beach profiles were developed to characterize the project study area as illustrated in Figure B–12. In this figure, green polygons represent the lot parcels and blue brackets show the spatial distribution of representative submerged profiles. Table B–4 defines the relationship between groin cells and the representative submerged beach profiles.

j. The next step in the development of the representative beach profiles for input into Beach-*fx* involved characterizing the upland dune and berm portion of the profiles. For this analysis only the July 2008 profile survey was used because the intent is to characterize the initial condition upper beach profile characteristics for initializing the life-cycle simulations performed within Beach-*fx*. The analysis involved first aligning the survey profile information within each groin cell such that the cross-shore position of the dune crest shared a common cross-shore position. Then an average profile was computed that yielded an average dune crest elevation within the groin cell as well as a representative average berm width within the groin compartment.

k. Finally, an idealized profile suitable for input to Beach-*fx* (horizontal upland, trapezoidal dune section with constant landward and seaward dune slopes, a horizontal berm section, and a constant foreshore slope down to datum) was generated. In a number of the groin cells, the upland, dune, and berm characteristics of the average upper beach profile was similar and in those cases a single idealized upper beach profile was generated. Figure B–13 through Figure B–16 show examples of the idealized upper beach profiles that define the Beach-*fx* reaches used in this feasibility study.

Table B–4
Representative submerged profile relationship to groin cells

Submerged Profile Name	Spatial Description
I1	0.75-mile segment of shoreline from Big Bay Creek toward the point
I2	0.6-mile segment of shoreline between I1 and the point
P1	Groin cell 28
P2	Groin cells 26 and 27
E1	Groin cells 24 and 25
E2	Groin cells 20, 21, 22, and 23
E3	Groin cells 16, 17, 18, and 19
E4	Groin cells 14 and 15
E5	Groin cell 13
E6	Groin cells 11 and 12
A	Groin cells 1, 2, 5, 7, and 10
B	Groin cells 3, 4, 6, 8, and 9
SP1	0.6-mile segment of shoreline extending north from groin 1
SP2	0.4-mile segment of shoreline north of SP2

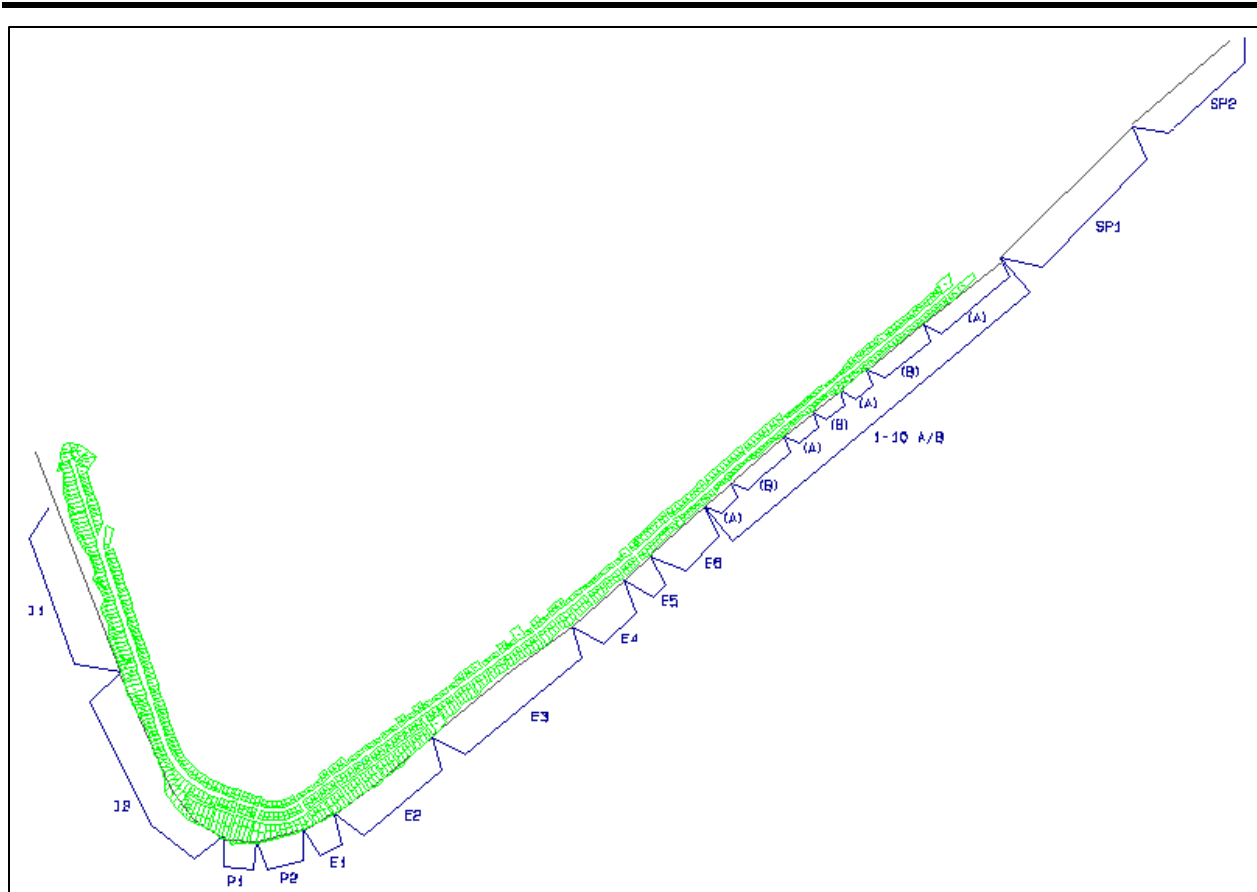


Figure B-12. Spatial distribution of representative submerged beach profiles

B-6. Reach determination

a. The Beach-*fx* analysis reaches are largely defined by the morphologically driven development of representative profiles discussed in the previous section. Due to subtle shoreline orientation differences and the requirement to simulate different upland widths to capture the first row of damage elements on the landward side Palmetto Boulevard, further subdivision of the Beach-*fx* analysis reaches was necessary. Specifically, Beach-*fx* Reaches I2, I3, and I4 were defined based on the same representative beach profile to account for differing shoreline orientations and upland width. Likewise, Beach-*fx* Reaches E2, E3, E4, and E5 were defined from common representative beach profiles. Figure B-17 shows the layout of the Beach-*fx* analysis reaches for the Edisto project. The blue polygons denote the reach boundaries, and the purple lines are the SBEACH reference lines that define the shoreline orientation within each reach.

b. The SBEACH reference line, defined by cartographic coordinates, is required to establish a common cross-shore frame of reference between the one-dimensional SBEACH coastal process model and Beach-*fx*. Damage elements within each reach are projected on to the SBEACH reference line to obtain the damage element's cross-shore position, which is subsequently used to determine the magnitude of the damage-driving

parameters (erosion, water depth, and wave crest elevation) and which are all tied to the SBEACH cross-shore coordinate system.

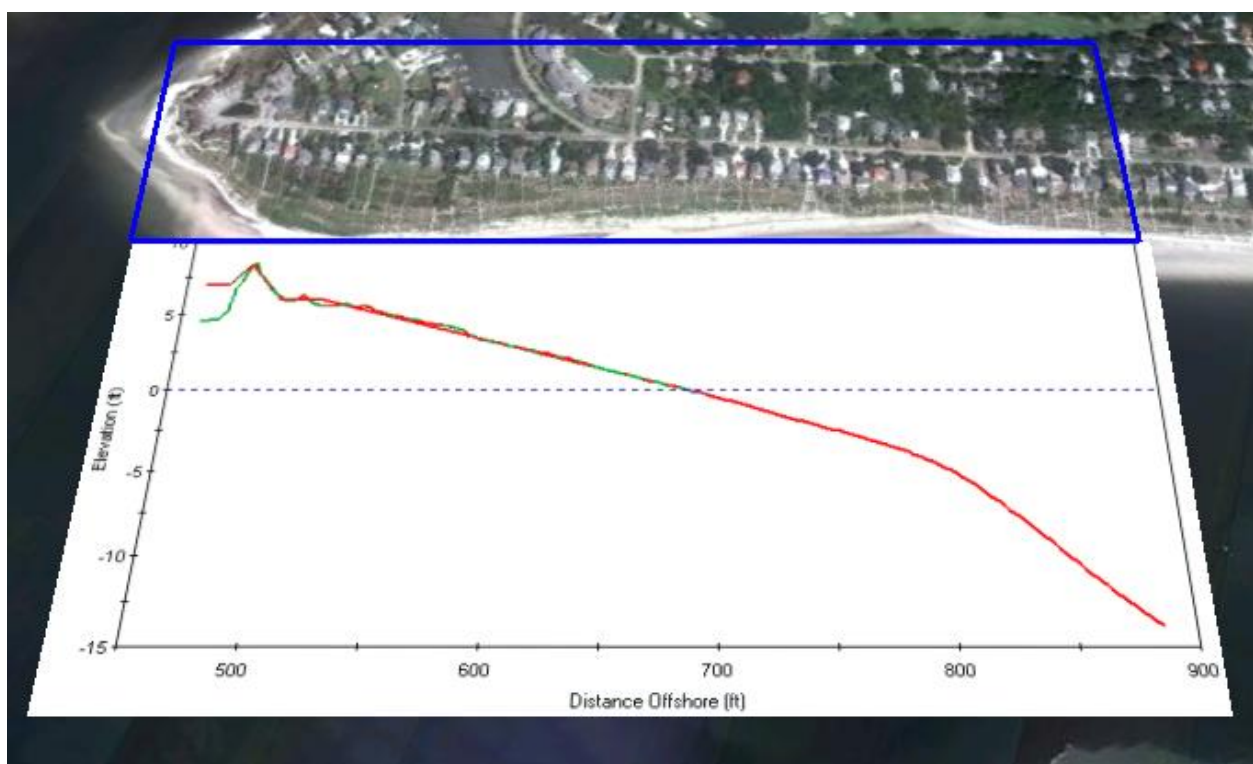


Figure B-13. Representative and idealized beach profile for Reach I1

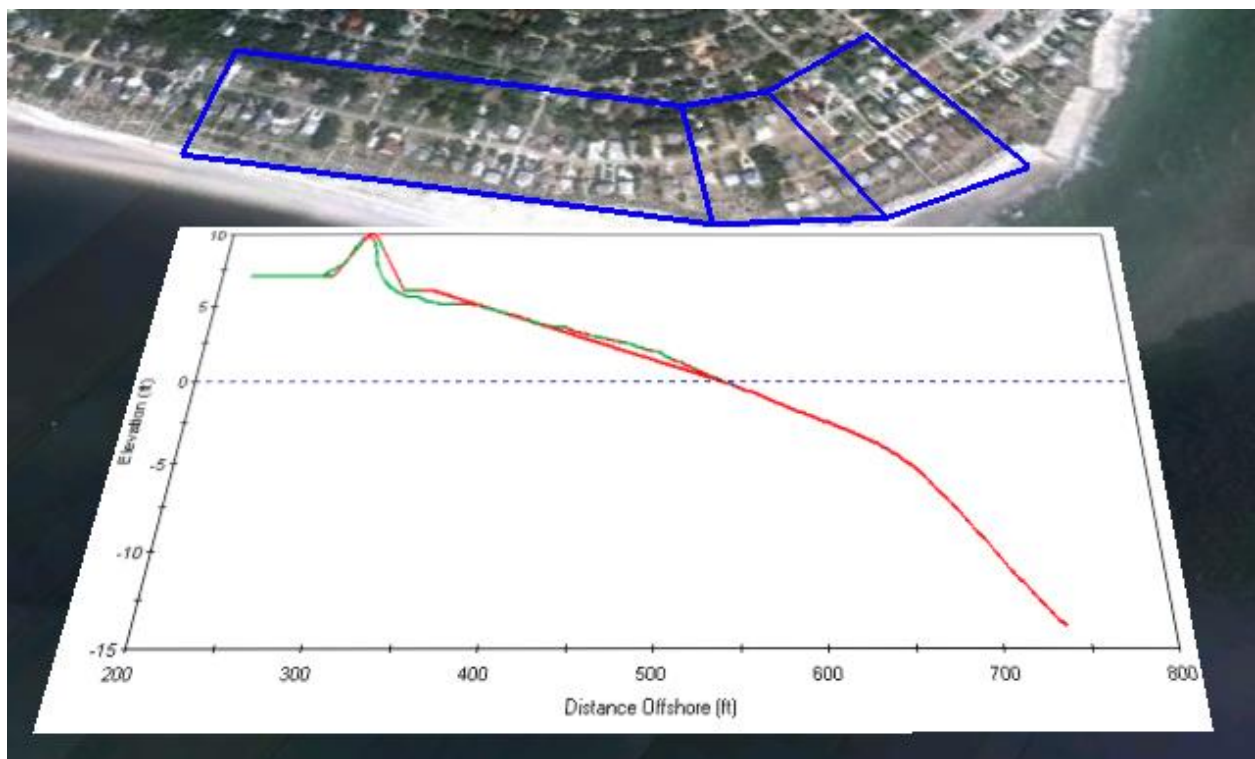


Figure B-14. Representative and idealized beach profile for Reaches I2-I4

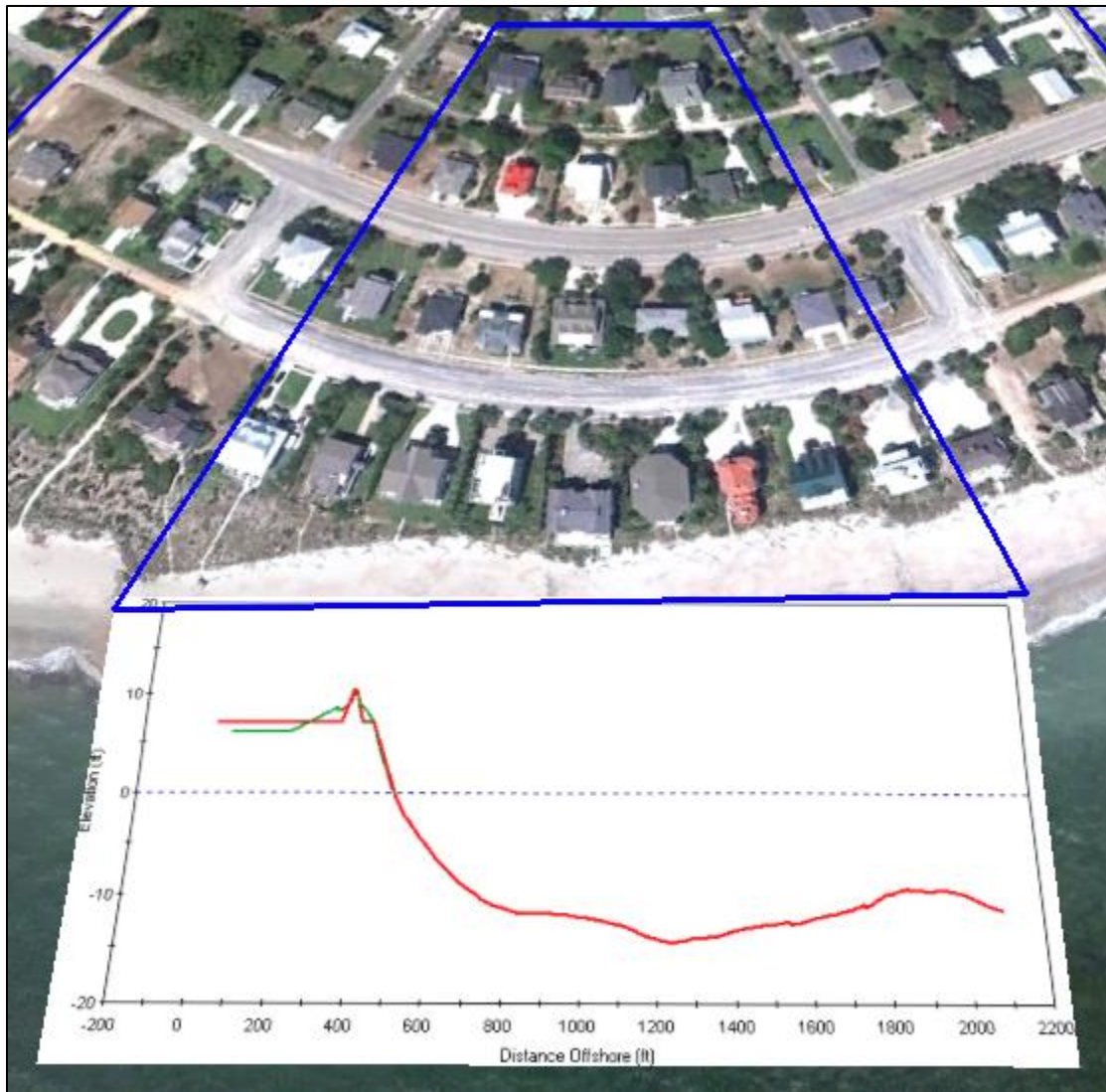


Figure B-15. Representative and idealized beach profile for Reach P2

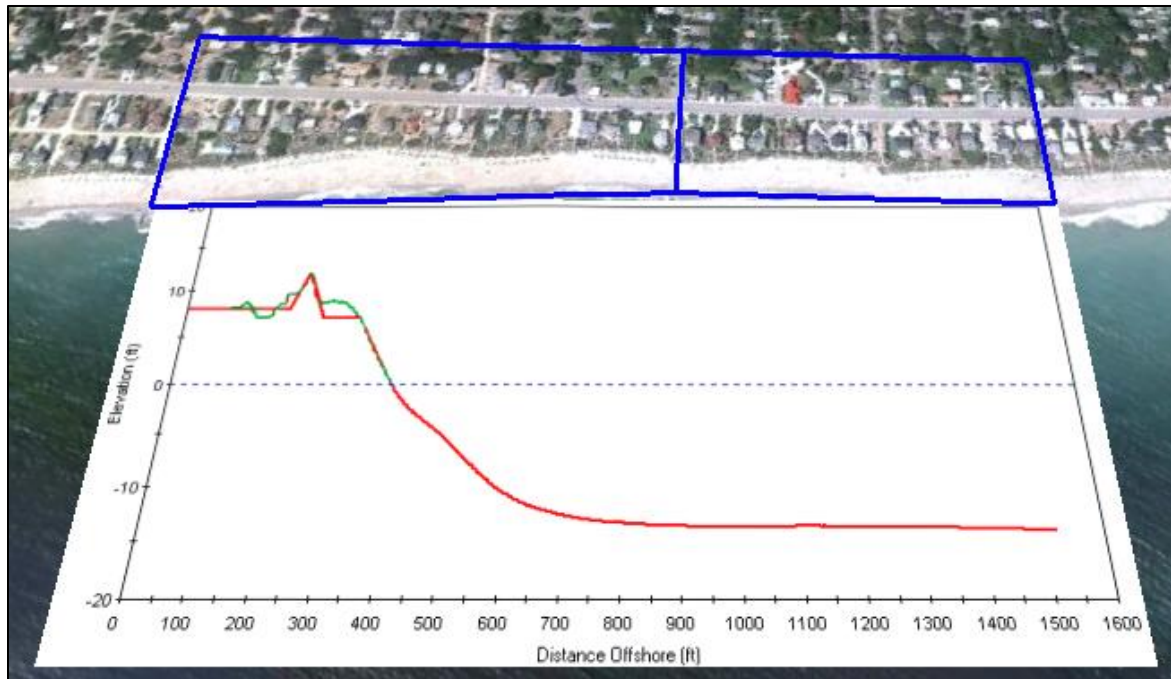


Figure B-16. Representative and idealized beach profile for Reaches E4 and E5

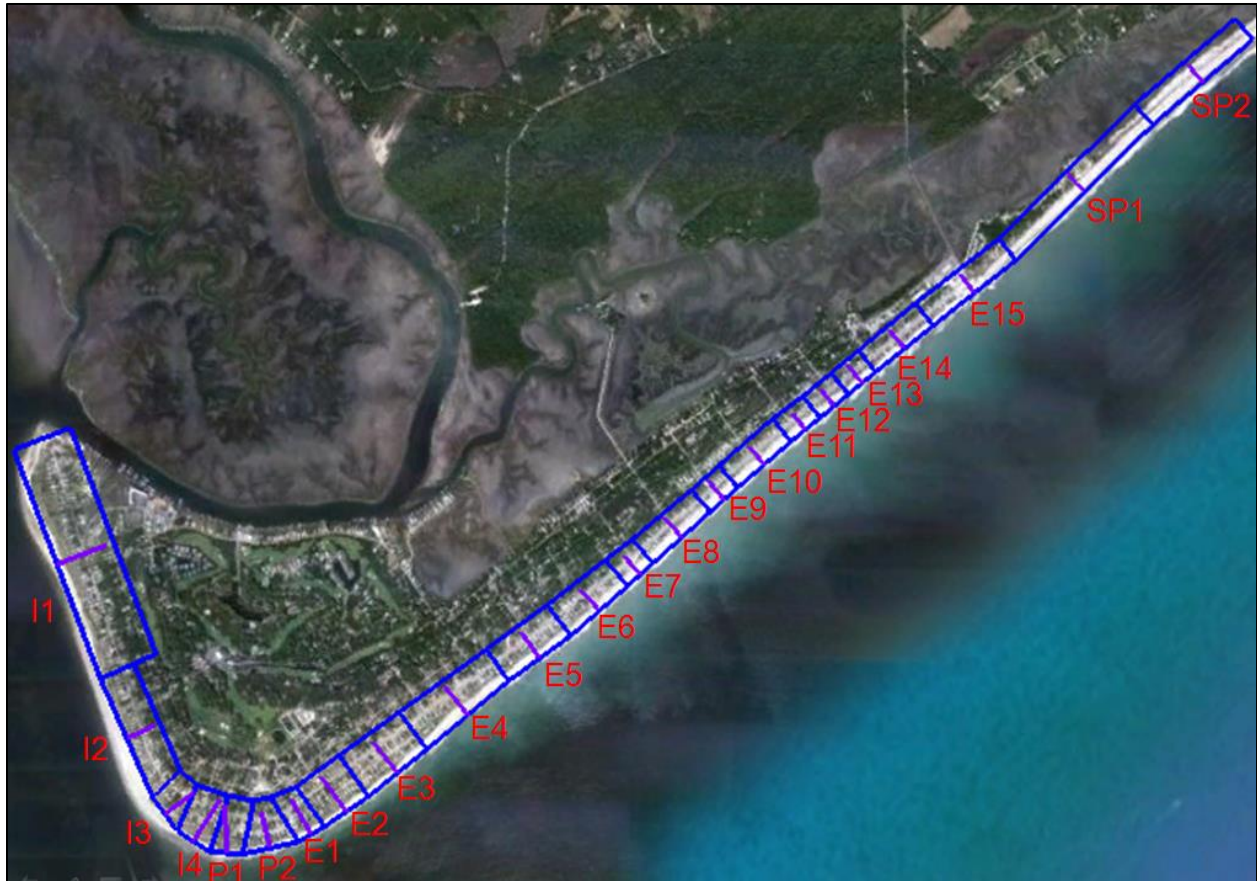


Figure B-17. Beach-*fx* reaches and SBEACH reference lines

B-7. Beach-*fx* coastal processes input data development.

a. Storm-induced beach profile responses.

(1) The availability of a large database of beach profile responses for each storm in the plausible storm suite is central to the operation of Beach-*fx*. This database is known to Beach-*fx* modelers as the shore response database (SRD). Two kinds of data are stored in the SRD for each storm/profile simulation: (1) changes in berm width, dune width, dune height and upland width; and (2) cross-shore profiles of erosion, maximum wave height, and total water elevation. The morphology changes (berm width, dune width, dune height, and upland width) are used to modify the pre-storm beach profile to obtain the post-storm profile. The damage-driving parameters (cross-shore profile of erosion, maximum wave height, and total water elevation) are used in the estimation of damages to damage elements within reaches associated with that representative profile.

(2) The SRD is a pregenerated set of beach profile responses to storms forming the plausible storm suite for a range of profile configurations that are expected to exist for different sequences of storm events and management action scenarios. The numerical model for simulating storm-induced beach change, SBEACH, was used to

estimate beach profile responses to each of the storms contained in the plausible storm suite. As discussed in paragraph B–3, the historically based storm suite includes 24 historical tropical storm events, 6 hypothetical tropical storm events, and 9 extratropical storm events. When combined with the statistical representation of astronomical tides, the number of storms increased by a factor of 12, resulting in a plausible storm suite involving 360 tropical storm events and 108 extratropical storm events.

(3) A companion range of beach profile configurations was developed to encompass all expected beach configurations encountered under each of the evaluated without-project scenarios. The most robust end of beach profile configurations considered was defined by the existing condition representative beach profile (paragraph B–4). The most vulnerable end of the beach profile configurations assumed that the dune feature was entirely removed and the upland was fronted with a zero berm width and foreshore slope down to the water’s edge. Profiles were developed at 10-foot increments on berm widths, 5-foot increments on dune widths, and 1-foot increments on dune heights between the most robust and most vulnerable beach profiles.

(4) This procedure generated a total of 2,335 unique beach profiles. The response of each of these beach profiles to the entire storm suite consisting of 468 plausible storm events was simulated using the SBEACH model. A total of 1,092,780 SBEACH simulations were performed, and the results were imported to populate the SRD used as input to the Beach-*fx* model. Due to the large size of the resulting SRD, the Edisto project was divided into three project domains:

- (a) Edisto South, covering Reaches I1, I2, I3, I4, P1, P2, E1, E2, E3, E4, and E5.
- (b) Edisto Central, covering Reaches E6, E7, E8 and E9.
- (c) Edisto North, covering Reaches E10, E11, E12, E13, E14, E15, SP1, and SP2.

b. Profile shoreline position changes.

(1) In the next step required to fully implement the Edisto Beach project, the Beach-*fx* model is calibrated such that the model reproduces, on average over multiple life-cycle simulations, the historical shoreline rate of change. To do this, an estimate of the historical shoreline rate of change must be developed. The available beach profile information was employed as input to make this required estimate. Since the beach profile data consist of distance and elevation pairs across the dune, berm, foreshore, and portions of the offshore and are collected at constant positions (monuments) along the length of the island (Figure B–18 from Fioto and Ward 2015), it is possible to use the profile data to analyze the evolution of the beach over time. The most common, easily understood, and useful shoreline positions are defined by the intersection of the sandy beach with the mean high-water tidal datum.

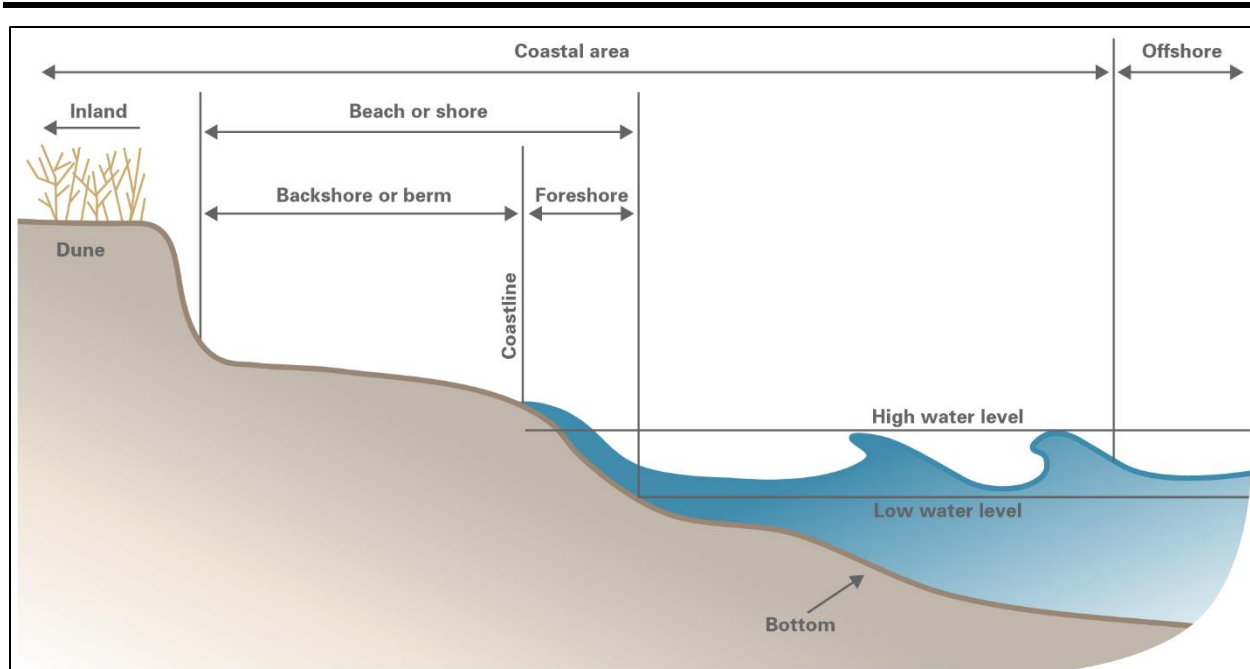


Figure B–18. Diagram of the features of a typical sandy beach

(2) This analysis uses shoreline positions defined by mean high water (MHW) to calculate shoreline change (erosion/accretion) amounts and rates. According to the benchmark sheet for Edisto Beach (ID 8667630), published by the NOAA’s National Ocean Service, the elevation of the MHW tidal datum is +2.48 feet (+0.756 meters) relative to the NAVD88.

(3) The magnitude of shoreline position change from one year to the next and the rate of change over longer periods are extremely important pieces of information for engineers, scientists, economists, etc.

(4) The 2008 Annual State of the Beaches Report states that Edisto Island has a low long-term erosion rate, but an extreme lack of sand (Harris and Eiser 2008). The report does not quantify the long-term erosion rate, but does contend that the low erosion rate is due to the presence of the extensive groin field. According to the report, the southern half of the developed portion of Edisto Beach has the widest oceanfront beach on the island, while the northern half was one of the most critically eroded sections of beach anywhere in the state until the 2006 renourishment.

(5) This analysis of shoreline change used the RMAP software package (RMAP 2024) to calculate the changes to the MHW and MLW (mean low water) contours between consecutive yearly beach profiles for each profile. Table B–5 provides results of an MHW shoreline change analysis. Results are provided for the yearly changes between 2004 and 2008 and the change rates. The MHW shoreline change rates were determined for the entire time period, identified as “With Fill,” and for the period not affected by the 2006 renourishment project, identified as “Without Fill.” Note from the

“2005–2006” and “With Fill” columns that the 2006 renourishment project had significant influence on the position of the MHW shoreline.

(6) More specifically, the 2006 renourishment was sufficient to counteract the short- and long-term erosion rates and result in positive (accretional) shoreline change rates at all but 9 of the 90 beach profiles. In addition, of the nine profiles with negative “With Fill” MHW change rates, seven of these are immediately downdrift of a groin. The MHW shoreline change rates from Table B–5 are in Figure B–19 where they were multiplied by a factor of 50 to accentuate the changes for graphing (pink and blue lines represent rates including and excluding the 2006 nourishment). It is clear from the saw-tooth shape of the change rates that the magnitudes are affected by the groin field.

(7) The magnitudes of the “With Fill” MHW change rates are larger at those profile locations immediately south of a groin and decrease while moving to the south within the same groin cell. Likewise, the magnitudes of the “Without Fill” MHW change rates are larger immediately south of a groin and decrease while moving south within the same groin cell.

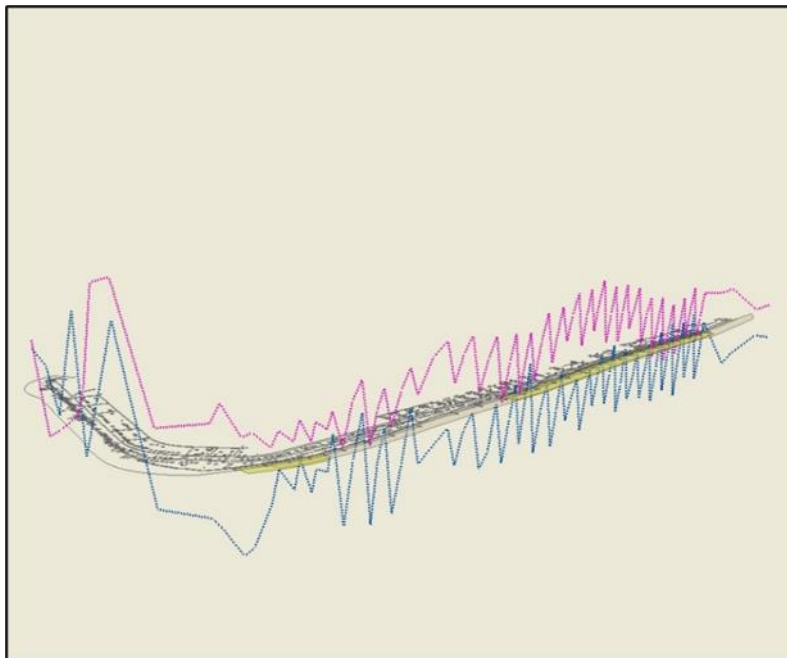


Figure B–19. Three-dimensional, mean high water shoreline change rates (feet/year) for Coastal Science & Engineering beach profiles

Table B-5
Mean high-water shoreline change for coastal science and engineering profile data

MHW Shoreline Change (ft)					MHW Shoreline Change Rate (ft/yr)	
Benchmark	(2004-2005)	(2005-2006)	(2006-2007)	(2007-2008)	With Fill	Without Fill
SCCC 2270	-36.58	43.37	4.26	-6.72	1.09	-18.78
SCCC 2250	-50.09	39.56	0.04	12.78	0.58	-14.73
SCCC 2230	-44.17	137.96	-23.11	-3.96	16.81	-20.60
SCCC 2210	-38.21	154.10	-38.95	-14.03	15.85	-23.12
1+75	-8.22	125.43	-55.00	17.12	19.98	5.09
1+300	-19.34	98.43	-38.17	-9.94	7.80	-13.12
1+525	-12.19	48.52	-14.10	-36.07	-3.49	-23.17
2+75	-1.07	129.40	-55.94	22.19	23.82	10.64
2+300	-1.76	99.23	-35.71	-7.51	13.66	-4.50
2+525	-4.97	52.68	-8.36	-37.32	0.51	-20.76
3+75	-6.07	116.49	-51.98	19.87	19.73	7.38
3+300	-4.80	70.94	-26.00	-12.15	7.05	-8.10
3+525	-17.47	28.46	-0.32	-40.21	-7.44	-27.47
4+75	-0.45	105.99	-53.72	17.43	17.44	8.53
4+300	-0.52	62.25	-27.35	-12.29	5.56	-6.36
4+525	-6.86	16.35	-3.78	-39.48	-8.51	-22.63
5+75	2.49	129.99	-54.41	9.74	22.12	5.92
5+300	0.22	94.64	-33.36	-21.86	9.98	-10.84
5+525	-5.46	48.14	-14.82	-47.16	-4.86	-25.88
6+75	4.27	132.97	-42.57	-1.01	23.59	1.30
6+300	-0.72	112.11	-29.12	-20.54	15.55	-10.57
6+525	2.88	63.17	-6.56	-47.86	2.93	-22.72
7+75	-0.60	146.38	-29.65	-1.00	29.00	-0.75
7+300	-1.73	118.17	-12.68	-23.25	20.28	-12.35
7+525	-2.86	71.02	15.49	-38.67	11.33	-20.54
8+75	-5.86	145.87	-27.45	11.89	31.35	3.47
8+300	-10.42	108.10	-1.91	-16.41	19.99	-12.60
8+525	-13.06	55.13	22.83	-37.64	6.87	-24.33
9+75	4.93	127.64	-14.90	8.13	31.69	6.14
9+300	-0.38	88.64	3.22	-11.28	20.20	-5.80
9+525	-6.91	42.31	30.05	-31.04	8.67	-18.43
10+75	-6.93	149.40	-16.88	13.97	35.15	4.06
10+300	-2.82	112.18	5.13	-7.52	26.94	-4.95
10+525	-4.77	67.52	27.73	-29.62	15.33	-16.82
11+75	-16.43	159.20	-26.22	13.34	32.72	-0.26

MHW Shoreline Change (ft)					MHW Shoreline Change Rate (ft/yr)	
Benchmark	(2004-2005)	(2005-2006)	(2006-2007)	(2007-2008)	With Fill	Without Fill
11+300	-17.94	132.93	-16.20	-10.84	22.15	-12.98
11+525	-23.62	100.37	-4.31	-29.85	10.73	-24.88
12+75	-19.54	166.25	-30.37	11.52	32.21	-2.48
12+300	-9.47	136.70	-19.74	-12.54	23.92	-10.26
12+525	-5.26	106.91	-14.30	-31.61	14.04	-18.02
13+75	-10.22	135.84	-25.27	11.90	28.27	1.64
13+300	-15.08	117.42	-16.36	-12.73	18.45	-12.72
13+525	-25.24	90.88	-5.69	-31.32	7.21	-26.30
14+00	-12.64	135.37	-25.47	10.31	27.10	-0.17
14+350	-15.61	98.93	-9.24	-7.80	16.70	-10.48
14+600	-29.00	54.34	12.19	-26.58	2.76	-25.52
15+65	-0.73	95.76	-31.33	20.33	21.17	9.86
15+245	-11.62	83.14	-27.30	1.98	11.64	-3.91
15+450	-26.66	55.65	-15.63	-16.72	-0.85	-19.60
16+75	-5.20	105.14	-22.07	10.58	22.28	3.10
16+300	-14.95	95.74	-26.62	-4.96	12.40	-8.78
16+525	-31.85	70.04	-24.74	-21.69	-2.08	-24.27
17+75	-13.70	144.20	-49.35	9.11	22.74	-1.22
17+300	-17.97	121.72	-21.25	-21.71	15.31	-18.43
17+525	-32.08	84.85	-16.37	-19.73	4.20	-23.39
18+75	-10.98	137.91	-32.70	4.30	24.82	-2.48
18+300	-7.64	117.30	-24.88	-8.86	19.12	-7.65
18+525	-21.28	94.73	-15.43	-19.94	9.59	-18.94
19+100	-14.27	137.34	-26.74	2.00	24.77	-5.02
19+525	-7.83	101.92	-1.37	-11.63	20.43	-9.12
19+955	-3.39	38.22	27.55	-24.19	9.62	-13.52
20+100	11.69	90.87	-29.49	2.15	18.95	6.00
20+350	-2.61	76.07	-3.07	-20.26	12.63	-11.23
20+600	-14.93	29.46	33.24	-39.78	2.01	-26.18
21+75	-2.16	76.44	-21.67	6.73	14.95	2.45
21+265	-8.33	64.89	-14.31	-8.69	8.45	-7.86
21+430	-28.96	47.16	3.17	-28.96	-1.91	-26.69
22+75	5.15	75.01	-17.91	14.43	19.31	9.39
22+268	0.28	68.85	-5.81	-10.06	13.42	-4.91
22+460	-18.06	41.78	10.80	-33.44	0.27	-24.33
23+100	1.95	48.24	-12.48	8.28	11.58	4.96
23+220	1.41	41.07	-2.46	-13.54	6.67	-6.18

MHW Shoreline Change (ft)					MHW Shoreline Change Rate (ft/yr)	
Benchmark	(2004-2005)	(2005-2006)	(2006-2007)	(2007-2008)	With Fill	Without Fill
24+100	1.57	48.17	-1.63	-10.07	9.58	-4.37
24+190	-0.18	39.56	-1.07	-21.86	4.14	-11.01
25+100	-9.45	70.01	-23.57	7.59	11.23	-0.19
25+200	-13.89	58.72	-18.47	-6.65	4.96	-9.18
26+115	-6.77	95.54	-54.72	1.22	8.88	-2.24
26+235	-12.30	86.10	-43.19	-13.58	4.29	-11.98
27+145	6.31	110.55	-26.73	-53.27	9.28	-23.97
27+290	8.92	69.15	13.74	-59.63	8.11	-26.05
28+130	-7.21	75.16	33.48	-28.94	18.26	-17.51
28+277	-8.59	47.83	32.39	-20.72	12.82	-13.98
SCCC 2135	46.56	-20.98	83.76	-59.06	12.66	-9.90
CSE 2130B	49.54	1.07	117.29	40.31	52.45	41.04
CSE 2130A	7.00	11.90	155.93	23.71	50.01	14.81
SCCC 2130	-13.01	-57.78	134.10	3.91	16.93	-3.53
SCCC 2120	30.19	-28.01	-30.62	52.89	6.16	39.17
SCCC 2115	21.11	-26.25	4.79	-15.58	-4.01	1.11
SCCC 2113	-19.80	-0.57	-68.11	44.94	-10.97	14.12
SCCC 2110	25.21	-24.93	65.20	12.61	19.67	16.93

(8) The record in the Office of Ocean and Coastal Resource Management (OCRM) beach profiles was too long to present all of the MHW shoreline change magnitudes, so only the shoreline change rates are presented in Table B–6. The shoreline change rates in Table B–6 are listed from north to south along the coastline of Edisto Island, from the county park to the South Edisto River. Since the beach profile records begin in 1988, the “With Fill” rates in Table B–6 include two beach nourishment projects, the first in April 1995 and the second in 2006. The “Without Fill” rates neglect the influence of these beach fill projects by excluding the shoreline changes directly attributable to the fill and the subsequent changes as the beach equilibrates. The “With Fill” change rates are missing from SCCC 2198 because no beach profile data exist for the periods surrounding the nourishment projects.

(9) Likewise, the “Without Fill” change rates for SCCC 2150 through 2113 are missing because at the time of this analysis, OCRM beach profile data were available only for 2006 and 2007. With the exception of SCCC 2198 (only 5 years of data), the shoreline change rates for SCCC 2230 through SCCC 2155, which represent the overwhelming majority of the Atlantic-facing shoreline, are calculated from at least 17 years’ worth of beach profiles. Such a long period of record results in a high level of confidence that the change rates being calculated capture the long-term morphologic processes affecting the island.

Table B-6
Shoreline change summary for Office of Ocean and Coastal Resource Management beach profile data

OCRM Benchmark	MHW Shoreline Change Rate (ft/yr)	
	With Fill	Without Fill
SCCC 2230	-0.16	-3.94
SCCC 2200	1.52	-2.13
SCCC 2198	N/A	-10.10
SCCC 2195	2.44	-0.21
SCCC 2193	3.36	-2.25
SCCC 2190	4.09	-1.58
SCCC 2185	5.52	-3.41
SCCC 2180	3.48	-4.18
SCCC 2178	-0.15	-8.49
SCCC 2173	2.92	-6.93
SCCC 2170	3.74	-1.77
SCCC 2165	3.11	-0.95
SCCC 2160	1.33	-3.20
SCCC 2155	3.38	-2.04
SCCC 2150	-14.79	N/A
SCCC 2145	-0.64	N/A
SCCC 2135	17.24	N/A
SCCC 2130	121.85	N/A
SCCC 2120	15.58	N/A
SCCC 2113	27.03	N/A
SCCC 2110	4.23	-10.56

(10) Examining the MHW “With Fill” rates versus the “Without Fill” rates reveals that the two beach nourishment projects, in 1995 and 2006, have performed well in offsetting the normal erosion rate and stabilizing the recreational beach. The long-term MHW shoreline change rate without the fill projects is uniformly erosional, but is not uniform in the magnitude of erosion, as it varies from -0.21 to -10.1 feet per year.

c. *Historical shoreline rate of change.* Based on the annual rates of shoreline change presented in the previous section and interpolation to the established Beach-*fx* analysis reaches, the long-term historical rate of change corresponding to each of the Beach-*fx* reaches was estimated and plotted as the red line in Figure B-20. As seen in the figure, there are extreme discontinuities between adjacent Beach-*fx* reaches that are not sustainable over the long term, which is of primary interest for purposes of Beach-*fx* modeling. To resolve the issue, smoothing was applied to the data and the resulting shoreline rates of change as depicted by the blue line were derived as a reasonable expectation of the future rate of shoreline change in the absence of

shoreline management activities. Table B–7 provides a list of the target shoreline rate of change values for each of the Beach-*fx* reaches.

d. History. Historically, the ocean-fronting shoreline within the Edisto Beach study area has been erosional with the rate of erosion generally decreasing from north to south. Near the point, the shoreline change rate decreases to nearly zero and transitions to accretion along the inlet-fronting shoreline. The natural variability in annual wave energy and storm occurrence can and does produce significant variations in this overall trend of shoreline change. Shore protection measures also have a direct influence on shoreline change rates. For example, following groin construction near the northern end of Edisto Beach, erosion accelerated south of the constructed groins and led to construction of more groins to the south. Likewise, after the nourishment projects in 1995 and 2006, the rate of shoreline accretion along the inlet-fronting increased as nourishment material was transported to the south and around the point.

e. Conclusions. Overall shoreline change trends within the Edisto Beach study area are well understood and as expected depend to a large degree on sediment supply from the north. If the sediment supply is reduced due to natural processes, such as increased over-wash into marsh or human intervention such as groin construction, erosion can be expected to the south. Likewise, if new sediment is introduced into the system through beach nourishment, a stabilization of the shoreline or even a transitioning to a pro-grading shoreline can be expected to the south of the placement area. Both of these scenarios have been observed and recorded in the past within the project study area.

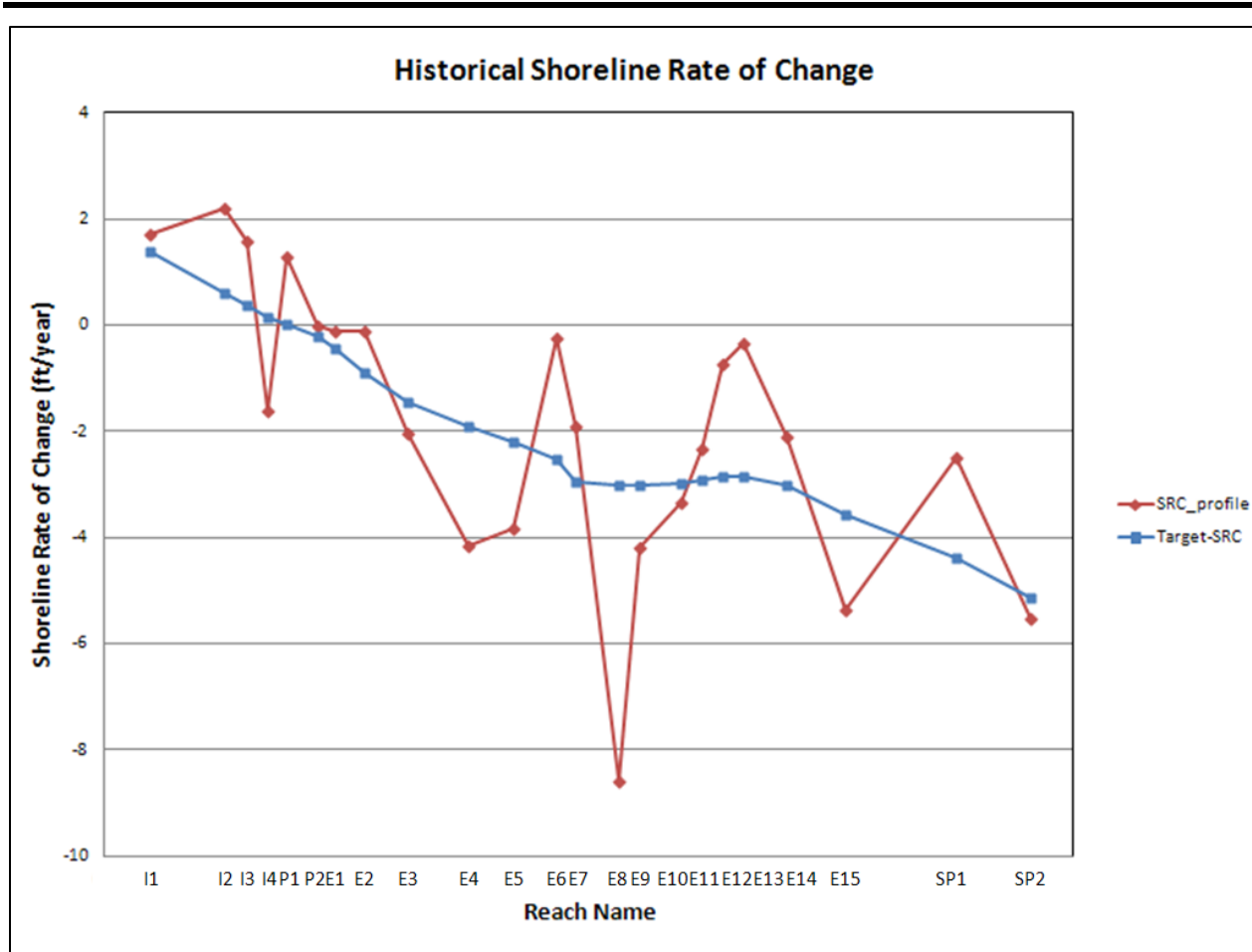


Figure B-20. Historical shoreline rate of change based on profile data and synthesized target shoreline rate of change

Table B-7
Target historical shoreline rate of change by Beach-fx reach

Reach	Target SRC (ft/yr)	Reach	Target SRC (ft/yr)	Reach	Target SRC (ft/yr)
I1	1.37	E3	-1.45	E11	-2.93
I2	0.62	E4	-1.91	E12	-2.85
I3	0.38	E5	-2.21	E13	-2.85
I4	0.16	E6	-2.52	E14	-3.03
P1	0.01	E7	-2.95	E15	-3.56
P2	-0.22	E8	-3.01	SP1	-4.38
E1	-0.43	E9	-3.03	SP2	-5.13
E2	-0.90	E10	-2.98	–	–

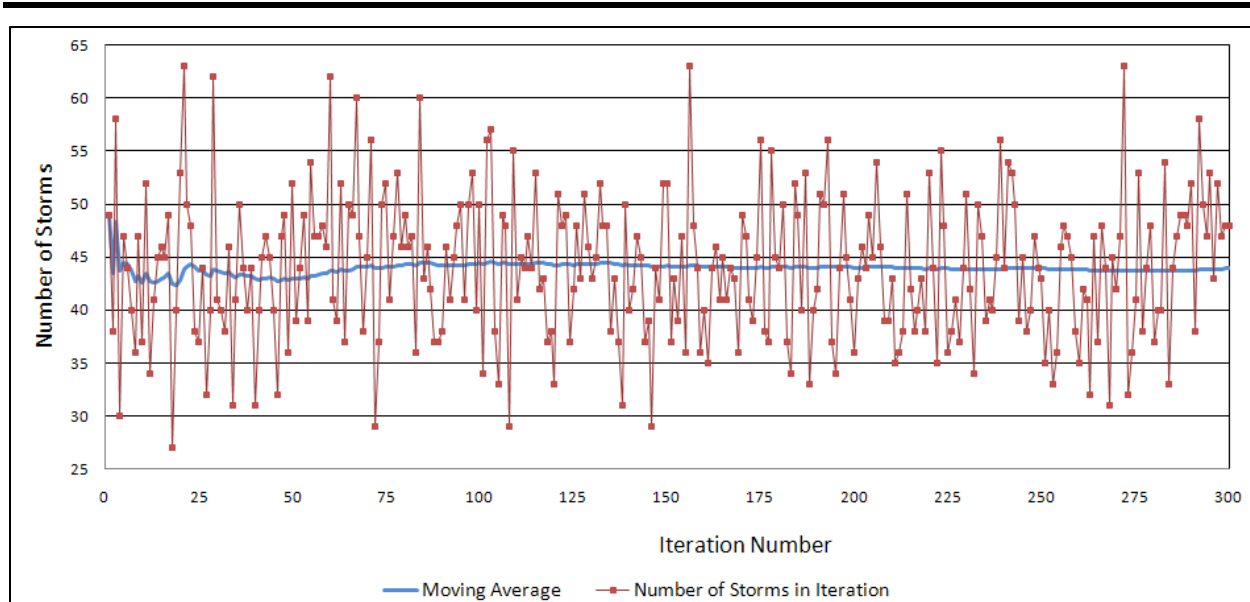
B–8. Beach-fx calibration

a. The calibration procedure for Beach-fx involves specification and tuning of a reach-level attribute known as the applied erosion rate. The applied erosion rate accounts for long-term shoreline change not attributed to storm-induced shoreline changes, which are captured within the model by the random sampling of storm events as the model progresses through the life-cycle simulation. The concept employed here is that there are two separable components of beach evolution. The first is cross-shore, transport-dominated shoreline change due to storm events, which is mostly recoverable due to post-storm, berm width recovery. The second is longshore transport-dominated shoreline change, which is driven by longshore sediment transport gradients, underlying geological setting, and other factors, such as RSLC. This second component of beach evolution is considered nonrecoverable.

b. The Beach-fx calibration concept is that the combination of these two drivers of beach evolution should, on average and over multiple simulated project life cycles, return the long-term average rate of shoreline change. Since the Beach-fx simulated life-cycle iteration employs a random sequence of storm events, the returned shoreline change rate differs for each life cycle simulated. The Beach-fx calibration task is to determine an appropriately applied erosion rate for each reach such that the computed average rate of shoreline change on a reach-by-reach basis is equal to the estimated target historical shoreline change rate over multiple life-cycle simulations.

c. For the Edisto Beach project, Beach-fx was calibrated across 300 iterations of a 55-year life cycle using an assigned depth of closure specification of negative 14 feet NAVD88. The depth of closure estimate was developed based on an analysis of the available beach profile data presented in paragraph B–7. The 55-year life-cycle duration stems from the use of the August 2008 beach profile survey to define the initial condition leading to a start year specification of 2009 and the specification of year 2014 as the base year for an economic analysis horizon corresponding to a 50-year project life. The use of 300 iterations was selected to obtain a stabilization of the model results in the context of capturing the expected variability in the environmental forcing. Evidence of the stabilization of results can be gaged by examining the moving average in various model outputs as compared to the individual iteration values.

d. An example of this is shown in Figure B–21 where the total number of storms per iteration is plotted along with the moving average number of storms across all iterations. Here it is seen that although the number of storms per iteration varies between a maximum of 63 storms and a minimum of 27 storms, the average number of storms stabilizes at approximately 44 storms after about 150 iterations of a 55-year life cycle.



**Figure B–21. Number of storms per iteration and moving average
number of storms per iteration**

e. After numerous calibration iterations, Beach-*fx* was calibrated to precisely reproduce the target historical shoreline rate of change (SRC), on average over 355-year life cycles. Figure B–22 shows the target historical SRC (blue line), the Beach-*fx* calculated average rate of shoreline change over 300 iterations (red stars), together with the calibration determined applied erosion rates on a reach-by-reach basis (green line).

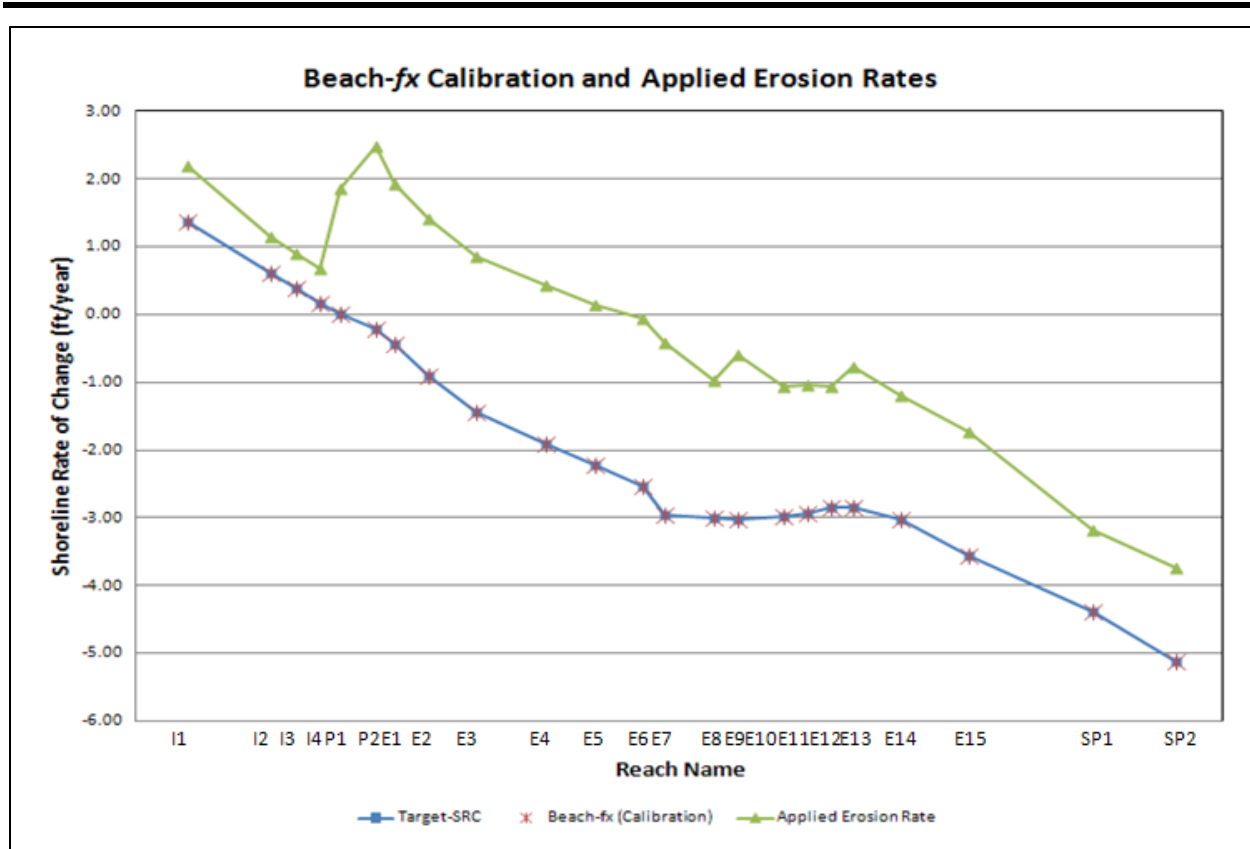


Figure B-22. Beach-fx calibration results and applied erosion rates

B-9. Structure inventory development

a. Structures, referred to as damage elements, were identified within the study area boundary and marked using the North American Datum (NAD) 1983 South Carolina State Plane U.S. Feet projected coordinate system. Damage elements were assigned deterministic values for use type, dimensions (along-shore length and cross-shore width), number of floors, foundation type, condemnation ratio, and number of rebuilds. Probabilistic values were assigned for first floor elevation, structure depreciated replacement value, content depreciated replacement value, and time to rebuild.

b. Damage element identification within the study area was completed using current high resolution, orthorectified, aerial imagery. Damage elements can include any damageable asset, including houses, mobile homes, garages, commercial structures, public structures, industrial structures, or critical infrastructure (roads, utilities, etc.). Any asset that can justifiably be assigned a damage function for inundation, wave attack, and/or erosion damage drivers can be considered a damage element. Locations were recorded using northing (Y) and easting (X) points.

c. Damage elements were grouped into beach profiles, reaches, and lots, depending on their respective location within the study area. As mentioned earlier in this

appendix, beach profiles are the basic unit of beach response. SBEACH calculates the response of the beach profile to different storms, tide conditions, phases, and sea level change scenarios. This response quantifies the inundation, wave attack, and erosion damage experienced by the inventory. Idealized, or simplified, beach profiles are used for SBEACH and Beach-fx due to the inherent complexity of natural beach profiles. Key morphological features (dune height, dune width, berm width, etc.) are maintained.

d. Reaches are contiguous stretches of the shoreline that share a common morphological makeup with a particular beach profile. They are the broadest socioeconomic category and are used to organize damage results, periodic nourishment volumes, emergency nourishment volumes, and berm width change rates. Each damage element was indexed to exactly 1 of the 23 study reaches.

e. Lots are organizational containers within the model to efficiently evaluate damage elements. The effects of morphology changes are transferred to individual damage elements via lots. Lots can also store information pertaining to armor failure thresholds and armor construction. Each damage element was indexed to exactly one lot.

f. Structures were categorized based on certain key characteristics and occupancy type. The type of foundation, number of floors, and use of the structure all inform the proper damage element type classification. The damage element type classification was then used to index the damage element to the relevant structure and content damage curves for each damage driver.

g. First-floor elevation was calculated using an NAVD88, LiDAR-derived bare-Earth DEM in combination with structure foundation height surveys. The LiDAR-derived bare-Earth DEM provides the ground elevation at each damage element. The DEM has a stated margin of error (95% confidence level) of ± 0.64 feet. Each damage element has a first-floor elevation above the ground elevation, so 10% of the structures in the study area were surveyed to determine their mean first-floor elevation. The margin of error at a 95% level of confidence for this mean was calculated using equation B-2.

$$95\% \text{ Confidence Interval} = \bar{x} \pm \left(1.96 \times \frac{\sigma}{\sqrt{n}} \right) \quad (\text{B-2})$$

where:

$\bar{x}$ = the sample mean

σ = the sample standard deviation

n = the sample size (equation B-2). The margin of error on the mean was calculated to be ± 0.5 feet. Assuming the joint distribution error and corresponding probability distribution functions are Normally distributed with a mean error of zero, the total margin of error for the structure elevation (ground elevation plus first floor elevation) is ± 1.14 feet. This error band is used to inform the minimum and maximum ends of the triangle probability distribution.

h. Depreciated replacement value (DRV) is based on tax assessor records using variables of interest relating to assessed value, date of construction, type of construction, number of floors, square footage, recent sales, and selling price. Nonstructure damage elements, including roads, utilities, and walkways, were evaluated based on an average value per linear foot. The DRVs for 10% of the structures in the study area were computed in Marshall and Swift (2024) and the margin of error on the sample mean was calculated to be $\pm 25\%$ of the structure value.

i. Content-to-structure value ratio was set at 50% with a lower bound of 40% and an upper bound of 60% based on guidance from EM 1110-2-1619 and IWR 96-R-12 (1996).

j. Damage functions for each damage driver (inundation, wave attack, erosion) were developed by the IWR using expert opinion elicitations on coastal storm damage relationships. The IWR results provided damage functions for most damage element types, though a few additional functions were developed by the project delivery team. Damage function application is based on damage driver type, damage element use type (single family residential, multi-family residential, etc.), foundation type (shallow piles, deep piles, slab, etc.), construction type (wood, masonry, etc.), and armor type (no armor, sheet pile, etc.). Damage functions are provided as a triangle distribution probability with minimum, maximum, and most likely percent damage output (Y) for each input (X) point along the function.

k. In addition to erosion-related impacts to damage element structure and content values, permanent erosion of land, referred to as land loss, was a quantifiable physical damage category. The land loss reduction benefit was determined for eroding reaches by calculating the amount of land lost during the study period multiplied by the value of nearshore upland. With a project in place, land that would be lost in the without-project future condition would be preserved instead. The design template should be in place for the period of analysis, preserved through the process of periodic renourishment. This benefit is based on the value of nearshore lands.

l. Nearshore land values were estimated by the USACE Savannah District from samples taken from recent land sales and calculated on cost per square foot. Margin of error on the sample mean was calculated to be $\pm 15\%$ of the square foot value. Erosion was calculated for each iteration based on the historical erosion rate and storm-induced erosion impact, then averaged to compute the mean land loss damage.

m. Recreation benefits of the study were completed using unit day values (UDVs). UDVs are used to determine the economic value of recreation using a point system that accounts for the following factors: recreation experience, availability of opportunity, carrying capacity, accessibility, and environmental (aesthetic) quality. Parking at Edisto Beach is sufficient to support recreation for the general public and is reasonably near and accessible to the project beaches. Along with designated parking areas for beach access, public parking along the rights of way of the Town's streets is permitted. The Town of Edisto Beach has 38 public beach access points along Palmetto Boulevard, Point Street, and Yacht Club Road.

B–10. Future without-project Beach-*fx* simulations

a. *Overview.* Two future without-project scenarios of Edisto Beach evolution were simulated using the calibrated Beach-*fx* model. The first scenario involved no action on the part of private landowners or local or state governmental agencies. This scenario illustrates the high vulnerability of developed properties, Palmetto Boulevard, and Edisto Beach State Park to losses due to continued coastal erosion, storm-induced inundation, and direct wave impact. The second future without-project scenario includes limited emergency actions, including emergency dune reconstruction as needed and armoring of Palmetto Boulevard if the highway becomes vulnerable to loss of function during the simulation. This future without-project is considered the most likely future as it best reflects the proactive shore protection posture of the Town of Edisto Beach, the state of South Carolina, and the South Carolina Department of Transportation.

b. *No-action scenario.*

(1) The calibrated Beach-*fx* model for Edisto Beach was configured for simulation of the no-action scenario of the without-project to estimate beach evolution and economic consequences of a 55-year future that involves no action on the part of private landowners or local or state governmental agencies. This without-project scenario is based on the Town of Edisto Beach's assertion that in light of current economic circumstances within the state of South Carolina, Colleton County, and the Town of Edisto Beach, the resources necessary to sustain historical shore protection measures are not anticipated to be available.

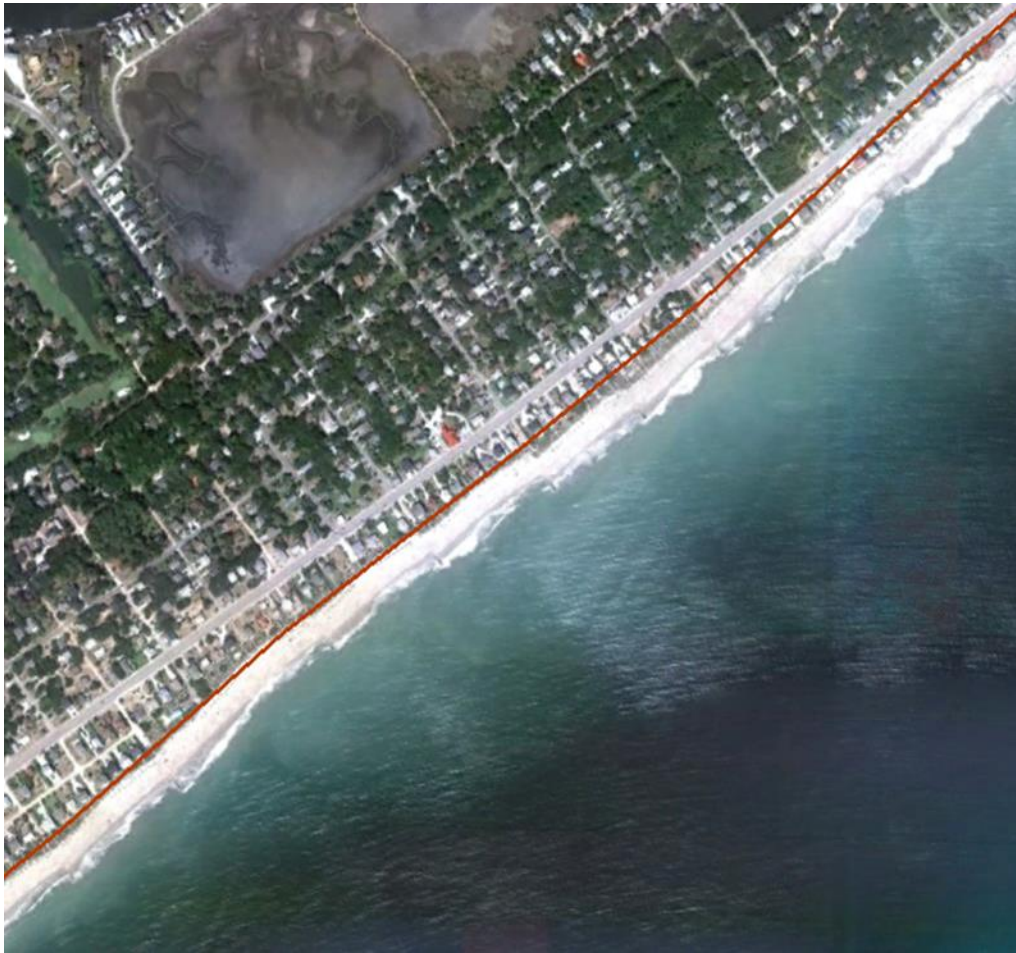
(2) The simulation again involved the simulation of 300, 55-year life cycles ending in the year 2064. The average shoreline position at the end of the without-project simulation is shown as the brown line in Figure B–23 through Figure B–26. Figure B–23, which spans the area between Beach-*fx* Reaches I1 and E3 (Groyne Cell 21), shows that along the inlet shoreline of Edisto Beach the shoreline is expected to be nearly stable or advance slightly. Slight ongoing erosion is predicted in the vicinity of the point. Along the ocean-fronting shoreline just north of the point, erosion begins to increase.



Figure B–23. Without-project forecast of average shoreline position in Year 2064 along the inlet shoreline and around the point to Groin Cell 21

(3) In Figure B–24, which covers the area between Beach-*fx* Reach E3 (Groin Cell 20) and Reach E8 (Groin Cell 12), the predicted Year 2064 shoreline indicates erosion increasing to the north. Based on this prediction, it is expected that wave swash will be under the existing homes beginning at approximately Groin Cell 17. Beginning at about Groin Cell 14, the homes appear to be highly vulnerable to complete loss. At Groin Cell 12 the shoreline is predicted to recede to Palmetto Boulevard, indicating that the developed properties on the ocean side of the highway will likely be destroyed.

(4) In Figure B–25, which covers the area between Beach-*fx* Reach E8 (Groin Cell 11) and Reach E15 (Groin Cell 1), the predicted Year 2064 shoreline reflects strong erosion indicating complete loss of all developed properties on the ocean side of Palmetto Boulevard. Beginning at approximately Groin Cell 7, the shoreline is predicted to coincide with Palmetto, indicating that the highway will be impassible. To the north, at Groin Cells 1 through 3, the shoreline is predicted to be at the upland side of Palmetto Boulevard, indicating that the highway will be completely lost and developed properties on the upland side of Palmetto Boulevard will be vulnerable to damages from coastal storms.



**Figure B–24. Without-project forecast of average shoreline position
in Year 2064 between Groin Cells 12 and 20**

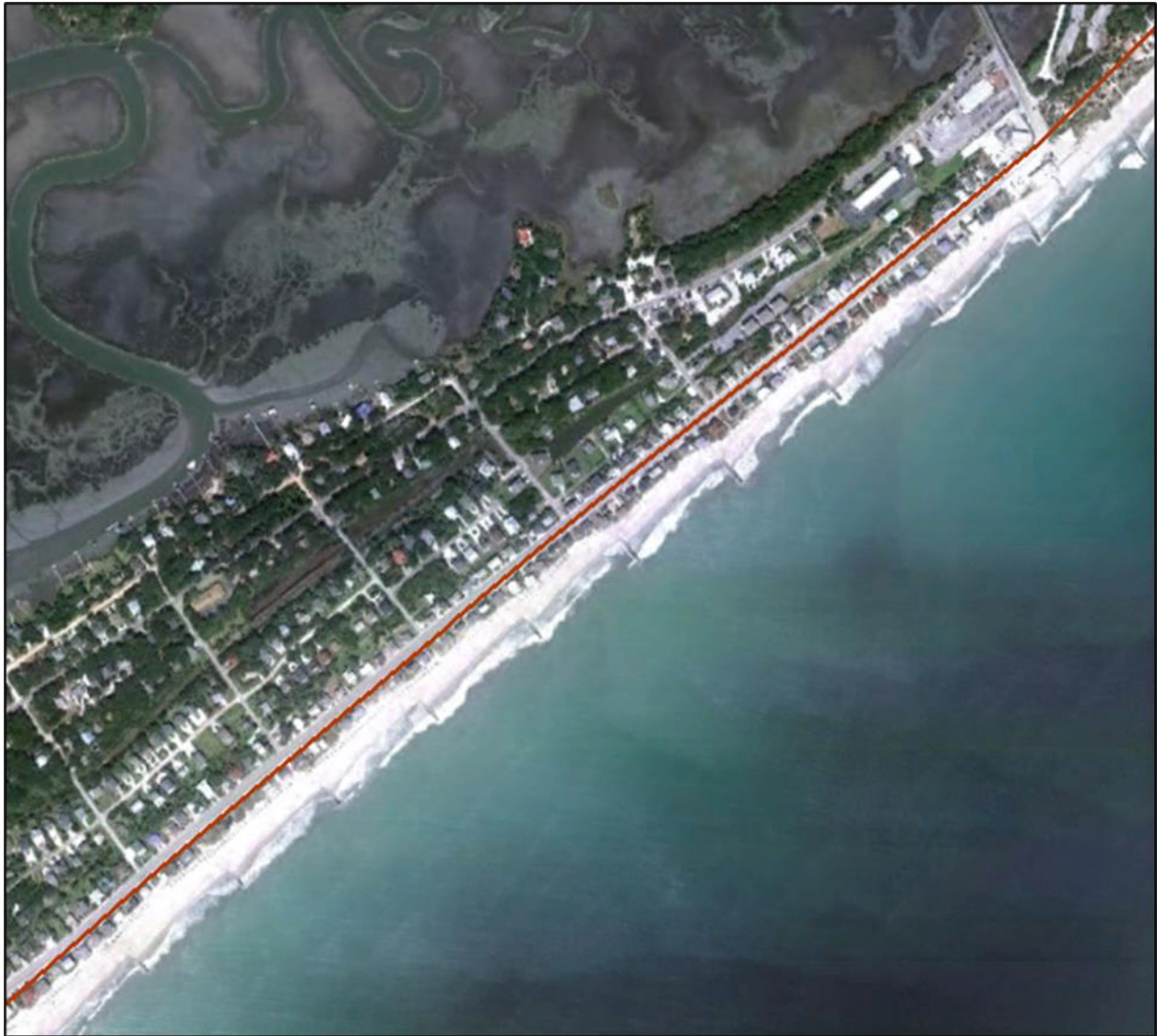


Figure B–25. Without-project forecast of average shoreline position in Year 2064 between Groin Cells 1 and 11

(5) Figure B–26, which encompasses the Edisto Beach State Park area north of the Town of Edisto Beach, shows extreme erosion. The predicted Year 2064 shoreline indicates that the barrier island will undergo extreme over-wash processes and migrate approximately one barrier island width into the upland marsh. The present camping area will be extremely vulnerable and it is likely that necessary infrastructure to support recreational use will be lost.



Figure B–26. Without-project forecast of average shoreline position in Year 2064 at Edisto Beach State Park

(6) The results of the no-action, without-project scenario as simulated with Beach-*fx* indicate an unfavorable future within the project study area. Significant losses to privately held developed properties are indicated. Publicly held infrastructure, including electrical power lines and water mains, will be extensively damaged and the use of Palmetto Boulevard will be lost. Indications are that Edisto Beach State Park will be subject to extreme losses due to coastal erosion, including the inability to support recreational camping within the state park. As this future without-project scenario is not considered the most likely future condition, estimated without-project damages are not provided here nor used during the economic analysis.

c. *Limited emergency action scenario.*

(1) The calibrated Beach-fx model for Edisto Beach was configured for simulation of the limited emergency action scenario of the future without project to estimate beach evolution and economic consequences of a 55-year future that involves emergency dune reconstruction actions as needed as well as armoring Palmetto Boulevard if the road becomes vulnerable to loss of function during the simulation. This without-project scenario is considered the most likely future as it best reflects the proactive shore protection posture of the community of Edisto Beach, the state of South Carolina, and the South Carolina Department of Transportation.

(2) The Town of Edisto Beach has indicated that they will take the shore protection actions that are within their means to protect existing infrastructure and to maintain recreational use of the beaches in their community. Since Palmetto Boulevard is both a state highway and the only hurricane evacuation route off the island, it is expected that the South Carolina Department of Transportation will act to maintain the road as an evacuation corridor by armoring the ocean side of the road if coastal erosion threatens the functionality of the highway.

(3) The emergency dune reconstruction action simulated in this future without-project scenario is implemented within the following constraints: on a reach-by-reach basis, if the simulated dune crest elevation falls below 9 feet NAVD88, an emergency dune nourishment action will be triggered. When the emergency nourishment action is triggered, a nourishment action is scheduled, assuming a thirty-day mobilization time, and the dune is nourished with a fill density of 10 cubic yard/feet of beach. The fill material is placed on the dune feature with a target dune elevation of 11 feet NAVD88. Any excess fill volume remaining after the target dune crest elevation is achieved is used to increase the dune crest width.

(4) Armoring of Palmetto Boulevard is triggered when the seaward edge of the berm erodes to within 10 feet of the road shoulder. Within Beach-fx, armoring functions only to prohibit erosion damages; direct wave-attack damages and inundation damages are still incurred.

(5) The simulation of this future without-project scenario involved the simulation of 300, 55-year life cycles ending in the Year 2064. The results indicate that, on average, approximately 1.61 million cubic yards of emergency nourishment fill material will be required to maintain the dune feature over the 55-year life-cycle simulation. The standard deviation in the average emergency nourishment fill volume is approximately 445,000 cubic yards. This metric can be viewed as the uncertainty in the estimated emergency nourishment fill volume needed over the 55-year life-cycle simulation. Table B-8 provides the number of emergency nourishment actions, total average emergency nourishment fill volume, and standard deviation.

Table B–8**Future without-project emergency dune reconstruction nourishment summary**

Reach	Number of Fill Actions	Fill Volume (yd ³)	Standard Deviation	Reach	Number of Fill Actions	Fill Volume (yd ³)	Standard Deviation
I1	N/A	N/A	N/A	E7	11.0	61,693	15,796
I2	N/A	N/A	N/A	E8	13.1	164,416	36,041
I3	N/A	N/A	N/A	E9	12.1	72,500	18,430
I4	N/A	N/A	N/A	E10	9.3	107,932	29,770
P1	6.1	32,471	10,302	E11	10.6	65,172	16,015
P2	10.9	96,197	26,833	E12	9.4	56,180	12,708
E1	7.4	36,712	14,946	E13	11.4	66,290	13,334
E2	8.0	69,897	27,149	E14	10.1	122,371	26,922
E3	9.1	111,117	38,035	E15	14.1	243,204	44,228
E4	4.8	83,787	42,903	SP1	N/A	N/A	N/A
E5	6.2	77,557	32,723	SP2	N/A	N/A	N/A
E6	11.8	145,509	39,248	–	–	–	–

(a) Since this future without-project scenario involves emergency dune reconstruction and armoring of Palmetto Boulevard, the estimated future without-project shoreline rate of change differs from the target historical rate of shoreline change indicated in Figure B–27.

(b) The average shoreline position at the end of the future without-project simulation is shown as the purple line in Figure B–28 through Figure B–31. In Figure B–28, which spans the area between Beach-*fx* Reaches I1 and E3 (Groin Cell 21), the shoreline along the inlet is nearly stable. Minor ongoing erosion is predicted along the ocean-fronting shoreline just north of the point.

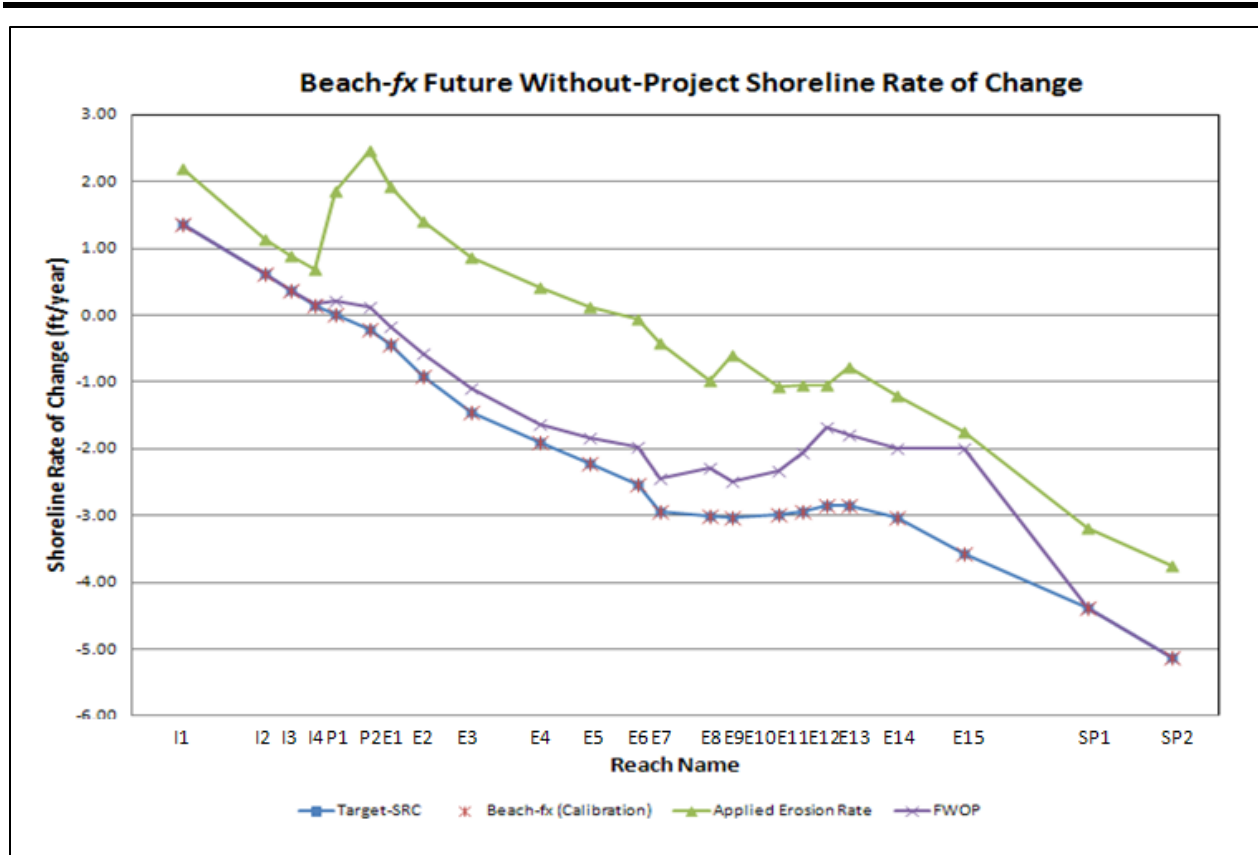


Figure B–27. Beach-fx calibration results and applied erosion rates

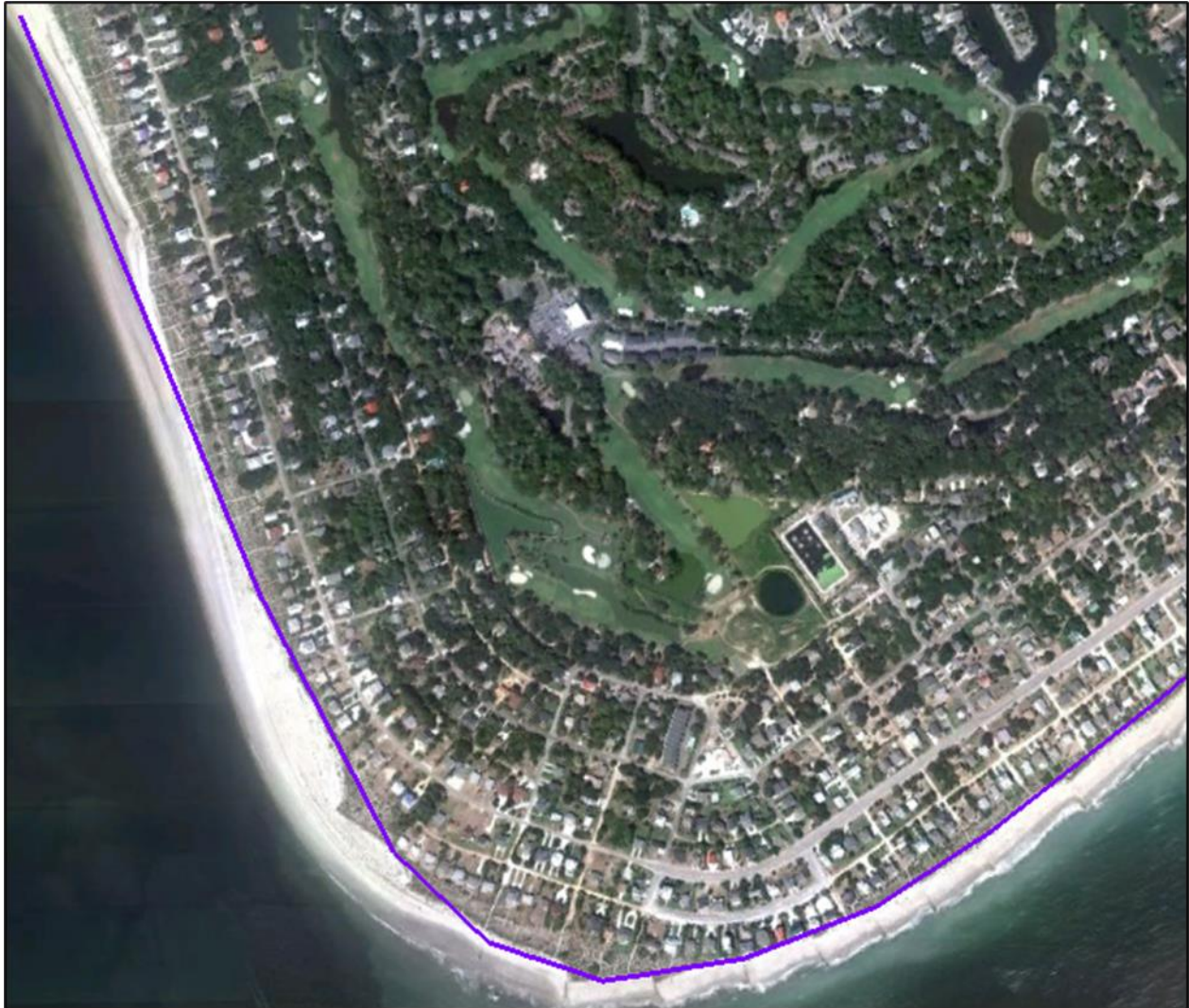


Figure B–28. Without-project forecast of average shoreline position in Year 2064 along the inlet shoreline to Groin Cell 21

(c) In Figure B–29, which covers the area between Beach-*fx* Reach E3 (Groin Cell 20) through Reach E8 (Groin Cell 12), the predicted Year 2064 shoreline indicates erosion increasing to the north, albeit slightly less than in the no action, without-project scenario due to periodic emergency dune reconstruction. Based on this prediction it is expected that wave swash will be under the existing homes beginning at approximately Groin Cell 16. At Groin Cells 13 and 12, the homes appear to be highly vulnerable to complete loss.

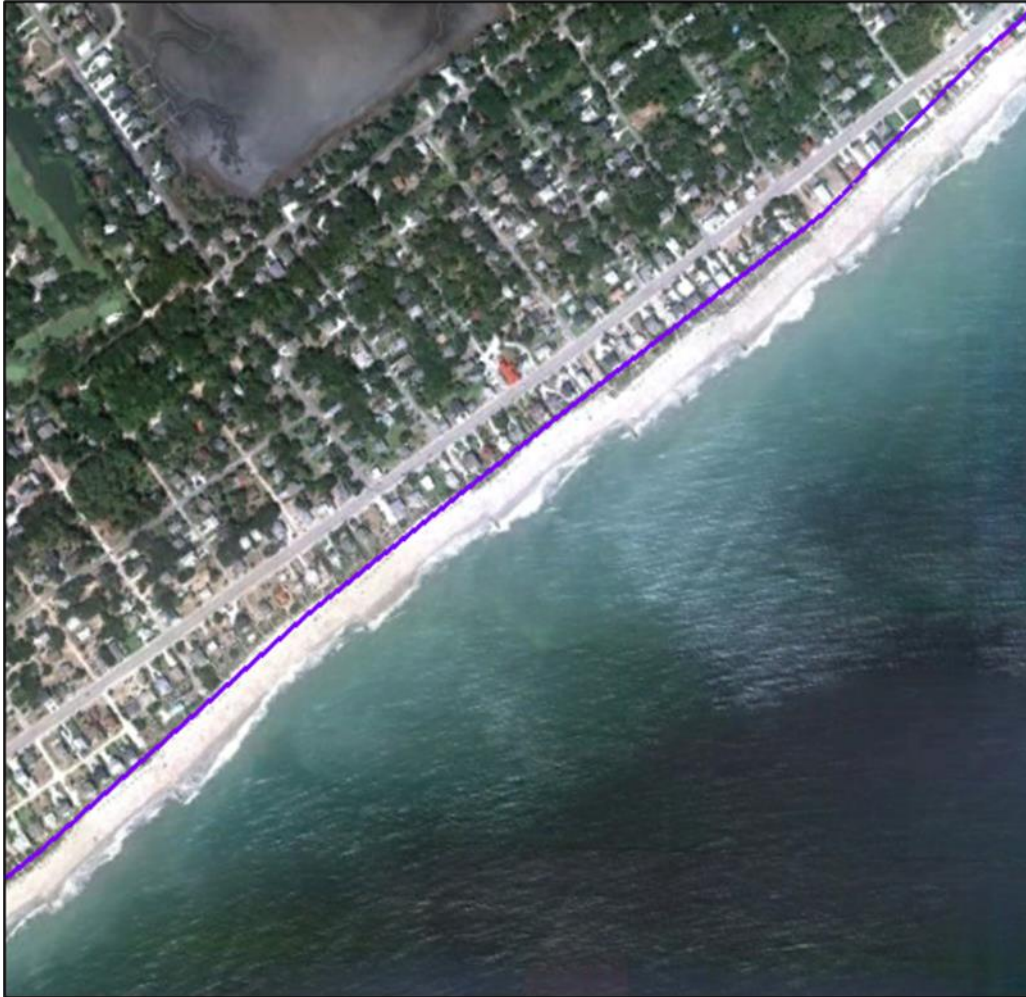


Figure B–29. Without-project forecast of average shoreline position in Year 2064 between Groin Cells 12 and 20

(d) In Figure B–30, which covers the area between Beach-*fx* Reach E8 (Groin Cell 11) and Reach E15 (Groin Cell 1), the predicted 2064 shoreline reflects erosion indicating likely a complete loss of all developed properties on the ocean side of Palmetto Boulevard. The shoreline is held just seaward of Palmetto Boulevard due to emergency dune reconstruction and armoring of the highway. However, most, if not all, developed properties on ocean side of Palmetto Boulevard in this segment of beach are predicted to be destroyed by coastal storms.

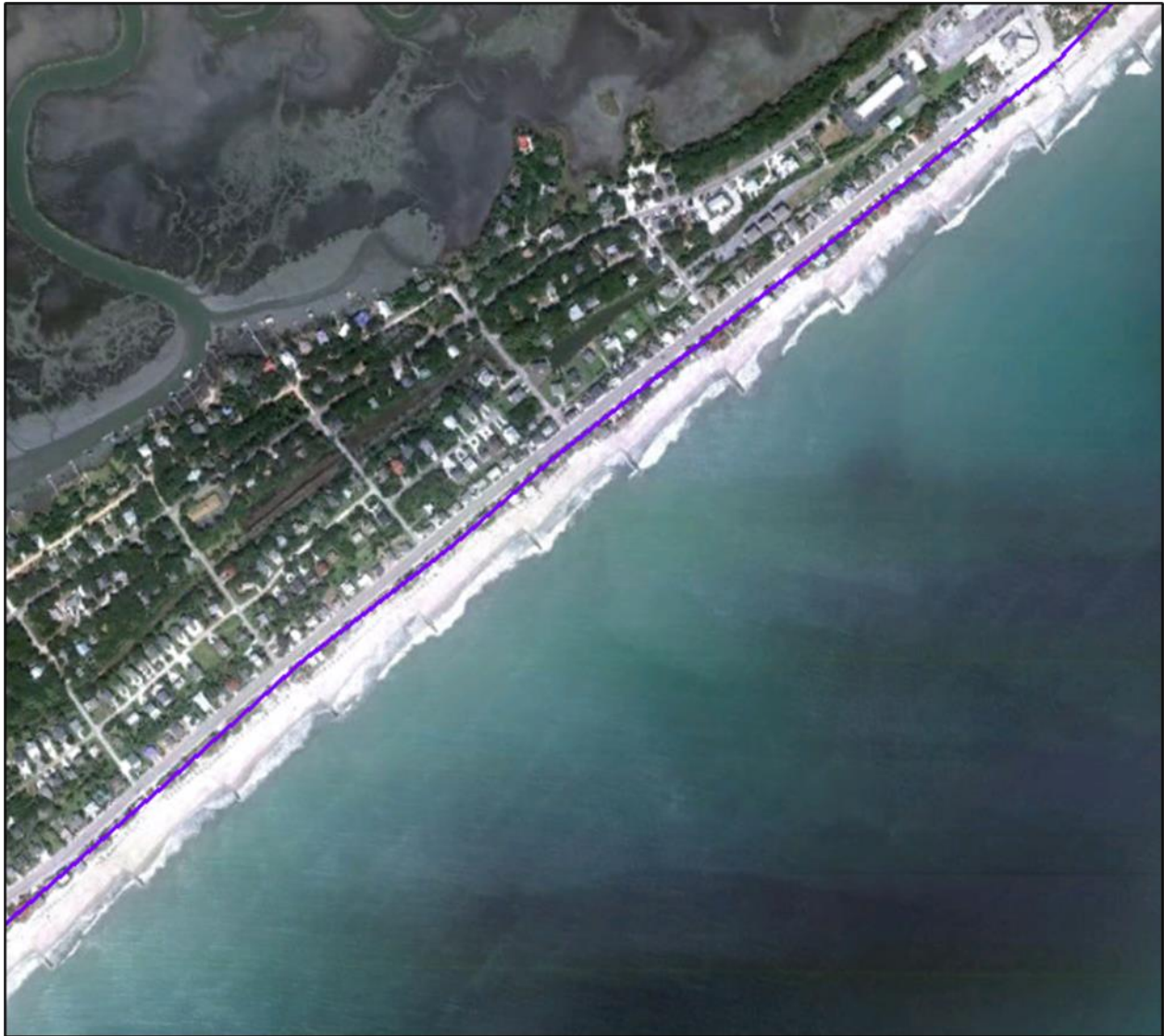


Figure B–30. Without-project forecast of average shoreline position in Year 2064 between Groin Cells 1 and 11

(e) Figure B–31, which encompasses the Edisto Beach State Park area north of the Town of Edisto Beach, shows the same extreme erosion as indicated in the no-action, without-project scenario since the emergency dune reconstruction actions are limited to the developed Town of Edisto Beach. The predicted Year 2064 shoreline indicates that the barrier island will migrate approximately one barrier island width into the upland marsh through barrier island over-wash processes. The present camping area will be extremely vulnerable and it is likely that necessary infrastructure to support recreational use will be lost.



Figure B–31. Without-project forecast of average shoreline position in Year 2064 at Edisto Beach State Park

(f) The results of the limited emergency action, without-project scenario as simulated with Beach-*fx* indicate that emergency dune reconstruction and armoring of Palmetto Boulevard will reduce erosion along the Atlantic-facing shoreline of Edisto Beach and preserve Palmetto Boulevard as the only hurricane evacuation route off the barrier island. However, considerable loss of privately held developed properties is likely, particularly in the north between Groin Cells 1 and 13. Edisto Beach State Park is subject to extreme losses due to coastal erosion, including the inability to support recreational camping within the state park.

(g) Total life cycle without-project damages to structures and contents are estimated at \$44.4 million. Costs associated with emergency dune renourishment over the 55-year life cycle are estimated at \$17.5 million, and average costs of armoring Palmetto Boulevard are estimated at \$2.2 million. The breakdown of damage categories by reach is presented in Table B–9.

Table B–9
Future without-project damages by reach

Reach	Beach-fx Reach	Avg Structure Damage	Avg Content Damage	Avg Total Damage	Avg Annual Damages	Avg Emergency Nourishment	Avg Annual Emergency Nourishment	Armor Cost	AA Armor Cost
1	I1	\$6,318,000	\$2,990,000	\$9,308,000	\$433,000	\$0	\$0	\$0	\$0
2	I2	\$3,063,000	\$1,115,000	\$4,177,000	\$194,000	\$0	\$0	\$0	\$0
3	I3	\$718,000	\$297,000	\$1,015,000	\$47,000	\$0	\$0	\$0	\$0
4	I4	\$1,043,000	\$417,000	\$1,460,000	\$68,000	\$0	\$0	\$0	\$0
5	PI	\$370,000	\$141,000	\$511,000	\$24,000	\$437,000	\$20,342	\$0	\$0
6	P2	\$636,000	\$272,000	\$908,000	\$42,000	\$1,350,000	\$62,843	\$0	\$0
7	E1	\$253,000	\$127,000	\$379,000	\$18,000	\$507,000	\$23,601	\$0	\$0
8	E2	\$703,000	\$289,000	\$991,000	\$46,000	\$854,000	\$39,754	\$0	\$0
9	E3	\$848,000	\$280,000	\$1,129,000	\$53,000	\$1,320,000	\$61,446	\$0	\$0
10	E4	\$1,419,000	\$645,000	\$2,065,000	\$96,000	\$727,000	\$33,842	\$0	\$0
11	E5	\$1,047,000	\$315,000	\$1,363,000	\$63,000	\$665,000	\$30,956	\$0	\$0
12	E6	\$336,000	\$145,000	\$481,000	\$22,000	\$1,552,000	\$72,246	\$0	\$0
13	E7	\$123,000	\$55,000	\$178,000	\$8,000	\$645,000	\$30,025	\$0	\$0
14	E8	\$1,311,000	\$641,000	\$1,952,000	\$91,000	\$1,835,000	\$85,420	\$383,000	\$17,829
15	E9	\$1,144,400	\$714,000	\$2,158,000	\$100,000	\$743,000	\$34,587	\$182,000	\$8,472
16	E10	\$2,151,000	\$1,058,000	\$3,209,000	\$149,000	\$951,000	\$44,269	\$455,000	\$21,180
17	E11	\$2,196,000	\$1,088,000	\$3,284,000	\$153,000	\$626,000	\$29,140	\$210,000	\$9,776
18	E12	\$388,000	\$184,000	\$572,000	\$27,000	\$504,000	\$23,461	\$160,000	\$7,448
19	E13	\$1,113,000	\$544,000	\$1,656,000	\$77,000	\$738,000	\$34,354	\$183,000	\$8,519
20	E14	\$3,637,000	\$1,791,000	\$5,428,000	\$253,000	\$1,284,000	\$59,770	\$414,000	\$19,272
21	E15	\$1,482,000	\$722,000	\$2,204,000	\$103,000	\$2,757,000	\$128,339	\$224,000	\$10,427
Total	–	\$30,599,000	\$13,830,000	\$44,428,000	\$2,067,000	\$17,495,000	\$814,395	\$2,211,000	\$102,923

(h) Beach-*fx* computes damages resulting from erosion, inundation, and direct wave impact. The future without-project damages in the Inlet Planning Reach (Beach-*fx* Reaches I1–I4) are distributed across the three damage drivers as follows: erosion, 9.9%; inundation, 12.3%; and wave attack, 77.8%. The vast majority of damages in the Inlet Planning Reach are driven by wave attack and inundation, indicating that beach elevation and a protective dune system in this planning reach is of primary importance. The future without-project damages in the Atlantic South Planning Reach (Beach-*fx* Reaches P1–P2 and E1–E6) are distributed across the three damage drivers as follows: erosion, 13.8%; inundation, 11.5%; and wave attack, 74.6%. Without-project damages in the Atlantic South Planning Reach are also dominated by wave attack-induced damages, with erosion-induced damages coming in a distant second. This section of beach will need both a protective dune system and a wider beach to provide storm damage reduction.

(i) Future without-project damages in the Atlantic North Planning Reach (Beach-*fx* Reaches E7–E15) are distributed across the three damage drivers as follows: erosion, 64.5%; inundation, 5.1%; and wave attack, 30.4%. Without-project damages in the Atlantic North Planning reach are dominated by erosion damages followed by wave attack damages. This section of the project will require increased beach width as well as a protective dune system to achieve storm damage reductions. The two state park reaches (S1–S2) contain no damage elements and are not included in Table B–9.

(j) To confirm that 300 life-cycle iterations were sufficient to determine the EAD of the future without-project condition, the moving average and percent change are calculated to evaluate the variation caused by including the next iteration. When the inclusion of additional iterations produces effectively no variation in the calculated mean EAD, then uncertainty in EAD cannot be further reduced with the addition of more iterations. Figure B–32 displays the moving average and percent change by iteration.

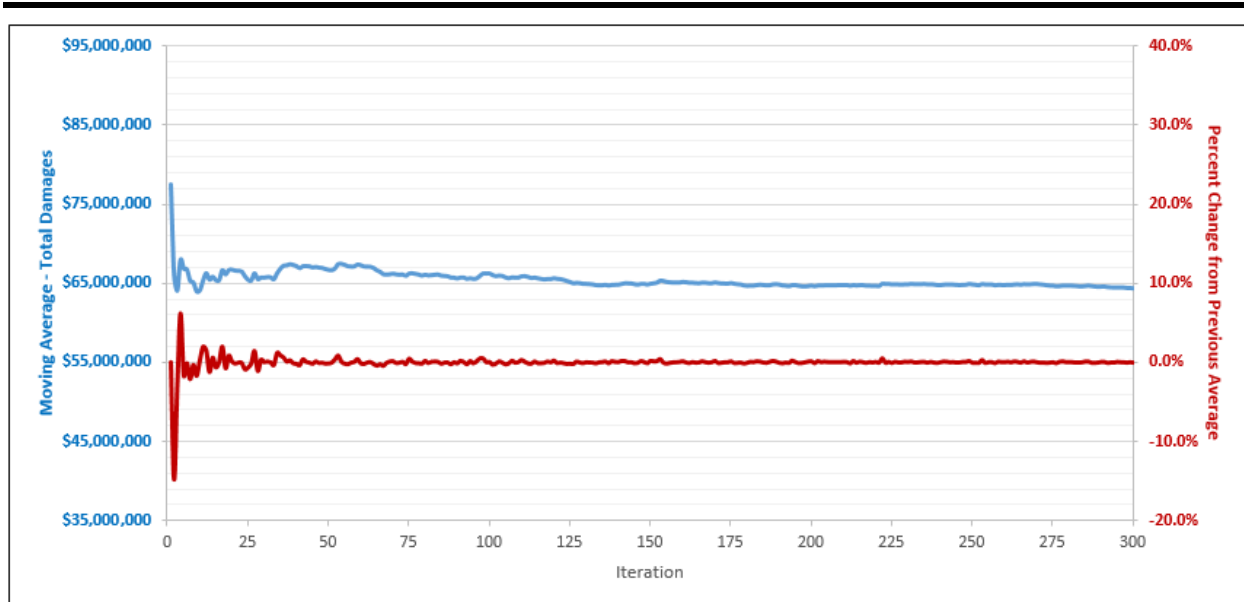


Figure B-32. Future without-project condition model equilibrium

(k) At approximately iteration 85, the percent change on the mean EAD approaches 0.0% with the inclusion of the next iteration result. This confirms that 300 iterations within Beach-fx are more than sufficient for this study area.

B-11. Alternative formulation

a. *Reduction alternatives.* Storm damage reduction alternatives were developed based on project site observations and known performance of past beach nourishment projects at Edisto Beach. Apparent during on-site inspection of the project beach was the lack of a significant dune feature seaward of the existing infrastructure. Developed properties seaward of Palmetto Boulevard are essentially constructed within the existing dune line. Little, if any, vegetation is present seaward of the developed properties north of Cheehaw Street (Groin 14, Beach-fx Reach E7), indicating that active swash processes propagate close enough to the structures to preclude the establishment of vegetation.

(1) South of this location, a wider vegetated buffer is present between the structures and the active shore face, although elevations within the vegetated buffer zone are typically just a few feet higher than the berm elevation. To provide meaningful storm damage reduction along the Edisto Beach project shoreline, a robust dune feature should be constructed along the entire project shoreline to serve as a barrier to storm surge and waves propagating landward toward developed properties and to provide a reservoir of sand for erosion forces associated with the storm events. To be effective, the dune feature should be exposed to active swash processes only during significant storm events and higher than typical water levels. Consequently, the project must also include a constructed berm feature to absorb long-term erosion processes and to ensure that the dune feature is in place at the time of significant storm events.

(2) The constructed berm feature will require periodic renourishment to restore its design dimensions and, depending on the intensity of the storms encountered since the previous nourishment, the dune feature may also require restoration to its design dimensions. A berm feature alone (without an accompanying dune feature) may serve to reduce wave energy and to some extent provide protection against erosion losses but cannot protect against inundation and direct wave impact damages during significant storms and elevated water levels driven by storm surge and wave setup.

b. Construction and monitoring. The construction and monitoring of the 2006 Edisto Beach restoration project provides valuable site-specific information about beach nourishment design and performance at Edisto Beach. This successful project was used as a guide for the development of alternative beach nourishment design templates that were evaluated using Beach-*fx* to identify the alternative that maximizes net benefits. Three alternatives were initially developed. All alternatives involved the creation of a protective dune along the inlet shoreline (Beach-*fx* Reaches I1–I4 and P1) and a design berm feature of varying widths along the Atlantic-facing shoreline.

c. Alternatives. Alternatives analysis in Beach-*fx* is performed by specifying specific periodic nourishment intervals, production rates, mobilization thresholds, and nourishment triggers for the future with-project condition and then evaluating the project performance of a specific combination of dune height, dune width, and berm width variables. As with the future without-project life-cycle simulation, Beach-*fx* references the SBEACH matrix of beach profile responses after each storm event and calculate the resulting inventory damages using the damage functions specified in the model. The difference is that unlike the future without-project condition, an initial beach profile design template is specified and then that template can be rebuilt with each periodic nourishment interval. The effectiveness of the beach profile design template in mitigating damages due to storm events informs the benefits of the alternative and can later be compared to the cost of constructing and maintaining that alternative to develop the benefit-cost ratio and AANB.

d. Beach profile. The beach profile is not static and will be damaged after each storm event. Longer periodic nourishment intervals can, therefore, result in higher residual risk if subsequent storm events occur before renourishment of the beach design template can take place. In addition to the impact on damage reduction, each beach profile template has different initial construction and renourishment costs due to differences in the initial volume of sand required and the erosion rates experienced.

e. Alternative 1. Identified as the “Medium” plan, this involved a 12-foot dune crest elevation and a 15-foot dune crest width along the inlet shoreline (Beach-*fx* Reaches I1–I4, and P1). Seaward and landward dune slopes were set at one on three. Along the Atlantic-facing shoreline, the design template involves a 14-foot dune crest elevation and a 15-foot dune crest width. The design template berm width transitions from 0 feet at Reach P1 to 50 feet at Reach E1.

(1) The design template berm width remains at 50 feet through Reach E6, where it transitions across Reach E7 to a width of 75 feet at Reach E8. The Alternative 1 design template berm width remains at a 75-foot width through Reach E15 and transitions to a width of 0 feet north of Groin 1. Alternative 1 is referred to as the “Medium” plan because it closely follows the observed added berm widths following the 2006 beach restoration project.

(2) The 2006 beach restoration project is viewed as an effective project that has performed well over the 7+ years since construction. Analyses indicated that periodic renourishment would be required at approximately 10-year intervals, and based on current conditions, this estimate of renourishment interval appears to be reasonably accurate.

f. Alternative 2. Identified as the “Minimum” plan, this involved a 15-foot dune crest width at a 10-foot NAVD crest elevation along the inlet shoreline. Along the Atlantic-facing shoreline, the design dune template involved a 15-foot dune crest width at a 12-foot NAVD crest elevation. The design template berm width transitions from 0 feet at Reach P1 to 25 feet at Reach E1. The design template berm width remains at 25 feet through Reach E6, where it transitions across Reach E7 to a width of 50 feet at Reach E8. The design template berm width remains at a 50-foot width through Reach E15 and transitions to a width of 0 feet north of Groin 1. Alternative 2 is referred to as the “Minimum” plan because it is believed that the dimensions of the Alternative 2 design template represent the minimum beach cross section that provides measurable storm damage reduction benefits at Edisto Beach.

g. Alternative 3. Identified as the “Maximum” plan, this involved a 15-foot dune crest width at a 14-foot NAVD88 crest elevation along the inlet shoreline. Along the Atlantic-facing shoreline, the design dune template includes a 15-foot dune crest width at a 16-foot NAVD88 crest elevation. The design template berm width transitions from 0 feet at Reach P1 to 75 feet at Reach E1. The design template berm width remains at 75 feet through Reach E6, where it transitions across Reach E7 to a width of 100 feet at Reach E8. The design template berm width remains at 100 feet through Reach E15 and transitions to a width of 0 feet north of Groin 1. Alternative 3 is referred to as the “Maximum” plan because it is believed that the dimensions of the Alternative 3 design template are the largest that could be justified through storm damage reduction benefits.

h. Summary. Alternatives 1, 2, and 3 were simulated with Beach-*fx*, and based on the results, a fourth alternative was developed to optimize the design template to maximize storm damage reduction and minimize project costs. Alternative 4’s design template is smaller than Alternative 3 (Maximum plan) but slightly larger than Alternative 1 (Medium plan). Dune crest elevation along the inlet shoreline is 14 feet NAVD88, the same as Alternative 3; whereas, the dune crest elevation along the Atlantic-facing shoreline is 15 feet NAVD88, between Alternative 1 and Alternative 3. The design template berm width for Alternative 4 is identical to Alternative 1 except for a longer transition zone at the southern end. The design template berm width transitions from 0 feet at Reach P1 to 50 feet at Reach E2. Table B–10 provides reach-by-reach design template dimensions for each of the design alternatives.

Table B–10

Dimensions of the four beach fill alternatives analyzed

Reach	Alternative 1: Beach and Dune Fill (medium)			Alternative 2: Beach and Dune Fill (minimum)			Alternative 3: Beach and Dune Fill (maximum)			Alternative 4: Beach and Dune Fill (bracketing)		
	Berm Width	Dune Height	Dune Width	Berm Width	Dune Height	Dune Width	Berm Width	Dune Height	Dune Width	Berm Width	Dune Height	Dune Width
I1	–	12	15	–	10	15	–	14	15	–	14	15
I2	–	12	15	–	10	15	–	14	15	–	14	15
I3	–	12	15	–	10	15	–	14	15	–	14	15
I4	–	12	15	–	10	15	–	14	15	–	14	15
P1	taper	12	15	taper	10	15	taper	14	15	taper	15	15
P2	25	14	15	13	12	15	38	16	15	13	15	15
E1	50	14	15	25	12	15	75	16	15	25	15	15
E2	50	14	15	25	12	15	75	16	15	50	15	15
E3	50	14	15	25	12	15	75	16	15	50	15	15
E4	50	14	15	25	12	15	75	16	15	50	15	15
E5	50	14	15	25	12	15	75	16	15	50	15	15
E6	50	14	15	25	12	15	75	16	15	50	15	15
E7	63	14	15	38	12	15	88	16	15	63	15	15
E8	75	14	15	50	12	15	100	16	15	75	15	15
E9	75	14	15	50	12	15	100	16	15	75	15	15
E10	75	14	15	50	12	15	100	16	15	75	15	15
E11	75	14	15	50	12	15	100	16	15	75	15	15
E12	75	14	15	50	12	15	100	16	15	75	15	15
E13	75	14	15	50	12	15	100	16	15	75	15	15
E14	75	14	15	50	12	15	100	16	15	75	15	15
E15	75	14	15	50	12	15	100	16	15	75	15	15
SP1	taper	–	–	taper	–	–	taper	–	–	taper	–	–

i. Upland construction baseline.

(1) As mentioned previously, the developed properties seaward of Palmetto Boulevard are constructed within the existing dune line. As such, the project design template must be offset seaward of the existing dune such that the landward toe of the constructed dune intersects the existing condition beach profile seaward of the existing infrastructure. To accommodate this requirement, a construction baseline was established and mapped to ensure the constructability of the proposed project. The location of the construction baseline is shown in Figure B–33 through Figure B–35.

(2) However, because there is an offset between the Beach-*fx* baseline (defined by the landward toe of the existing condition dune feature) and the construction baseline (located seaward of all habitable structures), the estimates of initial construction volumes calculated in Beach-*fx* are underestimated because the model has no provision for implementing an upland width offset at the time of project construction. That is, within Beach-*fx*, construction of a planned nourishment dune feature begins at the landward toe of the existing condition dune and extends seaward from that location according to the specified design template.

(3) The additional initial construction volume for each of the alternatives was computed externally from the model and added to the volume estimates generated within Beach-*fx*. The additional sand volume associated with the offset between the construction baseline and the Beach-*fx* baseline was estimated as follows:

(a) Step 1: Compute fill volume between the 2009 initial condition representative beach profiles and the design template referenced to the Beach-*fx* baseline on a reach-by-reach basis.



Figure B-33. Construction baseline, Reaches I1-I4, P1-P2, and E1

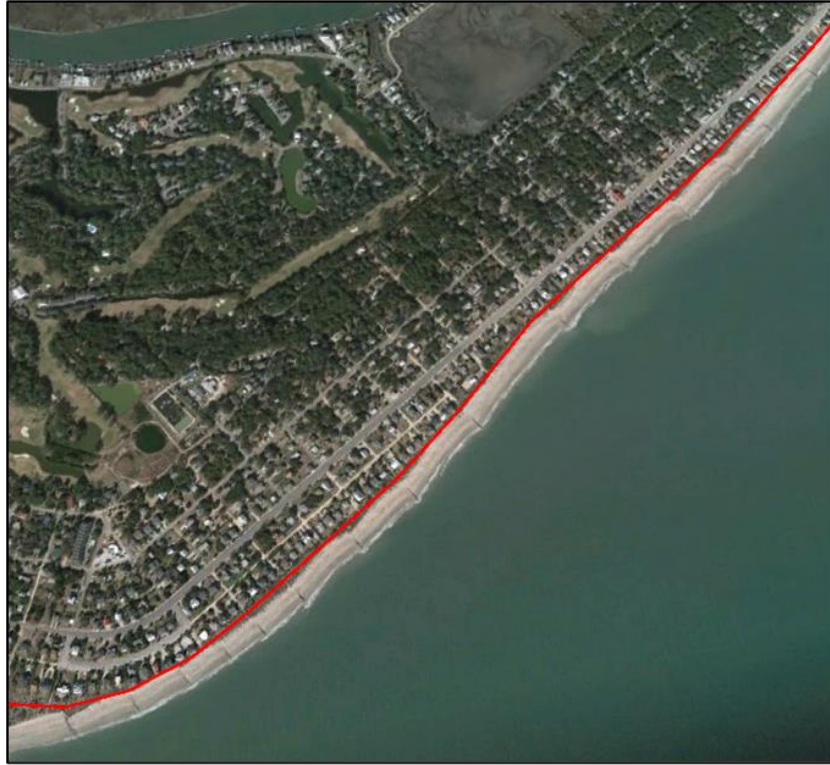


Figure B-34. Construction baseline, Reaches E1-E6 and P2

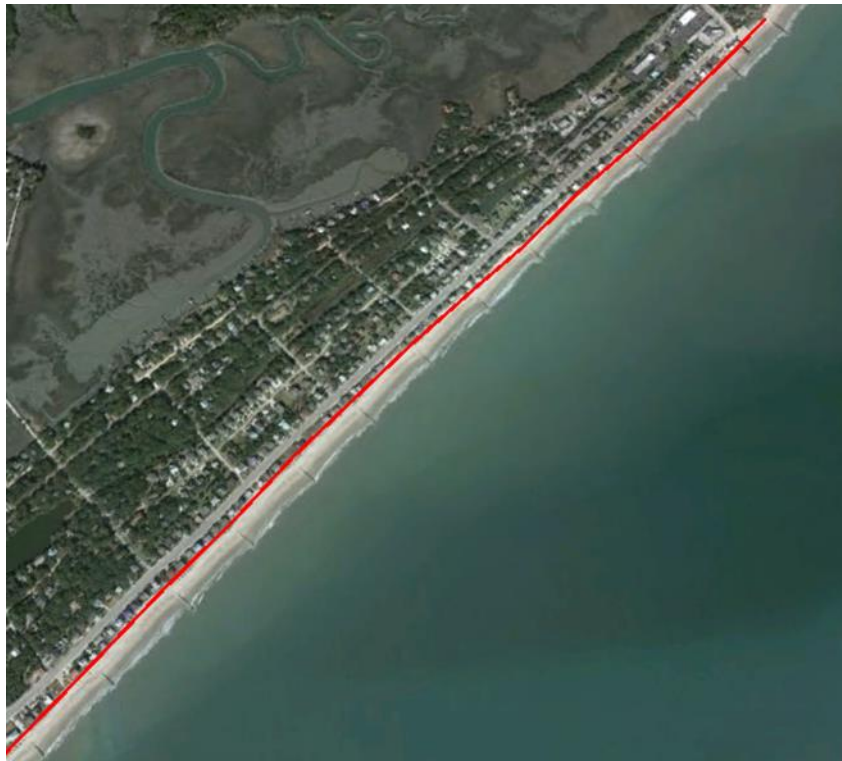


Figure B-35. Construction baseline, Reaches E7-E15

(b) Step 2: Compute fill volume between the 2009 initial condition representative beach profiles and the design template referenced to the construction baseline on a reach-by-reach basis.

(c) The fill volume associated with the offset between the construction baseline and the Beach-*fx* baseline is estimated as the total volume computed in Step 2 (paragraph B–11.i(3)(b)) less the total volume computed in Step 1 (paragraph B–11.i(3)(a)).

(4) This analysis indicated that the initial construction fill volume associated with the offset between the Beach-*fx* baseline and the construction baseline is approximately 364,000 cubic yards, 198,000 cubic yards, 443,000 cubic yards, and 388,000 cubic yards for Alternatives 1, 2, 3, and 4, respectively. The reach-by-reach fill densities and total construction baseline offset fill volumes for each of the four beach and dune fill alternatives are provided in Table B–11 through Table B–14.

j. Methodology and procedure for estimating required groin lengthening. This section provides background information related to the importance and function of the groin field at Edisto Beach and documents the procedure employed to estimate the amount of groin lengthening necessary to support the proposed project design template.

(1) *Introduction and background.* Existing coastal processes at Edisto Beach are driven by high-energy waves and water levels. As discussed in paragraph B–1, the construction of groins at Edisto Beach began in 1948 in an effort to reduce the rate of shoreline erosion and protect upland infrastructure, including commercial property (the Pavilion at the north end of Edisto Beach), Palmetto Boulevard (SC 174), and private property. Groins were constructed from north to south along the Atlantic-facing shoreline and as erosion continued to move down the beach, additional groins were constructed in an attempt to keep pace with the subsequent erosion moving downdrift (south).

(a) By 1958, a total of 17 groins had been constructed, covering approximately 65% of the Atlantic-facing shoreline of Edisto Beach. By 1975, 17 more groins had been constructed, for a total of 34 groins. Of the 34, 29 were along the Atlantic-facing shoreline and 5 were along the inlet shoreline. The chronology of groin construction at Edisto Beach is provided in Table B–1 (paragraph B–1). This groin field plays a central role in the stabilization of the Edisto Beach shoreline, and although long-term shoreline erosion persists along the ocean front, the groin field functions to reduce the rate of sand loss. Specifically, periodic beach profile monitoring surveys have shown that the rate of sand loss in Groin Cells 1 through 27 has been less than 1 cy/ft/yr in recent years, compared to an erosion rate of 1.5 cy/ft/yr in the southern part of the state park reach where there are no groins (CSE 2003).

(b) CSE has estimated that without groins 1 through 15, at least two rows of houses and Palmetto Boulevard would be destroyed by natural adjustment of the shoreline within 10 years (CSE 2003). This assertion is supported by observation of the more seaward location of the shoreline in Edisto Beach as compared to the state park

illustrated in Figure B–36. Therefore, the groin field at Edisto Beach is viewed as an essential element in the stabilization the beach, which, in turn, provides coastal storm damage protection to the upland infrastructure. The groin field also exerts a critical influence on current and future shoreline position in the Edisto Beach study area.

Table B–11
Construction baseline offset fill volume, Alternative 1

Reach	Beach-fx Baseline		Construction Baseline		Offset Volume
	Fill Density	Reach Volume	Fill Density	Reach Volume	
	cy/ft	cy	cy/ft	cy	
I1	5.6	21,280	5.6	2,1280	0
I2	2.9	6,127.7	2.9	6,127.7	0
I3	2.9	1,870.5	2.9	1,870.5	0
I4	2.9	1,841.5	2.9	1,841.5	0
P1	2.8	1,472.8	2.8	1,472.8	0
P2	5.1	4,498.2	5.1	4,498.2	0
E1	23.8	11,733.4	28.1	13,853.3	2,119.9
E2	16.5	14,338.5	16.5	14,338.5	0
E3	16.6	20,351.6	31.8	38,986.8	18,635.2
E4	2.6	4,544.8	2.6	4,544.8	0
E5	2.6	3,268.2	19.1	24,008.7	20,740.5
E6	2.4	2,952	15.1	18,573	15,621
E7	4.8	2,688	33.2	18,592	15,904
E8	18.4	23,128.8	39.8	50,028.6	26,899.8
E9	5.4	3,245.4	41.3	24,821.3	21,575.9
E10	2.7	3,121.2	41.1	47,511.6	44,390.4
E11	4.7	2,895.2	60.6	37,329.6	34,434.4
E12	17.2	10,320	82.2	49,320	39,000
E13	41.3	24,036.6	86.9	50,575.8	26,539.2
E14	33.8	40,898	82.1	99,341	58,443
E15	48.3	83,172.6	71.4	122,950.8	39,778.2
Total	–	28,7785	–	651,867	364,082

Table B-12
Construction baseline offset fill volume, Alternative 2

Reach	Beach-fx Baseline		Construction Baseline		Offset Volume
	Fill Density	Reach Volume	Fill Density	Reach Volume	
	cy/ft	cy	cy/ft	cy	
I1	2.6	9,880	2.6	9,880	0
I2	0	0	0	0	0
I3	0	0	0	0	0
I4	0	0	0	0	0
P1	0	0	0	0	0
P2	1.3	1,146.6	1.3	1,146.6	0
E1	1	493	1.3	640.9	147.9
E2	0.3	260.7	0.3	260.7	0
E3	0.3	367.8	2.4	2,942.4	2,574.6
E4	0	0	0	0	0
E5	0	0	4.3	5,405.1	5,405.1
E6	0	0	1.6	1,968	1,968
E7	1	560	5.6	3,136	2,576
E8	2.9	3,645.3	6.1	7,667.7	4,022.4
E9	1.6	961.6	7.3	4,387.3	3,425.7
E10	0	0	7.1	8,207.6	8,207.6
E11	0.2	123.2	26.6	16,385.6	16,262.4
E12	0	0	48.1	28,860	28,860
E13	7.2	4,190.4	52.8	30,729.6	26,539.2
E14	0	0	48.1	58,201	58,201
E15	14.3	24,624.6	37.4	64,402.8	39,778.2
Total	–	46,253	–	244,221	197,968

Table B–13
Construction baseline offset fill volume, Alternative 3

Reach	Beach-fx Baseline		Construction Baseline		Offset Volume
	Fill Density	Reach Volume	Fill Density	Reach Volume	
	cy/ft	cy	cy/ft	cy	
I1	10.2	38,760	10.2	38,760	0
I2	7.2	15,213.6	7.2	15,213.6	0
I3	7.2	4,644	7.2	4,644	0
I4	7.2	4,572	7.2	4,572	0
P1	6.6	3,471.6	6.6	3,471.6	0
P2	22.6	19,933.2	22.6	19,933.2	0
E1	58.3	28,741.9	62.6	30,861.8	2,119.9
E2	49.9	43,363.1	49.9	43,363.1	0
E3	50	61,300	65.1	79,812.6	18,512.6
E4	14.4	25,171.2	14.4	25,171.2	0
E5	14.4	18,100.8	53.5	67,249.5	49,148.7
E6	22.6	27,798	49.9	61,377	33,579
E7	26.9	15,064	68.1	38,136	23,072
E8	52.7	66,243.9	74.1	93,143.7	26,899.8
E9	30.7	18,450.7	76.3	45,856.3	27,405.6
E10	21	24,276	76.4	88,318.4	64,042.4
E11	39.7	24,455.2	95.6	58,889.6	34,434.4
E12	52.2	31,320	117.1	70,260	38,940
E13	76.3	44,406.6	121.8	70,887.6	26,481
E14	68.7	83,127	117.1	141,691	58,564
E15	83	142,926	106.1	182,704.2	39,778.2
Total	–	741,339	–	1,184,316	442,978

Table B–14
Construction baseline offset fill volume, Alternative 4

Reach	Beach-fx Baseline		Construction Baseline		Offset Volume
	Fill Density	Reach Volume	Fill Density	Reach Volume	
	cy/ft	cy	cy/ft	cy	
I1	8.9	33,820	8.9	3,3820	0
I2	6.7	14,157.1	6.7	14,157.1	0
I3	6.7	4,321.5	6.7	4,321.5	0
I4	6.7	4,254.5	6.7	4,254.5	0
P1	8.9	4,681.4	8.9	4,681.4	0
P2	7.3	6,438.6	7.3	6,438.6	0
E1	10.8	5,324.4	14.8	7,296.4	1,972
E2	23.5	20,421.5	23.5	20,421.5	0
E3	23.5	28,811	38.7	47,446.2	18,635.2
E4	4.8	8,390.4	4.8	8,390.4	0
E5	4.8	6,033.6	26.2	32,933.4	26,899.8
E6	4.6	5,658	22.2	27,306	21,648
E7	7	3,920	40.4	22,624	18,704
E8	25.5	32,053.5	47	59,079	27,025.5
E9	7.6	4,567.6	48.5	29,148.5	24,580.9
E10	5	5,780	48.3	55,834.8	50,054.8
E11	11.9	7,330.4	67.8	41,764.8	34,434.4
E12	24.4	14,640	89.4	53,640	39,000
E13	48.5	28,227	94.1	54,766.2	26,539.2
E14	41	49,610	89.3	108,053	58,443
E15	55.4	95,398.8	78.6	135,349.2	39,950.4
Total	–	383,839	–	771,727	387,887

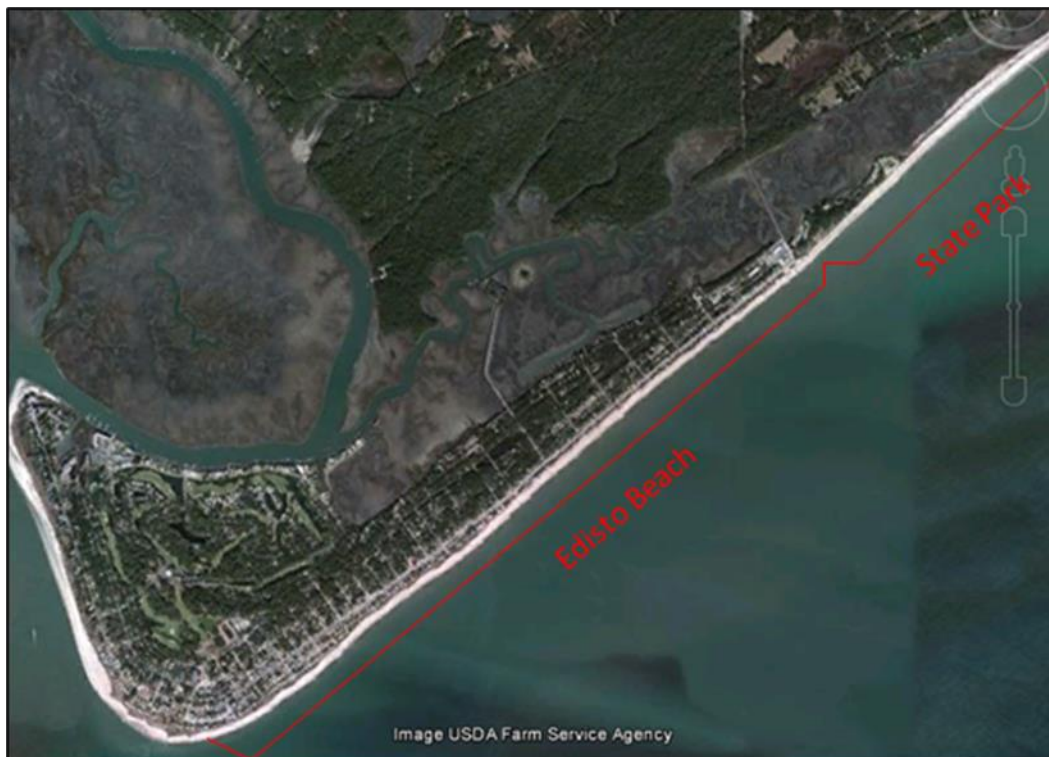


Figure B–36. Edisto Beach study area

(2) *Projects.* Locally funded beach nourishment projects were constructed along the Atlantic-facing shoreline in 1995 and 2006. The 1995 project involved the placement of 155,000 cubic yards of beach quality fill material, and the 2006 project involved the placement of 850,000 cubic yards of beach quality fill material. The subsequent downdrift migration of this fill material has resulted in the burial of the five groins constructed along the inlet-facing shoreline.

(3) *Change rates.* Long-term shoreline change rates along the inlet shoreline vary from nearly stable (plus 0.01 feet/year) at the point to accretional (plus 1.36 feet/year) at the northwest end of the inlet shoreline near Big Bay creek. Along the Atlantic-facing shoreline, long-term shoreline change rates vary from nearly stable (plus 0.01 feet/year) at the point to erosive (negative 3.56 feet/year) at the northern end of Edisto Beach. Shoreline change rates become even more erosive moving north into the state park (negative 4.38 to negative 5.13 feet/year).

k. Requirement for groin lengthening.

(1) Existing condition dune heights along the Atlantic-facing shoreline at Edisto Beach vary between 10 and 12 feet NAVD88, and the existing condition berm width varies between 35 and 105 feet. An analysis of the engineering and economic performance of a number of beach nourishment design alternatives has been conducted using Beach-*fx*, and the results of that analysis indicate that a design beach cross section involving dunes of varying crest elevations and berm widths varying

between 25 and 100 feet is needed to provide the desired coastal storm damage reduction.

(2) Since the distance between the construction baseline and the seaward edge of the alternative design template berm exceeds the distance between the construction baseline and the seaward edge of the existing condition berm along certain reaches within the project, the effective length of many of the existing groins needs to be increased to create and maintain beach width necessary to support the design template.

(3) The motivation for lengthening groins in the Edisto Beach study area is exclusively to provide necessary beach width to accommodate and maintain the alternative design template. Each of the four alternatives requires some amount of groin lengthening, though the alternatives involving larger cross sections and those involving higher dune crest elevations and wider berms require more groin lengthening than those alternatives involving smaller cross sections.

(4) The proposed groin lengthening is not provided as a means for trapping more sand and increasing beach width or significantly changing the rate of sand bypassing the groins. The amount of required groin lengthening was estimated using a technique that employed geometric considerations based on measured beach profile and groin structure survey data as opposed to a numerical simulation-based estimation approach. Use of the described geometric data-based estimation technique is justified and believed to be superior to a numerically based estimation approach because the available survey data represent actual on-site performance of the existing groin field and measured morphology response to those structures; whereas a numerically based estimation relies on theoretical representation of the groin field performance and estimated morphological response to the groin structures.

(5) If any amount of groin lengthening was determined to be necessary, a minimum increased length of 20 feet was specified. Groin lengthening beyond 20 feet was specified in even 10-foot increments for practical reasons.

I. Estimation of groin lengthening amount.

(1) The technique employed to estimate the amount of needed groin lengthening is based on the assumption that the representative existing condition beach profile is in dynamic equilibrium with, and held in place by, the existing groin and the groin's intersection with the seabed. Consequently, the amount of required groin lengthening was taken as approximately equal to the distance between the seaward edge of the existing condition representative beach berm and the seaward edge of the design template beach berm. Figure B-37 illustrates the technique for Alternative 4 and Beach-fx Reach E15. In this case, the distance between the seaward edge of the existing condition representative beach berm profile and the seaward edge of the Alternative 4 design template beach berm is 83.6 feet, so the recommended amount of groin lengthening is 80 feet, rounding to the nearest 10-foot increment.

(2) As a result, Groins 1 and 2, which are contained within Beach-fx Reach E15, are recommended for lengthening of 80 feet. This approach was applied for all Beach-fx reaches and all evaluated alternative design templates to estimate the amount of required groin lengthening. The resulting recommended amount of groin lengthening for each of the alternatives is listed in Table B–15. The total estimated amount of groin lengthening is 1,090 feet, 360 feet, 1,970 feet, and 1,130 feet for Alternatives 1, 2, 3, and 4, respectively. As seen in the table, the amount of required groin lengthening increases from south to north and reflects the greater setback distance between the shoreline and the developed infrastructure in the southern parts of Edisto Beach compared to the northern parts of Edisto Beach.

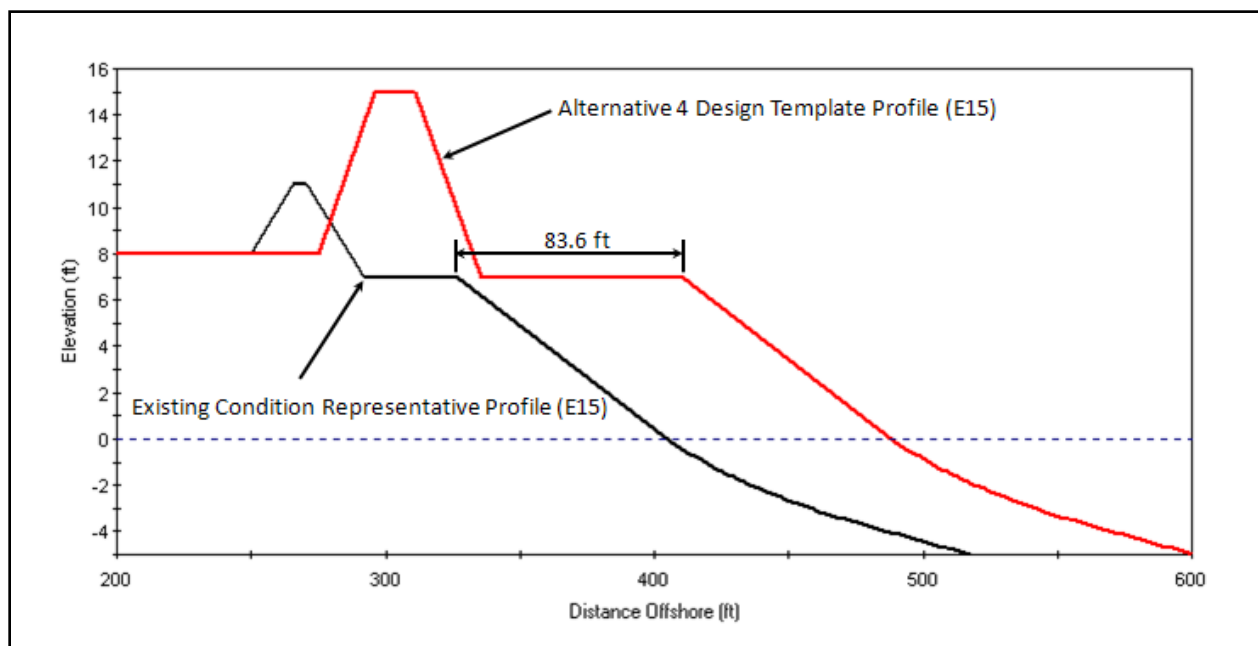


Figure B–37. Groin lengthening estimation technique

(3) Following is an assessment of groin lengthening influence on sand transport and shoreline change:

(a) The downdrift impacts of groins and groin fields depend to a large extent on the effective length of the groin or groins comprising the groin field. The effective length of a groin can be thought of as the length of groin extending beyond the shoreline. Likewise, the sand trapping capacity of a groin depends on the water depth at the seaward tip of the groin. As water depths at the seaward tip of the groin decrease, longshore transport rates bypassing the groin increase. The beach and dune fill project alternatives formulated for Edisto Beach are designed to reduce future storm damages to upland infrastructure by: (1) increasing the setback distance between the existing infrastructure and the shoreline, which allows increased wave energy dissipation; (2) providing a physical barrier to elevated water levels associated with the storm surge; and (3) providing a reservoir of sand to absorb the erosion forces associated with the storm waves and water levels.

(b) The design berm provides for the wave energy dissipation component of the protective system, the dune feature provides the physical barrier component to the elevated water levels, and the berm and dune together provide the reservoir of sand needed to absorb the storm associated erosion forces. However, construction of the beach and dune fill necessitates a seaward displacement of the entire Atlantic-facing project shoreline compared to existing conditions.

Table B-15
Recommended groin lengthening for all alternatives

Groin #	Groin Extension Length (ft)			
	Alt 1	Alt 2	Alt 3	Alt 4
1	80	40	110	80
2	80	40	110	80
3	80	50	120	90
4	90	50	130	90
5	90	60	130	100
6	90	60	130	100
7	80	40	110	80
8	50	20	90	60
9	40	–	80	50
10	40	–	80	50
11	40	–	80	40
12	40	–	80	40
13	30	–	70	40
14	20	–	60	30
15	20	–	50	20
16	20	–	50	20
17	20	–	50	20
18	20	–	30	20
19	–	–	20	–
20	20	–	40	20
21	30	–	70	30
22	20	–	60	30
23	20	–	50	20
24	20	–	60	20
25	30	–	70	–
26	20	–	40	–
Total	1,090	360	1,970	1,130

(c) Since the existing shoreline is already being held in place largely by the existing groin field, any constructed project that further displaces the shoreline in a seaward direction will be short-lived and can be expected to perform poorly without an

equivalent seaward displacement of the groin field that holds the shoreline in place. However, because the groin length extensions will be filled to capacity by the fill project and maintained over the life of the project, the delivery of sediment to downdrift beaches is expected to be largely unaffected by the recommended groin lengthening. That is, the effective length of the groins will not increase because not only are the groins being lengthened, the beach is also being renourished, causing the shoreline to be displaced seaward by the same amount that the groins are being lengthened.

(d) The equilibrium water depths at the seaward tips of the extended groins will be the same as the water depths at the seaward tips of the existing condition groins. Consequently, sand bypassing the post-project groin field is expected to remain the same as the existing condition. Overall sand volume delivery to the inlet shoreline of Edisto Beach and the shoal systems and islands of St. Helena Sound are expected to remain constant or to slightly increase due to the introduction of new sediments to the littoral system from the construction and maintenance of the coastal storm damage reduction project at Edisto Beach. The reasonableness of this expectation is supported by observations of increased sand volume and pro-gradation of the shoreline along the Edisto Beach inlet shoreline following nourishment projects in 1995 and 2006.

B-12. Alternative evaluations

a. This section describes the results of the Beach-*fx* life-cycle simulations of the beach and dune fill project alternatives formulated in the previous section. The details of each of the four project alternatives were specified in Beach-*fx* and 300, 55-year-long life-cycle simulations were performed for each of the alternatives. Each life-cycle simulation started in the year 2009 and involved emergency dune nourishment and armoring of Palmetto Boulevard actions as defined for the Limited Emergency Action Future Without-Project scenario between 2009 and 2014.

b. Starting in the year 2014, the alternative beach and dune project was constructed, and the physical and economic performance of the project was simulated for a 50-year project life concluding at the end of the year 2063 (start of 2064). In each year after initial construction of the fill alternative, the need for renourishment of the project was checked within the model simulation. If specific beach morphology and volume requirement thresholds were met, then a renourishment was scheduled and constructed. Project costs associated with beach nourishment and nourishment volumes were computed and stored, as were storm-induced damages to structures and contents.

c. For each alternative, net average annual project benefits were computed by comparing without-project damages and costs to with-project alternative damages and costs. For the without-project simulation, damages are taken as the sum of the computed structure and content damages, while costs are taken as costs associated with emergency dune nourishment actions and costs associated with armoring and repair of armoring along Palmetto Boulevard. Without-project damages and costs were computed on a reach-by-reach basis within Beach-*fx*. For the with-project alternatives, damages were again taken as the sum of the computed structure and content damages.

d. Emergency nourishment and armoring costs accrued during the first five years of the simulation were also recorded in the with-project condition. Figure B–38 displays the damage by year (temporal distribution) for Alternative 4 for both the without-project and with-project conditions. The damage that remains even in the with-project condition is referred to as residual damage or residual risk.

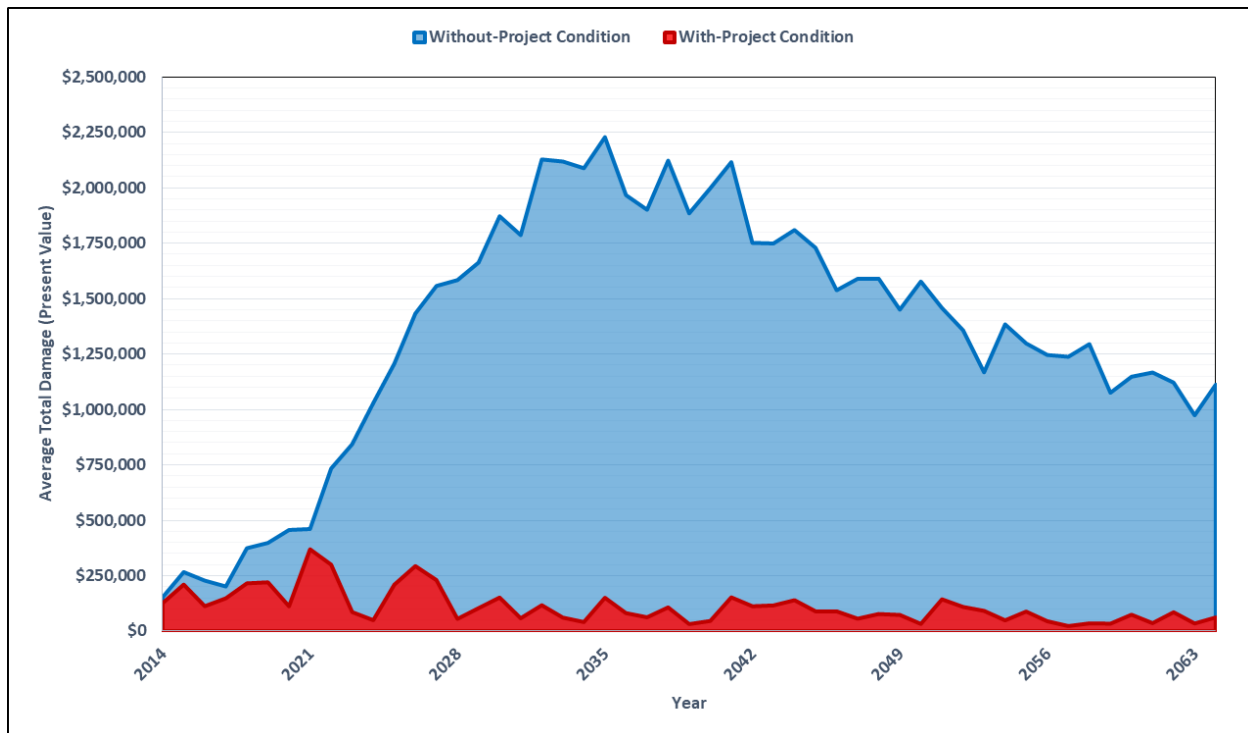


Figure B–38. Temporal distribution of future without-project and future with-project damages

e. Figure B–38 illustrates that some residual damages remain in the future with-project condition, but that Alternative 4 is very effective in mitigating damages related to storm events. Figure B–38 also shows that the damages in the without-project condition peak around year 2035 before steadily reducing as repetitive loss structures are removed from the inventory and no longer contribute to damage totals.

f. Future with-project condition damage distributions can also be displayed by iteration, by damage driver, and by damage element type to accurately showcase the damage figures beyond a simple average point value.

g. With-project benefits were computed as without-project damages, less with-project damages, plus without-project emergency nourishment costs, less with-project emergency nourishment costs, plus without-project armoring costs, and less with-project armoring costs.

h. In other words, with-project benefits include reductions in computed storm-induced damages plus avoided costs associated with emergency nourishment actions

and armoring of Palmetto Boulevard. With-project costs included project nourishment placement costs (including cost of sand volume associated with the offset between the Beach-*fx* baseline and the construction baseline), construction mobilization and demobilization costs, and groin lengthening costs. Total net average annual benefits of each alternative were computed as with-project benefits less with-project costs plus land-loss benefits. Land-loss benefits were computed based on a reduction of shoreline erosion rates on a reach-by-reach basis and apply only to those reaches that are erosional for the without-project condition. The individual damage and cost quantities employed average values computed across the 300 life cycles simulated.

i. Table B–16 lists the net average annual benefits for each of four beach and dune alternatives evaluated. Net benefits are given for each of the Beach-*fx* reaches as well as for the three larger planning reaches. Table B–16 shows that beach and dune Alternative 4 produces the maximum net average annual benefits of all the alternatives evaluated, with an average annual benefit of approximately \$1,600,000. Alternative 4 is identified as the optimized NED plan in that Alternative 4 is bracketed from the perspective of project size by Alternative 1 (a smaller project) and Alternative 3 (a larger project) and produces net average annual benefits exceeding those produced by Alternatives 1 and 3. Although other alternatives may generate greater net average annual benefits in specific individual reaches over the entire project and within the three planning reaches, Alternative 4 generates the greatest net average annual benefits.

Table B–16
Net average annual benefits

Reach	Net Benefits			
	Alt 1	Alt 2	Alt 3	Alt 4
I1	\$122,469	\$15,882	\$222,424	\$222,424
I2	\$57,558	\$7,021	\$107,922	\$107,922
I3	\$14,156	\$2,234	\$22,820	\$22,820
I4	\$19,108	\$2,416	\$33,788	\$33,788
P1	\$9,658	\$9,076	\$14,436	\$17,528
P2	-\$14,101	\$22,457	-\$1,185	-\$5,344
E1	\$3,472	\$13,017	-\$4,736	\$9,951
E2	\$21,848	\$22,470	\$11,313	\$21,978
E3	\$36,315	\$46,123	\$26,654	\$38,632
E4	\$81,740	\$28,222	\$98,315	\$93,723
E5	\$46,145	\$27,247	\$43,832	\$51,606
E6	\$58,933	\$66,524	\$53,368	\$59,216
E7	\$18,021	\$21,968	\$13,804	\$16,423
E8	\$130,028	\$104,432	\$121,698	\$133,471
E9	\$64,325	\$21,001	\$91,613	\$76,090
E10	\$135,694	\$70,100	\$145,367	\$151,388
E11	\$135,277	\$67,594	\$142,937	\$145,952

Reach	Net Benefits			
	Alt 1	Alt 2	Alt 3	Alt 4
E12	\$15,223	\$14,570	\$7,986	\$16,015
E13	\$60,498	\$46,982	\$59,520	\$61,747
E14	\$194,443	\$113,188	\$207,823	\$213,951
E15	\$126,759	\$120,963	\$112,765	\$130,192
Inlet Reach (I1–I4)	\$213,291	\$27,553	\$386,954	\$386,954
AS Reach (P1–P2, E1–E6)	\$244,010	\$235,136	\$241,997	\$287,289
AN Reach (E7–E15)	\$880,268	\$580,798	\$903,513	\$945,230
Total	\$1,337,569	\$843,487	\$1,532,464	\$1,619,473

B–13. Renourishment cycle optimization

a. Having identified the NED plan, the next step in the analysis was to determine the optimum renourishment cycle. Detailed analyses have shown that the total cumulative volume of fill material placed on a nourishment project over a 50-year project life is approximately the same regardless of the length of the renourishment cycle (EM 1110-2-1100). As such, optimization of the renourishment cycle effectively reduces to balancing the cost of frequent mobilizations and demobilizations for short-duration renourishments against the risk of storm-induced damages in the event the project needs renourishment for a prolonged period before a scheduled renourishment occurs.

b. As stated previously, the initial suite of model simulations was performed in a way that allowed renourishment to occur whenever it was determined to be required. That is, within the simulation the beach morphology was checked each year and if certain morphology conditions existed and a specified renourishment mobilization threshold volume was exceeded, then a renourishment was scheduled and constructed. The life-cycle results from these simulations were analyzed and the frequency distribution of the computed renourishment cycle was determined to be as shown in Figure B–39. This figure shows that required renourishment at Edisto Beach takes on a very broad distribution with renourishment needed in as short as 1 year and as long as 30-plus years. A mean renourishment interval of approximately 16 years was computed from the distribution shown in Figure B–40.

c. This broad distribution of renourishment intervals is an indication that the need for renourishment at Edisto Beach is primarily driven by the random occurrence of relatively strong storm events as opposed to a persistent background erosion rate or infrequent extreme events.

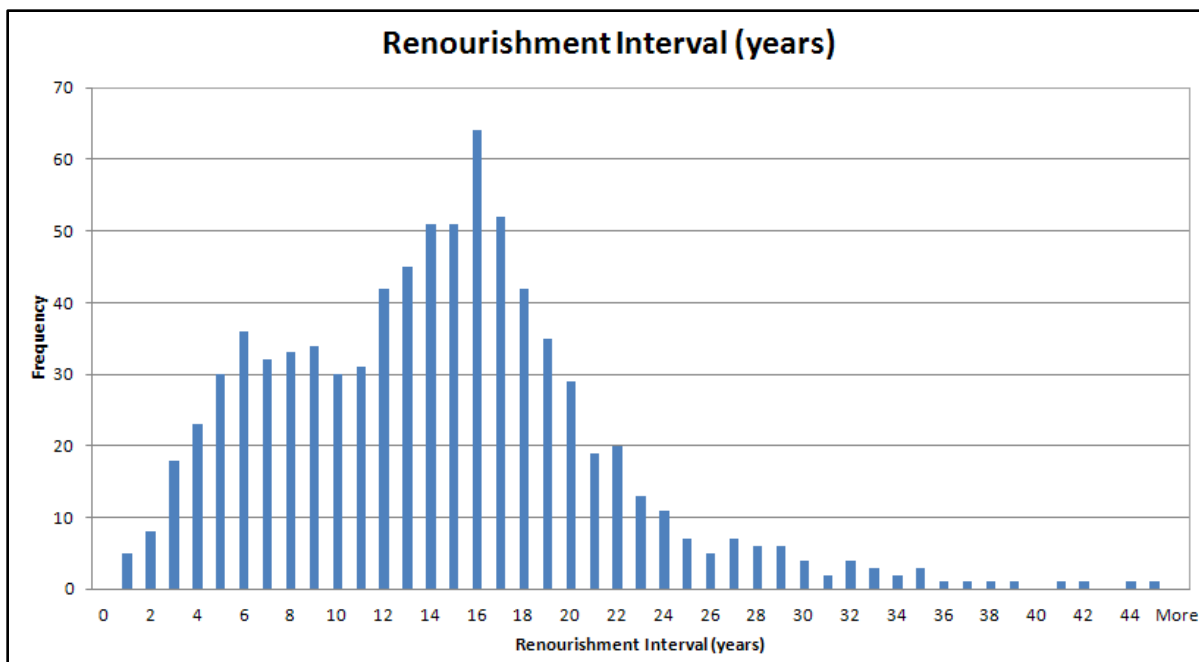


Figure B-39. Frequency distribution of renourishment cycle

d. A series of additional life-cycle simulations were performed to determine the optimum renourishment cycle for NED plan (Alternative 4) at Edisto Beach. Simulations were performed for renourishment cycles of 4, 6, 8, 10, 12, and 16 years. Net average annual benefits were computed for each of the simulated renourishment cycles and the results are shown in Figure B-40. This figure shows that net average annual benefits increase substantially between 4-year and 6-year renourishment intervals and again between 6-year and 8-year renourishment intervals. However, the results indicate that renourishment intervals of 8 years and longer produce approximately the same net average annual benefits. A notable decrease in net average annual benefits is seen between the 12-year renourishment cycle and the 16-year renourishment cycle.

e. The 12-year renourishment cycle generates net average annual benefits that exceed the 8-year renourishment cycle AANB by approximately \$4,500. However, this difference represents an increase of just 0.3% of the total AANB. Figure B-41 shows the cumulative probability distribution function of when renourishment at Edisto Beach is required. Figure B-39 shows that there is approximately a 40% probability that renourishment will be required at Edisto Beach at an interval of 12 years or fewer. Likewise, Figure B-41 shows that the probability of a renourishment being required at 8 years or less is approximately 23%.

f. Consequently, the risk of the project requiring renourishment before it is scheduled to occur is reduced by 17% with an 8-year renourishment cycle compared to a 12-year renourishment cycle. In light of the 17% risk reduction and the relatively small difference (0.3%) in AANB, an 8-year renourishment cycle for Edisto Beach is identified as the optimum renourishment cycle.

B-14. Project sensitivity to future sea level change

a. According to ER 1100-2-8162, the direct and indirect effects of future sea level change on the identified NED beach and dune fill alternative (Alternative 4) were evaluated using the Beach-*fx* model. The engineering and economic performance of the NED plan was evaluated under three scenarios of future sea level change consistent with official guidance as articulated in this ER. RSLC at Edisto Beach is one of rising sea levels. The historical rate of sea level change was determined to be +3.51 mm/year. The future low rate of sea level change reflects this historical rate of change. The future intermediate rate of sea level change was computed using modified NRC Curve I and equation 3 in this ER. The future high rate of sea level change was computed using modified NRC Curve III and equations 3 from this ER.

b. These relationships for future sea level change as defined in ER 1100-2-8162 are coded within Beach-*fx*, and sea level change is internally computed continuously throughout the simulated project life cycle. Figure B-42 provides a plot of the Beach-*fx* computed sea level change for each of the three sea level change scenarios. Figure B-42 displays the incremental sea level change across the simulation period (2009 to 2069) that was computed at 0.69 feet, 1.19 feet, and 2.78 feet, for the low, intermediate, and high rates of sea level change, respectively. The SLC curves adopted in support of the Edisto project and used as the basis for impacts on economic performance presented herein are slightly different than those presented within this case study (approximately 0.1 ft difference across all three SLC curves). The SLC curve plots and historic rates presented in this guidance have been updated to reflect current SLAT output for the tide gauge closest to Edisto Beach.

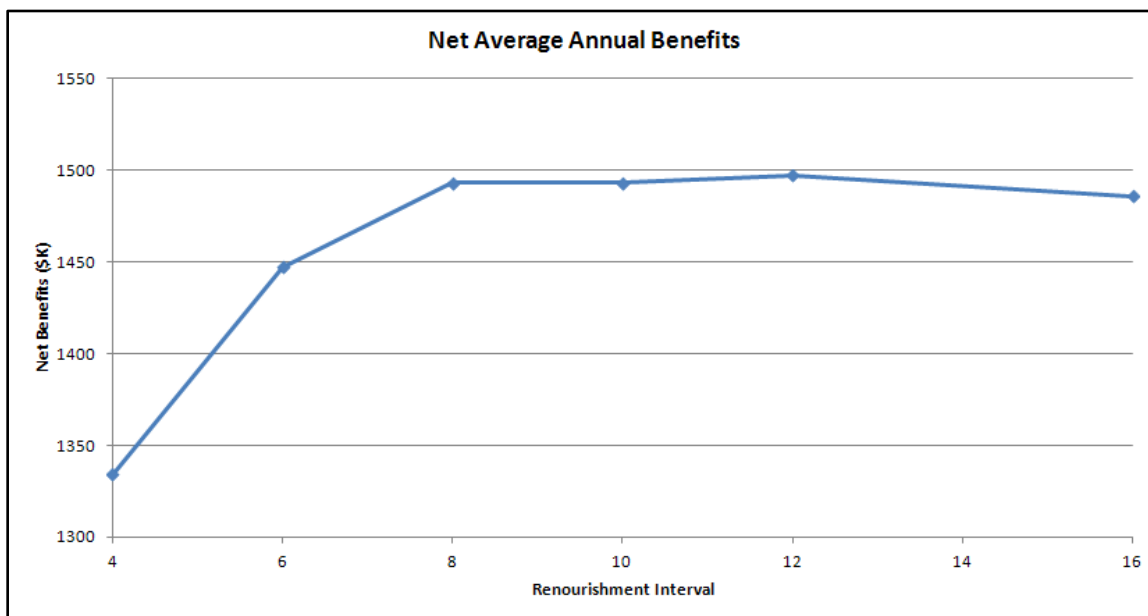


Figure B-40. Probability distribution of renourishment cycle

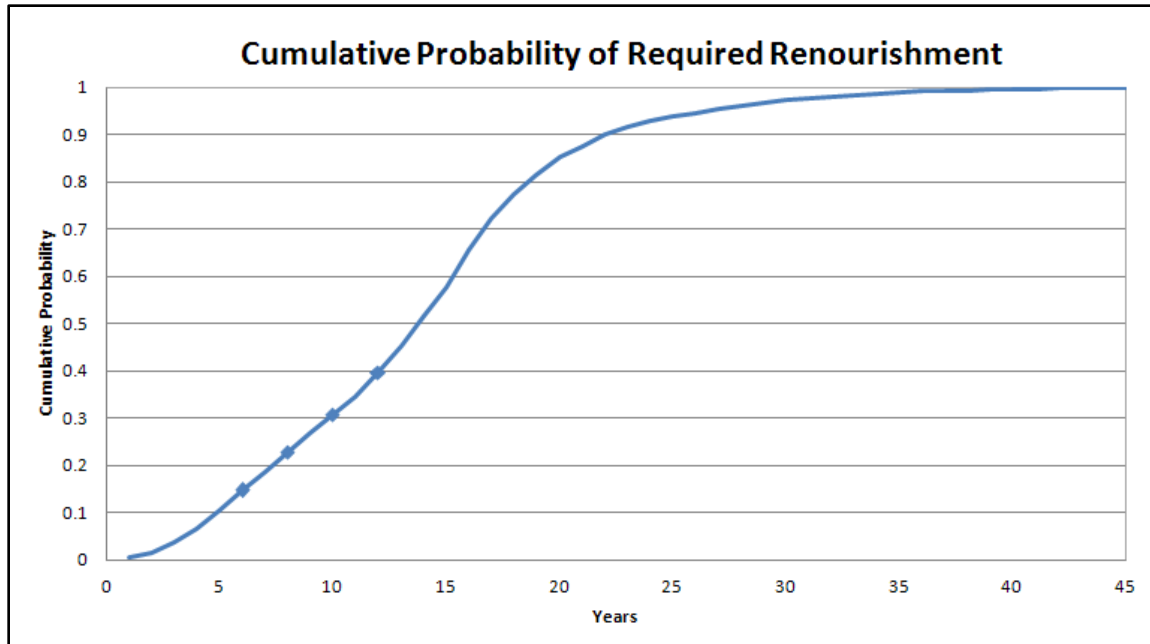


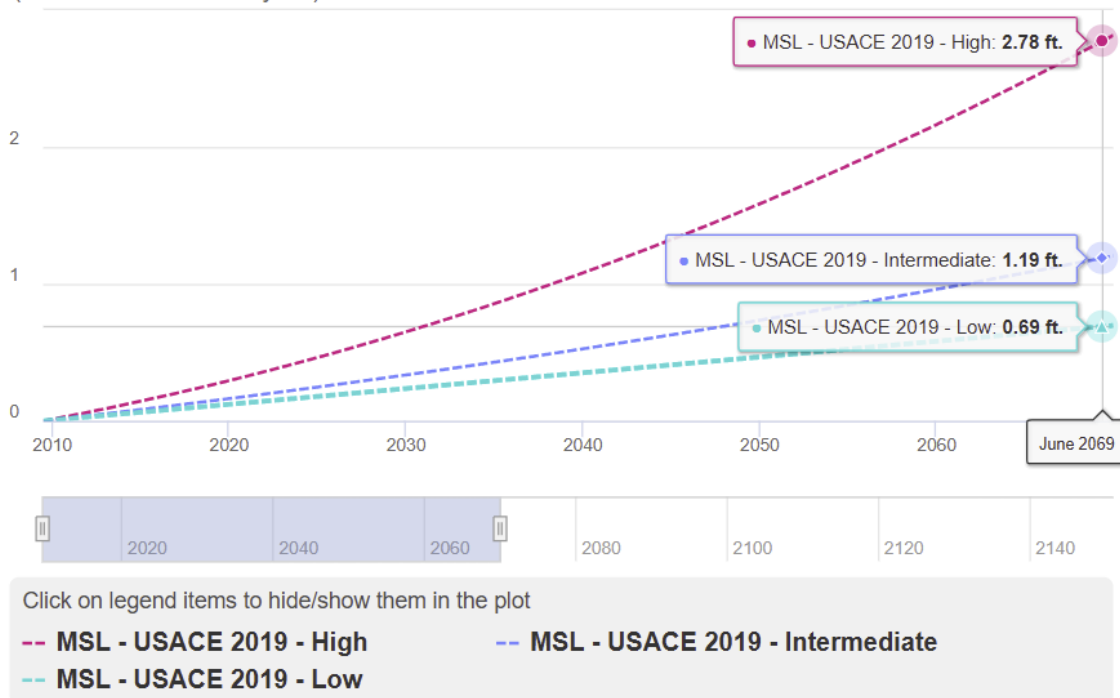
Figure B–41. Cumulative probability function of required renourishment

Sea Level Data and Projections: Charleston, SC

(8665530)

NOAA Tide Gauge

Feet of sea level change
(relative to 2009 start year)



SLC rate used in equation based projections: 3.51 mm/yr (1.15 ft/100 yrs)

SLC source: NOAA-NOS Tides & Currents Trend (Jan 1901 - Dec 2024)

MSL record span: 1901 to 2025 (124 years)

Missing data: The MSL record for this gauge has a gap of 5 or more years

Figure B–42. Relative sea level change functions

c. The effect of potential future sea level change on the economic performance of the NED alternative (Alternative 4) is illustrated in Figure B–43. This plot shows that average annual project costs, average annual project benefits, and net average annual benefits all increase with increasing rates of future sea level change. However, average annual benefits increase at a faster rate than average annual project costs for higher future rates of sea level change, resulting in net average annual benefits that are greater. These results indicate that, from an economic perspective, a federal storm damage reduction project at Edisto Beach is justified if accelerating rates of future sea level rise occur as some have predicted.

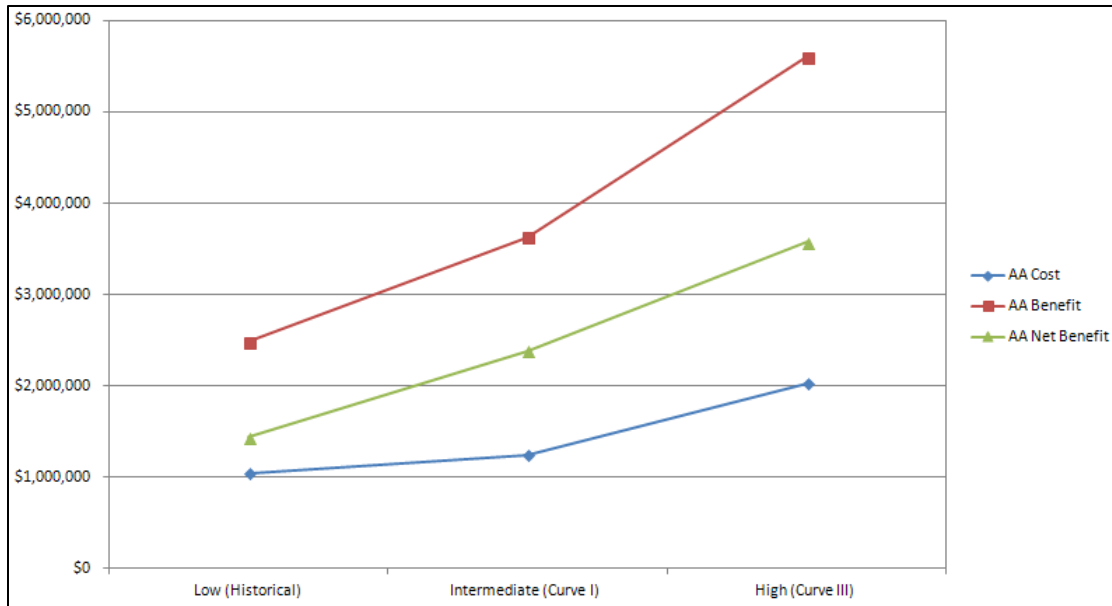


Figure B–43. Effect of future sea level change on economic performance of Alternative 4

d. From an engineering perspective, future sea level rise will require more sand volume to maintain the designed project features. Figure B–44 shows the estimated fill volume requirements for initial construction and the 8-year interval renourishment for each of the three future sea level change scenarios. The figure shows that the future sea level change scenario has little effect on the initial construction volume but has a large effect on the average renourishment volume. For the low rate of sea level change, the average 8-year renourishment volume is estimated at 220,400 cubic yards; whereas for the intermediate rate of sea level change, the average 8-year renourishment volume increases by more than double to 450,500 cubic yards. For the high rate of future sea level change, the average 8-year renourishment volume increases nearly six times, to 1,278,300 cubic yards.

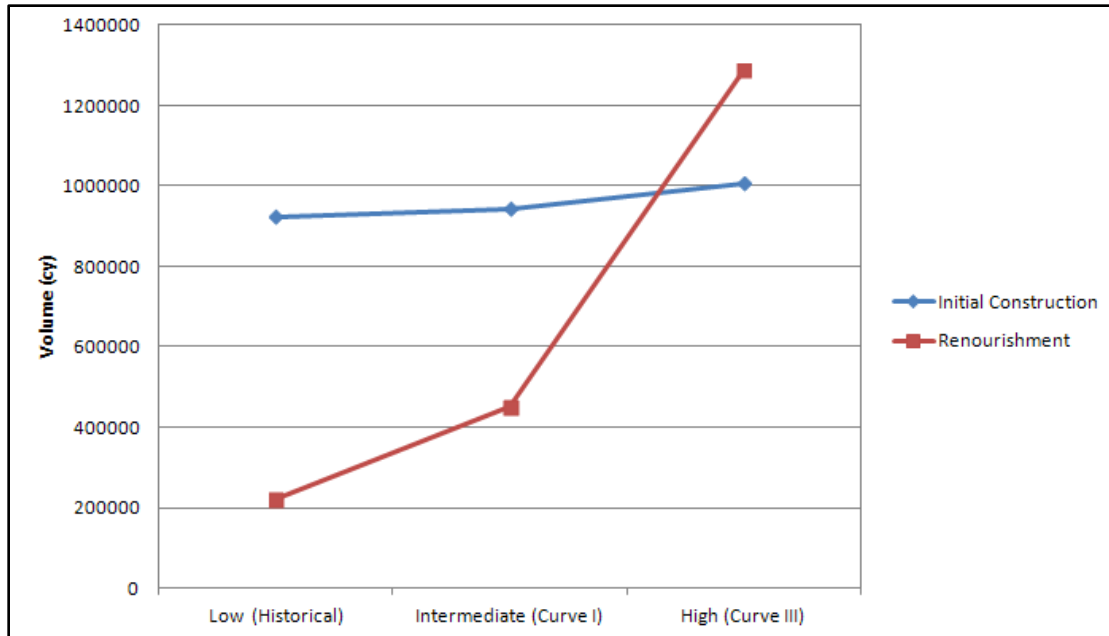


Figure B-44. Estimated fill volume requirements

B-15. Average annual net benefits distribution

a. Monte Carlo life-cycle simulation analysis allows evaluation of the economic justification risk of the proposed study through the investigation of each individual iteration output. As each iteration has a unique series of storm events with a corresponding unique calculation of inventory damages and periodic nourishment cost requirements, each iteration represents one possible future series of damages, costs, and net benefits. Risk analysis of the net benefits outputs provides the probability for a favorable outcome for the study. In other words, the probability for positive AANB after the 50-year period of analysis.

b. Figure B-45 shows the raw AANB output for all 300 iterations. Of the 300 iterations, 28 possible futures had negative AANB or failed economic justification. The remaining 272 iterations had positive AANB. Given the inputs to the model, the proposed alternative has a 90.7% probability of positive AANB with an average of \$1,619,473.

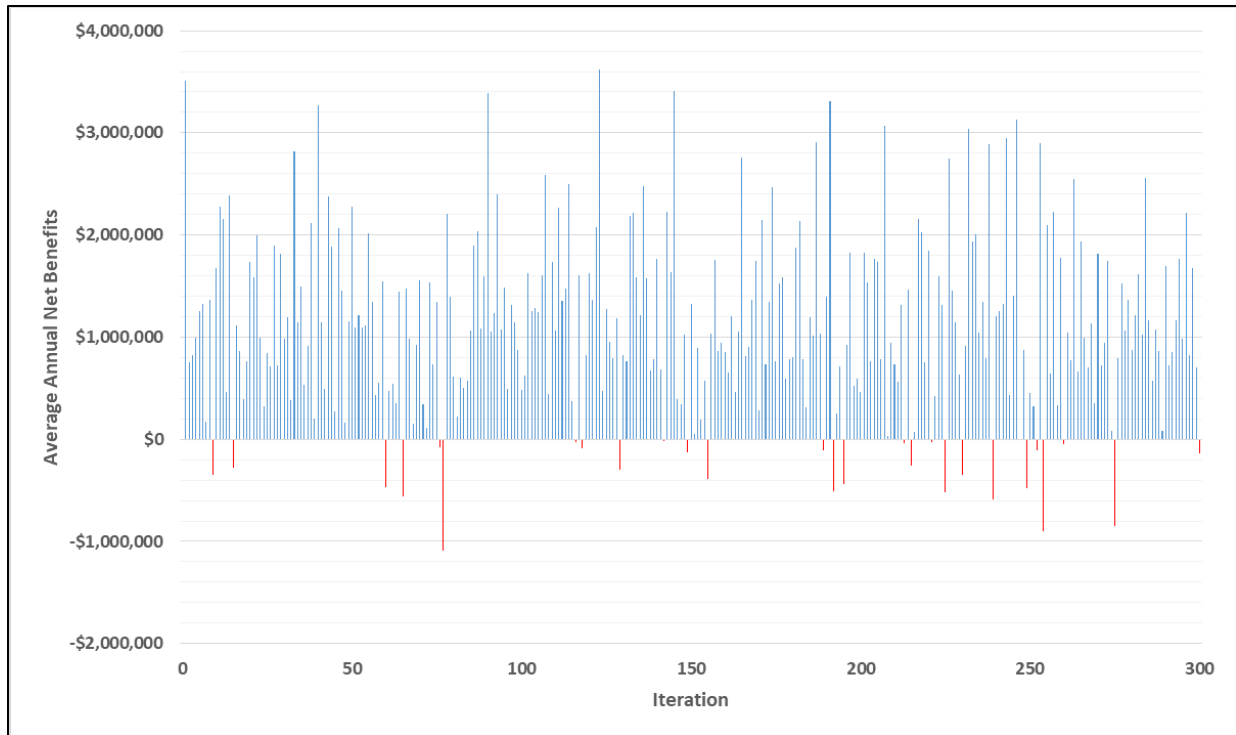


Figure B–45. Average annual net benefits distribution by iteration

c. Figure B–46 provides the same information, with the raw data provided as a frequency distribution with a cumulative probability function. Results follow a Normal distribution with a slight right-tail skew.

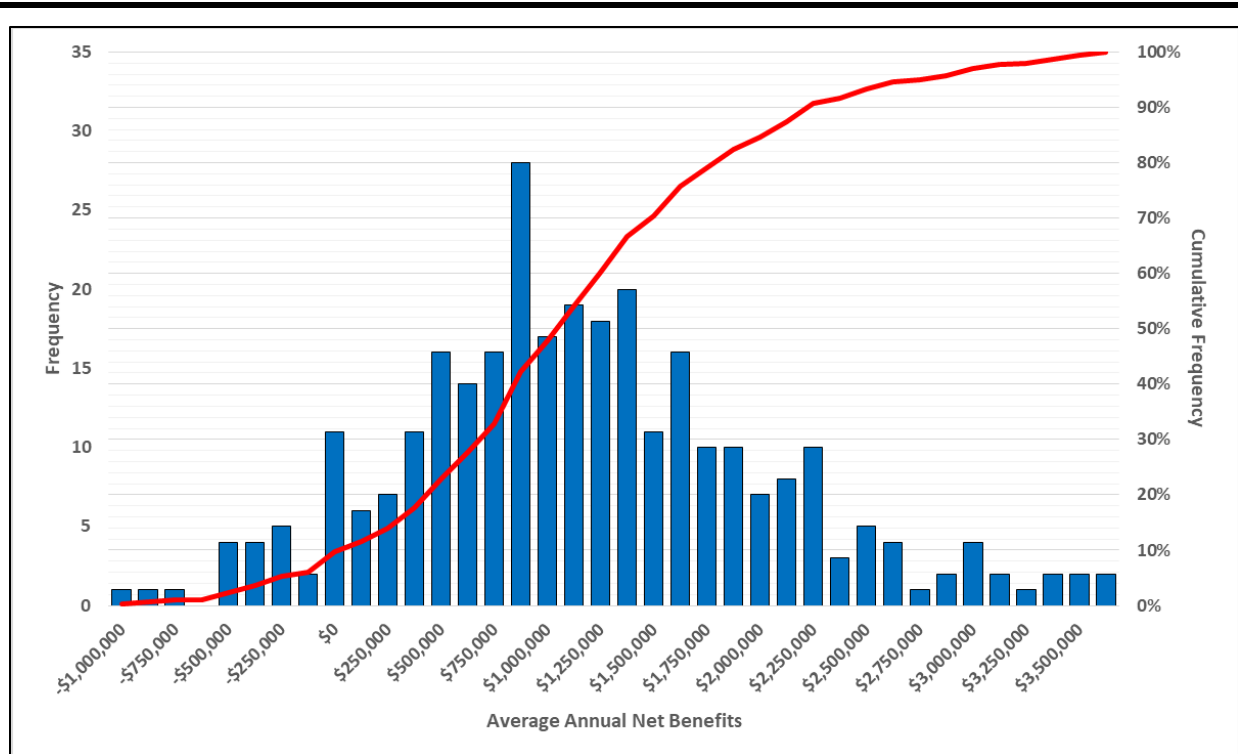


Figure B-46. Average annual net benefits frequency distribution

B-16. Uncertainty

The above analysis did not explicitly address epistemic uncertainty. Hydrodynamic and morphology model uncertainty and other contributions could be added within a life-cycle model by sampling from uncertainty distributions as discussed in Chapter 12. Although not available at the time of the above study, uncertainties associated with the storm hazard could be acquired from the CHS (USACE 2024a). This information includes confidence values. A Normal distribution of overall uncertainty could be constructed knowing the mean and standard deviation for each parameter. Then this distribution could be sampled and superimposed on the storm hazard at each time step of the life cycle. Other processes, like morphology, may contain considerable uncertainty so inclusion of some level of uncertainty evaluation is essential.

Appendix C

Hydrologic Engineering Studies Design

C–1. Introduction

a. Purpose.

(1) Content originally found in EP 1110-2-9 has been consolidated into this appendix. Current USACE planning policy for conducting Civil Works planning studies must be considered the final authority on any conflicting guidance found in this appendix.

(2) Current planning and management guidance for Civil Works studies requires careful technical activity management. Hydrologic engineering is a critical item, especially for feasibility investigations.

(3) Successful study completion requires management of time, money, and human resources to accomplish the necessary technical studies in an effective manner. Hydrologic engineering study products must satisfy study team and project sponsor needs. The technical studies must also be completed within available financial resources. It is important to plan the technical work at the beginning of the study to accomplish these requirements. Development of a hydrologic engineering management plan (HEMP) for the study can be a useful step toward accomplishing these objectives. This document describes activities involved in designing and preparing a HEMP for a USACE water resource investigation.

b. Applicability.

(1) This appendix is applicable to HQUSACE elements, major subordinate commands, districts, laboratories, and separate field operating activities having Civil Works responsibilities in the hydrology and hydraulics fields.

(2) The term hydrologic engineering is used throughout this appendix to encompass activities in hydrology, hydraulics, water control management, sediment analyses, water quality, or other water-related elements addressed in a USACE water resource investigation. Similarly, the HEMP plan includes all activities in these disciplines.

C–2. The hydrologic engineering management plan

a. General. The HEMP is a technical outline of the hydrologic engineering studies necessary to formulate a solution to a water resource problem. A HEMP could be an initial or detailed work outline. An initial HEMP is developed to define key issues and activities sufficient to address study time and cost. A detailed HEMP outlines significant technical activities in sufficient detail for the responsible engineer to perform the analysis.

b. *Use.* The use of a HEMP plan is threefold. It can provide the following:

(1) *Basis for time and cost estimates.* The HEMP should reflect the hydrologic information needs of the study team and define the methods for proceeding through the entire study process. Agreement between hydrologic engineering, planning, and project management on the study scope must be reached for an accurate cost estimate.

(2) *Technical guide for the hydrologic engineer.* Many feasibility-phase or preconstruction, engineering and design (PED) investigations require extensive hydrologic engineering effort. Even an experienced hydrologic engineer cannot foresee all facets of complex analyses without significant planning and input from others. A detailed technical outline allows work to be performed effectively and efficiently, without close supervision. The preparation and use of a HEMP provides inexperienced hydrologic engineers with a clear understanding of the analysis procedures and reporting process. A HEMP provides the basis for meaningful discussion and negotiation and helps in making decisions on refinements and changes as the study progresses.

c. *Personnel involved.* The HEMP is a hydrologic engineering document usually prepared by the principal engineer for the study. It is not required or approved by other disciplines, but must incorporate information needs of all disciplines. The hydrologic engineer plays the most important role in its development, but others have input as well. They include:

(1) *Senior personnel.* The HEMP may be prepared by an experienced engineer or a section chief to ensure that the time and cost estimates are adequate and that they address all study issues. Supervisors should review and critique the HEMP.

(2) *Project manager (PM).* Discussions on the effect of the HEMP to the overall project schedule, cost, budget, and all other project processes should be held with the PM who is responsible for the progress of the project.

(3) *Planning technical manager.* Discussions with the planning technical manager on the alternatives to be addressed, level of detail, combinations of different alternatives, study milestone dates, and other pertinent information should be held, agreed upon, and incorporated into the HEMP.

(4) *Economist.* The economist is important in establishing the type of analysis required. For example, significant agricultural damages require knowledge of the time of year and duration of the flood and typically use a continuous simulation analysis, while a study area with primarily urban damages could use an event analysis. Information on damage areas is necessary to ensure that planned hydrologic results support stage-frequency information where needed. This type of information is required to perform risk-based analysis for project alternatives in close coordination with economists and all other team members.

(5) Local sponsor. The sponsor always has useful hydrologic information on the study area. The sponsor may also have definite views on the type and size of flood damage alternatives most suitable for the investigation.

(6) Cost engineer. The cost engineer must have hydrologic engineering design information in sufficient detail to prepare project costs.

(7) Realty specialist. The real estate specialist needs stage-frequency, area inundated, project location, and other information to complete the analysis.

(8) Other team members. Structural, geotechnical, mechanical engineers, regulatory personnel, the recreational planner, and environmental specialist usually need specific information from the hydrologic engineer.

(9) Review authority. Controversial, complex, or costly hydrologic analyses should be discussed with Division and possibly HQUSACE hydrologic engineering personnel to confirm the approach and procedures proposed. This should be accomplished both informally and through the official technical review process.

d. Hydrologic engineering management plan. The HEMP should outline feasibility study hydrologic engineering activities so that time and funds needed may be estimated during the feasibility-phase study. Similarly, a HEMP can be prepared at the end of the feasibility-phase study to establish hydrologic engineering time and costs necessary for PED. It is assumed throughout this document that HEMPs will be prepared for both the feasibility and PED phases, but this does not mean that both HEMPs are always required. In fact, if a detailed HEMP can be developed during the feasibility phase, it should be done.

C-3. Scoping the investigation

a. General. A preliminary assessment should be made to gain an understanding of the key issues and concerns to be addressed in the analysis. This assessment leads to an initial HEMP, based on the main considerations of the study.

b. Study objective. The major study objectives should be defined: flood risk management, navigation, water supply, environmental restoration, water control management, hydropower, etc. Geographic scope of the study should be determined and key locations requiring hydrologic information specified. Preliminary hydrologic engineering requirements and strategies to accomplish these objectives may be postulated.

c. Type of study. Different types of studies require different levels of hydrologic planning. The various studies for which a HEMP may be developed are described in the following sections, with feasibility and preconstruction engineering and design forming the usual path for most USACE studies.

(1) *Feasibility-phase study.* As soon as feasibility funding is received, an initial HEMP may be developed to detail the timing and level of effort for hydrologic engineering activities for use by the hydrologic engineer throughout the study. Technical studies are detailed so that work activity durations may be established: milestone dates set, etc. Examples of HEMPs are shown in paragraphs C–6 through C–9. This phase results in a feasibility report with a series of engineering appendixes. The appendixes are in sufficient detail to allow the work effort to proceed directly to the design memorandum phase.

(2) *Preconstruction engineering and design.* The PED phase concentrates on the detailed design of the project. It is normally expected to consist of one or more design documents and plans and specifications to construct the project.

(a) *Design memoranda.* The project should move directly from the feasibility phase to PED, with a design memorandum to establish the detailed technical design necessary to construct the project. Consequently, a detailed HEMP should be prepared at the start of PED to outline the balance of the technical hydrologic engineering effort. This detailed HEMP should build on the initial HEMP prepared for the project management plan (PMP). PED hydrologic activities often include physical model testing, detailed hydraulic design, quantitative sediment transport analysis, two-dimensional flow analyses, etc., which provide the technical detail for final design of the project.

(b) *Plans and specifications.* The detailed HEMP prepared at the start of PED should include the hydrologic activities necessary for this phase of the project. Hydrologic engineering effort in the plans and specifications phase typically incorporates results of physical model tests into the hydraulic design, preparation of stage hydrographs to show potential high-water periods affecting construction, minor modifications in the hydraulic design based on additional detailed topographic and soils data obtained, and any changes found in the site conditions.

(3) *Regulatory.* An assessment of the impact of a proposal is necessary to obtain a permit for project construction in the floodplain. The hydrologic information needed to submit the permit for USACE projects should be readily available from previous work. A HEMP for regulatory purposes should seldom be necessary.

(4) *Water control management.* Establishing a water control management plan for a new project or updating an existing plan for new or changed purposes represents a major hydrologic engineering effort. The plan is described in a water control manual. Funding is usually from the operation and maintenance (O&M) program for an existing project. General investigations funding is appropriate for analyzing the addition of new project purposes to an existing project. A HEMP can be useful to establish time and cost estimates for O&M funding.

(5) *Water supply.* These investigations normally concentrate on potential reservoir storage reallocation for water supply or for drought operation planning. The HEMP recommendations are similar to those described in paragraphs C–3.c(1) and C–3.c(2).

Drought operation planning is usually done for an existing project with O&M funding. A HEMP may be useful for accurate estimates.

d. Key items to evaluate.

(1) *Major issues.* The HEMP should outline the information and methods necessary to address the major issues of the hydrologic engineering study. Methods and procedures needed to address complex or precedent-setting problems, sensitive environmental concerns, use of outside consultants (including USACE labs), local sponsor requirements, the need for new physical or analytical model development, adverse effects caused by a potential project, etc., should be scoped for budgeting purposes.

(2) *Level of detail.* Although the study phase usually establishes the overall level of detail, the interdisciplinary planning team must be queried to obtain their ideas on the hydrologic information they need. However, the hydrologic engineering effort often plays the largest role in determining the level of detail. Depending on the appropriate study costs, several iterations between the hydrologic engineer and the study team may be necessary to establish a level of detail proportional to the level of study funding. Adequate planning at the start of the study may result in lower overall hydrologic engineering costs.

(3) *Hydrologic information availability and requirements.* Databases should be examined to determine the rainfall, streamflow, topographic, and other records available for the particular study. The need for establishing a limited data collection program to address the objectives of the study should be determined. Existing federal and nonfederal projects (reservoirs, levees, water withdrawals, etc.) affecting the analysis should be determined.

(4) *Unusual features.*

(a) *Items requiring additional engineering effort.* Items peculiar to the study area that require additional hydrologic engineering effort must be addressed, especially if the work is necessary in the feasibility investigation.

1. Complex channel and floodplain hydraulics could require a two-dimensional, unsteady flow analysis, resulting in higher study costs compared to a simpler one-dimensional analysis.

2. Major quantitative sedimentation investigations may be necessary to firmly establish project feasibility. Reservoirs and extensive channel modifications may require significant quantitative sediment investigations during the feasibility phase.

3. Physical model testing may be required during feasibility to ensure the workability of a project, such as locating a replacement lock away from the main navigation channel or designing a supercritical flow channel for a highly populated area.

4. Lake stage-frequency analysis in closed basins that do not drain to a downstream watershed.
5. Major groundwater, snow hydrology, water quality, or other investigations.
6. Complex reservoir system problems in which political or environmental issues mandate extensive and unusual systems modeling.
7. Unstable rating relationships, complex interior flood control studies, multi-reservoir analyses, and other difficult water resource analyses must be recognized and evaluated during the early planning process leading to a HEMP.

(b) *Expert review.* Studies having unusual features and complex analyses may benefit from expert review. Committees on hydrologic engineering, river engineering, tidal hydraulics, and water quality are available for assistance on unusual features or challenges. Committee membership consists of selected senior hydraulic engineers from Districts and Divisions. ER 15-2-14 further describes these four committees.

(5) *Study boundaries.* The HEMP must distinguish between study boundaries and project boundaries in the development of estimates. Project effects often extend far upstream and downstream on the main stem of the study stream as well as up tributaries. Proposed projects may change the flood hydrology and sediment regime throughout the watershed, not just near the proposed project. Changes in water control management practices at USACE reservoirs can also affect interests remote from the reservoir site. The hydrologic analysis must include the evaluation of all positive and negative effects of a potential project or water control management change throughout the stream system or study area.

(6) *Likely alternatives.* The cost of the hydrologic engineering work effort in the HEMP should reflect the expected alternative(s) or combinations of alternatives. The major with-project scenarios must be developed by the study team for both preliminary and final scoping of the technical activities. The no-action case must also be determined for comparison to the with-project alternatives. Similarly, agreement should be reached among study team members, during the HEMP preparation, concerning the number of iterations (or sizes) to be evaluated for each alternative. Three or four sizes for each of two or three alternatives should be adequate for most studies to determine the optimal size.

e. *Major hydrologic engineering activities required.* USACE typically assesses with- and without-alternative conditions for the main study objective(s).

(1) *Flood risk management.* The HEMP should describe or reference the major study components: watershed hydrology, river hydraulics, frequency analysis, sedimentation analysis, storage operation, hydraulic design, etc., for both the with- and without-alternative condition. Analysis often involves discrete events, either actual or, more typically, hypothetical, and should include development of uncertainty relationships for risk-based analysis. ER 1105-2-100 contains additional information about this topic.

(2) *Water control management.* The HEMP should describe or reference the major study components: flood control capabilities, storage allocated for various project purposes, drought augmentation, operational analyses, data systems, forecasting, etc., for the existing and proposed method of regulation. Analyses usually involve discrete events and continuous record techniques. ER 1110-2-240 and EM 1110-2-3600 contain additional information in this area.

(3) *Water supply investigations.* The HEMP should describe or reference the major study components: existing project purposes and storage allocations of each, upstream and downstream demands, supply analysis, hydraulic data (uniform database), drought frequency analysis (volumes and durations), distribution system (pumping, conveyance, and storage), etc., for the existing and proposed reallocation of reservoir storage. Analysis may be for one or more severe droughts, although the full period of record can be used, similar to water control management methods. ER 1110-2-241 and ER 1110-2-1941 contain additional information.

f. *Primary hydrologic engineering investigation products.* The hydrologic engineering results needed by the study team may include the following information, as discussed by general study type:

(1) *Flood risk management.* The main product is the flood risk management effect of the selected alternative on the floods in the watershed. Supplemental investigation products could include discharge-frequency relationships, flood elevations, and areas inundated with and without a specified structural alternative (reservoir, channel, levee, diversion, pumping plant), stage-duration relationships, sizes of various alternatives for costing purposes, sedimentation analyses, residual flooding, flood forecasting and warning system, etc. Nonstructural alternatives may require only the without-project condition, since these alternatives affect the stage-damage relationship only and result in little, if any, change in hydrologic or hydraulic relationships.

(2) *Water control management.* The main product is a new or revised set of procedures for project operation and hydrologic forecasting, contained in a water control manual. A range of flows should be addressed and the effects on the watershed sediment regime could be a required product. Supplemental investigation products could include operation procedures, stage-duration and frequency, discharge-frequency, emergency operation procedures, gage data network, computer equipment needed, conservation and/or hydropower procedures, flood warning and preparedness procedures, and other required information.

(3) *Water supply investigations.* The main product is a reservoir storage reallocation to satisfy changed demands, such as decreased hydropower storage for increased water supply storage, or to satisfy as many critical demands as possible during time of drought. Water supply studies are usually performed for storage reallocations of existing reservoirs or for drought contingency planning for existing federal reservoirs. Drought contingency planning analysis is a separate study, but is usually included as an appendix to a reservoir water control manual. Supplemental investigation products could include both seasonal and annual: current and modified

condition discharge-frequency, reservoir storage-frequency, pool elevation-duration, flow- or storage-duration relationships, and power generation values.

C-4. Development of an initial hydrologic engineering management plan

a. *General.* A typical strategy first includes a preliminary assessment identifying the problems and issues within the study area. This assessment should result in the preparation of an initial HEMP, sufficient to scope time and funding requirements. The initial HEMP should include appropriate contingencies to establish total hydrologic engineering cost for inclusion in the initial project management plan. A senior hydrologic engineer should develop the initial HEMP, while the responsible engineer could expand this document into a detailed HEMP. The overall process is summarized in Figure C-1.

b. *Field inspection.* An early field inspection is recommended to become familiar with site-specific problems that should be incorporated in the HEMP. Field inspection should focus on any features causing analysis problems, ongoing changes in the study area, interviews with locals concerning past flood experiences or changes to the area since large past floods, contacting local agencies to obtain information on the area and on any plans for modifications that could affect the USACE analysis, and other items of interest.

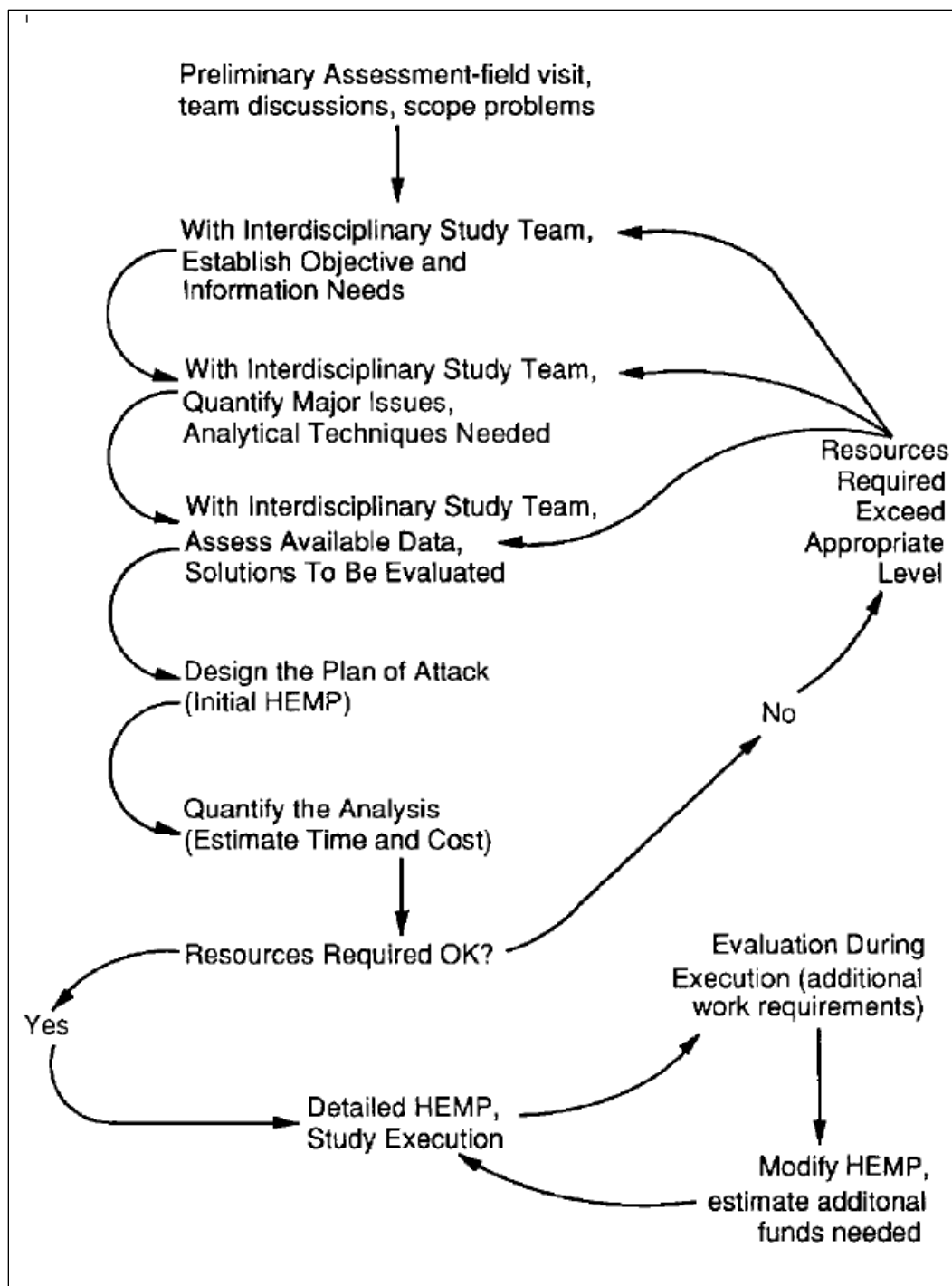


Figure C-1. Hydrologic study design

c. *Coordination.* Various coordination and information needs should be addressed in the HEMP.

(1) *Study team needs.* The various hydrologic information needs of the interdisciplinary planning team have been briefly described in paragraph C–2.c. Anticipated hydrologic information needs should be obtained from the study team for inclusion in time and cost estimates and schedule for the feasibility phase.

(2) *Sponsor needs.*

(a) The sponsor usually has valuable information about the study area. The sponsor may have some capability for obtaining necessary information pertinent to the project or for performing some of the hydrologic engineering necessary for the study, which could be a credit to the sponsor. The cost-sharing partner normally has specific views on the type of alternatives believed most suitable for the study area.

(b) An initial HEMP is useful in dealing with the local sponsor on necessary hydrologic engineering activities and in justifying the hydrologic engineering cost estimate, which the sponsor must cost share. Discussing the necessary hydrologic activities, summarized in a HEMP, with the sponsor is more likely to result in agreement on the effort involved than simply presenting the sponsor with a total cost.

d. *Collecting information.* The hydrologic engineer must evaluate the available data as well as estimate what additional data are necessary for conducting the study. The availability of meteorologic, hydrologic, hydraulic, sediment, water quality, and infrastructure data should be determined; sources and quality of such data evaluated; and any special needs for a limited data collection program determined. Topographic information necessary to develop accurate water surface profile information should be estimated, if not available from other sources.

e. *Basic analysis approaches.* The analysis approach must be based on the hydrologic information needs of the study team, unusual features of the study, the type of alternatives requiring investigation, the significance of the alternatives on the sediment regime, and other considerations.

(1) Selection of the appropriate hydrologic models and whether a qualitative or quantitative sediment analysis is needed must depend on the judgment and skills of the responsible hydrologic engineer. Models and procedures should be selected based on the reduction of uncertainty in the end product. If a sophisticated model or procedure does not give a significantly improved result and reduced uncertainty, a less sophisticated method is probably appropriate.

(2) Selection of new models or procedures could include an allowance for assistance by the HEC, the ERDC, or other consultants. Higher authority should approve new models and innovative, unusual procedures. Expert review by the appropriate HQUSACE-sponsored committee should also be considered for unusual or complex analyses.

f. Initial hydrologic engineering management plan preparation. Using information from the preliminary assessment, identify the major activities, including alternatives to be analyzed and the range of sizes to study. The initial HEMP can be used to estimate human resource requirements for each activity to establish a total hydrologic engineering study cost. An example of an initial HEMP for a flood control study is presented in paragraph C–6.

g. Time and schedule estimates. With the initial HEMP, determine the human resources required for each major technical hydrologic engineering study component (rainfall analyses, water surface profile, channel modifications, etc.) and for the complete hydrologic engineering effort. Estimate the human resources necessary for each discipline (hydrologic engineer, technician, supervisor, etc.). Estimate when necessary information must be furnished to or received from other study team members. Clearly indicate the number of alternatives to be evaluated and the number of sizes to analyze for each alternative. Determine if special training is necessary for the responsible engineer to effectively perform the study. Include any other factors having a significant impact on required time for the hydrologic engineering analysis, along with any assumptions on which the estimate is based. Include a reasonable contingency allowance.

h. Labor resources.

(1) Through an iterative process, come to agreement with all concerned on study objectives, analysis approaches, alternatives to be analyzed, sizes to study, and level of detail obtainable with funding constraints. Prepare written documentation on this agreement and include any problems, difficulties, or lack of engineering detail that may result from this reduced effort. Finalize these activities in the HEMP for inclusion in the initial PMP. Reference these changes and agreements in the HEMP or in separate documentation.

(2) The responsibility falls on the hydrologic engineer to ensure that the expended time and funding correspond with the agreed-upon amounts. Additional hydrologic work required by the interdisciplinary planning study team or sponsor during the feasibility or design phase must result in additional resources being made available by the project or study manager.

C–5. Development of a detailed hydrologic engineering management plan

a. General. As a project progresses, it may be necessary to further build on the initial HEMP and develop a fully detailed HEMP.

b. Document contents. The detailed HEMP for the feasibility phase should document the step-by-step analysis of data development, storm studies, routing operations, unit hydrograph development, model calibration, etc., while the detailed HEMP for the PED phase should document the hydraulic design, physical model testing requirements, hydrologic studies needed for water control manuals, detailed sediment investigations, multi-dimensional modeling efforts, etc. Paragraphs C–6 through C–9

give generic examples of different HEMPs. The document should be reviewed and approved by the immediate supervisor. Complex analyses may benefit from a higher level review. A courtesy copy should be provided to the study or project manager.

c. *Activity sequence.* For scheduling purposes, assign durations to each work item for all significant activities in the entire HEMP. Establish milestone dates for each major component. Use milestone dates for furnishing necessary information to various members of the interdisciplinary planning team, receiving information required from others to meet the study completion date, and meeting any other scheduled events.

d. *Hydrologic study management.* The detailed HEMP could be used on a day-to-day or week-to-week basis, especially for relatively new hydraulic engineers. Periodic updates and further detailing of tasks should be performed as work progresses or as additional effort becomes necessary to meet new interdisciplinary planning team or local sponsor requests. Additional work effort required by others that is not included in the HEMP becomes the basis for increases in hydrologic engineering time and cost allocations. This situation emphasizes the need for a HEMP and for the responsible hydrologic engineer to refer to it throughout the study.

e. *Study documentation and reporting.* Update the HEMP for final documentation of the completed technical study so that future, similar work will have a better planning basis. Recording the actual human resources expended for each major activity is particularly useful for estimating future studies. Use the completed HEMP for the outline of a technical hydrologic engineering appendix to the study documentation.

C-6. Example: Initial hydrologic engineering management plan for a flood damage reduction feasibility study

a. *Background.* This example was provided in EP 1110-2-9, dated July 1994. As a result, the example includes references to both planning study processes and hydrologic engineering software that have been superseded for modern planning studies. The example is provided here as a reference. Current methods and tools should be confirmed with the appropriate higher authority.

b. *Scenario.*

(1) The study objective is the development of a flood protection plan for a community experiencing periodic flooding from a stream draining a few hundred square miles. The preliminary phase of study was based primarily on flood insurance study data and an abbreviated hydrologic engineering analysis. The study has determined that a levee project is economically feasible. The community is willing to be the cost-sharing local sponsor and would like a minimum certifiable level of protection of a 1% AEP event. A gage with 15 years of discharge data is available at the site, with additional, short-record gages located elsewhere in the watershed.

(2) The feasibility phase establishes existing and future without-project conditions. Discussions with the interdisciplinary study team and local sponsor decided that three heights of levee will be studied, along with combinations of levee height and channel

improvements to develop the economic optimum plan. A number of alternatives will be evaluated. As all the levee alternatives are along a similar alignment, a detailed interior flood analysis will be evaluated for only the NED levee or levee and channel plan. The hydrologic engineer must prepare an initial HEMP for the hydrologic engineering cost estimate for the feasibility phase.

(3) This sample initial HEMP represents what one might develop at the end of the preliminary phase of study for a time and cost estimate for use in the initial project management plan.

c. Preliminary investigations/initial preparation. Finalize study objectives; confer with the study team members on hydrologic engineering information requirements, study constraints, development information needs, and field reconnaissance; prepare survey data request; prepare detailed HEMP.

d. Development of basin model Hydrologic Engineering Center.

(1) Calibration of runoff parameters. Using basin gage data, develop unit hydrograph and loss rate parameters for use in the study.

(2) Delineation of subareas. Subdivide study watershed based on the need for discharge-frequency information at specific locations: major tributaries, damage index points, routing reaches, project sites, etc.

(3) Subarea rainfall-runoff analysis of historic events. Develop historic storm events, and subarea loss rate and unit hydrograph data for ungaged areas.

(4) Channel routing characteristics. Determine based on information in paragraph C–7.e(4).

(5) Assemble, debug the HEC-Hydrologic Modeling System (HEC-HMS) model.

e. Hydraulic studies.

(1) Prepare water surface profile data. Code HEC-RAS model of study reach after receipt of surveys. Estimate n values, section locations, bridge routines applicable, effective flow areas, etc. Debug model.

(2) Calibrate HEC-RAS model to gage data and high-water marks from recent floods. Calibrate the HEC-HMS and HEC-RAS models to recorded events and high-water marks. Make preliminary selection of hydrologic and hydraulic model parameters for hypothetical flood event analysis.

(3) Develop storage-outflow relationships and flood wave travel time, by routing reach, for information required in paragraph C–6.d(4).

f. Calibration of models to historic events.

g. Frequency analysis for existing land-use conditions.

(1) Perform statistical analysis of gaged data for peak discharge-frequency relationship. Also estimate discharge-frequency relationships through available/applicable regression equations at key locations to use in later comparisons.

(2) Develop hypothetical frequency storm data from the NOAA and National Weather Service (NWS) technical publications. Develop rainfall pattern for each storm, including precipitation depth-area adjustments. Develop corresponding hydrograph for each hypothetical event throughout the basin using the calibrated hydrologic model of paragraph C–6.f.

(3) If judged appropriate, further calibrate model to reproduce the peak discharge calculated from the statistical analysis at the gage site. Emphasize the two-year through ten-year event since the data record is short. Make adjustments to loss rates and unit hydrograph coefficients for rarer events, as judged reasonable. Compare results to statistical and regression-derived peak discharge frequency relationships; further adjust coefficients as considered reasonable.

(4) Using the results of steps in paragraphs C–6.g(1) through C–6.g(3), adopt a discharge-frequency relationship at each needed location. Develop probability distribution of discharge uncertainty for use in risk-based analyses.

(5) Determine corresponding water surface profiles and inundated areas for selected frequencies at required locations. Furnish data to planning and economics.

(6) Adopt stage-discharge relationship at each required location for damage computations. At the gage site with 15 years of data, determine deviations about the adopted stage-discharge relationship. Further evaluate through sensitivity studies. Develop probability distribution of stage uncertainty for risk-based analysis.

h. Future without-project analysis. Determine future stage-discharge relationships, based on future watershed changes affecting the hydraulics. If necessary, adjust discharge-frequency and stage-discharge risk/uncertainty relationships. Furnish data to economics.

i. Levee alternative evaluations. For the preliminary levee alignment, develop revised discharge-frequency and stage-frequency relationships for each of the different levee heights. If judged necessary, determine revised stage-discharge risk/uncertainty relationship. Roughly size a “minimum-facility” interior flood control system for each. With the economist, perform risk and uncertainty studies to establish the claimable level of protection (risk-based) and average annual benefits resulting for each.

j. Levee and channel alternative evaluation.

(1) For various sizes of channel, reestablish stage-frequency relationships for each of the levee sizes. Evaluate the discharge and stage uncertainties for with-project conditions. Roughly size a minimum-facility interior system for each alternative, if

necessary. With the economist, perform a risk-based analysis to determine project benefits and claimable level of protection for each alternative. Perform qualitative sediment analyses for channel modifications to roughly determine dredging frequency for channel maintenance. After economic analysis to tentatively establish the NED plan (levee height) from among the alternatives, design TOL grade for controlled overtopping.

(2) If a channel modification is included in the NED plan, perform sensitivity tests to determine the importance of channel maintenance assumptions and costs on the NED plan. If a more conservative sedimentation analysis results in significant cost increases, possibly invalidating the NED plan, additional sediment analyses will be required in feasibility. Hydrologic engineering work for a quantitative sediment analysis is not included in this estimate. Adjust final levee grade for any sediment effects.

(3) As necessary, furnish hydrologic information as it becomes available to other study team members: stage-duration and frequency to environmental, data for Environmental Assessment Report, etc.

(4) Nonstructural analysis of emergency procedures in case of levee overtopping, evacuation, and flood warning.

k. Residual flooding and interior flood control.

(1) Establish residual flooding for remaining flood damages with the NED project. Evaluate higher levels of interior flooding protection compared to the minimum facility. Interior flood control measures are distinguished from minimum facilities in that these additional measures require net benefits and minimum facilities do not require incremental economic justification, only cost-effective design.

(a) Using the appropriate tools, evaluate gravity outlets larger than the minimum facility at each of the gravity drain locations.

(b) Evaluate interior excavated storage at the only site where it is currently thought feasible.

(c) Evaluate capacities of pumping plants at each of the sites.

(d) Evaluate interior improvements for the main drainage ways.

(2) Forward data to an economist and cost engineer for each increment. Supply hydrologic data for wetland determination and mitigation, as necessary.

l. Hydraulic studies. Some of the design work will have already been incorporated in the above activities.

(1) Levees. Levee design profile, controlled overtopping design, gravity drain design for minimum facility, etc.

(2) Channels. Channel geometry, bridge modifications, scour protection, channel cleanout requirements, channel and bridge transition design, etc.

(3) Drains. Size, slope, material, inlet/outlet, operation procedures, etc.

(4) Pumping. Capacities, start-stop pump elevations, sump design, outlet design, scour protection, operating floor elevations, etc.

m. Hydrologic engineering reporting requirements.

(1) Project management plan. Estimate major hydrologic engineering activities in the PED phase, prepare initial HEMP for PED work, prepare time and cost for hydrologic engineering, activity schedule.

(2) Hydrologic engineering appendix to the feasibility report. Using the detailed HEMP as appropriate, outline and write the text, prepare tables and figures.

(3) Environmental assessment report. Provide data to environmental section. Supply text, figures, plates, as needed.

C-7. Example: Generic detailed hydrologic engineering management plan for a flood damage reduction study

a. Background. This example was provided in EP 1110-2-9, dated July 1994. As a result, the example includes references to both planning study processes and hydrologic engineering software that have been superseded for modern planning studies. The example is provided here as a reference. Current methods and tools should be confirmed with the appropriate higher authority.

b. General. This sample detailed HEMP is appropriate for the hydrologic and hydraulic analysis associated with a typical USACE feasibility report for an urban watershed. It should be prepared at the end of reconnaissance-phase study or start of the feasibility phase. The intent of the hydrologic engineering analysis is to determine existing and future discharge-frequency and stage-discharge relationships at key points in the study area, along with flooded area maps by frequency. This analysis should be performed without project and for various flood reduction components that are considered feasible for relief of the flood problem.

c. Preliminary investigations. This initial phase includes a literature review of previous reports, obtaining the available data, and requesting additional information needed to perform the investigation.

(1) *Initial preparation.*

(a) Confer with the other disciplines involved in the study to determine the objectives, the hydrologic engineering information requirements of the study for other disciplines, study constraints, etc.

(b) Scope study objectives and purpose.

(c) Review available documents.

1. USGS reports.
2. Previous USACE work.
3. Local studies.
4. Hydrologic engineering analysis for reconnaissance report.
5. Initial project management plan.
6. Other.

(d) Obtain hydrologic (historic and design discharges, discharge-frequency relationships, etc.) and hydraulic (high-water marks, bridge designs, cross sections, etc.) data.

1. Local agencies (city/county highway departments, land-use planning, etc.).
2. State (state highway departments, planning agencies, water resource agencies, etc.).
3. Federal USGS, U.S. Soil Conservation Service, U.S. Bureau of Reclamation, etc.
4. Railroads.
5. Industries.
6. Other.

(e) Scope major hydrologic and hydraulic activities.

(f) Prepare detailed HEMP.

(g) Obtain study area maps.

1. County highway maps.
2. USGS topographic quadrangle maps.
3. Aerial photographs.
4. Others.

(h) Estimate location of cross sections on maps (floodplain contractions, expansions, bridges, etc.). Determine mapping requirements (orthophoto) in conjunction with other disciplines.

(2) *Field reconnaissance.*

(a) Interview local agencies and residents along the stream, review newspaper files, etc., for historic flood data (such as high-water marks, frequency of road overtopping, direction of flow, land-use changes, stream changes). Document names, locations, and other data for future reference.

(b) Finalize cross-section locations/mapping requirements.

(c) Determine initial estimate of n values for later use in water surface profile computations.

(d) Take photographs or slides of bridges, construction, hydraulic structures, and floodplain channels and overbank areas at cross-section locations. Consider dictating notes to a hand-held tape recorder to get a complete and detailed record.

(3) *Survey request.* Write survey request for mapping requirements and/or cross sections and high-water marks.

d. *Development of basin model using the Hydrologic Engineering Center Hydrologic Modeling System.* This phase of the analysis involves the selection of historic events to be evaluated, the development of runoff parameters from gaged data (and/or regional data from previous studies) and correlating these data to ungaged basins and the calibration of the basin model to historic flood events. This step assumes that at least some recording stream gage data are in or near the study watershed.

(1) *Calibration of runoff parameters.*

(a) Select historic events to be evaluated based on available streamflow records, rainfall records, high-water marks, etc.

(b) From USGS rating curves and time-versus-stage relationships for each event, develop discharge hydrographs at each continuously recording stream gage. Estimate peak discharge from flood crest gages.

(c) Develop physical basin characteristics (drainage areas, slope, length, etc.) for basin above each stream gage.

(d) Select computation time interval (t) for this and subsequent analyses. The computation interval must:

1. Adequately define the peak discharge of hydrographs at gages.
2. Consider type of routing and reach travel times.

3. Have three to four points on the rising limb of the unit hydrograph for the smallest subarea of interest.

4. Consider types of alternatives and future assessments.

(e) Using all appropriate rain gages (continuous and daily), develop historic storm patterns that correspond to the selected recorded runoff events for the basins above the stream gages.

1. Average subarea totals from isoheytal maps or total gage precipitation weightings.

2. Temporal distribution from weightings of nearby recording rain gages.

(f) Determine best-estimate unit hydrograph and loss rate parameters for each event at each stream gage by calibrating to recorded flood hydrographs.

(g) Make adjustments for better and more consistent results between events at each stream gage. Adjustments are made to:

1. Starting values of parameters and/or

2. Rainfall totals or patterns (different weightings of rain gages).

(h) Hold constant the most stable parameters, or relationships between parameters, and resimulate rainfall/runoff process to estimate other parameters.

(i) Adopt final unit hydrograph and base flow parameters for each gaged basin.

(j) Resimulate with adopted parameters held constant to estimate loss rates.

(k) Use adopted parameters of unit hydrographs, loss rates, and base flow to reconstitute other recorded events not used in the above calibration to test the correctness of the adopted parameters and to verify the calibration results.

(2) *Delineation of subareas.* Subareas are delineated at locations where hydrologic data are required and where physical characteristics change significantly.

(a) Index locations where economic damage computations are to be performed.

(b) Stream gage locations.

(c) General topology of stream system.

1. Major tributaries.

2. Significant changes in land use.

3. Significant changes in soil type.

4. Other.
 - (d) Routing reaches.
 - (e) Location of existing physical works (reservoirs, diversions, etc.) and potential location of alternate flood reduction measures to be studied.
 - (3) *Subarea rainfall-runoff analysis of historic events.*
 - (a) Subarea rainfall.
 1. Average subarea rainfall from isohyetal maps or weighting of total gage precipitation.
 2. Temporal distribution from weighing in line with information from nearby recording rain gages.
 - (b) Average subarea loss rates.
 1. From adopted values of parameter calibration.
 2. From previous studies of similar basins in the region.
 3. Others.
 - (c) Unit hydrograph parameters.
 1. From relationships based on calibration results at stream gages and physical basin characteristics.
 2. From previous regional study relationships of unit hydrograph parameters and physical basin characteristics.
 3. From similar gaged or known basins.
 4. From judgment, if no data are available.
 - (4) *Channel routing characteristics.*
 - (a) Modified Puls from water surface profile computations (HEC-RAS).
 - (b) Optimized from stream gage data (HEC-HMS).
 - (c) Adopted parameters from previous studies, experience, etc.
 - (d) Derived from reach hydraulics (Muskingum-Cunge) or other as appropriate.
 - (5) *Reservoir routing (if reservoirs are present).* This type of routing must be performed where storage has a significant effect on reach outflow values, with

reservoirs being the most notable example. However, one must also apply these techniques where physical features warrant, such as roads crossing a floodplain on a high fill, especially where culverts are used to pass the flow downstream.

(a) Develop surface area-capacity data (elevation-surface area-storage relationships).

(b) Develop storage-outflow functions based on outlet works characteristics.

(6) *Runoff hydrographs.* Using the subdivided rainfall-runoff model, including existing projects and the routing information of paragraph C–7.d, generate runoff hydrographs for previously selected runoff events at desired locations. Final calibration of the hydrologic model is described in paragraph C–7.e.

e. *Hydraulic studies.* These studies are used to determine water surface profiles, economic damage reaches, and Modified Puls channel routing criteria (if used). This example assumes that an evaluation was previously made that a steady flow-rigid boundary water surface profile analysis is appropriate.

(1) *Prepare water surface profile data.*

(a) Cross sections (tabulate data for each section).

1. Make cross sections perpendicular to flow (one-dimensional analysis) or establish grid based on two-dimensional analysis

2. Each cross section should be typical of the reach from half the distance to the next section both upstream and downstream of the current locations.

3. Develop effective flow areas. If Modified Puls routing criteria are to be determined from water surface profile analyses, the entire section must be used (for storage) with high n values in the noneffective flow areas.

(b) Refine n values from field reconnaissance and from analytical calculation and/or comparison with n values determined analytically from similar streams.

(c) Bridge computations. Estimate how high the selected floods will reach on each bridge and select either:

1. Normal bridge routine.

2. Special bridge routine.

(d) Develop cross sections above and below bridges to model effective bridge flow (use artificial levees or ineffective flow area options, as appropriate).

(2) *Proportion discharges.* Proportion discharges based on hydrologic analyses of historic storms and plot peak discharge versus river mile. Compute a series of water surface profiles for a range of discharges. Analysis should start below study area so that profiles converge to proper elevations at study limits. May want to try several starting elevations for the series of initial discharges.

(3) *Manual check.* Manually check all large differences in water surface elevations across the bridge, for example, greater than 3 feet.

(4) *Results.* The results are a series of rating curves at desired locations (and profiles) that may be used in subsequent analyses. Additional results are a set of storage versus outflow data by reach, which, along with an estimate of hydrograph travel time, allow the development of Modified Puls data for the hydrologic model.

f. Calibration of models to historic events.

(1) *General.* This study step concentrates on “de-bugging” the hydrologic and hydraulic models by recreating actual historic events, thereby gaining confidence that the models are reproducing the observed hydrologic responses. This effort continues from the activities described in paragraph C–7.d.

(2) *Calibration procedure.*

(a) Check historic hydrographs against recorded data; make adjustments to model parameters, and rerun the model.

(b) If no stream gages exist, check discharges at rating curves developed from water surface profiles at high-water marks.

(c) Adjust models to correlate with high-water marks.

(d) Adopt hydrologic and hydraulic model parameters for hypothetical flood event analysis.

(e) Quantify uncertainty of the stage-discharge relationship at each site where damage analysis is to be performed. As appropriate, use recorded gage data, comparison of profile to high-water marks, minimum deviation, and engineering judgement.

g. Frequency analysis for existing land-use conditions. The next phase of the analysis addresses how often specific flood levels occur at all required points in the study watershed. The procedures include developing discharge-and stage-frequency relationships at stream gages (when available) through statistical analysis using recorded peak discharges and at other required locations using available hypothetical storm data.

(1) *Statistical analysis.* Using the procedures described in Bulletin 17C, determine and plot analytical and graphical frequency curves at each stream gage. Adopt stage/discharge frequency relations at each gage. Regional relationships, regression analyses, and the results of hypothetical storm studies will be used to extend the records for rarer floods.

(2) *Hypothetical storms* (using HEC-HMS).

(a) Obtain hypothetical frequency storm data from NOAA and NWS technical publications, or from appropriate other sources. Where appropriate, develop the probable maximum storm.

(b) Develop a rainfall pattern for each storm. Include precipitation depth-area adjustments, where necessary.

(c) Develop a corresponding hydrograph for each hypothetical event throughout the basin using the calibrated hydrologic model.

(d) If deemed necessary, calibrate model of each frequency event to known frequency curves. Adjust loss rates, base flow, etc. as required, while remaining within reasonable limits for each parameter. The peak flow frequency at each ungaged area is assumed to be consistent with calibrated peak flow frequencies at gaged locations.

(e) If streamflow data are insufficient to develop analytical frequency curves, use the following procedure:

1. Obtain frequency curves from similar nearby gaged basins.
2. Develop frequency curves at locations of interest from previous regional studies (USGS, USACE, state, etc.).
3. Determine frequency hydrographs for each event from hydrologic model and develop a corresponding frequency curve at the locations of interest throughout the basin.
4. Plot all the frequency curves (including those using other methods if available) and, based on engineering judgement, adopt a frequency curve. The adopted curve may not be any of the developed curves, but simply the best estimate based on the available data.
5. Calibrate the hydrologic model of each frequency event to the adopted frequency curve. The frequency curve at other locations may be determined from the calibrated model results, assuming consistent peak flow frequencies.

(f) Quantify the uncertainty in the discharge-frequency relationship at all locations where damage computations will be performed. As appropriate, use gage data, regression equations, and calibrated models to determine equivalent length of record.

(g) Determine corresponding water surface elevations and profiles for selected frequencies from the rating curves developed by the water surface profile evaluations.

h. Future without-project analysis. Where hydrologic and/or hydraulic conditions are expected to significantly change over the project service life, these changes must be incorporated into the hydrologic engineering analysis. Urbanization effects on watershed runoff are the usual future conditions analyzed.

(1) From future land-use planning information obtained during the preliminary investigation phase, identify areas of future urbanization or intensification of existing urbanization.

(a) Types of land use (residential, commercial, industrial, etc.).

(b) Storm drainage requirements of the community (storm sewer design frequency, on-site detention, etc.).

(c) Other considerations and information.

(2) Select future years in which to determine project hydrology.

(a) At start of project operation (existing conditions may be appropriate).

(b) At some year during the project life (often the same year as whatever land-use planning information is available).

(3) Adjust model hydrology parameters for all subareas affected by future land-use changes.

(a) Unit hydrograph coefficients, usually reflecting decreased time-to-peak and decreased storage.

(b) Loss rate coefficients, usually reflecting increased imperviousness and decreasing infiltration characteristics.

(c) Routing coefficients, usually reflecting decreased travel times and storage capabilities.

(4) Operate the hydrology model and determine additional discharge-frequency relationships throughout the watershed that represent future, without-project conditions.

(5) Evaluate the need to adjust uncertainty parameters of stage-discharge and discharge-frequency relationships, compared to existing conditions.

i. Alternative evaluations. For the alternatives jointly developed with the members of the interdisciplinary planning team, modify the hydrologic and/or hydraulic models to develop the effects of each alternative (individually and in combination) on flood levels. Alternatives can be either structural (reservoirs, levees, channelization, diversions,

pumping, etc.) or nonstructural (flood forecasting and warning, structure raising or relocation, floodproofing, etc.). Considerably less hydrologic engineering effort is necessary for modeling nonstructural alternatives compared to structural.

(1) *Procedure.*

(a) Consider duplicating existing and future without-hydrologic engineering models for individual analysis of each alternative or component. The results provide existing and future, with-project information for each alternative to be evaluated.

(b) Most structural components are usually modeled by modifying storage-outflow relationships at the component location and/or modifying hydraulic geometry through the reach under consideration.

1. Reservoirs. Adjust storage-outflow relationships based on spillway geometry and height of dam.

2. Levees. Adjust cross-section geometry based on proposed levee height(s). Evaluate effect of storage loss behind levee on storage-outflow relationships and determine revised discharge- and stage-frequency relationships downstream, if considered significant. Develop uncertainty relationship for the revised stage-discharge function.

3. Channels. Adjust cross-section geometry based on proposed channel dimensions. Evaluate effect of channel cross section and length of channelization on floodplain storage, modify storage-outflow in reach, and determine revised downstream discharge-frequency relationships, if considered significant.

4. Diversions. Adjust hydrology model for reduced flow downstream of the diversion and to identify where diverted flow rejoins the stream (if it does).

5. Pumping. Adjust hydrology model for various pumping capacities to be analyzed.

(c) Evaluate the effects of potential components on the sediment regime. Refer to guidance given in EM 1110-2-4000.

1. Qualitatively for initial screening.

2. Quantitatively (where necessary) for final selection.

(2) *Nonstructural components.*

(a) Floodproofing/structure raises elevations of design events primarily.

(b) Flood forecasting. Development of real-time hydrology model, determination of warning times, etc.

(3) *Alternative evaluation and selection.*

(a) Alternative evaluation and selection is an iterative process, requiring continuous exchange of information between a variety of disciplines. An exact work flow or schematic is not possible for most projects, thus paragraph C–7.h could be relatively straightforward for one or two components or quite complex, requiring numerous reiterations as more cost and design information is known and project refinements are made. Paragraph C–7.h is usually the area of the HEMP requiring the most time and cost contingencies.

(b) For the selected alternative, provide hydrologic information to environmental engineers for use in studies concerning the effects of the recommended project.

j. *Hydraulic design.* This section and paragraph C–7.i are partly intertwined, as hydraulic design must be included with the sizing of the various components, both to operate hydrologic engineering models and to provide sufficient information for design and costing purposes. Perform hydraulic design studies proportional to the level of detail of the study process.

(1) *Reservoirs.* Dam height, spillway geometry, spillway cross section, outlet works (floor elevation, length, appurtenances, etc.), scour protection, pool guide-taking line, etc.

(2) *Levees.* Levee design profile, interior flood control requirements, etc.

(3) *Channels.* Channel geometry, bridge modifications, scour protection, channel cleanout requirements, channel and bridge transition design, etc.

(4) *Diversions.* May be similar to channel design activities, also includes diversion control (weir, gates, etc.). Where the diversions are tunnels, open-channel flow and pressure conduit hydraulic analyses may be necessary, depending on tunnel capacity and range of possible discharges.

(5) *Pumping.* Capacities, start-stop pump elevations, sump design, outlet design, scour protection, etc.

(6) *Nonstructural.* Floodproofing or structure raise elevations, flood forecasting models, evacuation plan, etc.

k. *Hydrologic engineering reporting requirements.* The last step must thoroughly document the results of the technical analyses in report form. Hydrologic and hydraulic information presented will range from extensive for feasibility reports to minimal for a typical engineering documentation report or design documentation report.

(1) *Project management plan.*

(a) Major hydrologic engineering activities in the PED phase.

- (b) Time and cost for hydrologic engineering.
- (c) Activity schedule.
- (2) *Hydrologic engineering appendix.*
 - (a) Text.
 - (b) Tables.
 - (c) Figures.
- (3) *Environmental impact statement/environmental assessment report.*
 - (a) Hydrologic information/data as necessary.
 - (b) Portions of text, selected figures, and tables.

C–8. Example: Generic detailed hydrologic engineering management plan for a water control management study

a. *Background.* This example was provided in EP 1110-2-9, dated July 1994. As a result, the example includes references to both planning study processes and hydrologic engineering software that have been superseded for modern planning studies. The example is provided here as a reference. Current methods and tools should be confirmed with the appropriate higher authority.

b. *Sample plan.* This sample HEMP can be used as a guide for hydrologic engineering components of a water control management study. This appendix assumes that a general plan for reservoir regulation has been established, which is the result of a water conservation study (paragraph C–9), a flood control study, or new project construction, and that a water control plan and water control manual are to be developed. The HEMP covers the development of data collection procedures, forecasting procedures, drought detection procedures, development of reservoir regulation rule curves, and other information. Reference is made to ER 1110-2-240 and EM 1110-2-3600. Not included in this appendix are the hydrologic engineering studies needed for a dam safety plan.

c. *Preliminary investigations.* This is a preparatory phase that includes scoping the project, deciding on and gathering data, coordinating, etc. Experience has shown that more than 50% of the study’s budget can be consumed by data gathering and preparation.

- (1) Do initial preparation/coordination.

- (a) Identify agencies/parties with which coordination is needed for data, operational requirements.

(b) Review existing documents.

1. Design documents.
2. Interagency agreements.
3. Studies by USACE.
4. Studies by other agencies.

(c) Visit existing project(s).

- (2) Obtain hydrologic data. See paragraphs C–7 and C–9.
- (3) Scope major hydrologic activities; choose models to be used.
- (4) Prepare detailed HEMP.

d. *Develop study models.* Since the water control management study is the final phase in project development, hydrologic/hydraulic models will likely have been developed in previous phases of the project. If additional modeling studies are required, then data must be obtained and model calibration undertaken. See paragraphs C–7 and C–9. Note that the development of an operational forecast model is treated separately, below.

e. *Develop seasonal flood control rule curve.* A seasonal guide curve specifying the upper limit elevation for flood control throughout the year may have been developed in earlier study phases; however, this may need review and refinement for actual regulation. Also, the object of the water management study might be to modify the existing rule curve to improve regulation or accommodate a revised policy. This analysis is one of evaluating flood potential throughout the year. If streamflow records are short, the analysis could be augmented by looking at precipitation records to ensure that an adequate sampling of runoff potential for a given calendar date is achieved.

(1) Obtain historic flood data for the project site, including secondary floods that occur off-season.

- (a) Obtain recorder traces from the USGS.
- (b) Convert traces to streamflow hydrographs.
- (c) Compute volumes appropriate to flood control storage.

(2) Review long-period rainfall and streamflow records in the region to augment project records. Estimate project inflow data through regional correlation techniques or a rainfall-runoff model.

- (a) Decide on a processing technique.

- (b) Enter data into database.
- (c) Perform calculations.
- (d) Display results; verify for reasonableness.
- (e) Use rainfall-runoff model if required.
- 1. Obtain precipitation records.
- 2. Calibrate model.
- 3. Apply precipitation, compute hydrograph.
- 4. Display, verify reasonableness.

(3) Compute volume of water stored for each flood, given the planned flood regulation plan.

- (a) Determine computational technique.
- (b) Prepare model, check with test data.
- (c) Establish regulation rules, assumptions of forecast knowledge.
- (d) Perform routings using newly derived inflow data.
- (e) Display results, review for reasonableness.
- (4) Construct seasonal flood control rule curve.

(a) Plot storage requirement as a function of date.

(b) Plot tentative envelope line representing rule curve. Incorporate limitations for rate of draft, etc.

(c) Identify outlier points, estimate probability of event. Decide whether to envelop or not.

(d) Identify impacts on other project functions and compare with rule curves used in project authorization.

(e) Decide on final rule curve.

f. Develop forecasting model. Assess whether the project under consideration warrants a forecast model as a part of the reservoir regulation activities, and whether staffing is available to maintain and operate the model. If that assessment is affirmative, then a model is needed for future operational application. It is likely that an existing study model can be used as a basis for development of the forecasting model. Assess

whether forecasting is also to be done for conservation operation purposes as well as flood control. If so, a continuous model capable of forecasting low-flow conditions might be the appropriate choice.

(1) Select computer program to be used. Factors to be considered are size and complexity of basin (reservoir projects, diversions, etc.); type of runoff regime (rain, snow, flash flood potential); and applications required (flood operations, low-flow forecasting).

(a) Review information and models; consult as necessary with users and experts.

(b) Obtain models, run tests.

(c) Evaluate resources needed for real-time application (computers, people, funds).

(d) Select model.

(2) Review historic and real-time data availability and obtain hydrometeorological data pertinent to forecasting and project operation. Process data for input to forecast model.

(a) Set up forecast database (likely HEC-Data Storage System [DSS]).

(b) Consider data types needed (precipitation, temperature, streamflow, etc.).

(c) Examine period of record and select flood events that should be used in calibrations.

(d) Obtain data and download to database.

(e) Perform data screening/data display to verify data.

(3) Choose likely [hydrometeorological](#) station candidates for real-time application.

(a) Review performance of existing real-time telemetry.

(b) Examine feasibility of providing future automation.

(c) Determine relative merit of stations as indices to forecasting runoff.

1. Compute correlations of precipitation versus runoff.

2. Examine locations so that a range of elevation and spatial coverage is obtained.

(d) Select likely best candidates.

(4) Configure model for operational forecast application. This might be an expansion or a simplification of the study model, if available.

(a) Develop model characteristics.

1. Project characteristics (storage tables, outflow limits, etc.).
2. Routing reaches (initial estimates of routing factors).
3. Basin configuration (combining points, diversions, balances, etc.).

(b) Estimate initial [hydrometeorological](#) station weightings.

(5) Perform calibration simulations with model with proposed operational data and operational model. Repeat process of calibration and [hydrometeorological](#) station selection until best model is configured.

(a) Flood runoff calibrations (rain/runoff model).

1. Decide on calibration procedure (trial/error, optimization).
2. Select events for calibration (or continuous simulation).
3. Make calibration simulation.
4. Make changes and repeat.

(b) Routing calibration (same process as above).

(6) Set up procedures for preparing forecast in real time.

(a) Initialization of forecast run.

(b) Estimating missing data.

(c) Estimating ungaged local inflow.

(d) Incorporation of quantitative precipitation forecasts.

(e) Estimating snowmelt.

(7) Test applications of forecast model with forecast/regulation personnel.

(8) Document forecast procedures.

g. Develop water management enterprise system. Develop a plan for the WMES and the WMES master plan. Developing forecasting procedures will reveal remote gaging needs, whether it be the installation of new stations or the automation of manually reported stations. The WMES also includes the computer facilities needed for

processing data and executing computer models. These developments are documented in the WMES master plan, which is submitted to higher authority for approval.

- (1) Establish field requirements for operational data.
 - (a) Determine frequency of reporting, backup requirements.
 - (b) Automate existing facilities, if needed.
 - (c) Establish new stations as necessary.
- (2) Develop plan for field data collection system-land-line, geostationary operational environmental satellite, line-of-sight radio, etc.
 - (a) Obtain manufacturer's specifications and costs.
 - (b) Coordinate with other water resource agencies collecting data in the region. Consult with the USACE Civil Works, Engineering Division, Hydrology, Hydraulics and Coastal Community of Practice Lead (CECW-EC).
 - (c) Estimate costs for maintenance at field equipment and receiving site.
 - (d) Prepare life-cycle cost analysis comparison; select best alternative.
- (3) Develop plan for WMES and USACE Water Management System computer processing.
 - (a) Consult with CECW-EC, HEC, Information Management, and other USACE offices.
 - (b) Obtain manufacturer's specifications and costs.
 - (c) Determine software requirements.
 - (d) Determine continuity of operations requirements.
 - (e) Perform life cycle cost analysis; select best alternative.
- (4) Write WMES Master Plan; submit for approval.
- (5) Establish capital and annual expenditure requirements; budget accordingly (Plant Replacement and Improvement Program and annual funds).

h. Develop flood control operation guidance. The water control plan for flood control operations should include several items to assist regulators and project operators in making regulating decisions. These may require new hydrologic and hydraulic studies, or they may simply require development of existing material for presentation in the water control manual.

- (1) Regulating outlet and spillway gate opening sequences, limitations.
 - (a) Consult with hydraulic design engineers, project personnel.
 - (b) Perform or request special hydraulic analysis as required.
 - (c) Document procedures with text and diagrams as necessary.
- (2) Spillway gate regulation procedures.
 - (a) Review guidance and design documents regarding gate regulation schedules.
 - (b) Prepare a gate regulation diagram.
 - 1. Select a design flood for recession volume analysis.
 - 2. Perform routings.
 - 3. Plot required outflows versus inflow and storage.
 - (c) Test diagram by routing with different floods (inflow design flood, historic floods). Adjust as necessary.
 - (d) Finalize guidance plots; prepare documentation.
- (3) Miscellaneous guidance curves for flood control operations. There may be need for guidance curves and rules that can be used in the flood control operating plan in lieu of or in conjunction with a flood-forecasting operation. One application could be as a backup to a forecasting system in cases where communications and power are lost. Examples might be indices to runoff, given precipitation magnitudes, procedures for changing outflows, given reservoir rate of fill, and rules for operating several dams controlling a single control downstream control point.
 - (a) Determine need for guidance, considering factors such as:
 - 1. Remoteness of project; communications and transportation viability.
 - 2. Accuracy and viability of operational forecasting procedures, models.
 - 3. Basin runoff response.
 - (b) Perform hydrologic study using models or manual analysis.
 - (c) Test guidance on historic floods, design floods.
 - (d) Prepare plots, narrative text.

(4) Examples of flood regulation. Examples of flood control regulation provide a useful form of guidance for water control manuals. These can be plots and/or tabulations, accompanied by explanatory text.

(a) Select floods for possible examples, considering factors such as magnitude of flood; unusual runoff timing or shape of hydrograph; seasonal considerations; and ability to demonstrate use of guidance procedures.

(b) Perform reservoir routings with historic data, using guidance materials that have been prepared.

(c) Plot hydrographs; prepare narrative material.

(5) Emergency instructions to dam tenders when communication is lost. This is a mandatory requirement for dams subject to rapid flood runoff requiring gate operations. It is particularly important for dams in remote areas. Coordination is needed with Dam Safety Plan emergency procedures.

(a) Review operating guidance determined in above paragraphs.

(b) Review staffing, travel conditions, and timing for dam tenders with project personnel. Visit project if necessary.

(c) Prepare procedures for nonguided operation:

1. Staffing requirements.
2. Data gathering; project monitoring.
3. Determination of outflows.
4. Alert procedures.

i. Develop guidance for conservation operation. Although not as critical as flood operation guidance because timing and possible emergency conditions are not as problematic, there may be special procedures that are required for low-flow operations. Examples might be (1) procedures for determining instream flow releases as a function of reservoir status, and (2) municipal and industrial release schedules, etc. This activity might be a part of the Drought Contingency Plan studies, paragraph C–8.k.

(1) Determine need for guidance.

(a) Review studies used for project formulation and design.

(b) Review historic records; select drought periods for examination.

(c) Consult with affected agencies/parties.

(d) Decide on need for further analysis and need for special guidance.

(2) Perform hydrologic study as required, using models or manual procedures. Refer to paragraph C–9.f and C–9.g for details.

(3) Test guidance on varying hydrologic conditions, considering real-time conditions such as forecasting accuracy or slippage in implementing actions.

(4) Prepare plots, narrative description for water control manual. Refer to paragraph C–8.l.

j. Development of guidance for hydroelectric operations. The design of the hydro plant for the project and the nature of the hydroelectric system involved determines how the project is to be operated for power production, so additional hydrologic engineering studies for the water control plan may not be extensive. Possible items: (1) rate-of-change restriction studies, including possible river fluctuation studies; (2) block-loading schedules; (3) plant characteristic charts; and (4) unit operating procedures. Refer to EM 1110-2-1701.

(1) Identify need for guidance. See paragraph C–8.i.

(2) Perform analysis. Refer to paragraphs C–9.f and C–9.g for details.

(3) Prepare material for water control manual. Refer to paragraph C–8.l.

k. Develop drought contingency plan. If the project has a water conservation operating objective, then a drought contingency plan is required in the water control manual. Reference is made to ER 1110-2-1941.

l. Prepare water control plan and manual. The final step in the water control management study is the documentation of the plan of operation in the water control manual. If the plan that has been developed represents a significant change in operation from previous operational policy, then public coordination is required if it hasn't already taken place. Refer to ETL 1110-2-251.

(1) Ascertain need for public meeting on water control plan. (Refer to the Water Resources Development Act). Prepare materials and hold meeting if required.

(2) If meeting is required, prepare briefing material and hold meeting.

(a) Decide on presentations to be made.

(b) Review studies, availability of briefing material.

(c) Prepare new visual aids as necessary.

(d) Hold meeting.

(e) Carry out follow-up actions as necessary.

(3) Establish manual content, organization, and work program.

- (a) Prepare outline of manual.
- (b) Assemble material; identify critical items to be worked on.
- (c) Assess human resources requirements, time schedule, funds.
- (d) Consider contractor assistance.
- (e) Obtain agreement and approval of content and approach. Review with division office.
- (4) Prepare manual.
- (a) Decide on plotting methodology.
- (b) Prepare required plots and graphics.
- (c) Prepare narrative material.
- (d) Conduct first draft review by in-house personnel, division office.
- (e) Prepare final manual for approval.

C-9. Example: Generic detailed hydrologic engineering management plan for a water supply investigation

a. Background. This example was provided in EP 1110-2-9, dated July 1994. As a result, the example includes references to both planning study processes and hydrologic engineering software that have been superseded for modern planning studies. The example is provided here as a reference. Current methods and tools should be confirmed with the appropriate higher authority.

b. Sample plan. This sample detailed HEMP can be used as a guide for the hydrologic and hydraulic analysis needed for conducting a water conservation or a storage reallocation study. Examples might include a water supply allocation plan for a new reservoir project; a drought operating plan for an existing reservoir project or system of projects; and a system-wide reevaluation of a master water control plan considering the potential of benefit reallocation. The analysis typically employs a reservoir simulation model, such as the Hydrologic Engineering Center (HEC-ResSim).

c. Other criteria. Flood control criteria might affect conservation purposes; therefore, flood routing studies may be required, and perhaps a rainfall-runoff model should be employed. The goal is to evaluate nominated alternatives of storage quantities, firm water supply yields, seasonal rule curves, and instream flow goals by model simulation. The alternatives are judged by comparing regulated flood frequency curves, low-flow frequency curves, reservoir elevation-duration curves, etc.

d. Preliminary investigations. This is a preparatory phase that includes scoping the project, deciding on and gathering data, coordinating with agencies affected by the study, etc. Experience has shown that more than 50% of the study's budget can be consumed by data gathering and preparation.

(1) Initial preparation.

(a) Identify agencies/parties with which coordination is needed for data, operational requirements.

(b) Identify problem; scope study objectives.

(c) Review existing documents.

1. Design documents.

2. Interagency agreements.

3. Previous studies by USACE.

4. Studies by other agencies.

(2) Choice and application of models. Research, obtain consultation (HEC and others) on which models to employ and how they can be applied.

(3) Obtain historic hydrologic/meteorological data.

(a) Project operating records, such as reservoir elevations, outflows.

(b) USGS streamflow records.

(c) NWS precipitation and temperature data.

(d) Evaporation records.

(e) Snow data.

(f) Irrigation withdrawals.

(g) Consumptive use (municipal and industrial) records.

(4) Obtain project requirements and alternatives to be investigated.

(a) Municipal and industrial requirements.

(b) Instream flow requirements.

(c) Mandatory reservoir requirements.

- (d) Alternative storage capacities.
- (e) Future irrigation demands.
- (5) Scope major hydrologic activities; choose models.
- (6) Prepare detailed HEMP.
- e. *Develop hydrologic data for analysis.*
 - (1) Load raw data into study database (usually HEC-DSS).
 - (a) Develop naming conventions/database structure. Ascertain computer requirements. Consult with experts where necessary.
 - (b) Download data from various sources.
 - (c) Rough-check data viability by display or screening processor.
 - (2) Process hydrologic data for model input (streamflow data). This is a very important and potentially time-consuming step in which a uniform database of streamflow is derived to be used as inflows in reservoir regulation models. Although monthly data are typically used, a shorter time period (daily or more frequent) may have to be developed if flood regulation analysis is involved. Take full advantage of HEC-DSS linked programs to evaluate and process data.
 - (a) Perform cross-station checks, such as double-mass plots, to reveal record inconsistencies. Evaluate and correct as necessary.
 - (b) Decide on period of record to be used, considering data availability and incorporation of significant events.
 - (c) Estimate missing data.
 - (d) Compute storage changes at upstream reservoirs. Adjust historic records to remove effects of historic upstream reservoir regulation.
 - 1. Convert elevations to storage, if necessary, for each project.
 - 2. Calculate adjusted streamflow at downstream stream gages.
 - 3. Accumulate storage adjustments where more than one reservoir is involved.
 - 4. Produce tabulations, statistical summaries for checks.
 - (e) Process/estimate irrigation depletion data. Correct historic records for changes in irrigation diversions.
 - 1. Review irrigation records; determine magnitude of impact on study.

2. Decide on scope of effort.
3. Compute year-by-year historic diversions and return flows.
4. Estimate future irrigation depletions.
5. Compute changes in irrigation: future-historic.
6. Prepare final computation of corrected flows.
7. Produce display statistics of final irrigation quantities to be used.

(f) Prepare final computations; display and check data and formatting for model input.

1. Perform checking computations to check data validity (compute locals).
2. Prepare final production of statistical summaries, displays.
3. Prepare final formatting for model input.

(3) Derive synthetic (stochastic) monthly flow record (optional). In addition to the historic period of monthly flows, it may be desirable to employ a synthetic flow sequence, derived by stochastic techniques. This is particularly important when multiple-year critical periods are encountered.

(a) Consult with HEC and others on best available programs and techniques to derive a stochastic flow record, given the basin configuration and analysis requirements.

- (b) Review technical publications; find examples of applications.
- (c) Obtain computer programs; run on test data.
- (d) Determine scope and cost of work and benefits of using stochastic record.
- (e) Prepare input to stochastic flow generator as program requires.
- (f) Analyze output to ascertain its statistical soundness. Revise input as needed.

(4) Derive historic or synthetic flood flows. A water supply investigation or storage allocation study may require analysis of flood control criteria. If so, historic and/or synthetic flood hydrographs are needed. Examples of synthetic floods of different spatial and temporal flood patterns that appear plausible but have not been observed historically. Derivation of such events employ a runoff model. (See paragraph C-7 for generic HEMP guidance.)

f. Preliminary analysis. Prior to extensive analysis with hydrologic models (or in place of, for small-scale studies) it may be desirable to perform manual, short-cut water conservation analyses. These include procedures such as (1) mass-curve analysis for

identifying critical periods and estimating firm yields; (2) limited scope sequential accounting using appropriate tools; and (3) low-flow frequency or flow-duration analysis.

- (1) Obtain references and consultation on use of methodology as necessary.
 - (a) Search literature as necessary.
 - (b) Consult with HEC and others.
 - (c) Decide on methodology to use.
- (2) Obtain or construct computer programs, or other computational procedures.
 - (a) Obtain or develop computer code.
 - (b) Test programs.
 - (c) Develop display routines.
- (3) Prepare data input for analysis procedure. See paragraph C–9.e.
- (4) Obtain project demands, requirements.
 - (a) Analyze requirements for validity.
 - (b) Prepare input for program being used.
- (5) Perform analysis with nominated alternatives to be considered.
 - (a) Verify technique on current data, if possible.
 - (b) Execute technique with each alternative.
- (6) Display results, document as necessary.

g. Develop models for detailed study. This phase of the analysis involves the setting up, calibrating, and testing of computer models that are to be used in the analysis. An important part of this effort also is the processing of basic data that are needed for the model simulations. The following outline assumes that a reservoir simulation model with a monthly time-step (for example, HEC-ResSim) is to be employed for analysis of seasonal or multi-year reservoir operations. Also, for flood control evaluations, a precipitation-runoff model coupled with a short-term reservoir system and basin routing model might be needed. This might be HEC-HMS and HEC-ResSim.

(1) A prescriptive model can be used in multiple purpose reservoir systems to help define the most desired operation, given relative economic values (or penalties) assigned to operating constraints and objectives.

- (a) Consult with HEC and/or other users of these model.
- (b) Obtain program; develop understanding of its application, requirements, end products.
- (c) Decide on use of program and scope of effort.
- (d) Formulate basic model structure to represent reservoir system. Decide on location of reservoirs, nodes.
- (e) Define penalty functions at nodes and reservoirs.
 - 1. Coordinate with economics personnel for penalty function derivation.
 - 2. Obtain indicators of economic impact versus flow or stage: flood damage, power values, recreation, instream flow (fishery, water quality), navigation, irrigation, and municipal and industrial.
 - 3. Plot penalty functions and input into model.
- (f) Execute model with first estimate of current conditions and constraints.
- (g) Check results (seasonal reservoir elevations) against real-world conditions. Modify penalty functions to obtain intuitively “calibrated” model. Repeat model adjusting as necessary.
- (h) Perform production runs.
 - 1. Sensitivity analysis of economic functions.
 - 2. Potential future changes in operating constraints.
- (2) Develop descriptive model for monthly routing, such as HEC-ResSim. A descriptive model performs a period-by-period simulation of reservoir operation following rule curves and other operating variables. The rule curves represent alternative possibilities of operation, reflecting both existing and nominated alternative conditions. Either may have emerged from the results of the prescriptive model simulation performed above.
- (a) Consult as necessary on use and application of model. Obtain program and test.
- (b) Formulate basic model structure to represent reservoir system. Decide on location of reservoirs, control points.
- (c) Prepare reservoir model characteristics.
 - 1. Reservoir elevation/area/storage capacity tables.

2. Powerhouse characteristics.
3. Tailwater ratings.
4. Outlet constraints (maximum, minimum, rate of change, etc.).
- (d) Prepare river system characteristics.
 1. Irrigation diversion relationships.
 2. Routing characteristics (for flood hydrograph routing).
 3. Natural lake characteristics elevation/storage/outflow.
 4. Evaporation.
- (e) Prepare reservoir rule curves (existing conditions if for an existing system).
 1. Seasonal flood control curves.
 2. Seasonal proportional draft curves.
 3. Other rule curves.
- (f) Develop procedures for output display and summary.
- (g) Execute model for initial test runs. Check results for reasonableness against known conditions.
 1. Historic regulated flows at control points.
 2. Historic reservoir elevations.
 3. Historic power generation.
 4. Low flow (7-day volumes, 10-day volumes, etc.) and flood frequency curves.
- (h) Adjust model parameters to obtain a calibrated and tested model. Repeat above steps as necessary.
- (3) Develop daily (or shorter period) flow routing model.
- (a) Determine basin configuration for model.
 1. Control/damage points.
 2. Routing reaches.
 3. Diversions.

4. Lake characteristics.

5. Backwater effects.

- (b) Derive routing characteristics for individual reaches.

1. Choose routing method and model.

2. Obtain data to derive routing coefficients.

3. Determine local inflows.

4. Perform routings.

5. Adjust model characteristics as necessary until calibration is achieved.

- (c) Verify total basin model by simulating historic records and comparing with observed data.

h. Alternative evaluations. This task is the heart of the water management study, requiring repeated simulations with nominated alternatives of reservoir operating policies and procedures. It should be performed in close coordination with the interdisciplinary planning team. Alternatives may have been suggested by the prescriptive model, if used, or by the team's agreed-upon formulations.

- (1) Define nominated alternative; prepare operating rule curve or other criteria accordingly.

- (2) Execute model; display and examine results.

- (a) Seasonal reservoir elevations number of refill failures, frequency curves, etc.

- (b) Instream flow frequency, duration.

- (c) Flood frequency curves.

- (d) Power generation.

1. Firm energy.

2. Average energy.

3. Capacity.

- (3) Repeat with alternatives; perform sensitivity tests as necessary.

- (4) Prepare model results for briefings, public meetings, etc., as necessary.

- (5) Document results.

Appendix D

Managed Overtopping of Levee Systems

D–1. Introduction

a. This document is applicable for all USACE riverine levee and floodwall systems and supersedes expired ECB 2019-08 and technical guidance ETL 1110-2-299. It provides a methodology for configuring the engineered capacity exceedance related to flood overtopping at a specific location or locations along the levee system. This guidance does not address overtopping of the entire system on those occasions when the overall system capacity is exceeded. The procedures noted here are general and can be applied to any phase of study, as directed by ER 1110-2-1150. The level of detail in the overtopping analysis depends on the study phase.

b. A levee system (sometimes referred to as just “levee” in this document), is composed of one or more components that, collectively, provide flood risk reduction to a defined area, referred to as a leveed area. A levee system includes all components that are interconnected and necessary to exclude floods from the leveed area. Components include embankment and floodwall sections, closure structures, pumping stations, culverts, and interior drainage works. The configuration of the levee system could also be influenced by reservoir operations and channelization features. The 1986 version of ETL 1110-2-299 conceptually introduced the overtopping reach in a system that is located adjacent to a low hazard, lower risk inundation area. The ETL also presented the notion of system superiority for reaches of higher economic and life safety risks.

c. The ETL then provided a methodology for establishing the TOL profile incorporating these overtopping and superiority concepts. However, the ETL did not include the risk analysis concepts that USACE requires today, nor did the ETL delve into the composition of a levee cross section (base levee and appropriate increments) and the evaluation of the economic and the engineered performance of overtopping features. The ETL only tangentially touched on resiliency of the overtopping reach and how resiliency might contribute to the design of the overtopping configuration. Resiliency, as used in this document, refers to the ability of an engineered structure to withstand overtopping flows at a specified location.

d. Configuring a levee system using methods presented in this document includes setting TOL profile, determining overtopping reach length and depth, considering resiliency measures in the overtopping reach (see paragraph D–6.b for definition of overtopping discharge) and managing residual risk. It is important to note that this document addresses the engineering aspects of managing overtopping along levee segments and must be used in concert with the current planning guidance. Resiliency measures provided by various forms of surface hardening, armoring, or resistance to overtopping scour define one of the driving parameters in sizing the overtopping reach and establishing overtopping rates and overtopping volumes. Resiliency in the overtopping reach provides a higher degree of confidence in levee performance at the point of overtopping to assist floodplain managers.

e. The methodology in this document is consistent with the flood risk framework concepts and guidance presented in ER 1105-2-101 and EM 1110-2-1913, and the ideas and concepts of the USACE Flood Risk Management Program (USACE 2024f) and the USACE Levee Safety Program (USACE 2024g). USACE established the Flood Risk Management Program to advance the goals of flood risk identification, communication, response, and management services across all levels of government to save lives and reduce property damage in case of floods and coastal storms. The mission of the USACE Levee Safety Program is to ensure levee systems provide benefits to the nation by working with sponsors and stakeholders to assess, communicate, and manage the flood risks to people, property, and the environment.

f. The level of study effort should align with SMART (Specific, Measurable, Attainable, Risk Informed, Timely) Planning Principles, which promote balancing the level of uncertainty and risk with the level of detail of the study. The level of detail required to make planning decisions will grow over the course of the study, as the study team moves from an array of alternatives to a single recommended alternative.

D–2. Managed overtopping and life safety

a. Flood risk is a fact of life for those living in areas subject to flooding. No matter what project or action is proposed, evaluated, adopted, and implemented for life safety, complete containment or total mitigation is never possible. Further, information used to estimate flood risk, formulate and evaluate plans, and consequent results of the analyses are uncertain. All measured or estimated values in project development are inaccurate to various degrees, reflecting both inherent natural variability in flooding phenomena, cyclical rainfall patterns, and uncertainty in estimating various parameters (such as estimation of n value) relevant to project works and their performance.

b. The risk framework (composed of risk assessment, risk management, and risk communication) is an explicit means of better understanding both the flooding and associated consequences and thus should support development of robust strategies for managing flood risk. Implementing the concept of “superiority” is one such strategy for managing risk. Superiority simply means providing higher levees at all points except where initial overtopping is desired when practicable in a given system. Not all systems may be able to accommodate superiority measures. Superiority is an increment of the levee height that increases the likelihood that when the system approaches capacity, flooding will occur at a specified overtopping section. Designs that include superiority can initiate overtopping in the least hazardous location.

c. Water surface profiles and distribution of overtopping volumes and the subsequent consequences need to be examined to understand where superiority should exist and where overtopping reaches should be located. Documenting overtopping consequences in the leveed area is an element of the strategy in flood hazard and emergency action plans, including a local flood warning system. Overtopping reaches may provide relief only up to the capacity of the superiority reach, and beyond that capacity the system is assumed to have been overwhelmed. The

overtopping reach is to provide a known initial exceedance location and to provide some warning/evacuation time before total system exceedance.

d. A levee system is expected to exclude flood waters from the floodplain up to a given flood elevation. A levee system may be overtopped during a future event that is high enough to exceed the top of the levee. As a result, there is always a residual flood risk that must be acknowledged and managed. Note that residual risk can consist of more than just the risk due to water overtopping the levee; the potential for a breach either prior to or subsequent to overtopping also contributes to incremental risk of a levee system. Superiority is a levee feature that can be used to reduce and manage residual risks.

D–3. Risk-based levee design concepts

a. *Levees and floodwalls.* These are flood risk management structures meant to keep flood waters out of a floodplain area. These structures have upper limits beyond which larger floods cannot be contained or managed. Since the structure will someday experience floods that overtop and flood the interior, overtopping becomes a design concern. The hazard of inundation by overtopping can be significant for flood risk management levees or floodwalls, and the consequences can be costly and potentially catastrophic. The rate of breach of a levee or floodwall is difficult to predict when sudden breach is a possibility, even when flood-fighting practices are employed. Sudden breaching in an urban setting could cause a catastrophe. Proper design to predict and manage overtopping helps manage residual risk.

b. *Flood overtopping.* Flood overtopping of a structure into a previously leveed area is a hazard inherent in any levee or floodwall system. This risk varies with the overtopping probability of the structure and can be significant even for areas with design levels for rare floods. Levees are evaluated at specific locations currently referred to as index locations. After the TOL elevation is defined at the index location, that elevation is used to set the profile as described in paragraph D–4.

c. *Risk assessments and top of levee design.* When a risk assessment is applied in a flood risk management study, the goal is a comprehensive approach in which the values of key variables, parameters, and components of flood risk management studies are subject to probabilistic assessment. Risk assessments are used as a part of decision making for levee systems throughout planning and design. Risk assessments during a study evaluate variables and their uncertainties to determine which variables are critical to decision making. Critical variables then become the focus of plan formulation and evaluation.

(1) Not all variables are critical to decision making. If one of the measures identified during the planning phase is a levee system, then a risk assessment will be used to further evaluate and refine design aspects of the system, including various levee heights, reliability, benefits, and costs. However, take care to avoid attempts to reasonably maximize net economic benefits during initial plan formulation.

(2) Once a levee height is determined, a risk assessment can then be used to select the most appropriate location or locations for managed overtopping by seeking the least hazardous reaches. After design, risk assessment information, including inundation mapping for the leveed area, can be used to further manage residual risk. For policies and procedures regarding risk assessments and managing residual risk, see ER 1105-2-101, EM 1110-2-1619, and other applicable levee safety policies.

d. Levee system features for capacity exceedance. Levee systems should have features to accommodate capacity exceedance. The concept is that, where feasible and practical, the system should be designed and constructed to account for that capacity exceedance. If capacity exceedance results in overtopping or breaching of the levee system, the breaching will occur in a predictable location, allowing orderly floodplain evacuation and minimizing reconstruction requirements (time and cost) after an exceedance or breach. Levee superiority is the concept of designing one levee reach at elevation different than another adjacent reach so that one overtops before the other in a predictable fashion.

(1) Levee superiority is one physical means for such accommodation; hardening sections to withstand overtopping helps minimize levee reconstruction efforts. Superiority and its accompanying increased resiliency (resistance to erosion from overtopping) at overtopping sections may be used. Identifying a feasible and practical overtopping location includes the requirement that with reasonable confidence, the location be designated and maintained for the life of the levee system.

(2) Additional superiority measures, such as additional overtopping reaches or increased overtopping resiliency, may be added if the evaluation of such measures produce benefits in excess of costs or are required to realize the benefits identified. Overtopping resiliency measures, if used, should be formulated and included in the recommended plan when justified by net economic benefits or tolerable risk guidelines for life safety.

D-4. Top of levee or floodwall elevation and profile

a. Analysis. In the analysis of a levee system, the system is expected to perform up to the TOL or floodwall grade and is hereafter referred to as TOL. The TOL should include other increments added to a particular base levee to account for settlement, superiority, maintenance access, etc. (Figure D-1).

(1) As the crest elevation changes over time, and it will for a variety of reasons (settlement, subsidence, erosion, etc.), the prior computed overtopping probability, amount of superiority, and other performance indices (economic, life safety, and engineering) also change. Typical levee elevation increments that collectively define the TOL elevation are described in the following sections; the list may not be all inclusive (see paragraph D-4.e).

(2) Once the TOL elevation is set at the index location, that elevation is used to set the profile using the associated water surface profile. The final levee system configuration, including the TOL and overtopping reach, is used to determine system performance and economic indices (see paragraphs D–8.a and D–8.d).

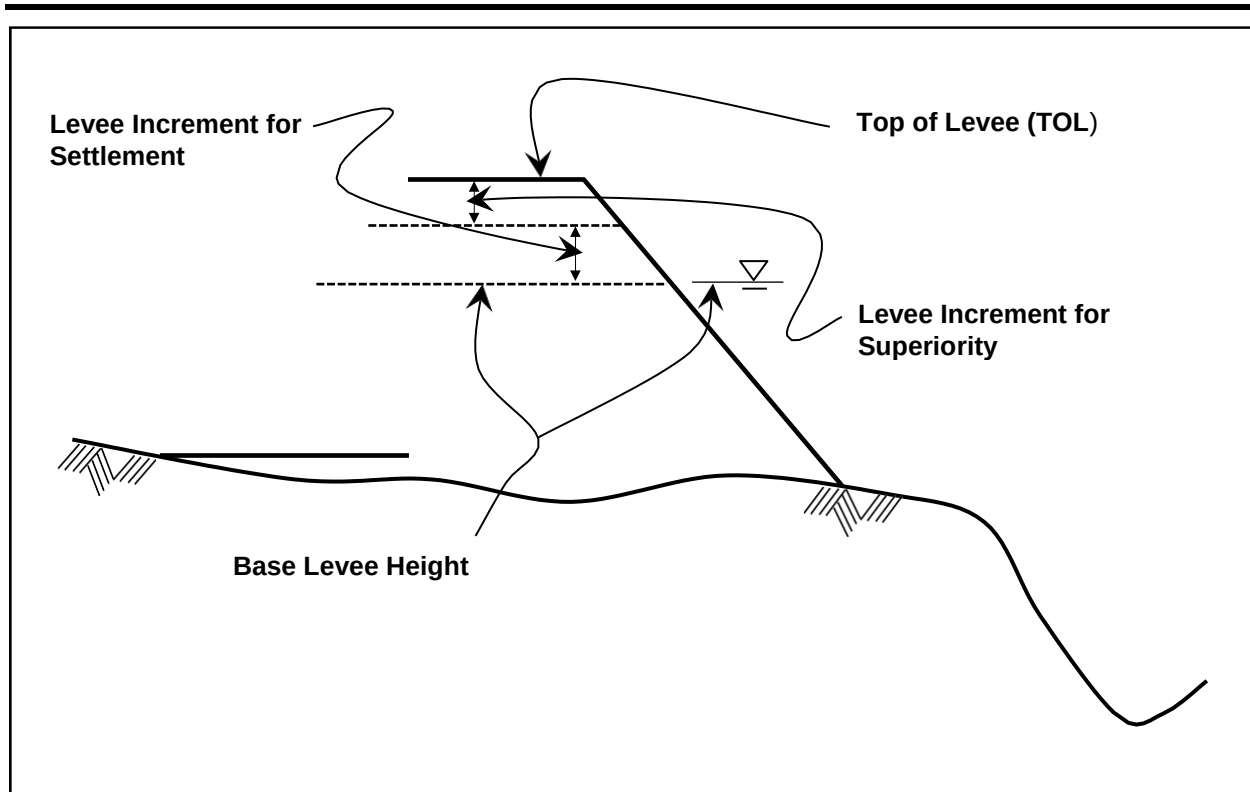


Figure D–1. Arriving at top of levee elevation (not all possible increments are shown)

b. Base levee height. The base levee heights are typically based on the range of still-water stages as defined by the stage-discharge or stage-frequency functions. For lake and coastal/estuarine levee systems or for riverine levee situations where fetches are sufficiently long, a proper analysis of wind conditions that can accompany extreme water levels must be done, and an increment for wave setup should be included in the base levee height.

(1) Wave overtopping is defined here as the condition in which individual waves break on the levee slope. Broken wave bores advance up the slope, onto and across the levee crest, and down the landward side. The potential exists for wave overtopping to occur while the still-water level is below the levee crest.

(2) The final exceedance frequency and other performance indices of the base levee result from an iterative planning process that evaluates a range of levee system configurations. This planning process includes the variations of TOL and the overtopping reach (base levee height and overtopping width).

c. *Increment for superiority.* Superiority in overtopping is designing adjacent levees or levee reaches to overtop one before the other. Superiority simply means providing higher levees at all points except where initial overtopping is desired. A more complex example involves two separate levees across the river from each other; one surrounding highly urbanized areas, the other mostly agricultural area, but both having similar levee elevations. Decision making could allow overtopping into the agricultural area before the urban area to reduce potential consequences.

(1) Superiority could be used to reduce this potential of a “chain failure” of adjoining independent levees when breach of one levee may trigger the breach of the next by providing relief structures at the upstream end of the adjoined systems. Another example involves a situation with flank or tie-back levees along tributaries to the river. The hydrology for the tributary may provide higher water surface profiles than a river.

(2) In addition, the tributary may be flashy with short warning times and potential dangers from quick overtopping. Superiority along the tributary reaches over the mainstem reaches may be appropriate. For long or complex levee systems, multiple overtopping reaches should be considered.

d. *Increment for settlement.* The geotechnical engineer will provide estimates of settlement in the foundation and embankment, embankment shrinkage, cracking, and geologic subsidence. The increase in risk due to lowering of the TOL must be accounted for in a levee system with site geology that is undergoing long-term regional settlement and/or on coastlines where future sea level change is occurring (see ER 1100-2-8162 for sea level change guidance). For the levee system to be reliable, the top of the flood protection must be able to provide the required design height over the service life of the project. To ensure reliability of the system, and to account for local settlement caused by the weight of levees or from general lowering of an area due to regional subsidence and/or sea level rise, levee systems could be initially constructed to a height sufficient to maintain the required design elevation for the service life of the project.

(1) Alternatively, levee systems could also be constructed to the design level for current conditions with allowance for raising in the future to meet “design” heights as settlement and/or subsidence occurs. The alternative selected should be based on which alternative is more cost effective over time. If future raising is not proposed and the future levee height is the intended final “settled/design” height, then the final height should be recognized in the planning study as the most likely future condition.

(2) Project benefits and engineering performance could reflect the initial constructed height and the final height, assumed to the “most likely future” condition, via an equivalent annual damage analysis.

e. *Other increment considerations.* In certain situations, other increments may be included as a separate levee increment. These increments could include superelevation through curved channels, debris blockage at bridges, impacts to water surface elevations due to ice flow, impacts due to sediment loads, and impacts to water surface

due to vegetation in channels and overbanks inside of levees. Generally, all increments (paragraph D–4.a through D–4.e) are deterministically defined and their associated uncertainties should be captured in the probabilistic risk and performance.

f. Establishing general top of levee. TOL profiles are typically set using computed water surface profiles either from an unsteady or steady-state hydraulic model, such as HEC-RAS as depicted in Figure D–2. The procedures presented here use an unsteady hydraulic analysis as the primary method for generating these profiles, as unsteady analyses speak directly to defining overtopping flows, establishing an overtopping length, and defining residual flooding volumes.

(1) If a steady hydraulic analysis is used to generate the profiles, define overtopping location and establish an overtopping length, then defining residual flooding should be made as a separate assessment depending on knowledge of the flow hydrographs. Figure D–2 illustrates the hydraulic routing of a specific flood hydrograph and the resultant water surface profile traced from the maximum stage generated at each channel cross section. To understand the hydraulic nature of the system, one should produce a set of profiles using a range of target hydrographs and note that all profiles may not be parallel (Figure D–3).

(2) The selected profile should be the one identified through the planning process, and the levee height should equal the stage of a particular hydrograph at the point of evaluation/index location.

g. Detailed top of levee profile evaluation procedure. The identification of the TOL profile is an iterative process, defined as follows:

(1) Water surface elevations are computed for a number of frequency-based events (for example, 0.02, 0.01, and 0.002 exceedance events) at an index point location and defined as a range of base levee heights; one such instance is shown in Figure D–1.

(2) Various increments, as discussed in the sections above, are added to each of the alternative base levee heights to create initial TOL heights at the index point; also illustrated in Figure D–1.

(3) The initial TOL heights at the index point must be translated into TOL profiles to support the planning analysis process. This translation is done by determining the exceedance frequency events that match the initial TOL heights at the index location, then evaluating the corresponding flows in a hydraulic model to generate the range of TOL profiles. Note that the events used to determine the TOL profiles should be larger than the events used to determine base levee heights because the TOL height is greater than the base levee height, given that various levee increments are usually added.

(4) The final levee alignment and levee height is identified through an iterative process where alternative levee heights and alignments are assessed until the recommended levee height is found. The recommended levee system is likely to be somewhere between the frequency-based events that were originally evaluated.

D-5. Selection of superiority and managed overtopping reaches

For initial overtopping, the overriding concern is choosing the least hazardous location to minimize consequences for initial inundation of the interior leveed area. A least hazardous location could be a golf course, an oxbow lake, a ponding area, the least developed area, or a downstream reach. In some cases, overtopping may be partially controlled in open spaces or by routing to ponding areas. In other situations, internal dikes or high ground may control overtopping volumes. Long-term development restrictions through acquisition of real estate may need to be considered as part of the project to ensure the overtopping design is not compromised.

D-6. Overtopping reach

a. *Overtopping design.* Several design types can be used to manage initial overtopping. One is to use a sloping superiority profile from reach to reach to force overtopping in a desired location. Other designs use notches, openings, or weirs in the structure. The invert for these features are at the base levee height elevation of the structure and could include an increment for settlement depending on how settlement is accounted (paragraph D-4.d). Examples are railroad or road openings and rock weirs.

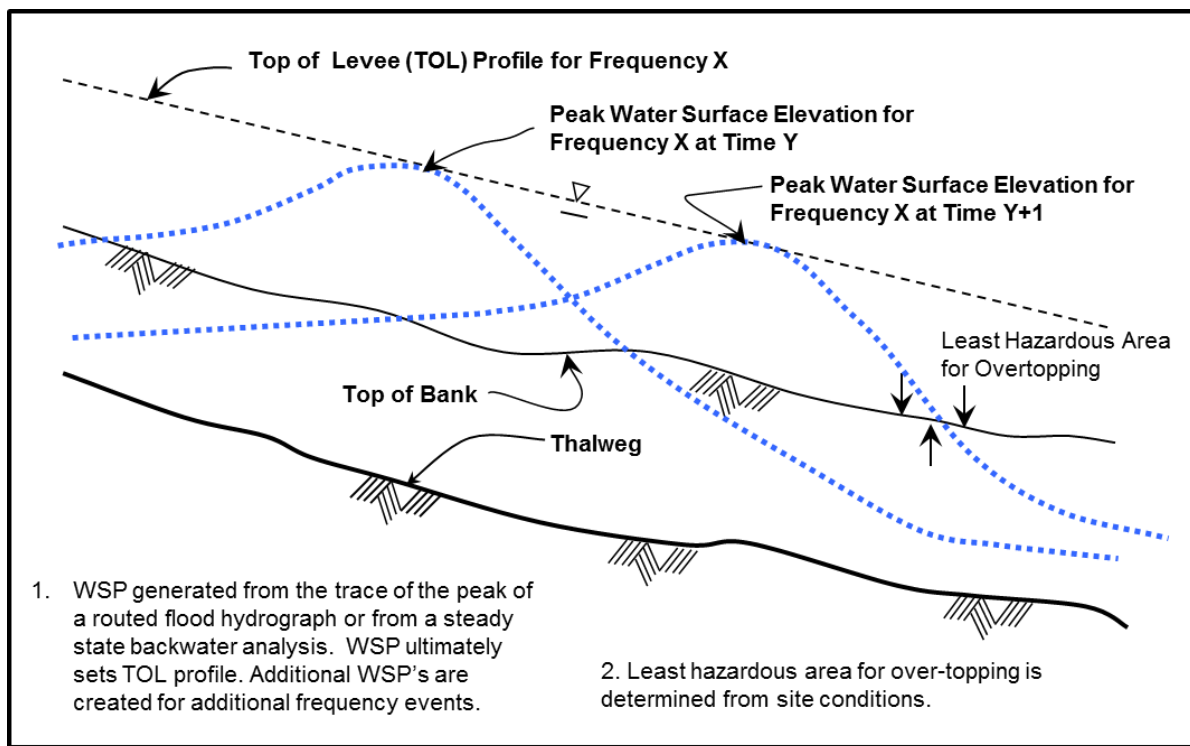


Figure D-2. Top of levee profiles

b. *Overtopping discharge.* The evaluation of overtopping measures and associated residual risks should be performed for a range of overtopping events, with the goal of identifying cost-effective resilience and risk-management measures (see ER 1105-2-101). The overtopping discharge for a particular flood event is the portion of the containment hydrograph (hydrograph that maps to the TOL) that spills over the overtopping reach and is usually expressed in cfs/feet. The allowable overtopping discharge (AOD) is the maximum recommended overtopping flow in cfs/feet for a given resiliency type, as discussed in paragraph D-7.

(1) The AOD should be selected so that the overtopping reach can pass the peak overtopping discharge without levee breach. Given that the AOD is the maximum recommended overtopping flow in cfs/feet based on resiliency type (and not based on an event size), an initial length and depth are selected, and the overtopping discharge hydrograph is routed through the “notch” to determine if the overtopping discharge exceeds the AOD for the selected resiliency type.

(2) If yes, then adjust resiliency type, length, depth, or all, and then reassess. The total volume of flow that spills via the overtopping reach may also be of interest for residual floodplain mapping.

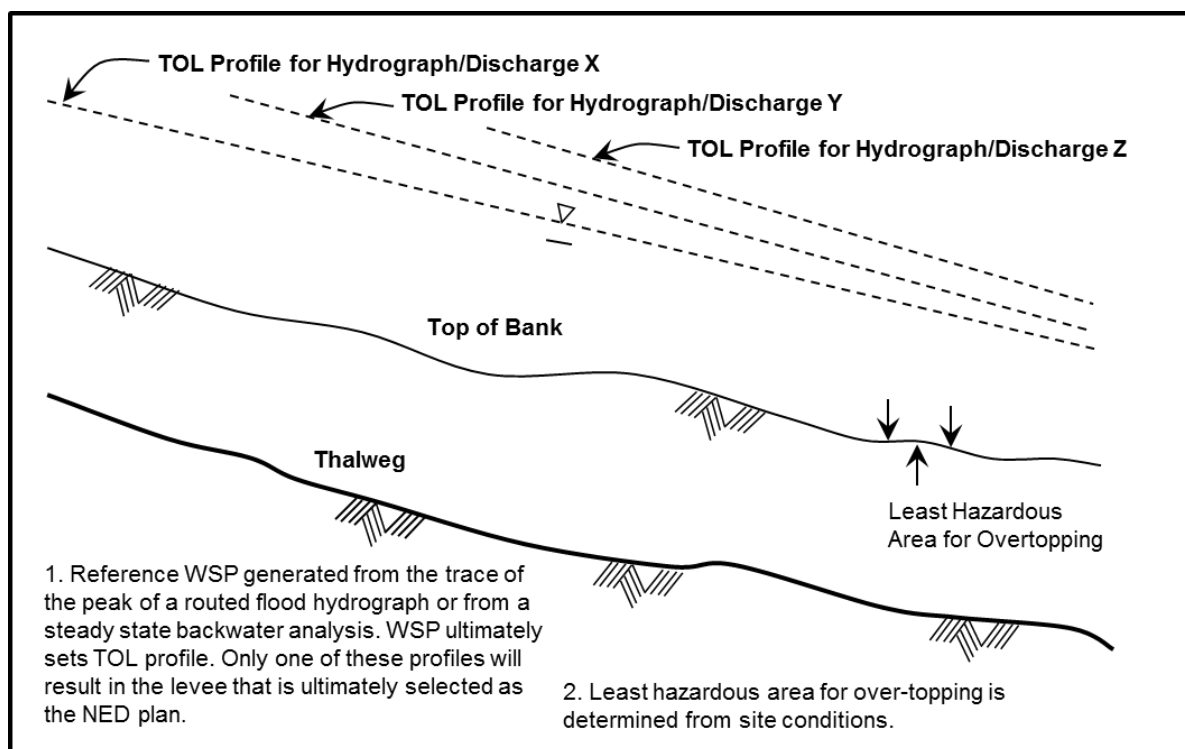


Figure D-3. Pairing of profiles to each levee height

c. *Overtopping reach length and depth.* Determination of the overtopping length and depth is based on one or more project design constraints. Typical project design constraints include:

(1) Maximum desired overtopping discharge or volume. This may be a function of the interior floodway flow or volume capacity.

(2) Maximum possible overtopping length. This may be constrained by features such as bridge or roadway crossings as well as interior floodplain or floodway width.

(3) Desired time to exceedance of TOL. A longer overtopping length passes more flow before the system capacity is exceeded, and thereby provides a longer window for emergency response activities.

(4) Desired height increment for levee superiority. There may be an optimal superiority increment height that balances cost and overtopping discharge.

(5) Other site-specific constraints. For example, estimate the time and volume needed to fill the leveed area “bathtub” and determine if it will fill at all, or size and design the overtopping section so that it can survive long enough for the leveed area to fill enough to reduce the chance of catastrophic breach.

d. Selection. Selection of the overtopping discharge, length, depth, and resiliency type is interrelated and iterative. The process of determining the final overtopping reach design (Figure D–4) depends on solving for unknown design parameters using known or assumed design parameters. The known versus unknown design parameters vary across projects.

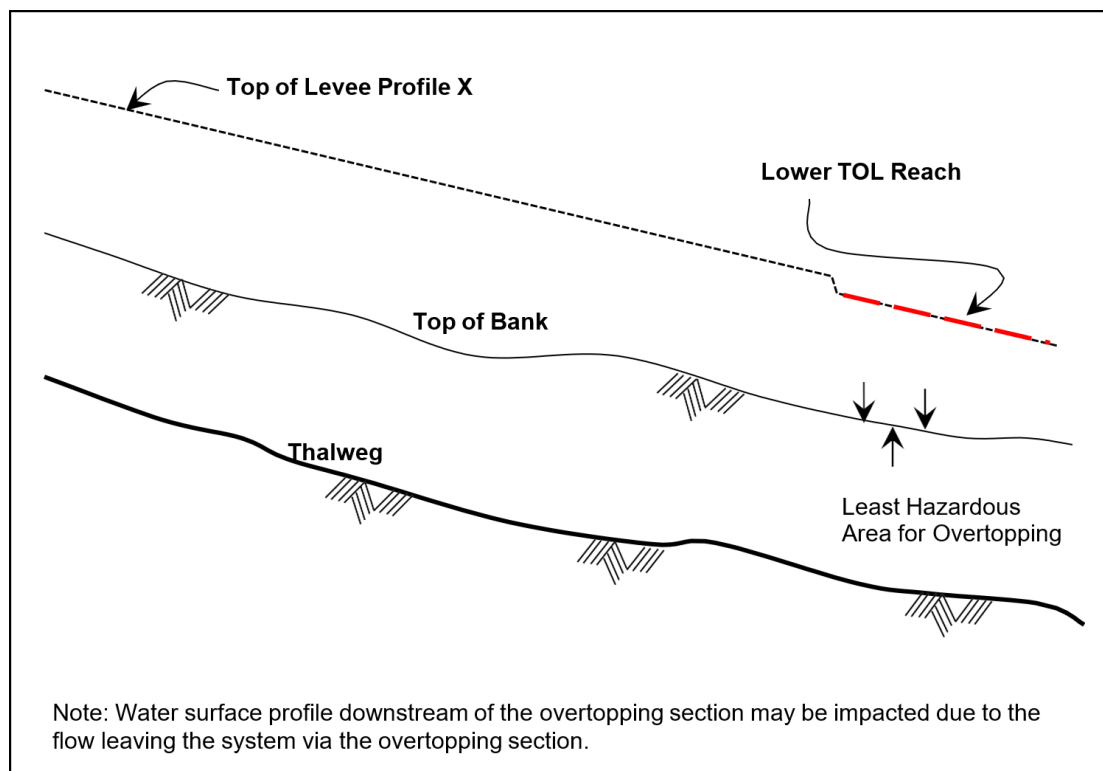


Figure D–4. Final overtopping reach design

D-7. Scour protection at superiority location

A resilient component or system is capable of absorbing energy during loading without experiencing permanent deformation, extensive damage, cumulative degradation, or catastrophic failure. Relatively high resilience is required in systems that must survive occasional large overloads for a long duration and then recover to their original state after the overload event. Resiliency can be incorporated into a levee and floodwall design by constructing a scour protection apron on the protected side of the levee or floodwall for the purpose of minimizing erosion during flood events that exceed the top of wall elevation. Resiliency types can be found in EM 1110-2-2502.

D-8. System performance, cost evaluation, and residual risk

a. *System performance.* System performance should be provided as defined in ER 1105-2-101 and presented using AEP, Assurance (formerly Conditional Nonexceedance Probability [CNP]) and LTEP. For completeness, these performance indices should be presented for both the superiority reach and for the overtopping reach. The performance for the superiority reach provides general project alternative information and separately, performance for the overtopping reach and information for the “least hazardous” inundation area.

b. *Cost effectiveness and incremental cost analyses.* Cost effectiveness seeks to identify the least cost alternative for each level of output. Incremental cost analysis allows an assessment of whether the increase in output is worth the additional cost. The ability to identify cost-effective plans is useful for evaluating and reducing economic and life-safety risks. For example, if two overtopping plans can pass the overtopping flows given the resiliency treatment for each but one plan costs more, the more expensive plan should be dropped from further consideration (See ER 1105-2-100 for more on cost effectiveness and incremental analyses).

c. *Residual floodplain mapping or formal flood easement.* In addition to the performance indices, residual overtopping flood mapping should be presented for floodplain management purposes or for definition of formal flood easements. Residual flooding for the planned overtopping reach should present flood boundaries for residual flood hydrographs, such as the hydrograph starting at the initiation of overtopping at the overtopping reach and ending at the initiation of overtopping of the TOL.

d. *Adjustments.* Adjust with-project damages, net benefits, and life safety consequences. When an overtopping reach is included in a project, there may be additional areas of inundation, and therefore, consequences (damage or life loss). If project consequences increase substantially with an overtopping reach, it may be necessary to iteratively evaluate the overtopping design and associated consequences to find a balance between residual flooding and benefits (See ER 1105-2-100).

D–9. Procedure summary

A summary of the entire process is provided here:

a. Establish base levee height at an index point using stage-frequency for a range of flows, and then add appropriate levee increments for superiority, settlement, and other factors to determine TOL. This process is described in paragraph D–4 and illustrated in Figure D–1.

b. Perform hydraulic water surface analysis (steady or unsteady) for the range of hydrographs mapping to the range of levee heights of interest. This creates a set of potential levee profiles.

c. Evaluate this range of levee profiles without consideration of overflow features and select an initial levee profile design.

d. Determine overtopping reach location, discharge, length, and depth. These factors are interrelated, and one or more may represent a key project design constraint. Considerations include:

(1) Maximum desired overtopping discharge or volume.

(2) Maximum AOD (see paragraph D–6.b) based on preferred resiliency type.

(3) Maximum possible overtopping length.

(4) Desired time to exceedance of TOL.

(5) Desired height increment (for levee superiority, settlement, and other factors such as time and volume to fill the leveed area).

e. Incorporate overtopping reach design into initial levee profile to create the final levee profile design. Ensure resilient features are well defined.

f. Determine residual overtopping flood mapping.

g. If appropriate, adjust alternative evaluation based on residual overtopping mapping.

h. Evaluate cost effectiveness of superiority enhancement and overtopping resiliency measures. Check impact on alternative selection resulting from cost of levee protection in overtopping reach and changes in residual damages, if applicable.

i. *Note.* Determining the TOL levee height, overtopping length, and overtopping depth (superiority increment) is an iterative process that may take several cycles through the entire procedure to reach a final design.

Glossary of Terms

Section I

List of Acronyms

Term	Definition
AANB	Average Annual Net Benefits
ADCIRC	Advanced Circulation
AEP	Annual Exceedance Probability
AMS	Annual Maximum Series
AOD	Allowable Overtopping Discharge
ASCE	American Society of Civil Engineers
BMAP	Beach Morphology Analysis Package
BMS	Block Maxima Series
CDF	Cumulative Distribution Function
CHL	Coastal and Hydraulics Laboratory
CHS	Coastal Hazards System
CI	Confidence Interval
CL	Confidence Limits
CNP	Conditional Nonexceedance Probability
CSE	Coastal Science & Engineering
CSRM	Coastal Storm Risk Management
CSS	Composite Storm Set
DEM	Digital Elevation Model
DRP	Dredging Research Program
DRV	Depreciated Replacement Value
DSWL	Dynamic Still-Water Level
EAD	Expected Annual Damage
EALL	Expected Annual Life Loss
EAP	Emergency Action Plan
EC	Engineer Circular
ECB	Engineering and Construction Bulletin
EM	Engineer Manual
EMA	Emergency Management Agency
EP	Engineer Pamphlet
ER	Engineer Regulation
ERDC	Engineer Research and Development Center
ERDC-CHL	Engineer Research and Development Center – Coastal and Hydraulics Laboratory
ERL	Effective Record Length
ETL	Engineer Technical Letter

Term	Definition
FEMA	Federal Emergency Management Agency
FSS	Full Storm Set
GPD	Generalized Pareto Distribution
HEC	Hydrologic Engineering Center
HEC-DSS	HEC-Data Storage System
HEC-FDA	HEC-Flood Damage Reduction Analysis
HEC-FIA	HEC-Flood Impact Analysis
HEC-HMS	HEC-Hydrologic Modeling System
HEC-RAS	HEC-River Analysis System
HEMP	Hydrologic Engineering Management Plan
HQSACE	Headquarters, U.S. Army Corps of Engineers
HSE	Health and Safety Executive
HURDAT	Hurricane Database
IACWD	Interagency Advisory Committee on Water Data
IBTrACS	International Best Tracks Archive for Climate Stewardship
ICOLD	International Commission on Large Dams
IID	Independent, Identically Distributed
IWR	Institute of Water Resources
JPM	Joint probability method
JPM-OS	Joint Probability Method with Optimal Sampling
LiDAR	Light Detection and Ranging
LPIII	Log-Pearson Type III
LTEP	Long-Term Exceedance Probability
MCLC	Monte Carlo Life Cycle
MGC	Meta-Gaussian Copula
MHW	Mean High Water
MLM	Maximum Likelihood Method
MLW	Mean Low Water
MMS	Monthly Maximum Series
MMSLA	Monthly Mean Sea Level Anomaly
MSE	Mean Squared Error
NACCS	North Atlantic Coast Comprehensive Study Report
NAVD	North American Vertical Datum
NCEP/NCAR	National Centers for Environmental Prediction/National Center for Atmospheric Research
NED	National Economic Development
NLR	Nonlinear Residual
NOAA	National Oceanic and Atmospheric Administration
NRC	National Research Council
NSI	National Structure Inventory

Term	Definition
NTR	Nontidal Residual
NWS	National Weather Service
O&M	Operation and Maintenance
OCRM	Ocean and Coastal Resource Management
PAI	Protective Action Initiation
PAR	Population at Risk
PBL	Planetary Boundary Layer
PCHA	Probabilistic Coastal Hazard Analysis
PDF	Probability Density Function
PDS	Partial Duration Series
PED	Preconstruction, Engineering and Design
PM	Project Manager
PMP	Project Management Plan
POT	Peaks Over Threshold
RMAP	Regional Morphology Analysis Package
RSLC	Relative Sea Level Change
SAFCA	Sacramento Area Flood Control Agency
SCCC	South Carolina Coastal Council
SOP	Start of Project
SRC	Shoreline Rate of Change
SRD	Shore Response Database
SRR	Storm Recurrence Rate
TOL	Top of Levee
TWL	Total Water Level
UDV	Unit Day Values
USGS	U.S. Geological Survey
WES	Waterways Experiment Station
WIS	Wave Information Study
WMES	Water Management Enterprise System
WRDA	Water Resources Development Act

Section II

Terms

Annual Exceedance Probability

The annual probability that a variable's magnitude exceeds a given threshold. For example, the per-year chance that the interior floodplain in an impact area will be inundated to some depth due to channel overflow, or in the case of a leveed area, due to levee overtopping or failure. In each case, the AEP can be computed with commonly available risk assessment software, considering the uncertainty in the hydrologic, hydraulic and if necessary, levee system response curve inputs.

Annual Probability of Failure

Per-year chance of unintentional occurrence of an event, such as flooding combined over all modes of failure; with due regard for nonmutually exclusive failure modes.

Assurance

The probability that a target stage will not be exceeded during the occurrence of a flood of specified exceedance probability considering the full range of uncertainties. This term was selected to replace "conditional nonexceedance probability" (CNP).

Baseline

The condition, data, or other representations of the topic of interest that will be used for comparison of changes that may occur without intervention or remediation to changes and impacts estimated to result from an intervention or remediation alternative. It is composed of the existing conditions and future projected conditions chosen to be relevant for the comparison.

Conditional Nonexceedance Probability

See Assurance.

Conditional Probability

The probability of event occurrence based on the assumption that another event (or multiple events) has occurred. For example, the probability of an event-tree branch that is determined based on the fact (condition) that a previous event (or several events) has occurred. At any node of an event tree, the sum of the discrete conditional probabilities associated with each of the events (branches) immediately following that node should equal 1.

Consequence

The outcome of an event, including immediate, short- and long-term, and direct and indirect losses and effects. Losses may include human casualties, monetary and economic damages, and environmental impact; may also include less tangible, and therefore, less quantifiable effects, including political ramifications, decreased morale, reductions in operational effectiveness, or other impacts. Less tangible effects, such as the changes in quality of life, should be quantified to the extent practicable, if needed.

Consequence is the harm that results from a single occurrence of a hazard. Consequence is measured in terms of indices such as economic damage, acreage of habitat lost, crops values damaged, and lives lost.

Deterministic

The description of a process with an outcome that is always the same for a given set of inputs (Hartford and Baecher 2004). Specifically, not random or not treated as random.

Discharge

The volume of water that passes a point in a given period of time. Rate of discharge is usually measured in cubic feet per second (cfs).

Economic Consequences

The direct losses from a flood hazard such as failure of a flood risk management measure and other economic impacts on the regional or national economy. Items in this category include ones commonly computed for the NED account in any USACE flood risk management study.

Economic Damages

Damages to private and public buildings, contents of buildings, vehicles, public infrastructure such as roads and bridges, public utility infrastructure, agricultural crops, agricultural capital, and erosion losses to land.

Economic Risk

Measure of the probability and severity of economic consequences. One metric of economic risk is the result of integrating the damage-frequency function to yield the mean or EAD.

Event

An occurrence or outcome with its own specific characteristics.

Exceedance Probability

The probability that a specified magnitude of a variable will be exceeded. Exceedance probability is most often specific to the annual chance exceedance probability, which is the likelihood of exceedance in any year.

Expected Annual Damage

A commonly used metric of economic risk. The result of integrating the annual damage-frequency function. See Economic Risk.

Expected Value

The average or central tendency of a random variable.

Exposure

Describes who and what may be harmed by the flood hazard. Exposure incorporates a description of where the flooding occurs at a given magnitude and what exists in that area. Tools such as flood inundation maps provide information on the extent and depth of flooding; structure inventories, population data, crop data, and habitat acreage provide information on the population and property that may be affected by the flood hazard. People and property threatened by flood hazard.

Extreme Event

An event with a very low annual chance of exceedance or nonexceedance.

Failure

Inability of a project to function as intended, for example, loss of structural integrity of a flood risk management project resulting in flooding. Failure is generally used in the context of an uncontrolled release of the reservoir pool via the dam structure. For levees, the term for failure is a breach of the levee.

Failure Mechanism

A mechanism described by the physical processes and states that must occur according to natural laws for the failure mode to progress from initiation (cause) through to the realization of ultimate failure effect of interest (Hartford and Baecher 2004).

Flood Risk

Measure of the probability and severity of undesirable consequences related to flooding.

Flow

Referred to herein as discharge, or volume of water that passes a point in a given period of time.

Frequency

The rate of occurrence of an event measured in terms of the number of a particular type of event occurring in a particular time period of interest, as a fraction of the total chances to occur. An annual period is commonly used. Thus, the resulting parameter represents an annual rate, typically referred to as annual probability. In this document, frequency and probability are used interchangeably.

Hazard

A potential source of harm (fire, earthquake, flood, etc.) to a valued asset (human, animal, natural, economic, or social) or a situation with a potential to cause loss. The hazard is what causes the harm, and in this manual, is a flood. The flood hazard is described in terms of probability, stage, velocity, extent, and depth.

Impact Area

A portion of a floodplain usually delineated based on flooding characteristics and/or land uses.

Index Location

A selected location used to represent hydrologic, hydraulic, geotechnical, and economic characteristics for a reach of a stream (index point, location of interest, area of interest).

Joint Probability

The probability that two or more variables will assume certain values simultaneously or within particular time intervals (ICOLD 2005).

Likelihood (probability)

A measure of the chance that a particular outcome or consequence will occur. It can also describe the degree of belief that an unknown parameter holds a certain value. A probability provides a quantitative description of the likelihood of occurrence of a particular event. Probability is expressed as a value between 0 and 1.

Long-Term Exceedance Probability

The probability that a variable exceeds some value during a specified period. For example, 30-year risk refers to the probability of one or more exceedances of the capacity of a measure during a 30-year period; formerly long-term risk.

Monte Carlo Simulation

A method for calculating the probability distribution of a function of random variables by simulation of pseudorandom outcomes. It seeks to simulate stochastic processes by random selection of values in proportion to their joint PDF.

Nonstructural Measures

Measures that reduce consequences of flooding without affecting flood characteristics such as flood depths, velocities, or durations.

Parameter

A quantity in a function that helps define the relationship of known input and unknown output. An example is Manning's roughness coefficient in energy loss calculations, which determines the relationship between a specified discharge rate and the unknown energy loss in a specific channel reach.

Parameter Uncertainty

Uncertainty in a parameter due to limited understanding or lack of accuracy with which parameters can be estimated.

Performance

a. The system's response to the hazard. Performance refers to the system features and the capability to accommodate the flood hazard as a single event or load. In this manual, it is termed "system performance" (also termed "engineering performance"). Performance also refers to a metric that describes the uncertainty of the system accommodating a specific event (assurance) and the full range of events (AEP and LTEP).

b. In that light, in addition to the levee failure probability functions, performance can also be described by the interior-exterior functions for leveed areas; unregulated-regulated transforms for reservoirs and diversions; and stage-discharge functions (rating curves) for channels. These, too, should be considered “system performance.” When the structural integrity of a system or system component is discussed, such as the system response curve, the reference is termed “structural performance.” When the economics of a system is discussed, the reference is termed “economic performance.”

c. The performance of a feature is described by various elements such as flood risk management, reliability, capability, efficiency, and maintainability. Design and operation affect system performance.

Population at risk

The number of people located within the maximum flood inundation boundary for a given scenario upon initiation of that flood event.

Probability (likelihood)

a. The chance of a random event occurring, often estimated by the ratio of past outcomes that satisfy the conditions of the event to the number of total past outcomes.

b. Frequency or fraction; the ratio of outcome from a repetitive experiment of some kind, such as flipping coins. Such a number is called an “objective” probability because it exists in the real world and is, in principle, measurable by doing the experiment.

c. A measure of the degree of confidence in a prediction, as dictated by the evidence, concerning the nature of an uncertain quantity or the occurrence of an uncertain future event. This measure has a value between 0 and 1.0.

d. Subjective probability; quantified measure of belief, judgment, or confidence in the likelihood of an outcome, obtained by considering all available information honestly, fairly, and with a minimum of bias. Subjective probability is affected by the state of understanding of a process, judgment regarding an evaluation, or the quality and quantity of information. It may change over time as the state of knowledge changes (ICOLD 2005).

Probability Density Function

The noncumulative form of a probability distribution. A function describing the relative likelihood that a random variable will assume a particular value in contrast to taking on other values (ICOLD 2005).

Quantile

Value of a variable that has a certain cumulative probability or exceedance probability.

Reliability

The likelihood of successful performance of a given project element over a specified period. It may be measured on an annualized basis or for some other specified period of interest. Mathematically, reliability equals 1 minus the probability of unsatisfactory operation.

Risk

Generally, the probability and severity of undesirable consequences. In the context of flood risk, it includes the probability that an area will be flooded, resulting in undesirable consequences as well as the consequences themselves. As used in this manual, risk is the function of five factors: hazard, performance, exposure, vulnerability, and consequences. Risk involves exposure to a chance of injury or loss.

Risk Analysis

A process that identifies the potential for adverse events and their consequences. Risk analysis can be defined by a framework that comprises three tasks: risk assessment, risk management, and risk communication.

Risk Assessment

A systematic, evidence-based approach for estimating and describing the likelihood and consequences of existing and future without-action outcomes, and risk reduced by any proposed risk-management action. A risk assessment includes explicit acknowledgment and incorporation of the uncertainties in the risk.

Risk Communication

The open, two-way exchange of information about risk. The process begins with information about how stakeholders can participate in the multi-stakeholder risk-management process. The information exchanged includes (1) characterization of current and possible future risks, including information concerning the uncertainty in the risk assessment; (2) the feasible life-safety risk management measures; and (3) the monetary and nonmonetary benefits and costs of alternative actions, with emphasis on life safety.

Risk Management

The process of evaluation, assessment by consideration of available options, prioritization of recommendations, selection of options, implementation, and monitoring and reviewing actions taken. The risk management choice should acknowledge the need to manage the residual risk and the effectiveness and acceptability of options available to do so.

Sensitivity Analysis

Evaluation to determine how much impact each input value or assumption has on the output values of a study. An analysis to determine the rate at which an output parameter varies, given unit change in one or more input parameters (ICOLD 2005).

Stationary

The assumption that the statistical characteristics of hydrologic time series data are constant through time. (*Stationarity* is the corresponding noun.)

System

An integrated combination of people, property, environment, and processes that work in a coordinated manner to achieve a specific desired output under specific conditions. Systems should be defined based on the analytical objectives, which may lead to different types of definitions such as functional systems, management systems, and engineering systems.

System Performance

The capability of the system to accommodate the flood hazard as a single event or load.

System Response Curve

A function that defines the probability of failure of an element as a function of the level of applied load. Also known as a fragility curve or geotechnical failure relationship.

Uncertainty

A measure of knowledge incompleteness due to inherent deficiencies in acquired knowledge. Also, a characterization of the degree to which the state of a system is unsettled or in doubt, such as the uncertainty of the outcome. In a quantified risk assessment, uncertainty is a representation of the confidence in the state of knowledge about the models and parameter values used. The two general categories of uncertainty are epistemic and aleatory. Epistemic uncertainties are associated with a lack of knowledge; while aleatory uncertainties are attributable to randomness.

Variability

The heterogeneity of values within a population; the inherent randomness of natural or social systems (aleatory uncertainty).